MAKINO

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F-SERIES Professional 6

MAINTENANCE DRAWING

MAKINO

VERTICAL MACHINING CENTER

F-SERIES

Professional 6

MAINTENANCE DRAWING
(ELECTRICAL)

F3 F5



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V2631I-C1E-YB

F3-PRO 6 / F5-PRO 6

SCHEMATIC & LAYOUT DIAGRAM

DOCUMENT No. : V2631I-C1E-Y

VERSION : B

MAKINO

MAKINO ASIA PTE LTD

DRAWING No. V2631I-C1E-Y

HISTORY OF MODIFICATION

VERSION No.	SEQUENCE	DATE	DRAWING INFORMATION No.	CHANGE DRAWING No.	CONTENT	RELATED DRAWING No.	PREEPARED BY
Ver.1.00	———	07/03/2018	V2631I-C1E-YA	———	FIRST RELEASE	———	ANISHA
Ver.2.00	———	28/06/2018	V2631I-C1E-YB	———	SECOND RELEASE (MTC CHANGE)	———	ANISHA

REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH			HISTORY OF MODIFICATION	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH					PAGE	A002
	----.--.--	DRAWN	ANISHA						
	----.--.--	DATE	6/28/2018						

THE MEANING OF THE DRAWING NUMBER FOR MAINTENANCE DRAWING IS AS FOLLOWS.

V26 31I- C 1 E- Y B

(1) (2) (3) (4) (5) (6) (7)

(1)MODEL NUMBER

V26: F3/F5 PRO6

(2)CONTROLLER TYPE

31I: 31I-MB

(3)DRAWING TYPE

C: CIRCUIT DRAWING

(4)CIRCUIT TYPE

1: STD M/C, 9:CUSTOM CIRCUIT

(5)LANGUAGE

E: ENGLISH, C:CHINESE

(6)MACHINE SPEC

X: STD SPEC, Y:CE SPEC

(7)VERSION

PAGE GROUP REFERENCE

SHEET	DESCRIPTION
A	COVER, MODIFICATION HISTORY, LEGEND
B	TABLE OF CONTENTS
C	SCHEMATIC DRAWING
D	DI/DO ADDRESS TABLE
E	ASSY DRAWINGS

REVISION			
	----.--.--	APPROVED	AVINASH
	----.--.--	CHECKED	AVINASH
	----.--.--	DRAWN	ANISHA
	----.--.--	DATE	28/6/2018

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TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
LEGEND	V2631I-C1E-Y1	Ver.	B
		PAGE	A003

PAGE NUMBER	DESCRIPTION	VERSION	CHANGED BY	DATE
A001	COVER PAGE	B	ANISHA	28/6/2018
A002	HISTORY OF MODIFICATION	B	ANISHA	28/6/2018
A003	LEGEND	B	ANISHA	28/6/2018
B001	TABLE OF CONTENT1/9	B	ANISHA	28/6/2018
B002	TABLE OF CONTENT2/9	B	ANISHA	28/6/2018
B003	TABLE OF CONTENT3/9	B	ANISHA	28/6/2018
B004	TABLE OF CONTENT4/9	B	ANISHA	28/6/2018
B005	TABLE OF CONTENT5/9	B	ANISHA	28/6/2018
B006	TABLE OF CONTENT6/9	B	ANISHA	28/6/2018
B007	TABLE OF CONTENT7/9	B	ANISHA	28/6/2018
B008	TABLE OF CONTENT8/9	B	ANISHA	28/6/2018
B009	TABLE OF CONTENT9/9	B	ANISHA	28/6/2018
C001	INPUT POWER SOURCE	B	ANISHA	28/6/2018
C002	TB0 DITRIBUTION	B	ANISHA	28/6/2018
C003	CP11 (R1/S1/T1)	B	ANISHA	28/6/2018
C004	CP12 (R2/S2/T2)	B	ANISHA	28/6/2018
C005	CP14,CP21 (A230H,230H)	B	ANISHA	28/6/2018
C006	CP OVERLOAD CONTROL	B	ANISHA	28/6/2018
C007	CONTROL VOLT(24VDC)	B	ANISHA	28/6/2018
C008	E-AUXILIARY CONTACT DISTRIBUTION	B	ANISHA	28/6/2018
C009	CE RELAY CONTACT	B	ANISHA	28/6/2018

REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH			TABLE OF CONTENT1/9 図面リスト	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH					PAGE	B001
	----.--.--	DRAWN	ANISHA						
	----.--.--	DATE	6/28/2018						

PAGE NUMBER	DESCRIPTION	VERSION	CHANGED BY	DATE
C010	CONTROL VOLTAGE (C24HB)	B	ANISHA	28/6/2018
C011	CONTROL VOLTAGE (E24H/E24HS/E24HM)	B	ANISHA	28/6/2018
C012	MAIN PCB POWER INPUT	B	ANISHA	28/6/2018
C013	OPTION PCB 1,2 - POWER INPUT	B	ANISHA	28/6/2018
C014	OPTION PCB 3,4 - POWER INPUT	B	ANISHA	28/6/2018
C015	POWER OUT	B	ANISHA	28/6/2018
C016	POWER SUPPLY MODULE	B	ANISHA	28/6/2018
C017	SPINDLE AMPLIFIER MODULE	B	ANISHA	28/6/2018
C018	X, Y, Z - SERVO AMPLIFIER MODULE	B	ANISHA	28/6/2018
C019	ATC - SERVO AMPLIFIER MODULE	B	ANISHA	28/6/2018
C020	4th - SERVO AMPLIFIER MODULE	B	ANISHA	28/6/2018
C021	5th - SERVO AMPLIFIER MODULE	B	ANISHA	28/6/2018
C022	FEEDBACK SCALE (SDU)	B	ANISHA	28/6/2018
C023	Z AXIS MOTOR BRAKE	B	ANISHA	28/6/2018
C024	CONTROLLER UNIT	B	ANISHA	28/6/2018
C025	I/O LINK	B	ANISHA	28/6/2018
C026	I/O LINK	B	ANISHA	28/6/2018
C027	NC POWER ON CONTROL (C24HA2)	B	ANISHA	28/6/2018
C028	EMERGENCY STOP	B	ANISHA	28/6/2018
C029	REFERENCE AND OVERTRAVEL	B	ANISHA	28/6/2018
C030	SCREW CONVEYOR	B	ANISHA	28/6/2018

PAGE NUMBER	DESCRIPTION	VERSION	CHANGED BY	DATE
C031	NOZZLE COOLANT & WW GUN (FRONT TANK)	B	ANISHA	28/6/2018
C032	NOZZLE COOLANT & WW GUN(REAR TANK)	B	ANISHA	28/6/2018
C033	BASE / FIXTURE COOLANT (OPT)(FRONT TANK)	B	ANISHA	28/6/2018
C034	BASE / FIXTURE COOLANT (OPT)(REAR TANK)	B	ANISHA	28/6/2018
C035	SHOWER COOLANT (OPT)	B	ANISHA	28/6/2018
C036	TSC 1.5MPa & 3MPa PWR (OPT)	B	ANISHA	28/6/2018
C037	TSC 1.5MPa & 3MPa CONTROL (OPT)	B	ANISHA	28/6/2018
C038	TSC 1.5MPa CONTROL (OPT)	B	ANISHA	28/6/2018
C039	TSC 3.0MPa CONTROL (OPT)	B	ANISHA	28/6/2018
C040	CYCLONE FILTER PRESSURE SW (OPT)	B	ANISHA	28/6/2018
C041	SECONDARY TRANSFER PUMP (OPT)	B	ANISHA	28/6/2018
C042	LUCC DRUM (OPT)	B	ANISHA	28/6/2018
C043	LIFT UP CHIP CYR PWR (OPT)	B	ANISHA	28/6/2018
C044	LIFT UP CHIP CYR CONTROL (OPT)	B	ANISHA	28/6/2018
C045	OIL SKIMMER (OPT)	B	ANISHA	28/6/2018
C046	OIL MIST (OPT)	B	ANISHA	28/6/2018
C047	MIST COLLECTOR (OPT)	B	ANISHA	28/6/2018
C048	COOLANT LEVEL SENSOR - MAIN TANK	B	ANISHA	28/6/2018
C049	COOLANT LEVEL SENSOR - SEC TANK (OPT)	B	ANISHA	28/6/2018
C050	AIR BLOW & TSA (OPT)	B	ANISHA	28/6/2018
C051	MASTER SOLENOID	B	ANISHA	28/6/2018

REVISION							
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE: F3/ F5 Pro 6
	----.--.--	CHECKED	AVINASH		TABLE OF CONTENT3/9 図面リスト	V2631I-C1E-Y	Ver.: B
	----.--.--	DRAWN	ANISHA				PAGE: B003
	----.--.--	DATE	28/6/2018				

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C052	ATC MASTER SOLENOID	B	ANISHA	28/6/2018
C053	POWER PACK (TOOL CL/UCL)	B	ANISHA	28/6/2018
C054	COOLANT CHILLER (OPT)	B	ANISHA	28/6/2018
C055	SPINDLE SWITCHING CONTROL	B	ANISHA	28/6/2018
C056	SPINDLE TEMPERATURE CONTROLLER	B	ANISHA	28/6/2018
C057	TOOL CLAMP/UNCLAMP SOL	B	ANISHA	28/6/2018
C058	TOOL CLAMP/UNCLAMP LIMIT SW	B	ANISHA	28/6/2018
C059	ATC ARM MOTOR	B	ANISHA	28/6/2018
C060	ATC TOOL EXTRACTOR	B	ANISHA	28/6/2018
C061	ATC ARM ENCODER	B	ANISHA	28/6/2018
C062	ATC POT CONTROL	B	ANISHA	28/6/2018
C063	ATC SHUTTER	B	ANISHA	28/6/2018
C064	ATC PANEL	B	ANISHA	28/6/2018
C065	AWM & ATLM INTERFACE (OPT)	B	ANISHA	28/6/2018
C066	AWM & ATLM INTERFACE (OPT)	B	ANISHA	28/6/2018
C067	ATLM SOLENOIDS (OPT)	B	ANISHA	28/6/2018
C068	AWM MARPOSS (OPT)	B	ANISHA	28/6/2018
C069	AWM RENISHAW (OPT)	B	ANISHA	28/6/2018
C070	AWM BLUM (OPT)	B	ANISHA	28/6/2018
C071	ATLM TP MARPOSS (OPT)	B	ANISHA	28/6/2018
C072	ATLM TP RENISHAW (OPT)	B	ANISHA	28/6/2018

PAGE NUMBER	DESCRIPTION	VERSION	CHANGED BY	DATE
C073	ATLM TP BLUM (OPT)	B	ANISHA	28/6/2018
C074	ATLM LASER RENISHAW (OPT)	B	ANISHA	28/6/2018
C075	ATLM LASER BLUM (OPT)	B	ANISHA	28/6/2018
C076	ATLM INTERFACE BOX (OPT)	B	ANISHA	28/6/2018
C077	USER M-CODE (OPT)	B	ANISHA	28/6/2018
C078	S/G DOOR LOCK W INTERLOCK	B	ANISHA	28/6/2018
C079	ATC DOOR LOCK (STD) W INTERLOCK (OPT)	B	ANISHA	28/6/2018
C080	LIGHT CURTAIN (OPT)	B	ANISHA	28/6/2018
C081	4TH AXIS CONTROL (PNEUMATIC)	B	ANISHA	28/6/2018
C082	4TH AXIS CONTROL (HYDRAULIC)	B	ANISHA	28/6/2018
C083	5TH AXIS CONTROL (PNEUMATIC)	B	ANISHA	28/6/2018
C084	5TH AXIS CONTROL (HYDRAULIC)	B	ANISHA	28/6/2018
C085	RWH POWER PACK (OPT)	B	ANISHA	28/6/2018
C086	TABLE CHUCK CL/UCL (OPT)	B	ANISHA	28/6/2018
C087	CONTROL VOLTAGE DISTRIBUTION OPE	B	ANISHA	28/6/2018
C088	MACHINE LIGHT	B	ANISHA	28/6/2018
C089	AUTO GREASE	B	ANISHA	28/6/2018
C090	MTC FAN	B	ANISHA	28/6/2018
C091	MTC AIR CONDITIONER (OPT)	B	ANISHA	28/6/2018
C092	AIR DRYER (OPT)	B	ANISHA	28/6/2018
C093	TWO HAND PUSH BUTTON (OPT)	B	ANISHA	28/6/2018

PAGE NUMBER	DESCRIPTION	VERSION	CHANGED BY	DATE
C094	STRIKER SWITCH (OPT)	B	ANISHA	28/6/2018
C095	AIR SEAT CHECK (1-8 SENSOR) (OPT)	B	ANISHA	28/6/2018
C096	AIR SEAT CHECK (SOL & CHECK SW) (OPT)	B	ANISHA	28/6/2018
C097	HYDRAULIC FIXTURE (OPT)	B	ANISHA	28/6/2018
C098	HYDRAULIC FIXTURE SOL/PS-1 (OPT)	B	ANISHA	28/6/2018
C099	HYDRAULIC PANEL (OPT)	B	ANISHA	28/6/2018
C100	ROBOT INTERFACE-1 (OPT)	B	ANISHA	28/6/2018
C101	ROBOT INTERFACE-2 (OPT)	B	ANISHA	28/6/2018
C102	ROBOT INTERFACE-3 (OPT)	B	ANISHA	28/6/2018
C103	EXTERNAL HYDRAULIC PUMP I/F(OPT)	B	ANISHA	28/6/2018
C104	OPE PCB - POWER INPUT	B	ANISHA	28/6/2018
C105	24V FOR OPE PCB & IO MODULE	B	ANISHA	28/6/2018
C106	LOADER ACCESS DOOR	B	ANISHA	28/6/2018
C107	LOADER ACCESS DOOR-RH SIDE	B	ANISHA	28/6/2018
C108	OPERATOR PANEL EMG CIRCUIT	B	ANISHA	28/6/2018
C109	OPERATOR PANEL CONTROL CIRCUIT	B	ANISHA	28/6/2018
C110	OPERATOR PANEL CONTROL CIRCUIT	B	ANISHA	28/6/2018
C111	OPERATOR PANEL CONTROL CIRCUIT	B	ANISHA	28/6/2018
C112	OPERATOR PANEL CONTROL CIRCUIT	B	ANISHA	28/6/2018
C113	OPERATOR PANEL CONTROL CIRCUIT	B	ANISHA	28/6/2018
C114	OPERATOR PANEL CONTROL CIRCUIT	B	ANISHA	28/6/2018

REVISION					MAKINO	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH		MAKINO ASIA PTE LTD	TABLE OF CONTENT6/9 図面リスト	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH					PAGE	B006
	----.--.--	DRAWN	ANISHA						
	----.--.--	DATE	28/6/2018						

PAGE NUMBER	DESCRIPTION	VERSION	CHANGED BY	DATE
C115	OPERATOR PANEL CONTROL CIRCUIT	B	ANISHA	28/6/2018
C116	OPERATOR PANEL CONTROL CIRCUIT	B	ANISHA	28/6/2018
C117	MANUAL PULSE GENERATOR (PORTABLE)	B	ANISHA	28/6/2018
C118	SIGNAL / PATROL LIGHT	B	ANISHA	28/6/2018
C119	DCS ZBRK CONNECTION	B	ANISHA	28/6/2018
C120	DCS SMC CONNECTION	B	ANISHA	28/6/2018
C121	DCS SG/ATC CONNECTION	B	ANISHA	28/6/2018
C122	DCS EMG CONNECTION	B	ANISHA	28/6/2018
C123	AUXILLARY COMPUTER SYSTEM(SBC)(OPT)	B	ANISHA	28/6/2018
C124	VISION BTS (OPT)	B	ANISHA	28/6/2018
C125	SPINDLE ACTIVE CARE (OPT)	B	ANISHA	28/6/2018
C126	ACU & ETHERNET ROUTER (OPTION)	B	ANISHA	28/6/2018
C127	CONTROL VOLTAGE DISTRIBUTION OPE (C24HB)	B	ANISHA	28/6/2018
C128	I/O CONFIGURATION	B	ANISHA	28/6/2018
D001	DI/DO ADDRESS TABLE - IO MODULE 1	B	ANISHA	28/6/2018
D002	DI/DO ADDRESS TABLE - IO MODULE 1	B	ANISHA	28/6/2018
D003	DI/DO ADDRESS TABLE - IO MODULE 2	B	ANISHA	28/6/2018
D004	DI/DO ADDRESS TABLE - IO MODULE 2	B	ANISHA	28/6/2018
D005	DI/DO ADDRESS TABLE - IO MODULE 3	B	ANISHA	28/6/2018
D006	DI/DO ADDRESS TABLE - IO MODULE 3	B	ANISHA	28/6/2018
D007	DI/DO ADDRESS TABLE - IO MODULE 4	B	ANISHA	28/6/2018

REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

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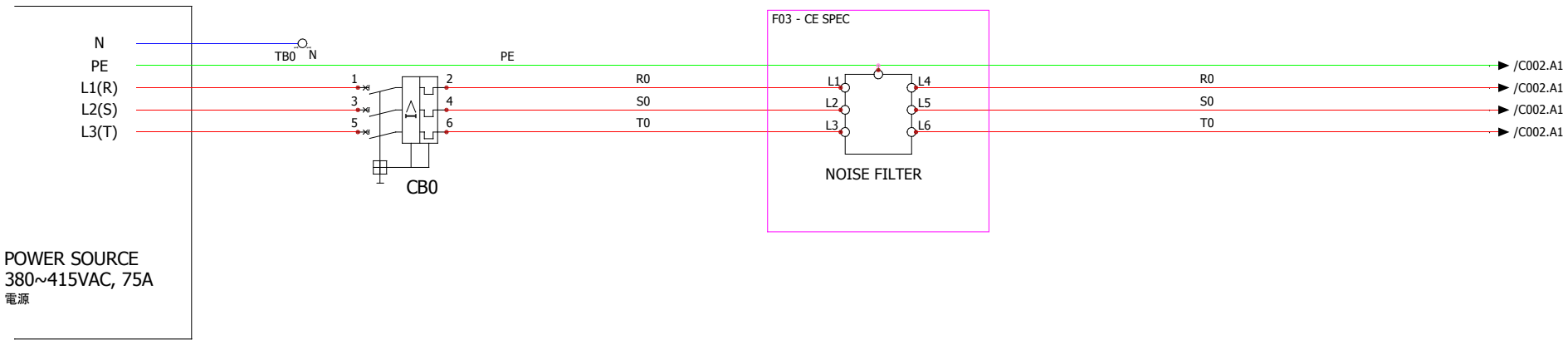
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TABLE OF CONTENT7/9 図面リスト	V2631I-C1E-Y	Ver.	B
		PAGE	B007

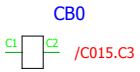
PAGE NUMBER	DESCRIPTION	VERSION	CHANGED BY	DATE
D008	DI/DO ADDRESS TABLE - IO MODULE 4	B	ANISHA	28/6/2018
D009	DI/DO ADDRESS TABLE - IO MODULE 5	B	ANISHA	28/6/2018
D010	DI/DO ADDRESS TABLE - IO MODULE 5	B	ANISHA	28/6/2018
D011	DI/DO ADDRESS TABLE - IO MODULE 6	B	ANISHA	28/6/2018
D012	DI/DO ADDRESS TABLE - IO MODULE 6	B	ANISHA	28/6/2018
D013	DI/DO ADDRESS TABLE - IO MODULE 7	B	ANISHA	28/6/2018
D014	DI/DO ADDRESS TABLE - IO MODULE 7	B	ANISHA	28/6/2018
D015	DI/DO ADDRESS TABLE - IO MODULE 9	B	ANISHA	28/6/2018
D016	DI/DO ADDRESS TABLE - IO MODULE 9	B	ANISHA	28/6/2018
D017	DI/DO ADDRESS TABLE - IO MODULE 11	B	ANISHA	28/6/2018
D018	DI/DO ADDRESS TABLE - IO MODULE 11	B	ANISHA	28/6/2018
D019	DI/DO ADDRESS TABLE - IO MODULE 11	B	ANISHA	28/6/2018
D020	DI/DO ADDRESS TABLE - IO MODULE 11	B	ANISHA	28/6/2018
E001	MTC LAYOUT DETAILS	B	ANISHA	28/6/2018
E002	MTC LAYOUT DETAILS	B	ANISHA	28/6/2018
E003	MTC COMPONENT DETAILS	B	ANISHA	28/6/2018
E004	MTC COMPONENT DETAILS	B	ANISHA	28/6/2018
E005	TB10/TB20/TB30 LAYOUT	B	ANISHA	28/6/2018
E006	CONNECTOR PIN DETAILS	B	ANISHA	28/6/2018
E007	CONNECTION PIN DETAILS	B	ANISHA	28/6/2018
E008	TB40 SPINDLE TERMINAL BOX	B	ANISHA	28/6/2018

REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH			TABLE OF CONTENT8/9 図面リスト	V2631I-C1E-Y	Ver.	B
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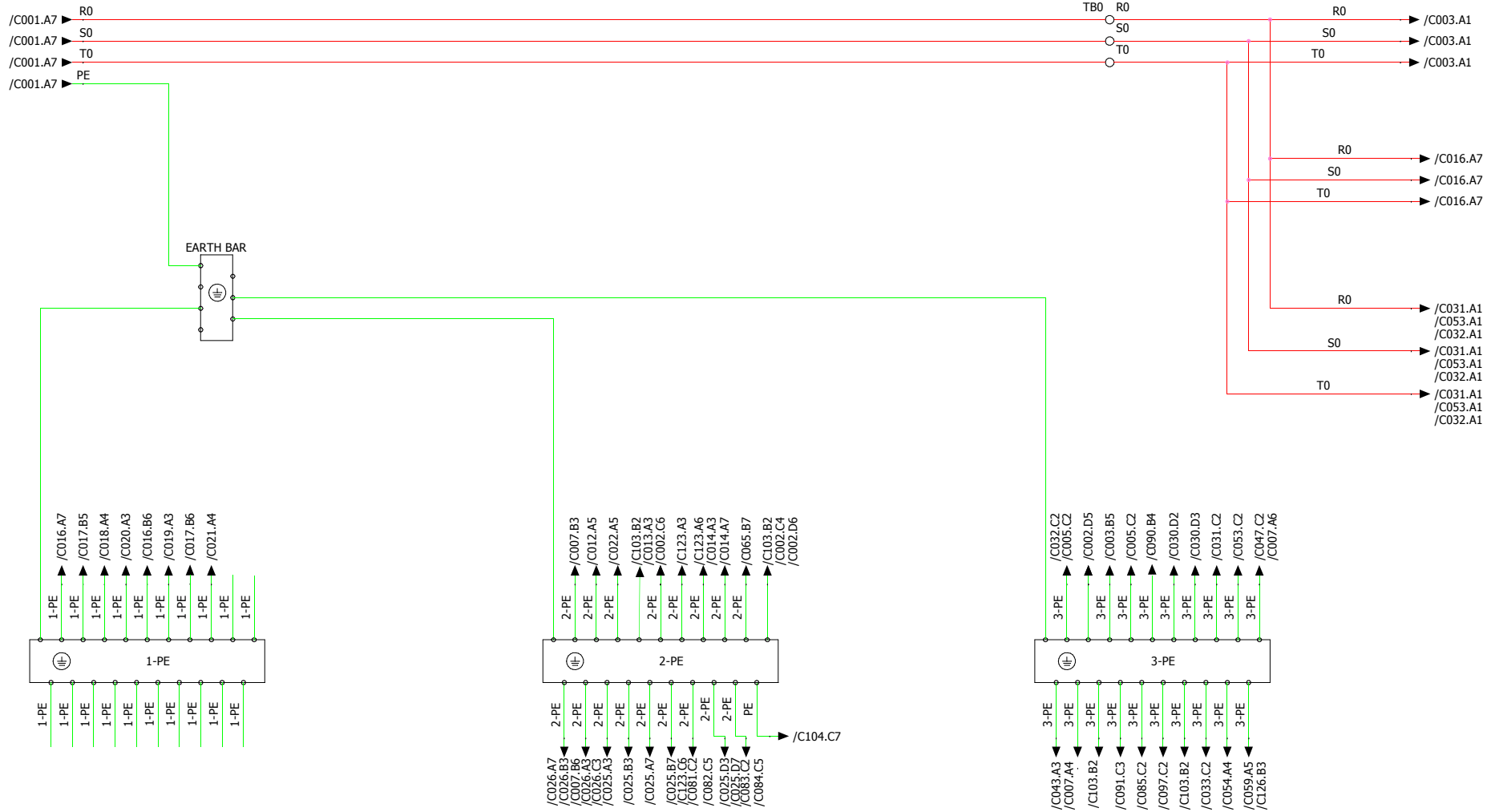
PAGE NUMBER	DESCRIPTION	VERSION	CHANGED BY	DATE
E009	TB50 COOLANT TERMINAL BOX	B	ANISHA	28/6/2018
E010	M12 JUNCTION BOX 1	B	ANISHA	28/6/2018
E011	M12 JUNCTION BOX 2	B	ANISHA	28/6/2018
E012	TB70 ATC TERMINAL BOX	B	ANISHA	28/6/2018
E013	RELAY PCB LAYOUT	B	ANISHA	28/6/2018
E014	RELAY PCB PARTS LIST	B	ANISHA	28/6/2018
E015	OPTION PCB LAYOUT & PARTS DETAILS	B	ANISHA	28/6/2018
E016	TB90 SEC TANK TERMINAL BOX	B	ANISHA	28/6/2018
E017	SWING PANEL TB100/TB110 LAYOUT	B	ANISHA	28/6/2018
E018	OPERATOR PANEL A LAYOUT	B	ANISHA	28/6/2018
E019	OPERATOR PANEL A PARTS LIST	B	ANISHA	28/6/2018
E020	OPERATOR PANEL FRONT B LAYOUT	B	ANISHA	28/6/2018
E021	OPERATOR PANEL B REAR LAYOUT	B	ANISHA	28/6/2018
E022	OP PANEL REALYS & TB-160	B	ANISHA	28/6/2018
E023	OPERATOR PANEL B PARTS LIST	B	ANISHA	28/6/2018
E024	OPERATOR PANEL B PARTS LIST	B	ANISHA	28/6/2018

REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
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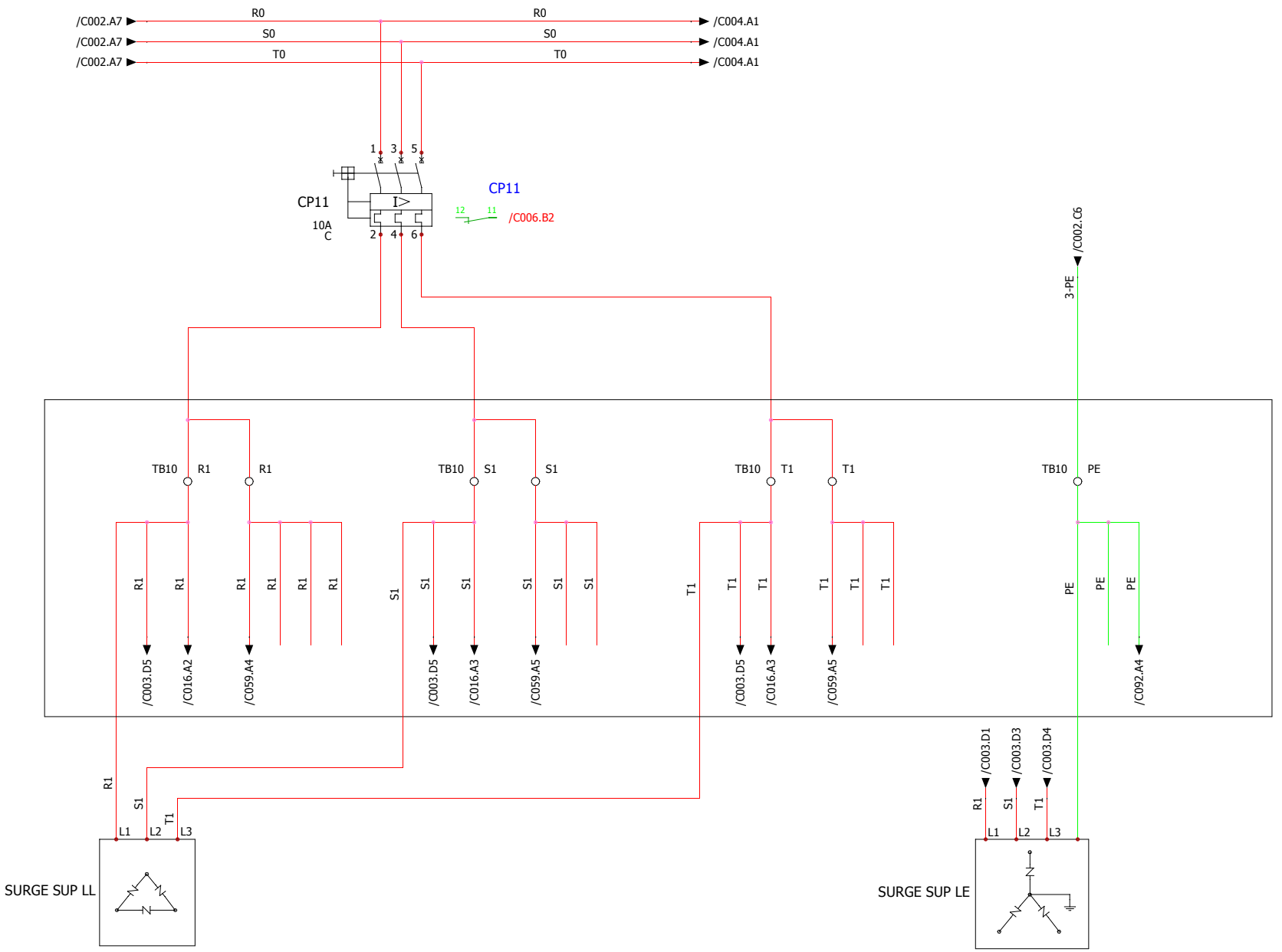


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
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		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	6/28/2018					



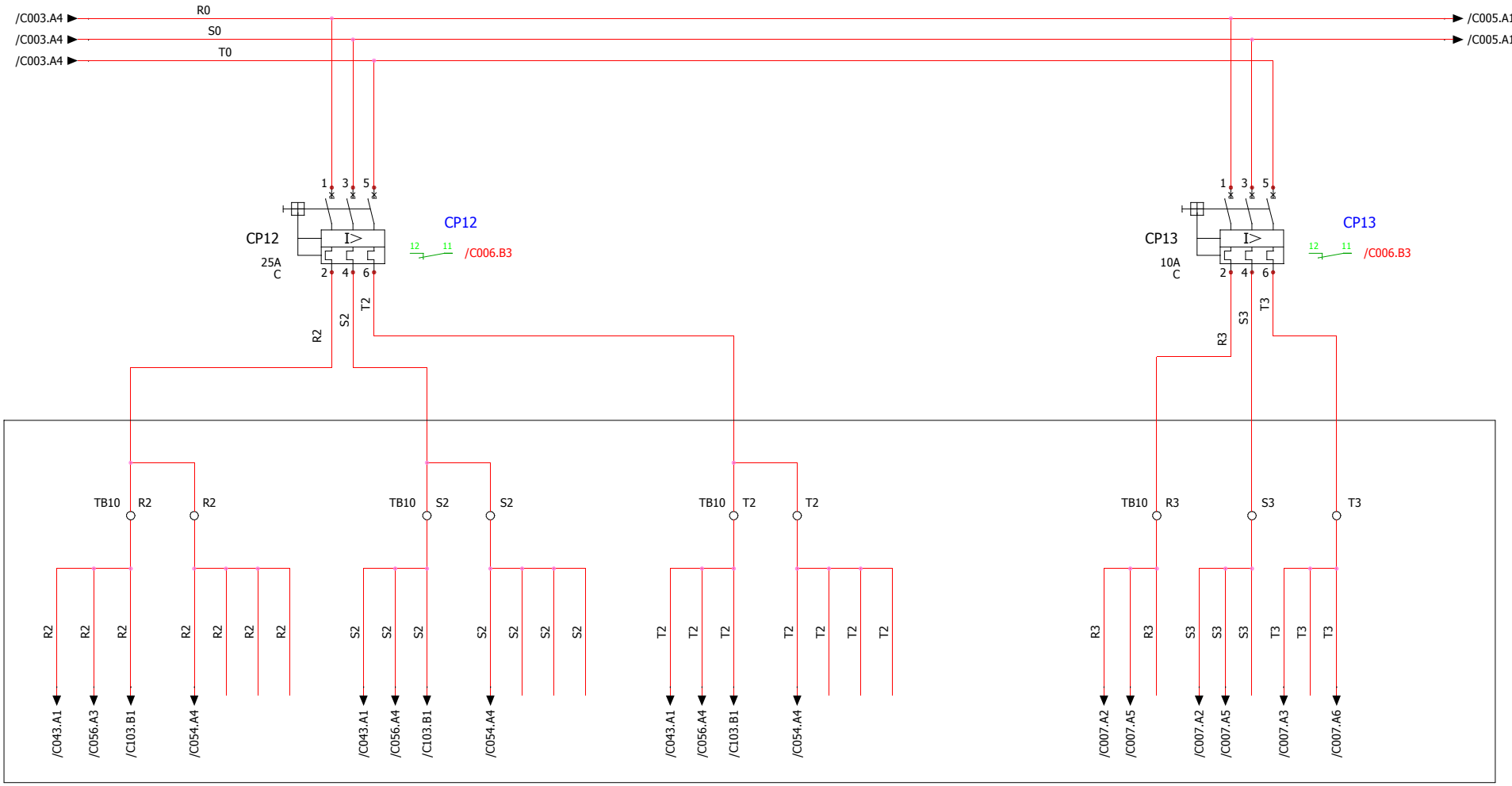
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	TB0 DITRIBUTION	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C002
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

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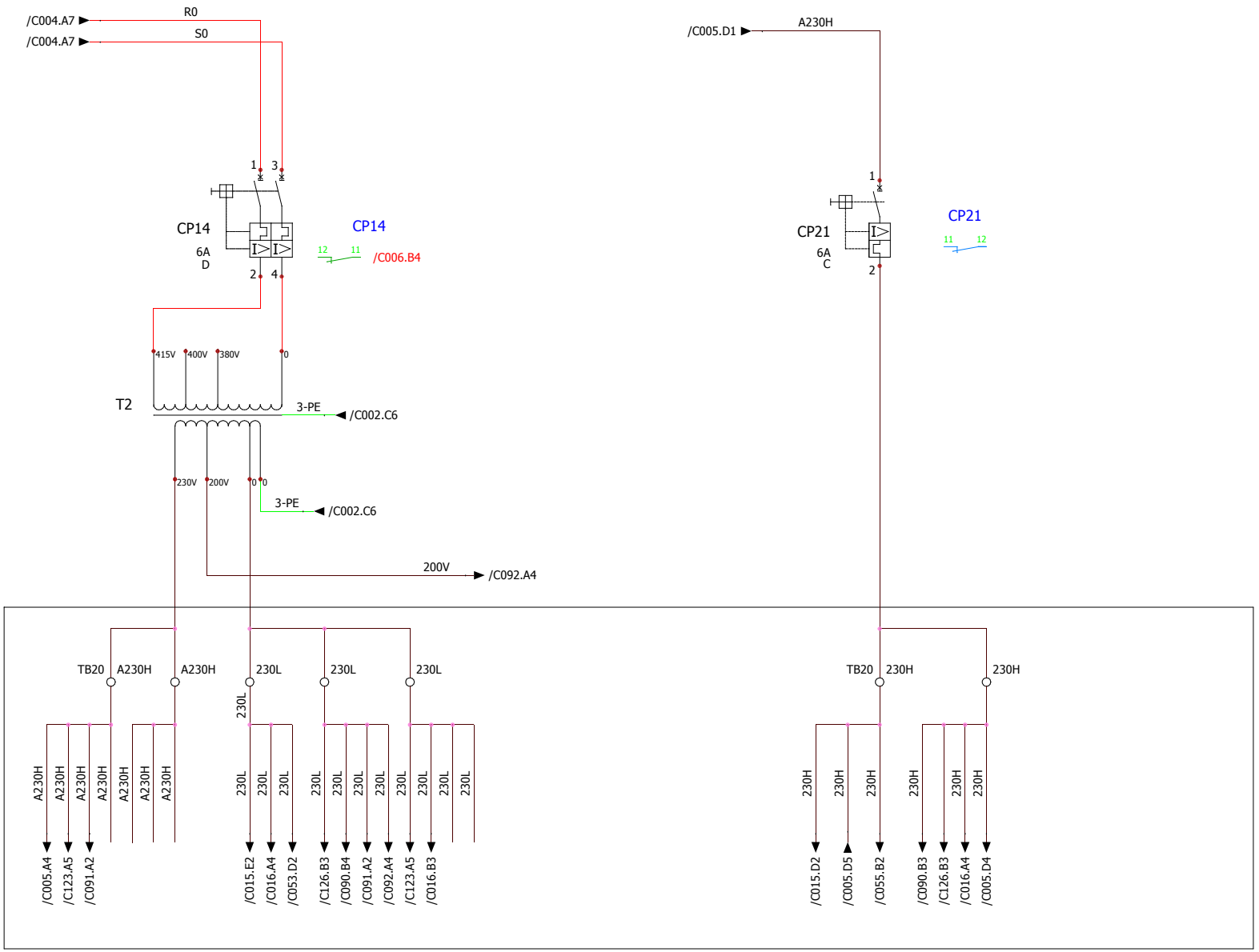


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	CP11 (R1/S1/T1)	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C003
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	6/28/2018				

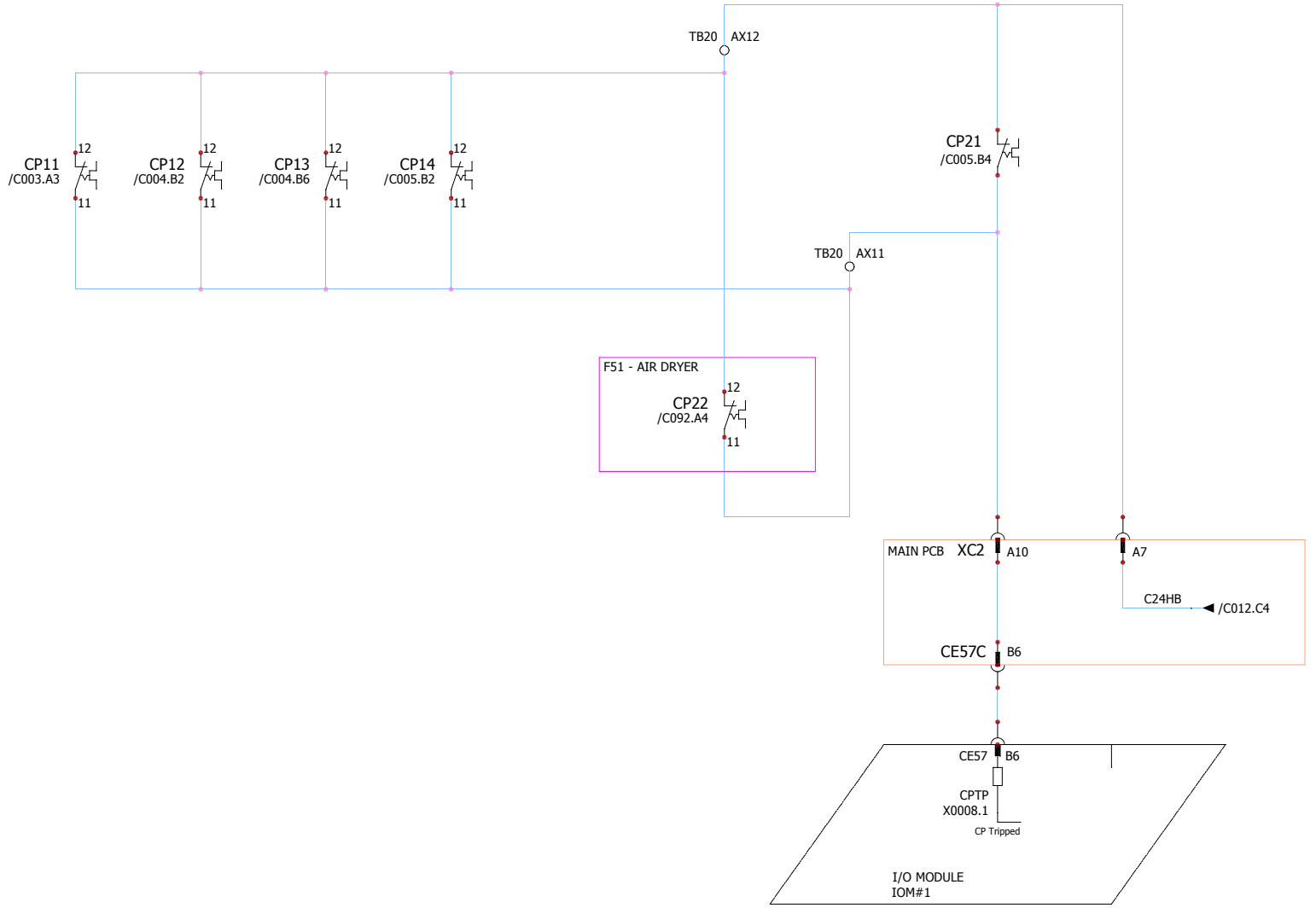
MAKINO
MAKINO ASIA PTE LTD



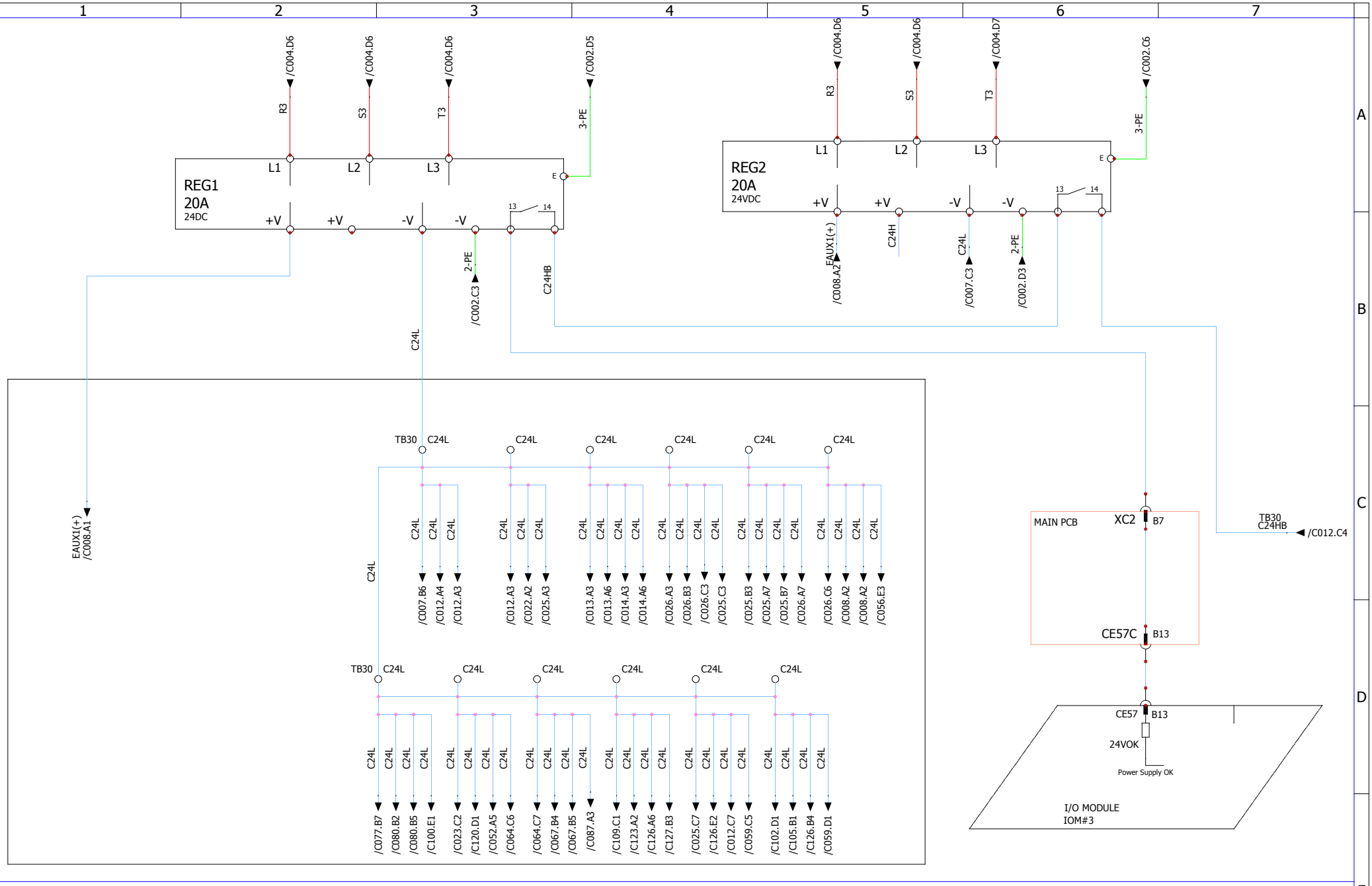
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	CP12 (R2/S2/T2)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C004
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



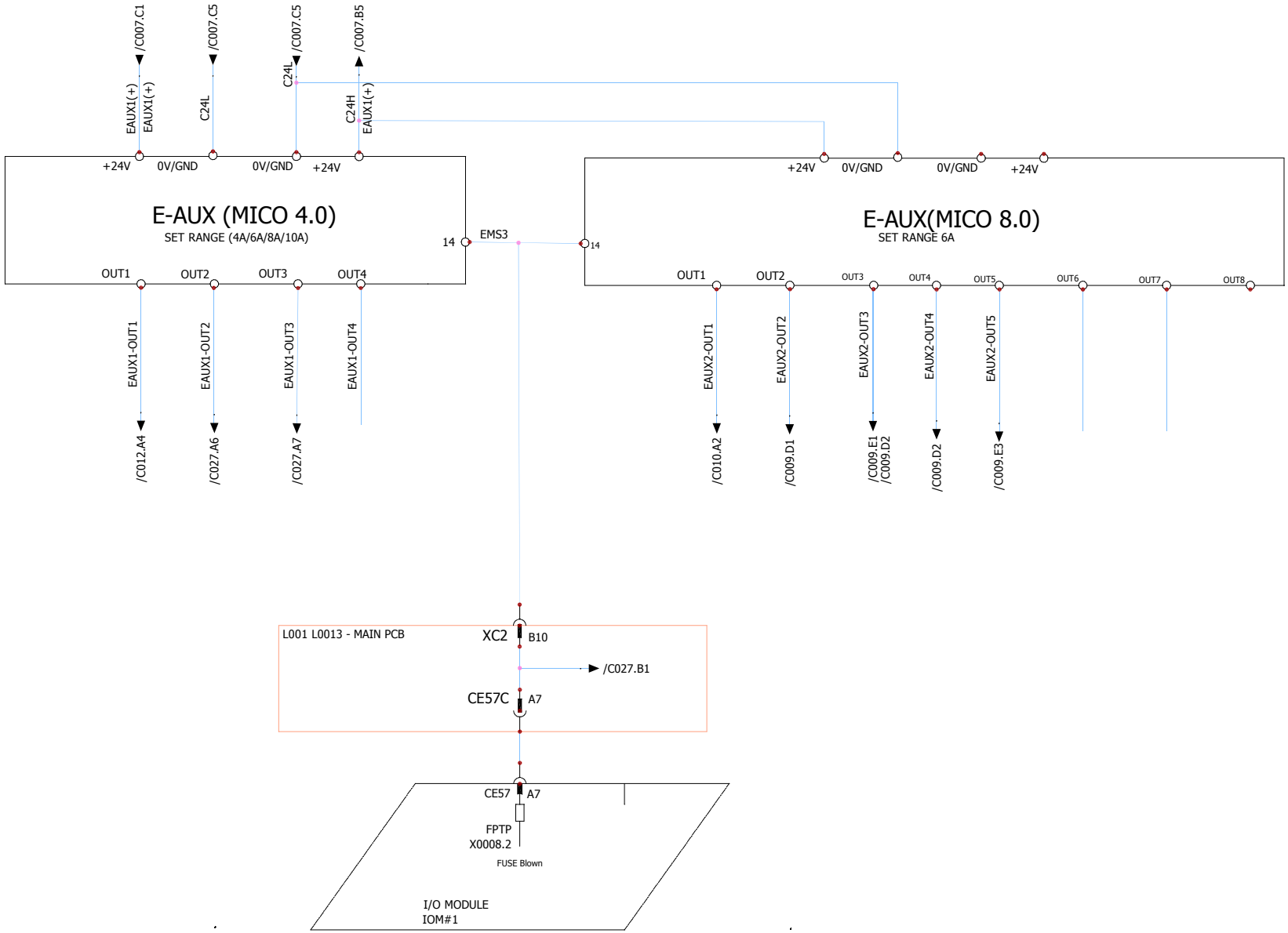
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	CP14,CP21 (A230H,230H)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C005
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	CP OVERLOAD CONTROL	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C006
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

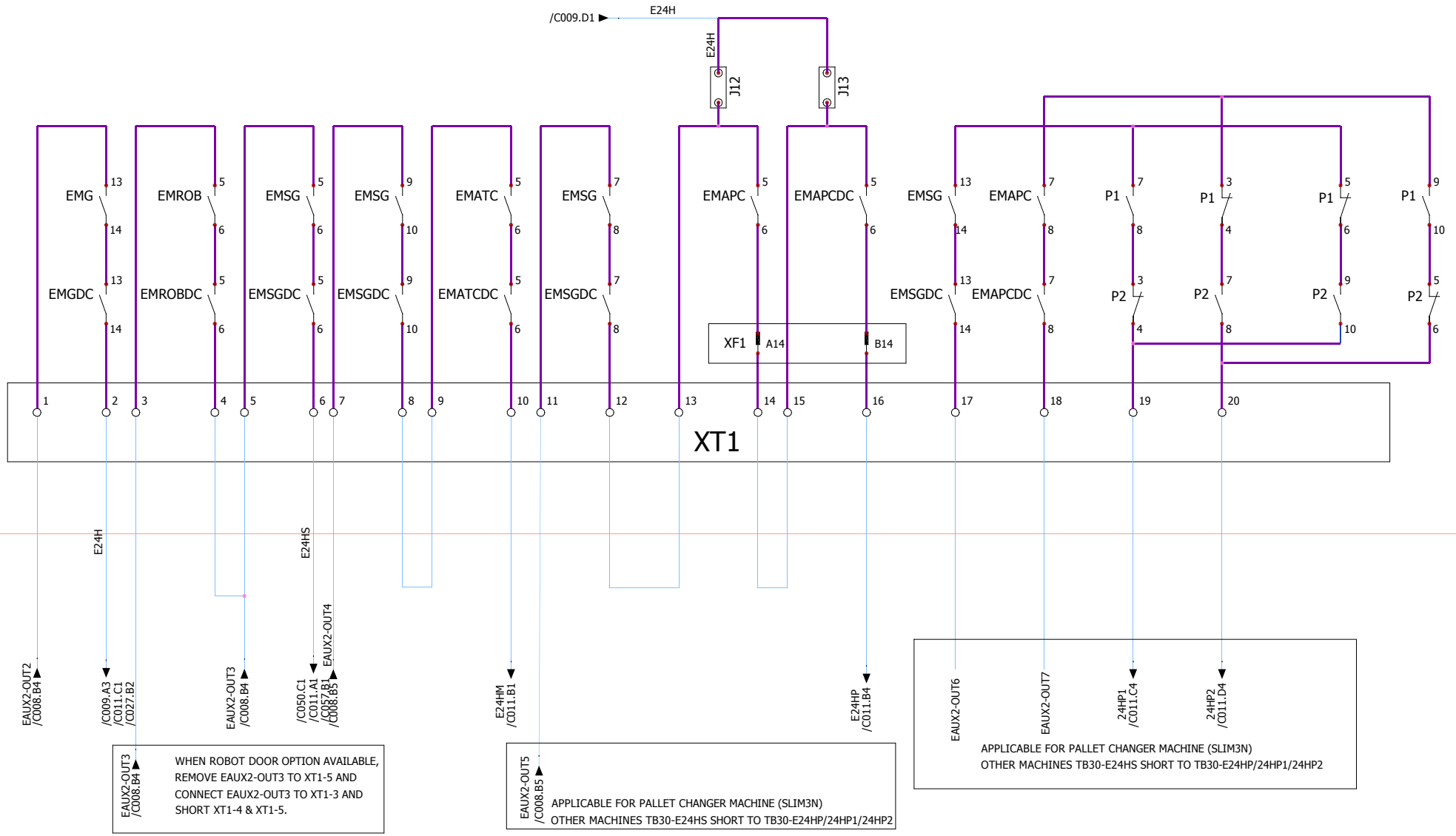


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	CONTROL VOLT(24VDC)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C007
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

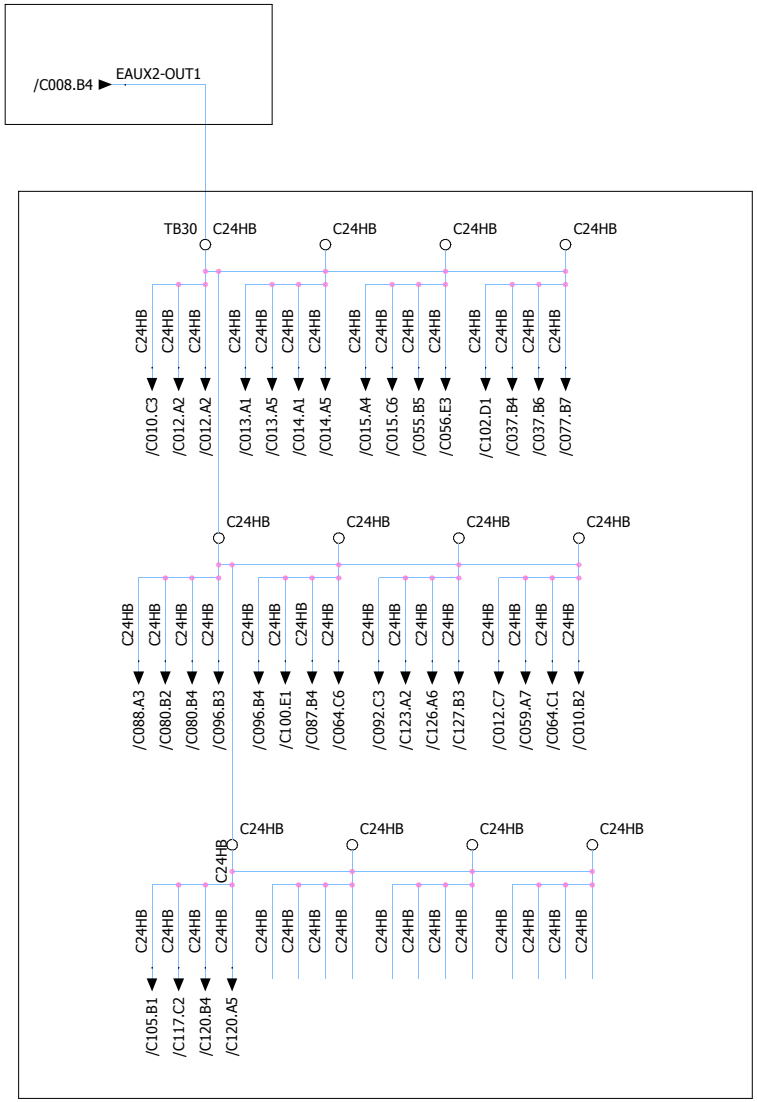


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	E-AUXILIARY CONTACT DISTRIBUTION	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C008
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	6/28/2018					

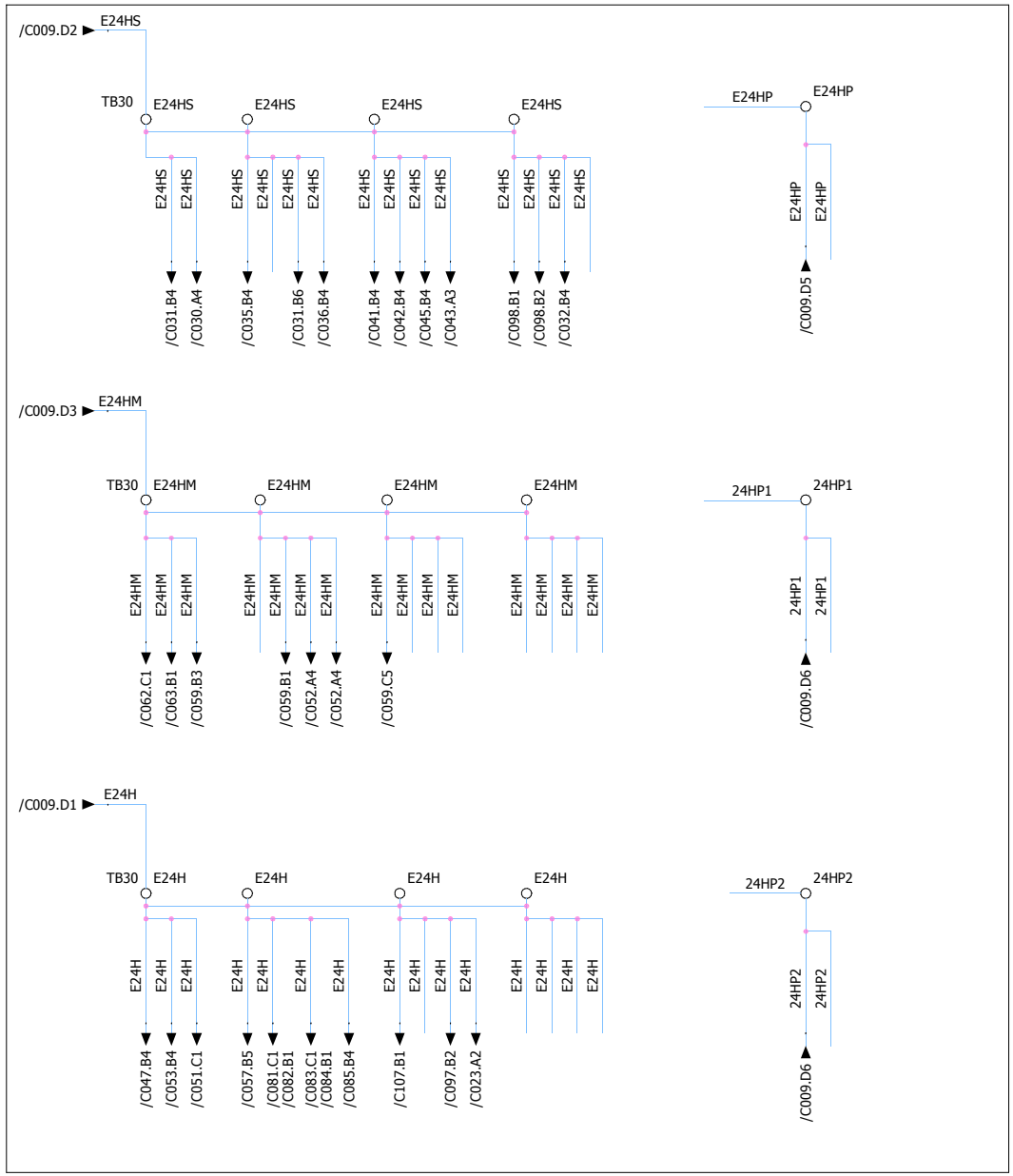
L001 L0013 - MAIN PCB



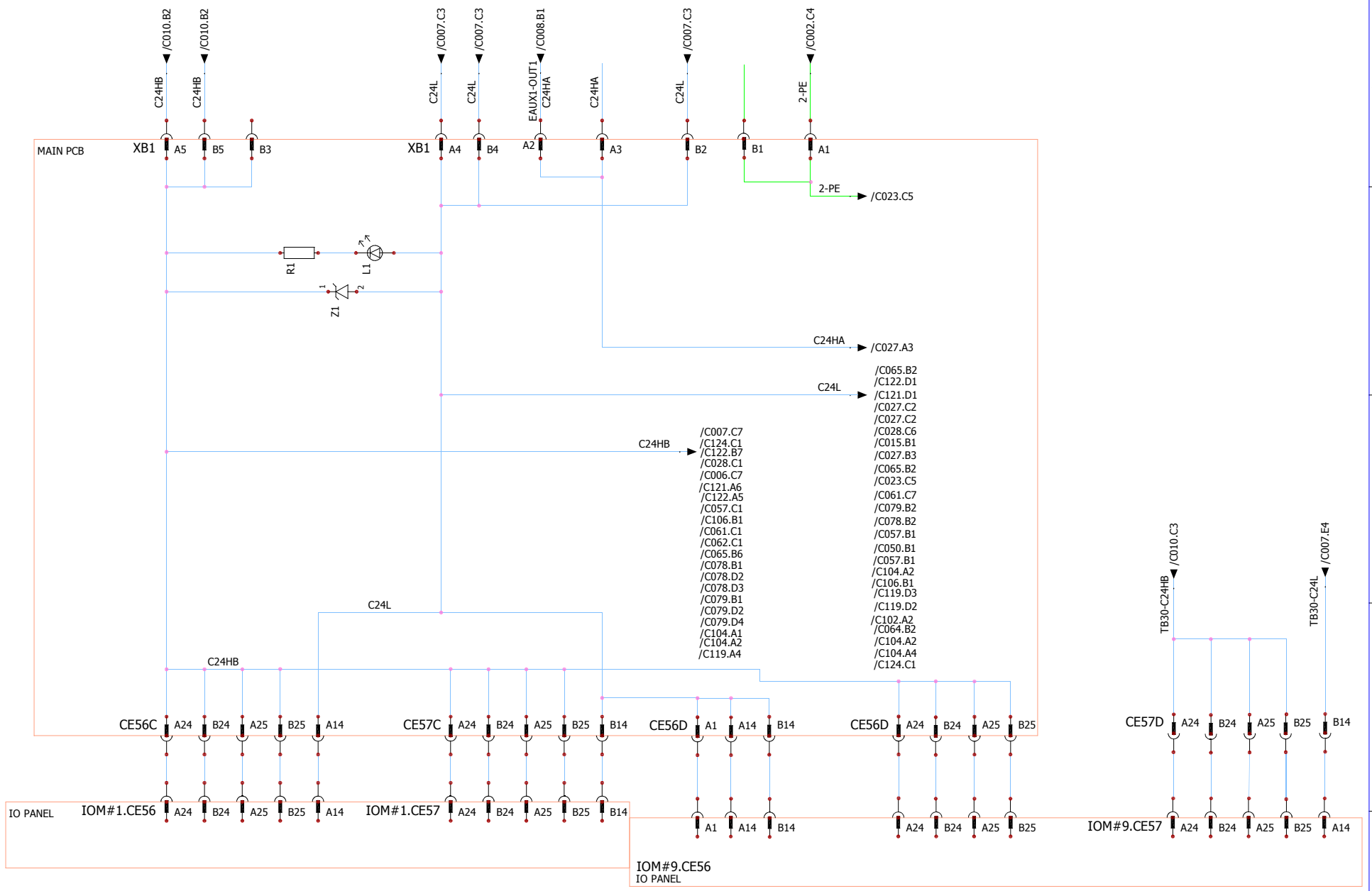
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	CE RELAY CONTACT	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C009
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



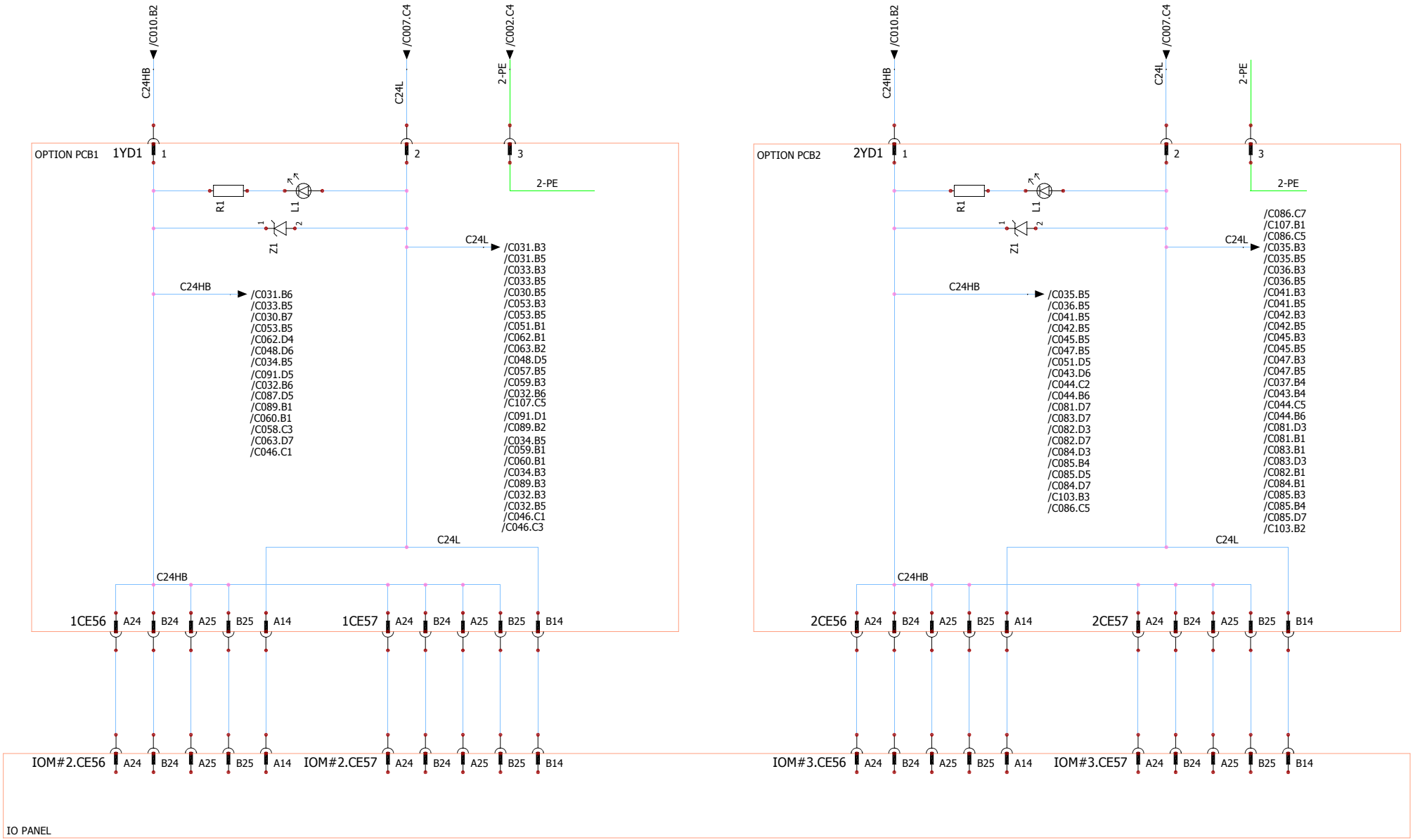
REVISION							
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE: F3/ F5 Pro 6
	----.--.--	CHECKED	AVINASH		CONTROL VOLTAGE (C24HB)	V2631I-C1E-Y	Ver.: B
	----.--.--	DRAWN	ANISHA				PAGE: C010
	----.--.--	DATE	6/28/2018				



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	CONTROL VOLTAGE (E24H/E24HS/E24HM)	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	C011
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					



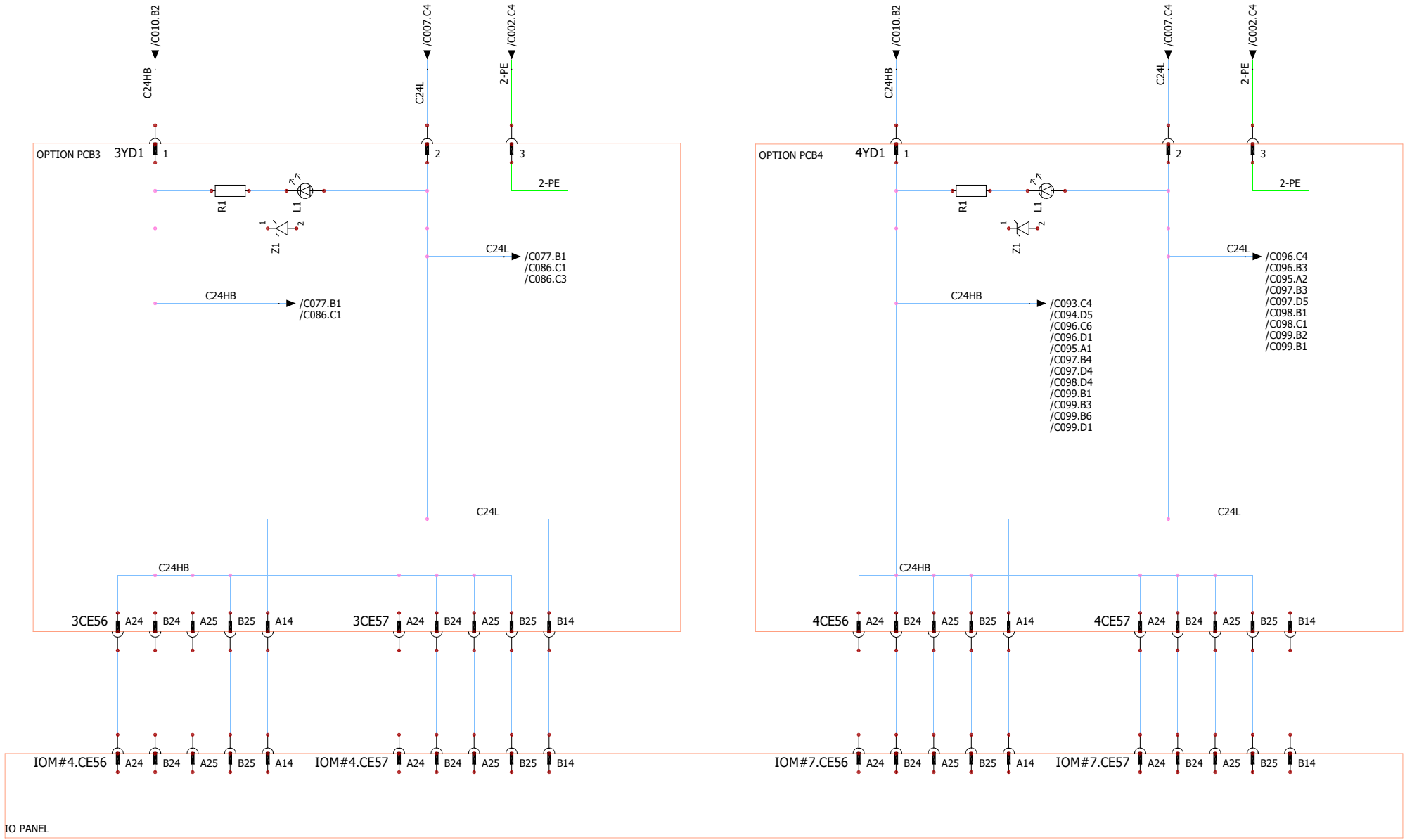
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	MAIN PCB POWER INPUT	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C012
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



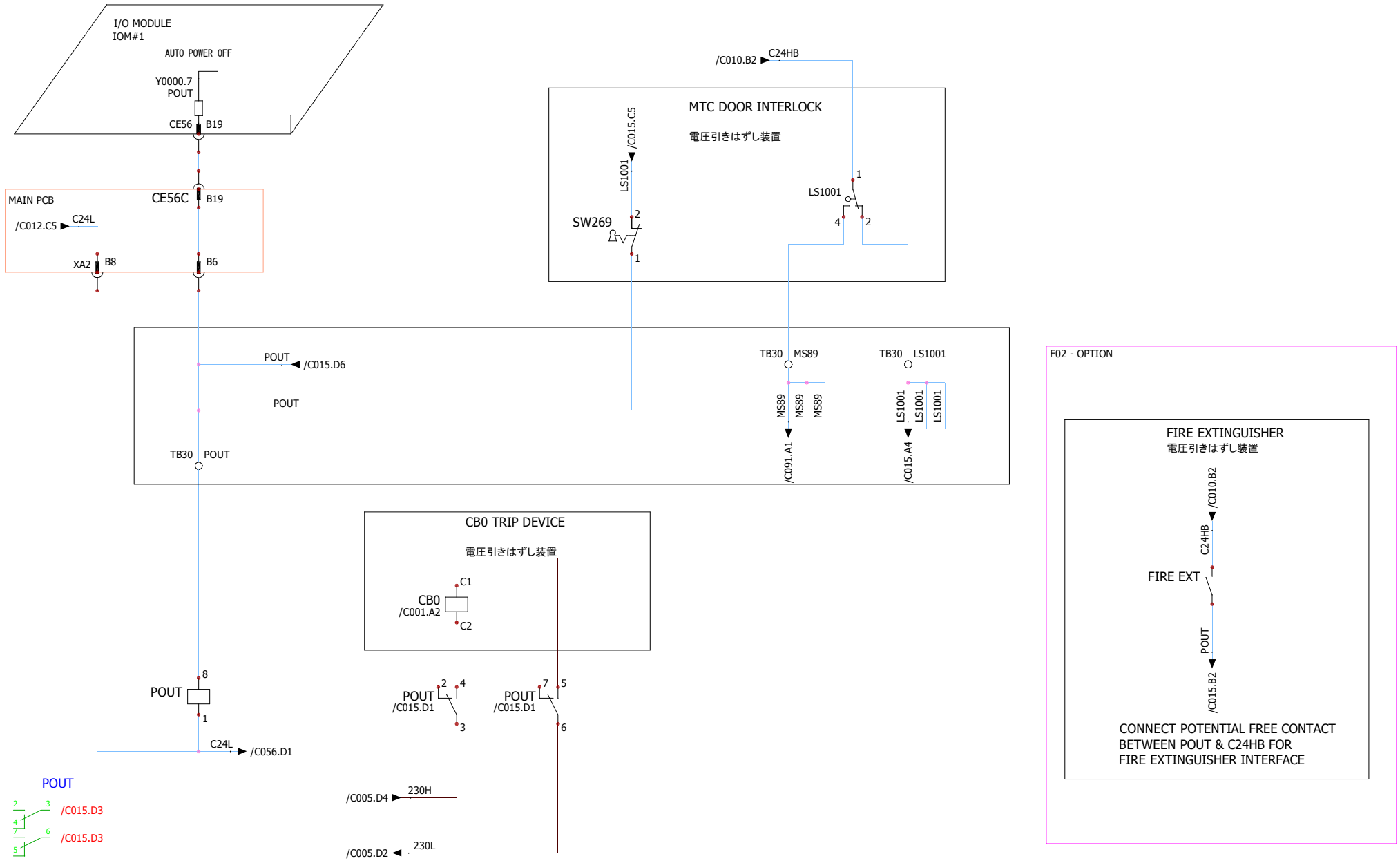
REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	6/28/2018	

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MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
OPTION PCB 1,2 - POWER INPUT	V2631I-C1E-Y	Ver.	B
		PAGE	C013



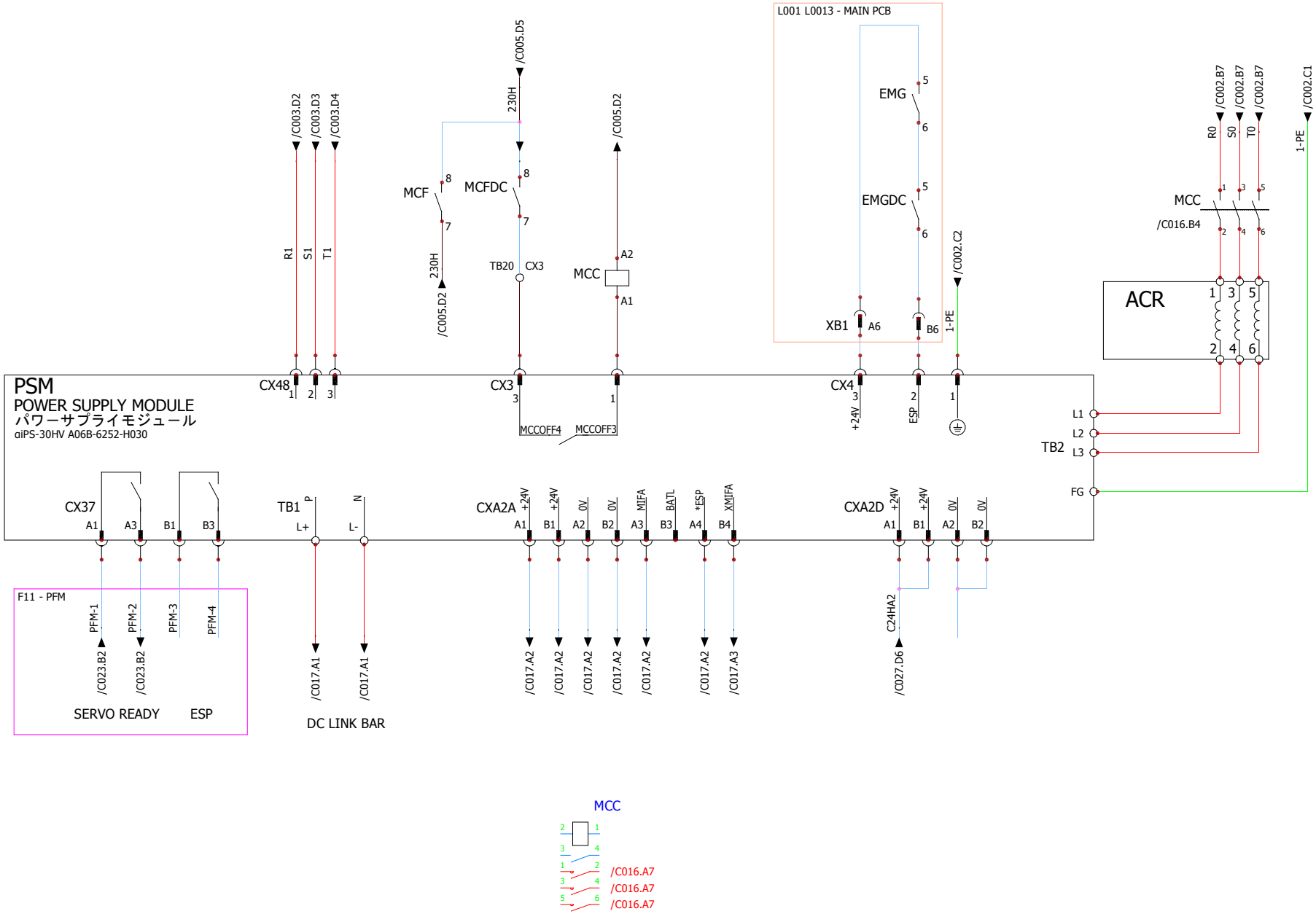
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	OPTION PCB 3,4 - POWER INPUT	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C014
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

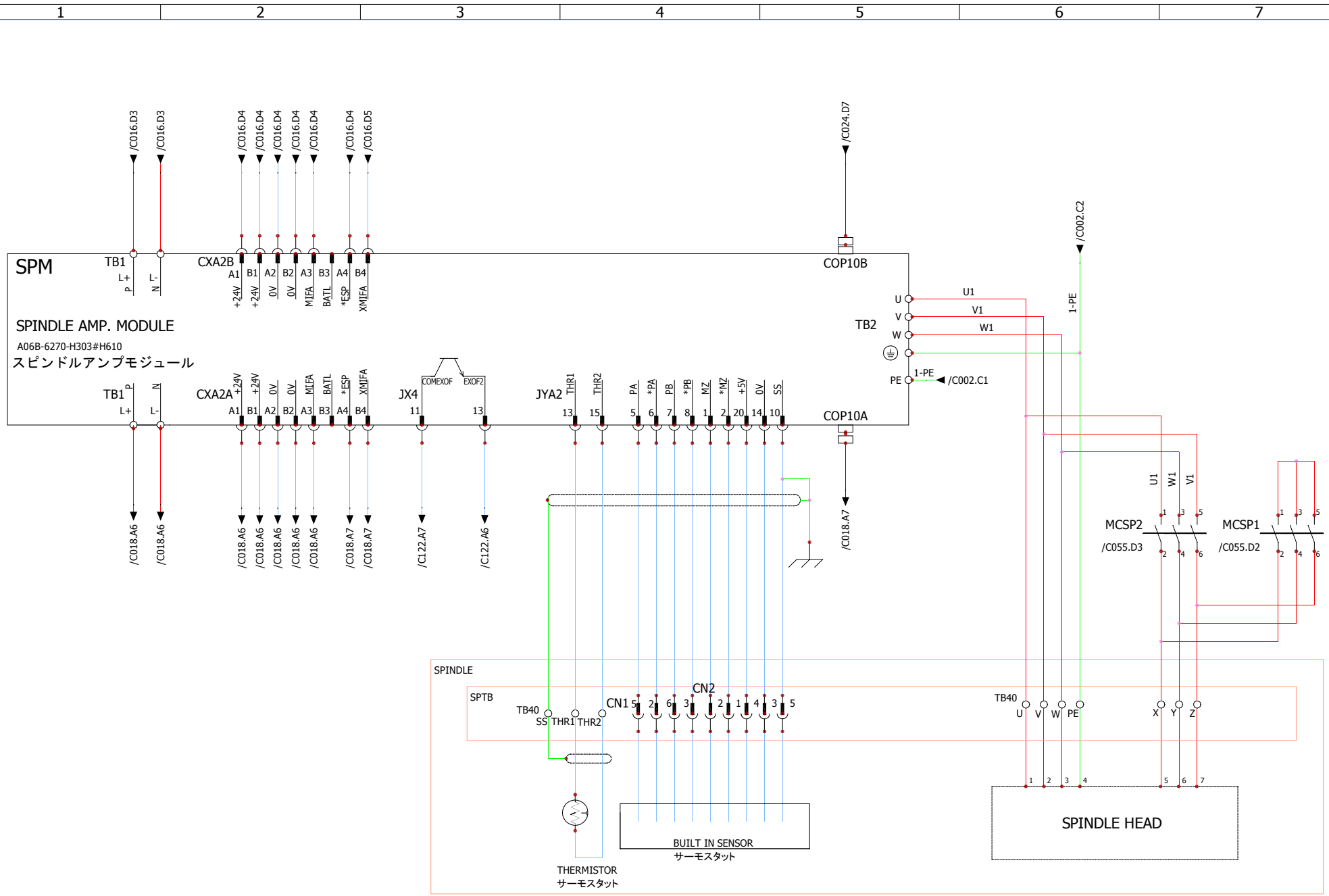
MAKINO
MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
POWER OUT	V2631I-C1E-Y	Ver.	B
		PAGE	C015

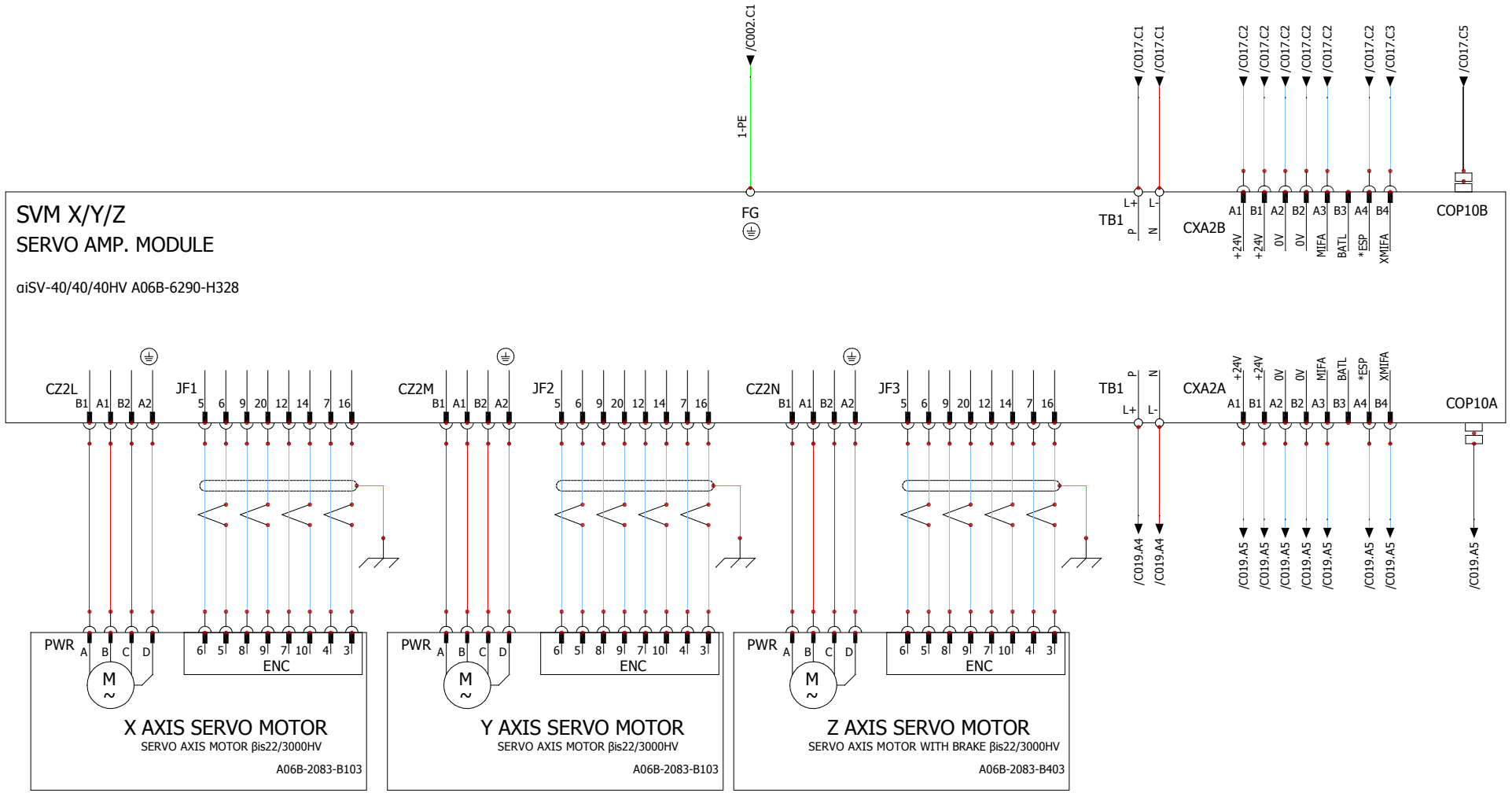


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	POWER SUPPLY MODULE	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C016
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

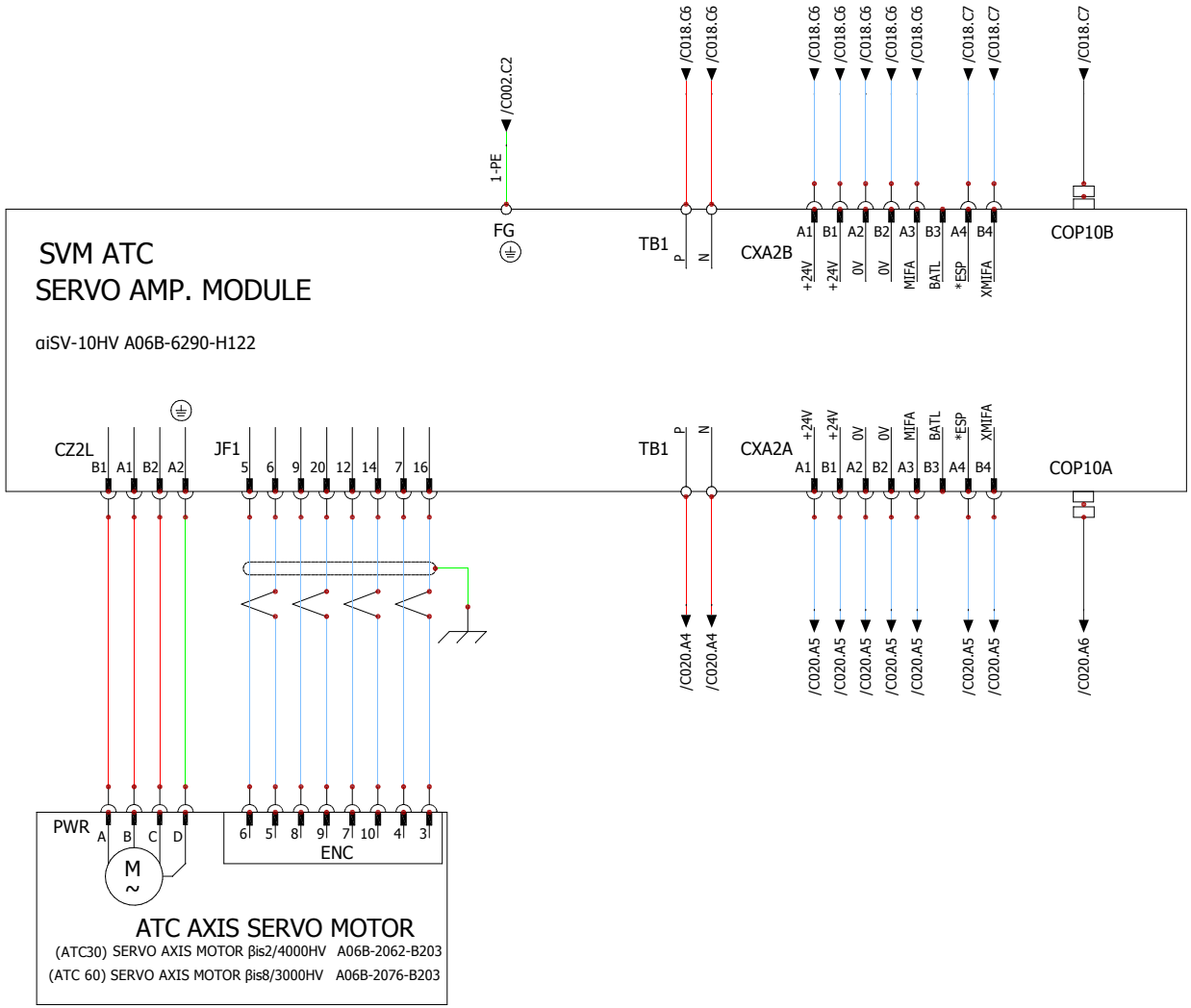
MAKINO
MAKINO ASIA PTE LTD



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	SPINDLE AMPLIFIER MODULE	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C017
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	6/28/2018					

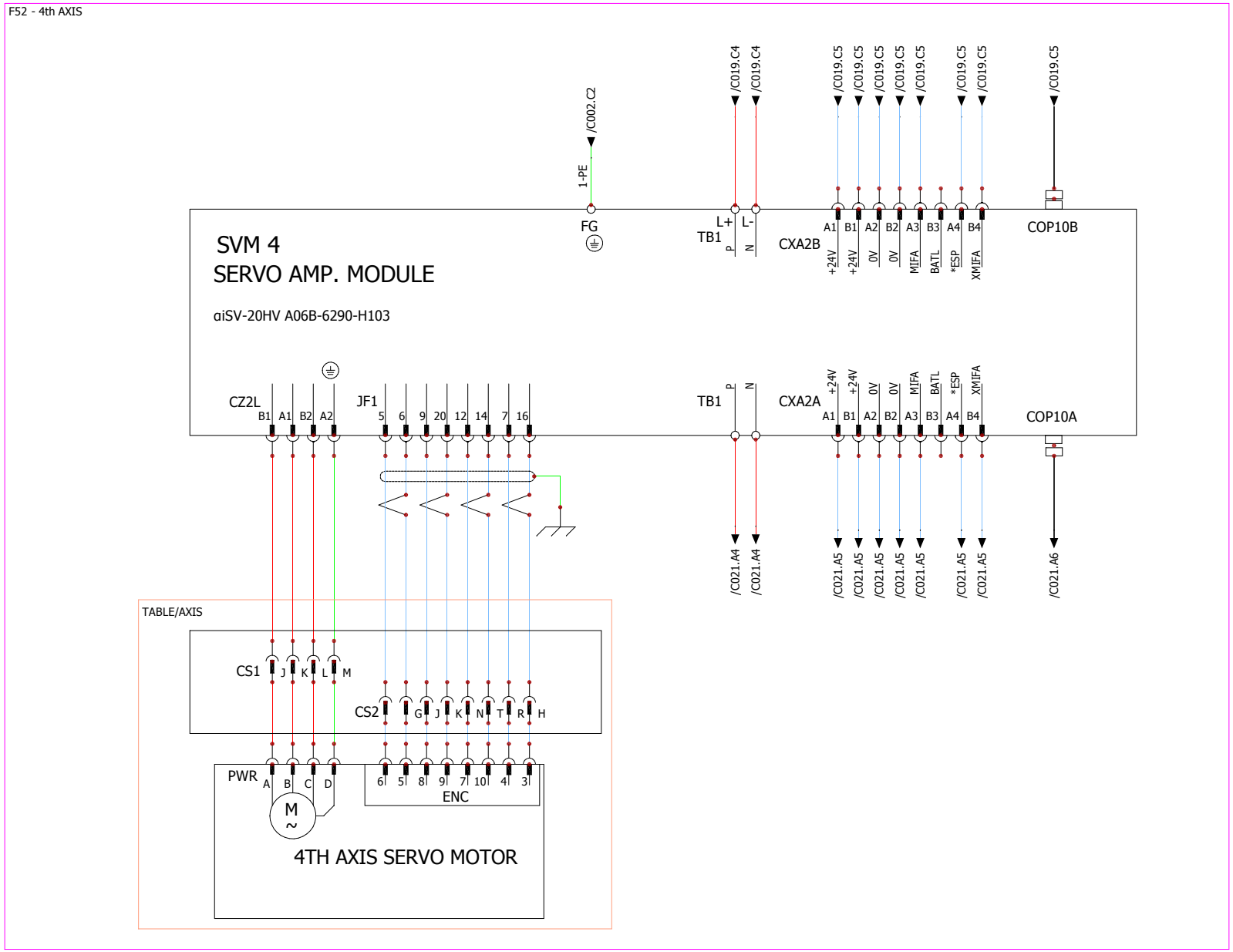


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	X, Y, Z - SERVO AMPLIFIER MODULE	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	C018
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	6/28/2018					



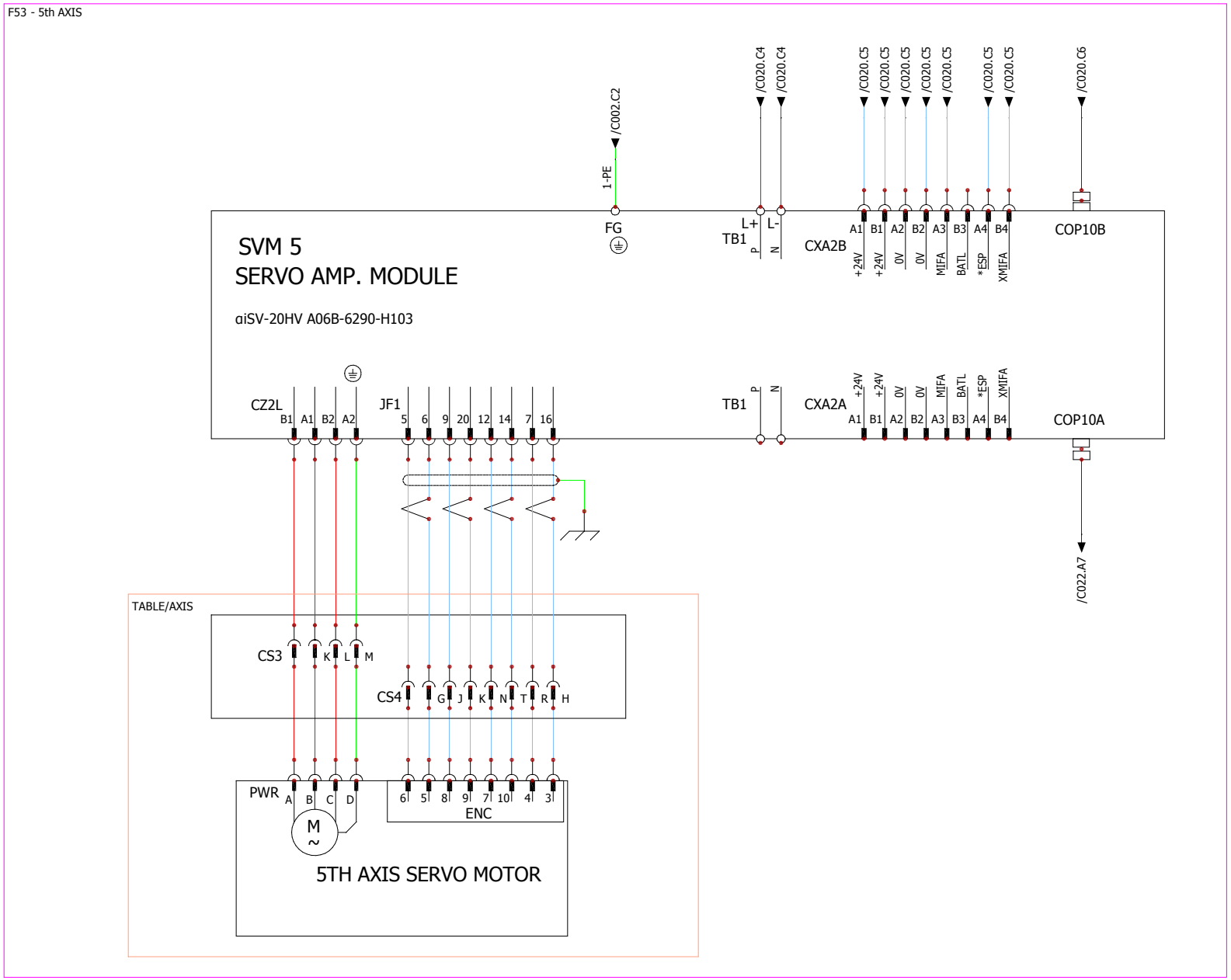
REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH			ATC - SERVO AMPLIFIER MODULE	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH					PAGE	C019
	----.--.--	DRAWN	ANISHA						
	----.--.--	DATE	28/6/2018						

F52 - 4th AXIS

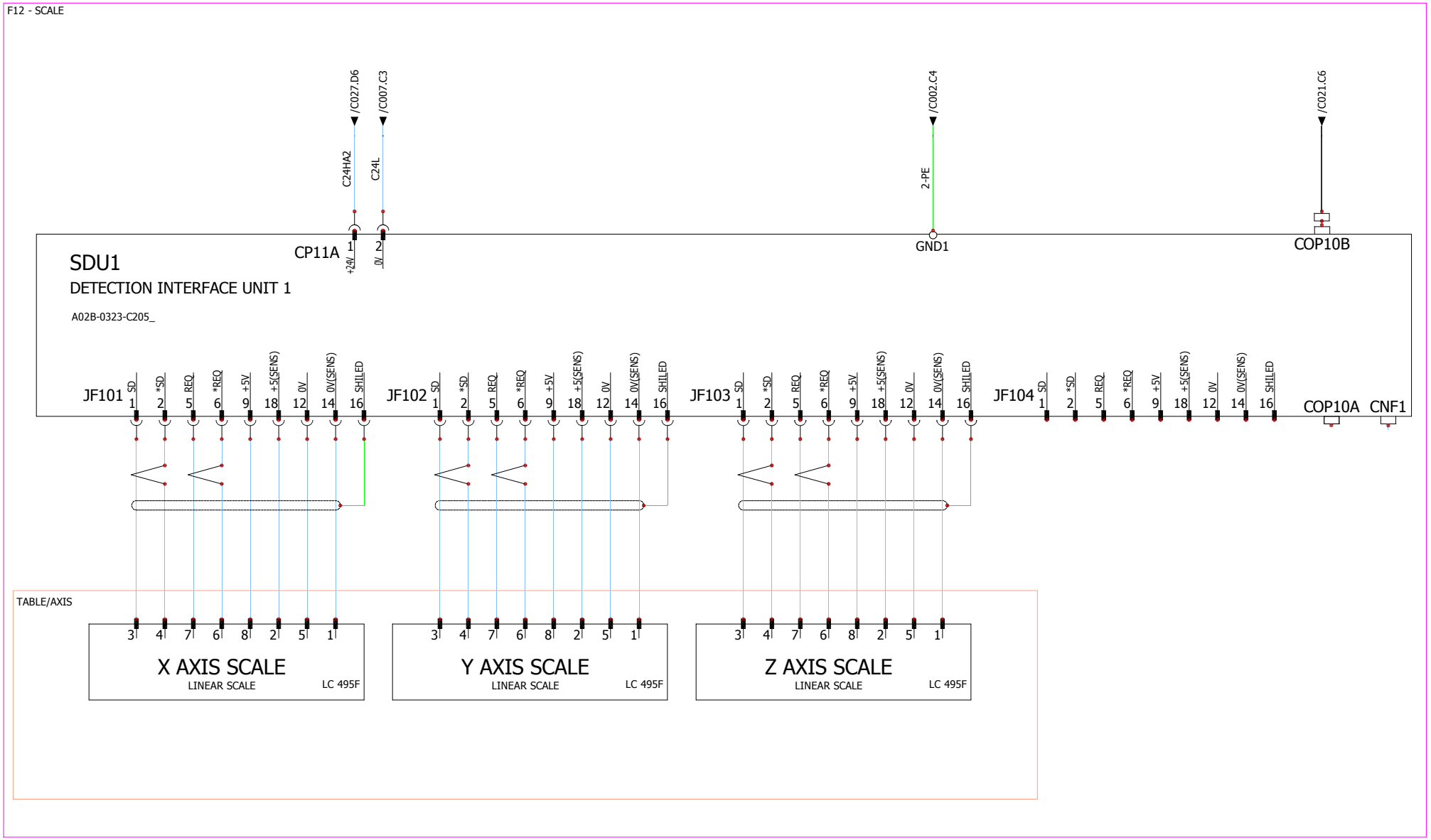


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	4th - SERVO AMPLIFIER MODULE	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C020
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

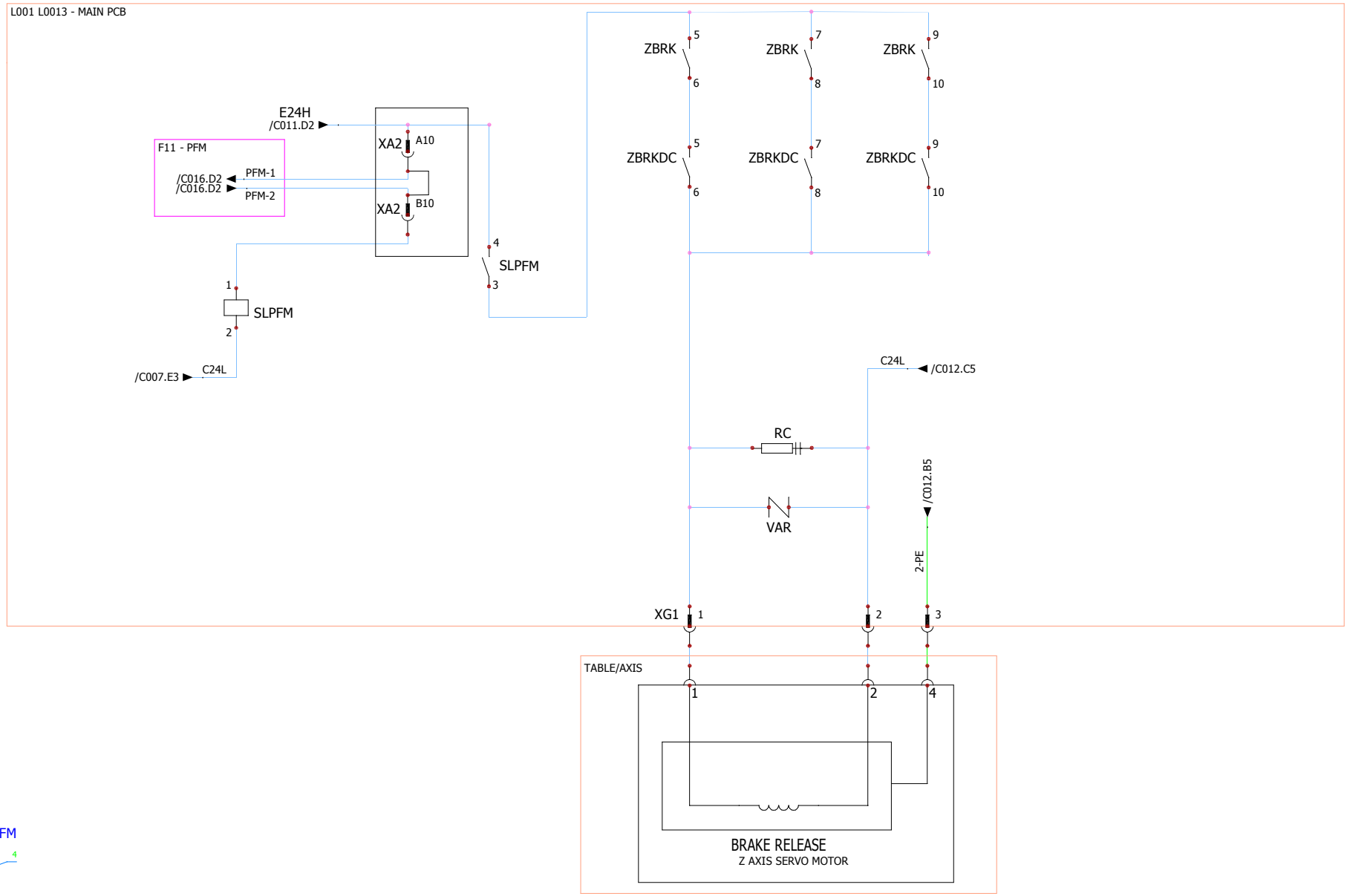
F53 - 5th AXIS



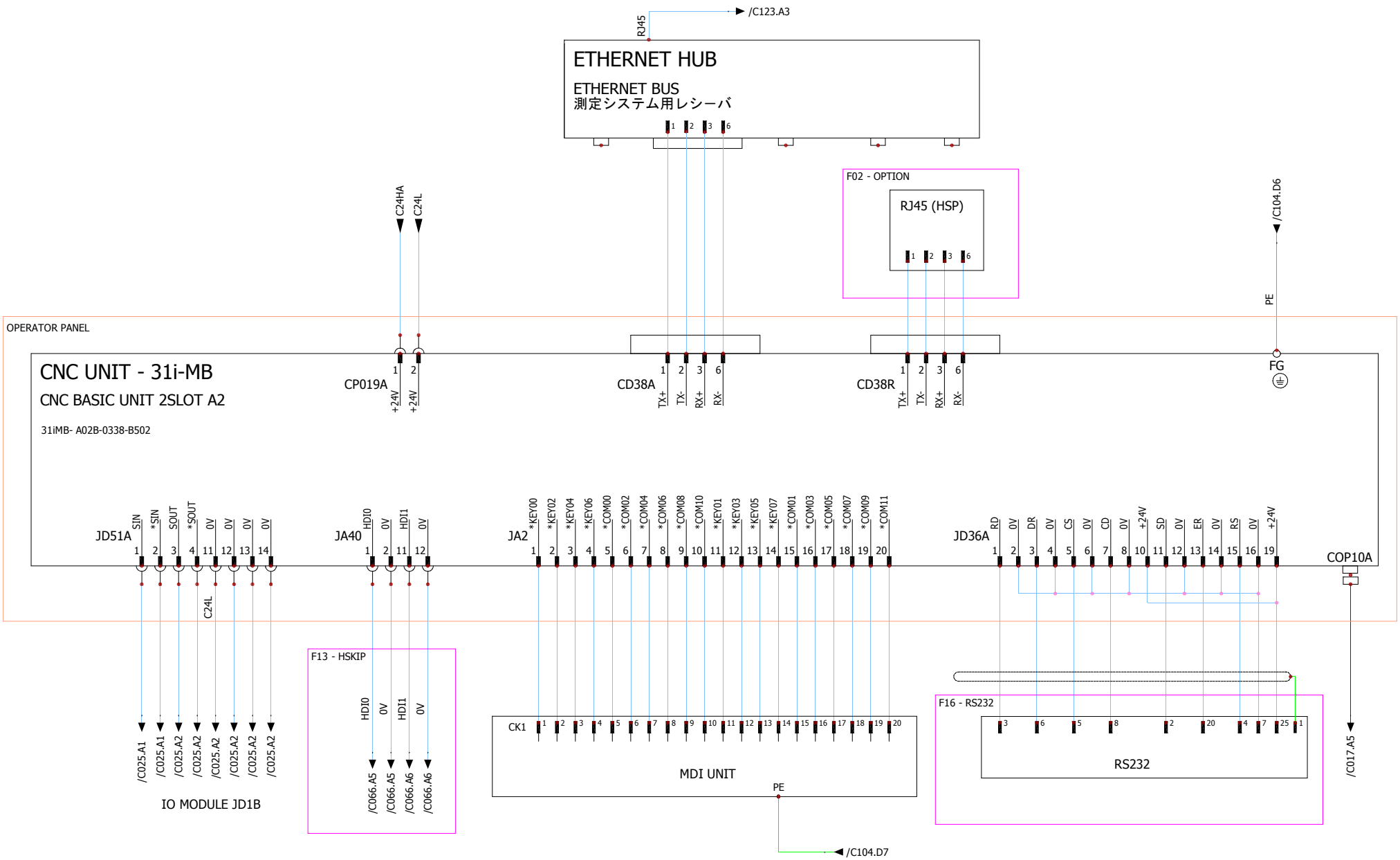
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	5th - SERVO AMPLIFIER MODULE	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C021
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



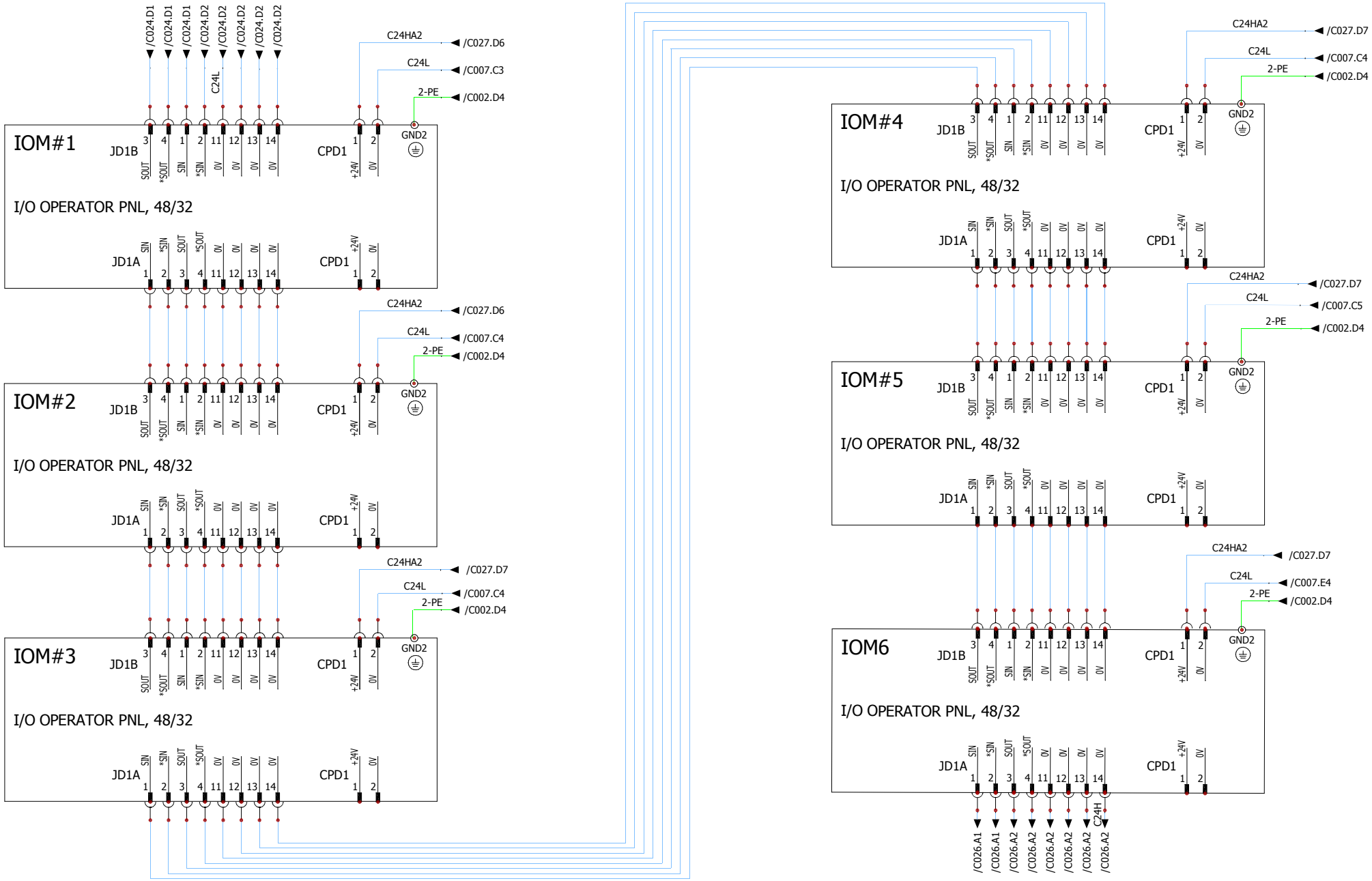
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	FEEDBACK SCALE (SDU)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C022
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	Z AXIS MOTOR BRAKE	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	C023
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					

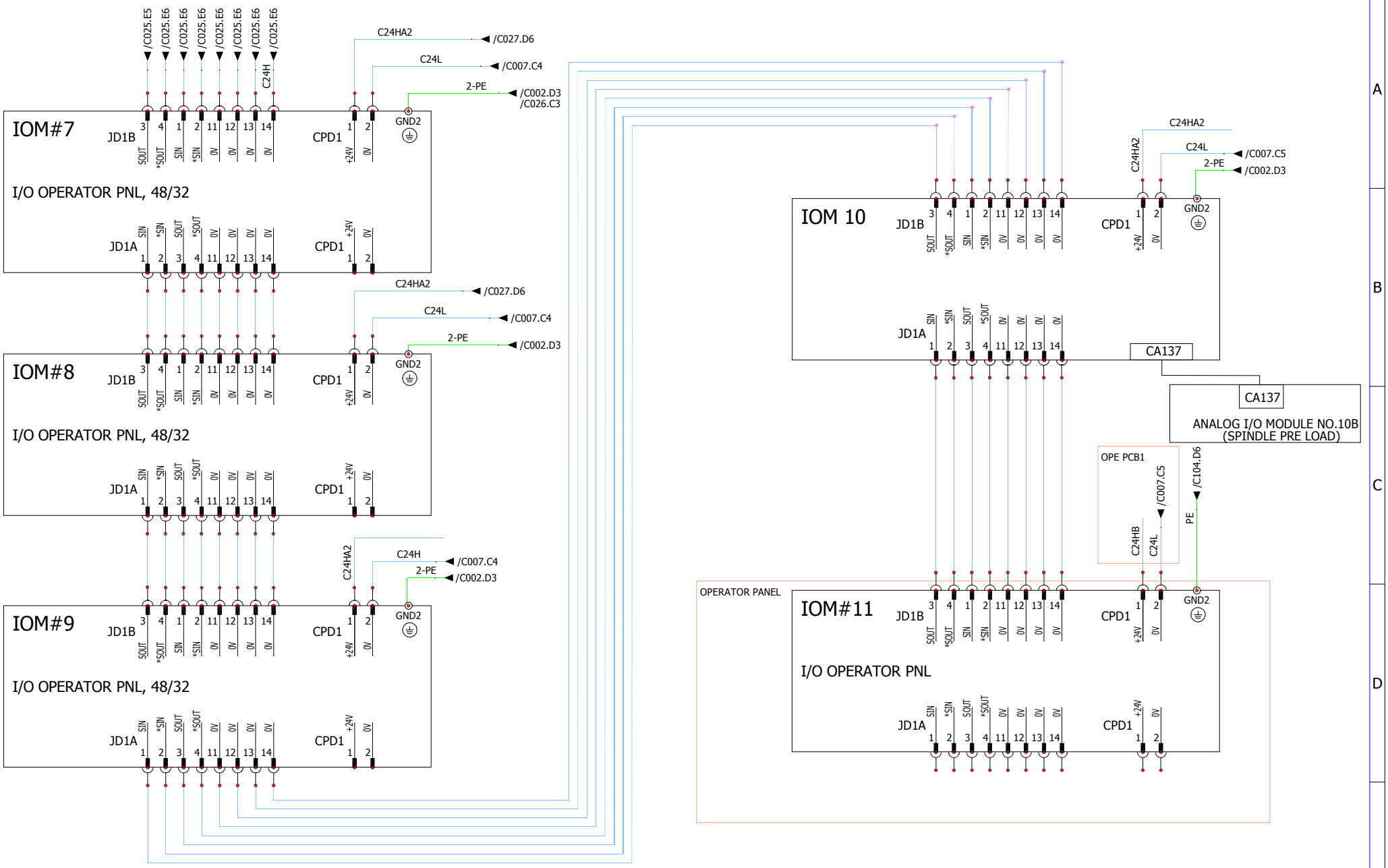


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	CONTROLLER UNIT	V2631I-C1E-Y	Ver.	B
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	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

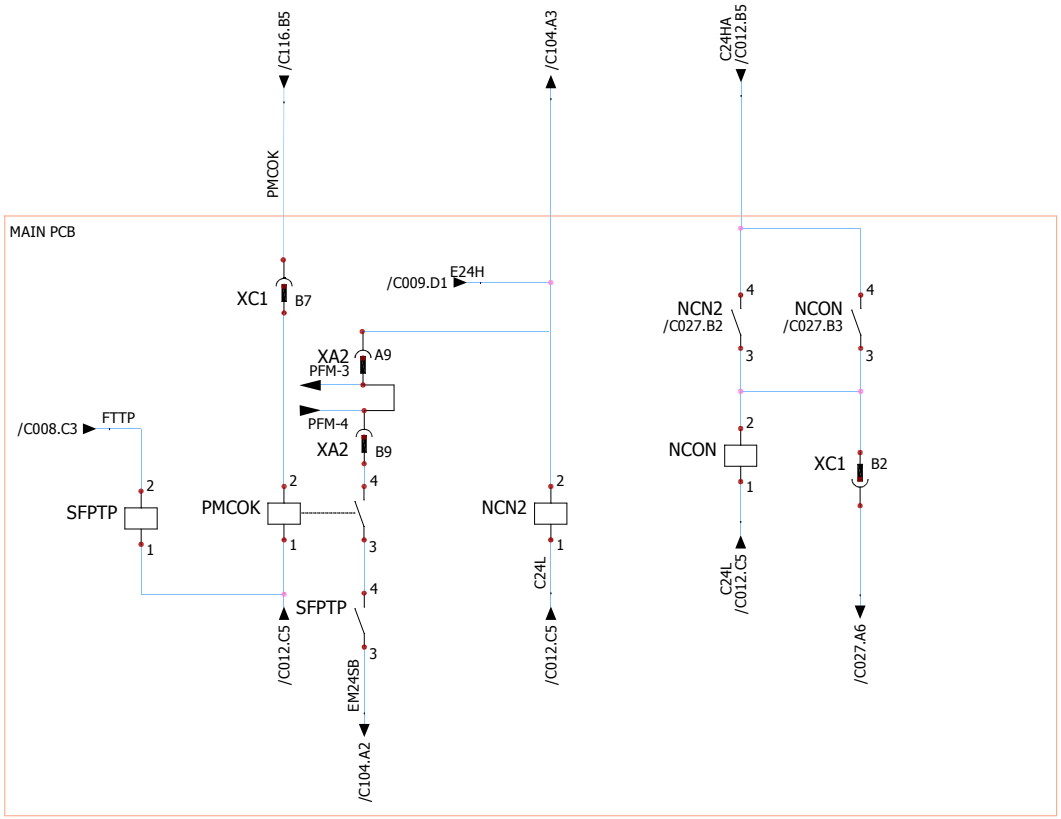


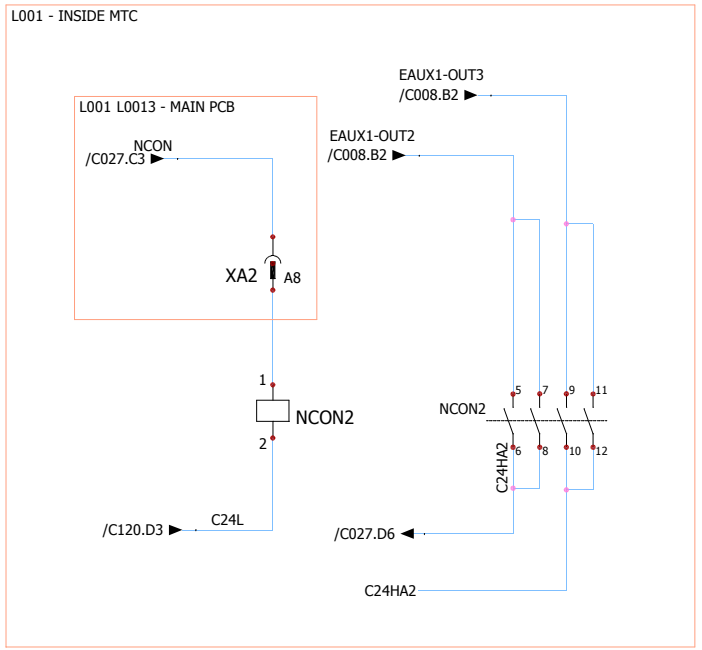
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	I/O LINK	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C025
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

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MAKINO ASIA PTE LTD

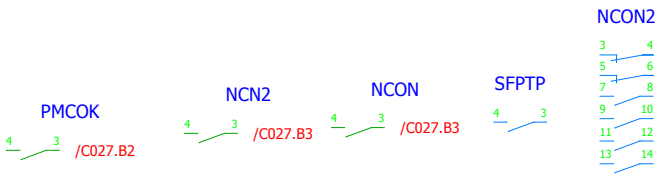


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	I/O LINK	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C026
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



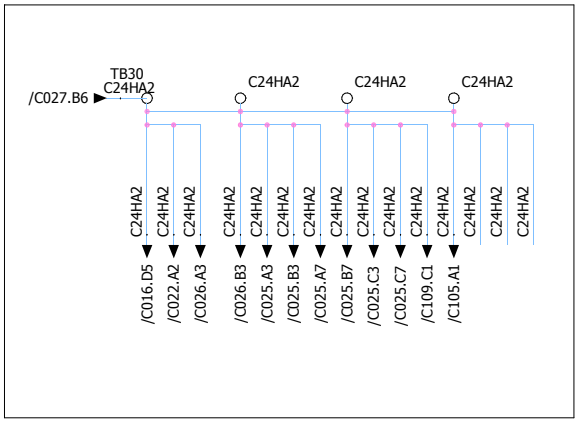


EMERGENCY BUTTONS



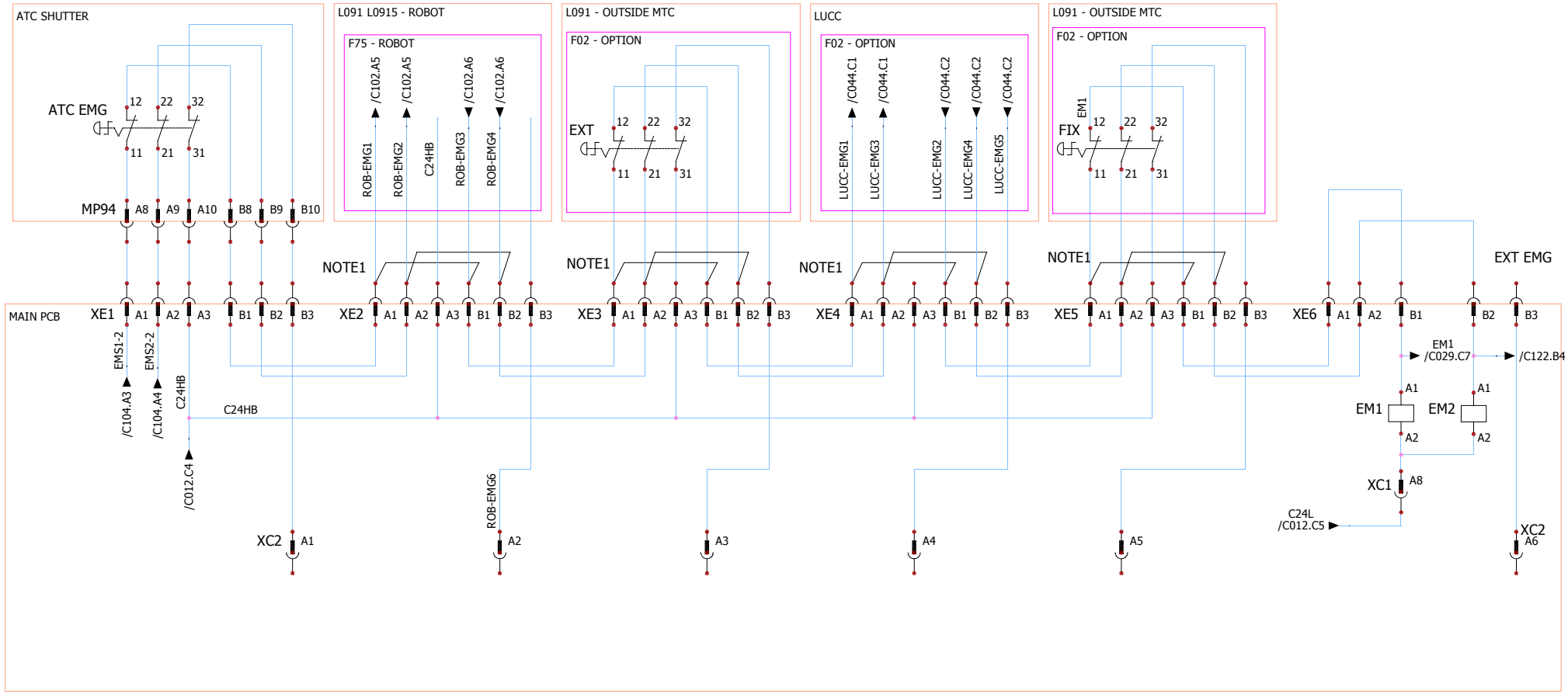
RELAY PCB JUMPER SETTING

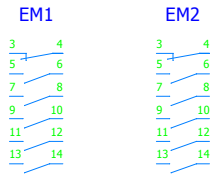
J1 = SHORT	J8 = SHORT
J2 = SHORT	J9 = SHORT
J3 = SHORT	J10 = SHORT
J4 = SHORT	J11 = SHORT
J5 = SHORT	J12 = SHORT
J6 = SHORT	J13 = SHORT
J7 = SHORT	



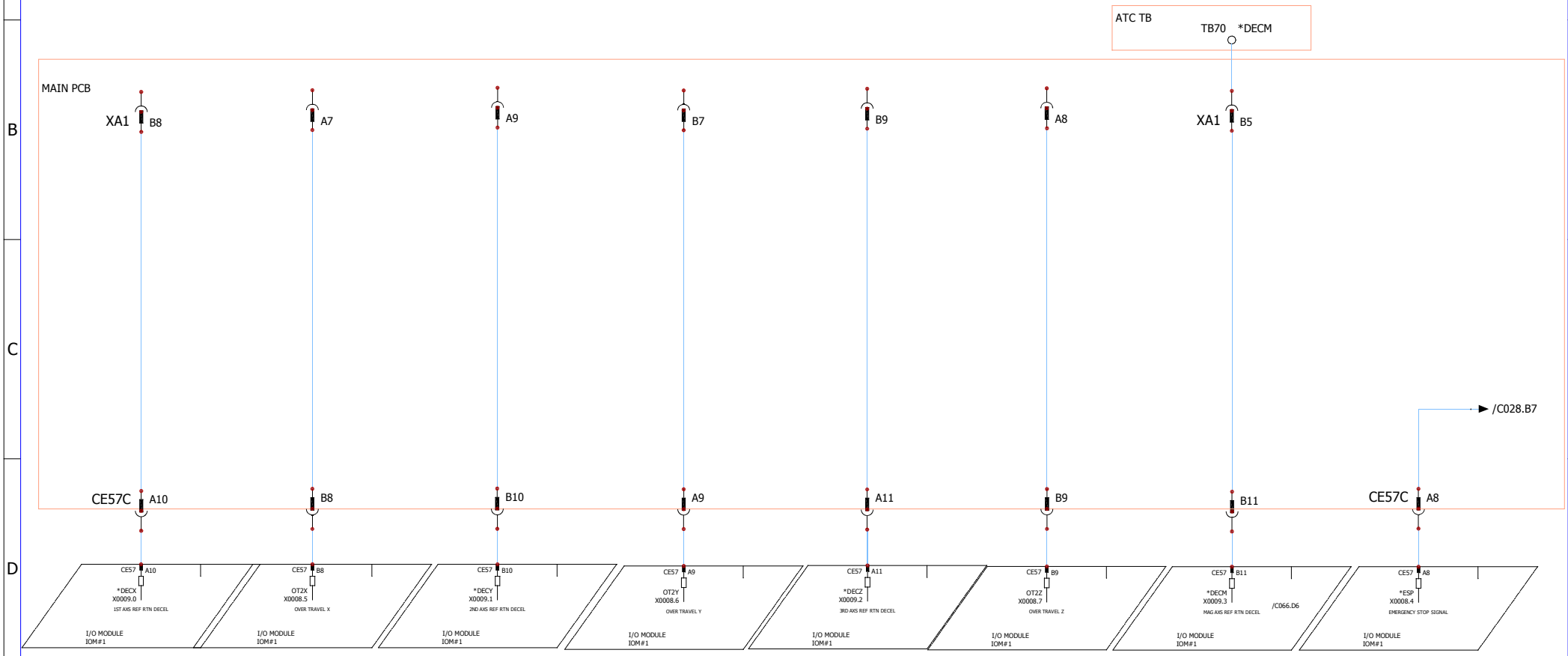
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	NC POWER ON CONTROL (C24HA2)	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	C027
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					

NOTE1: SHORT IN CASE THE RESPECTIVE OPTION IS NOT PRESENT

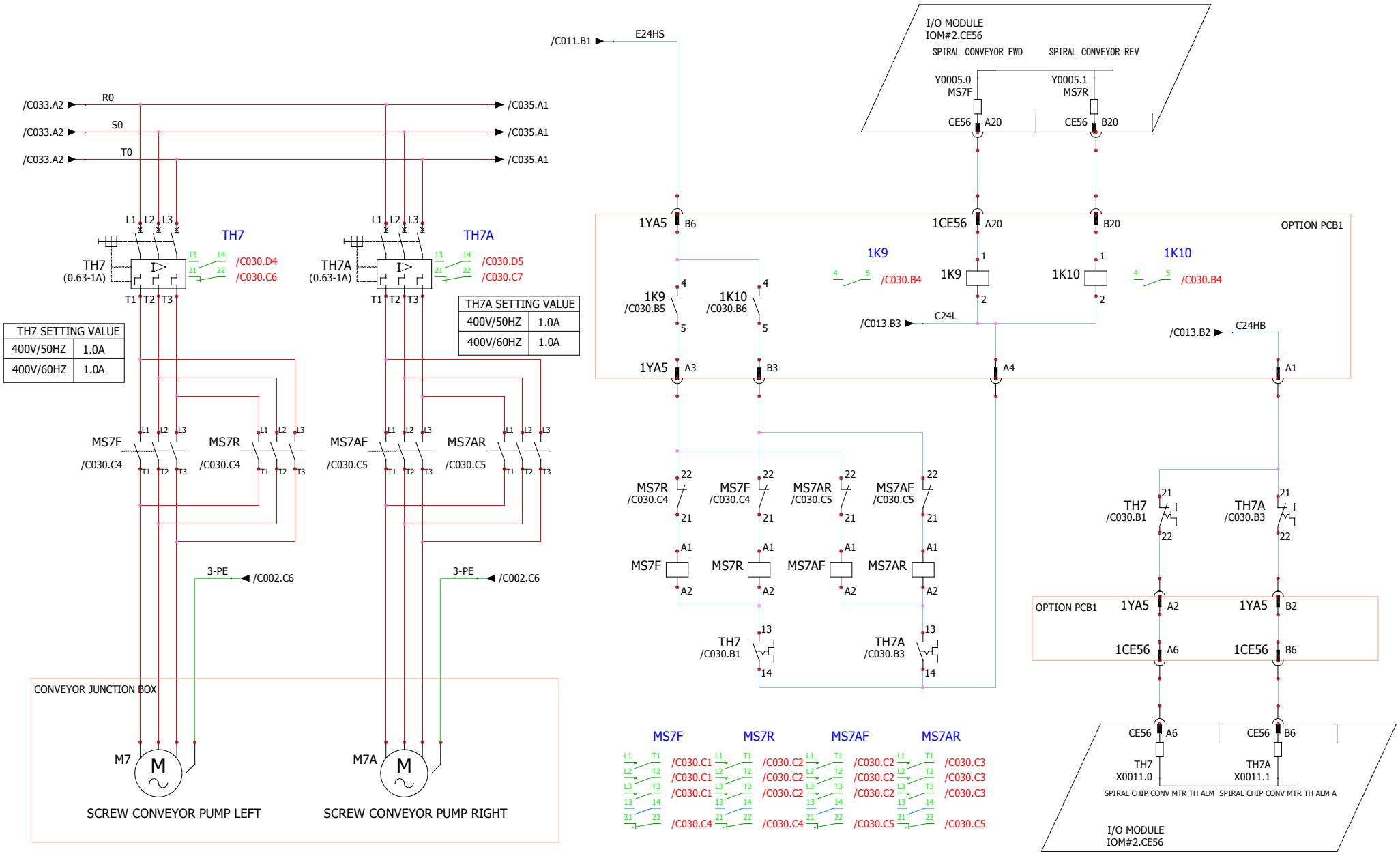




REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	EMERGENCY STOP	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C028
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



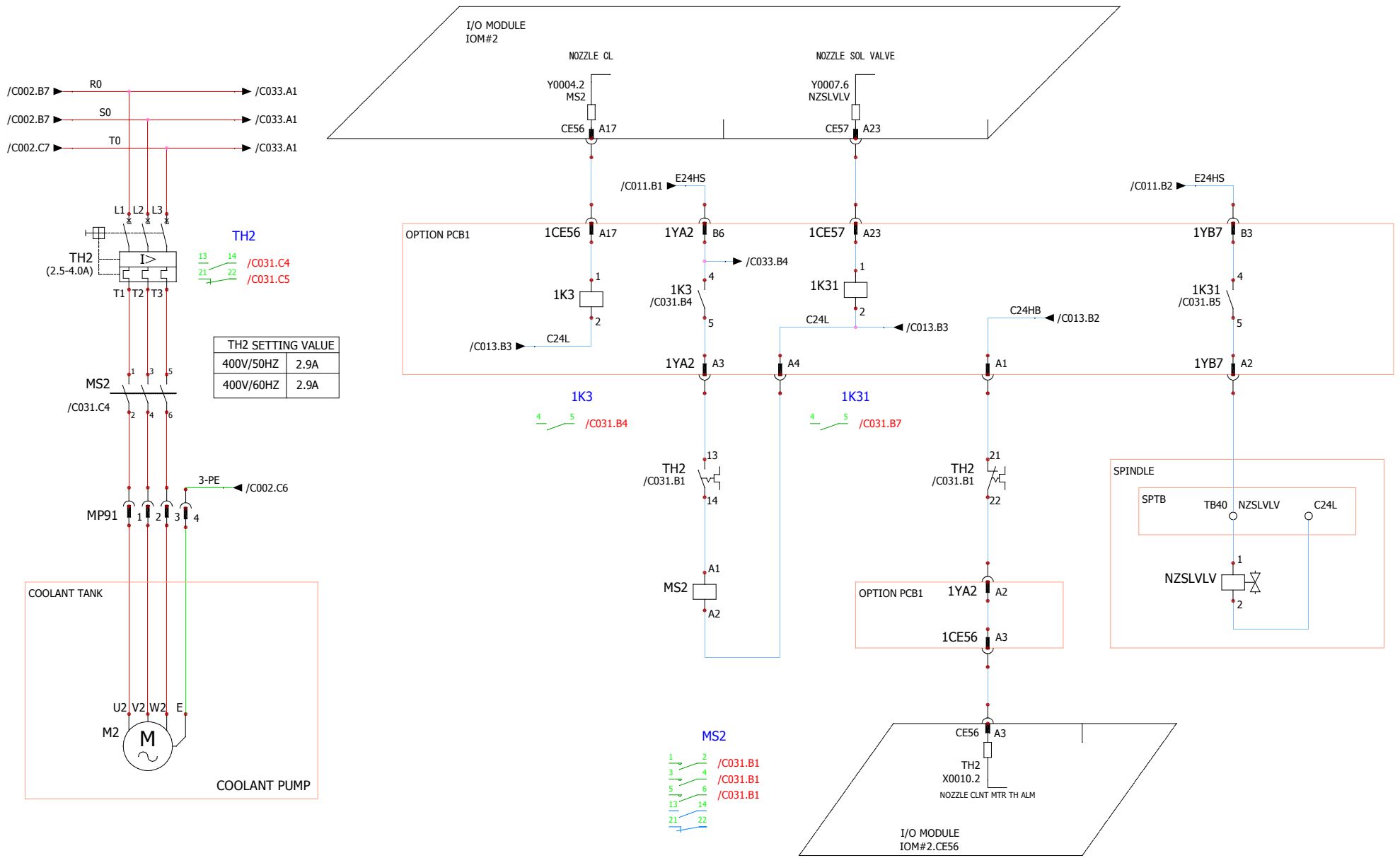
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	REFERENCE AND OVERTRAVEL	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C029
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



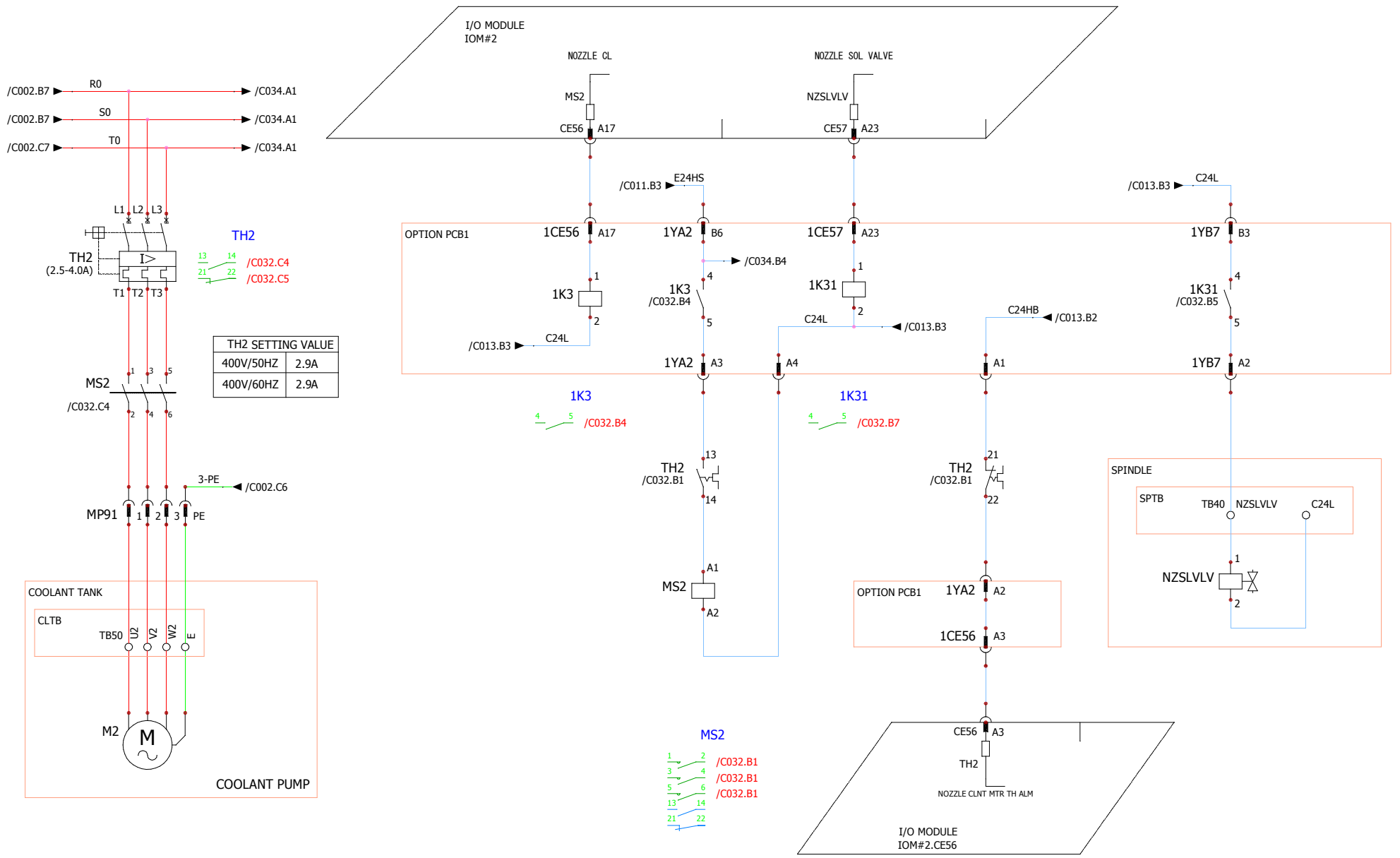
REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

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TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
SCREW CONVEYOR	V2631I-C1E-Y	Ver.	B
		PAGE	C030



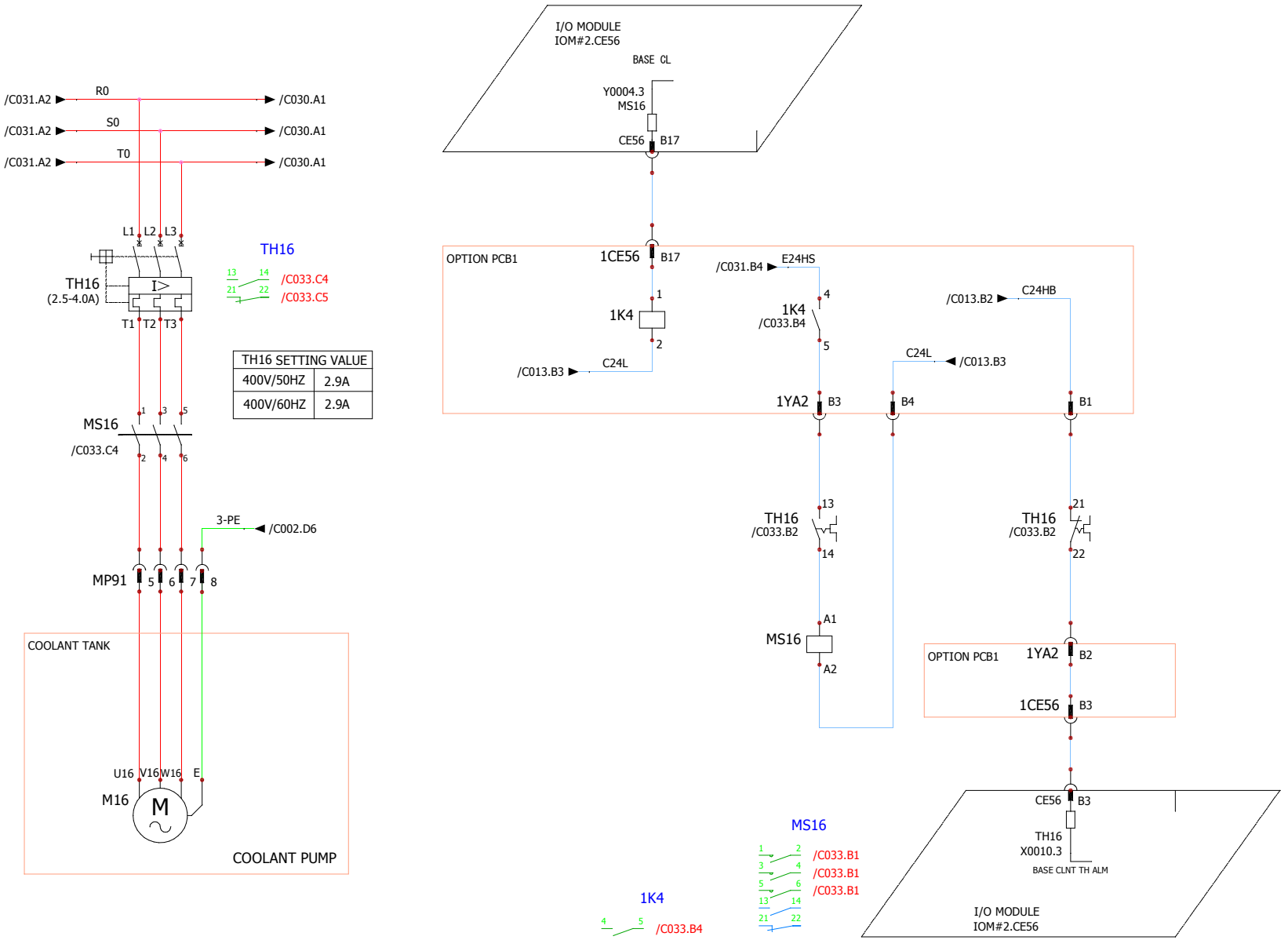
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	NOZZLE COOLANT & WW GUN (FRONT TANK)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C031
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	NOZZLE COOLANT & WW GUN(REAR TANK)	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C032
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

MAKINO
MAKINO ASIA PTE LTD

F22 - BASE COOLANT

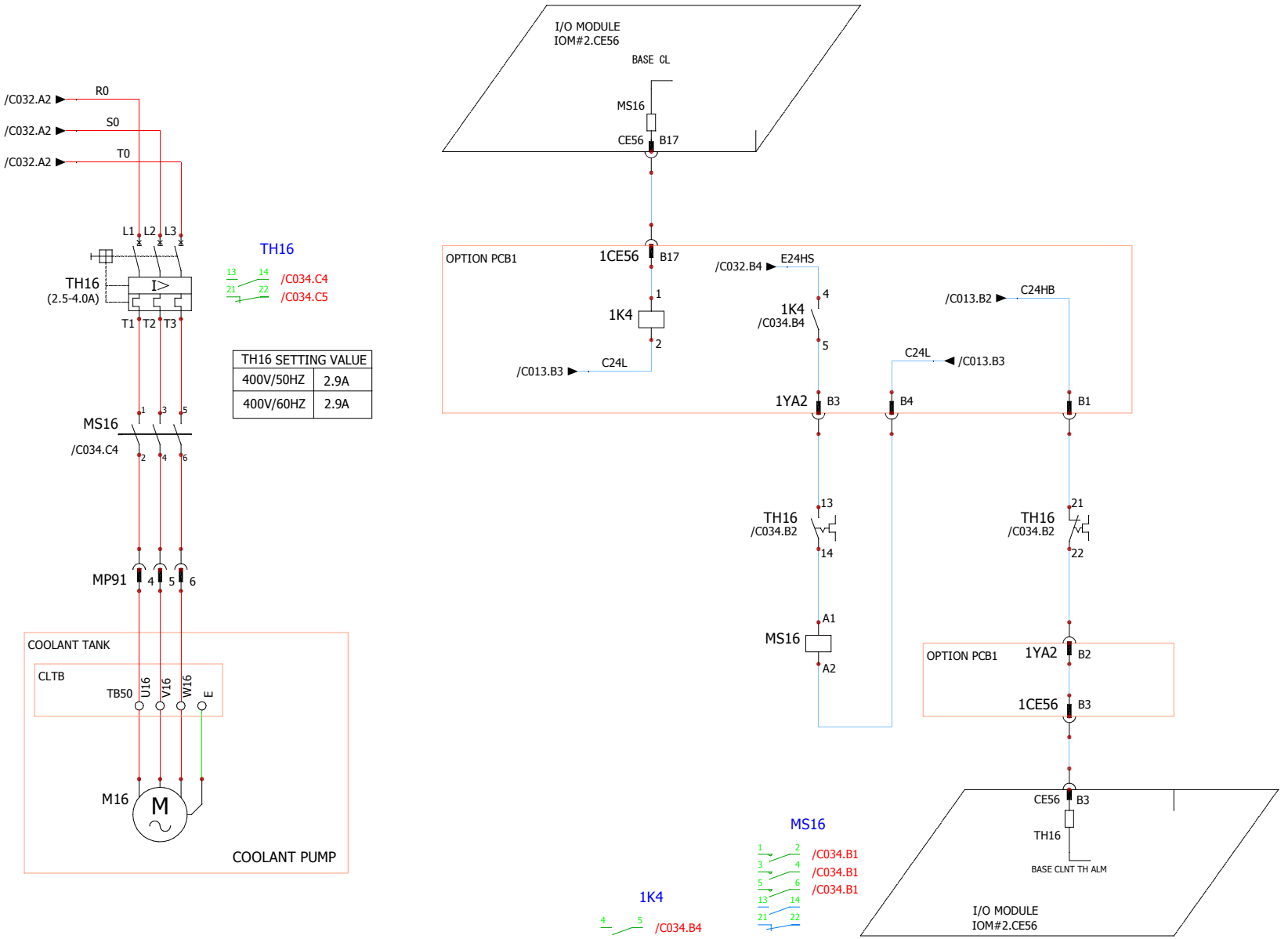


REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

MAKINO
MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
BASE / FIXTURE COOLANT (OPT)(FRONT TANK)	V2631I-C1E-Y	Ver.	B
		PAGE	C033

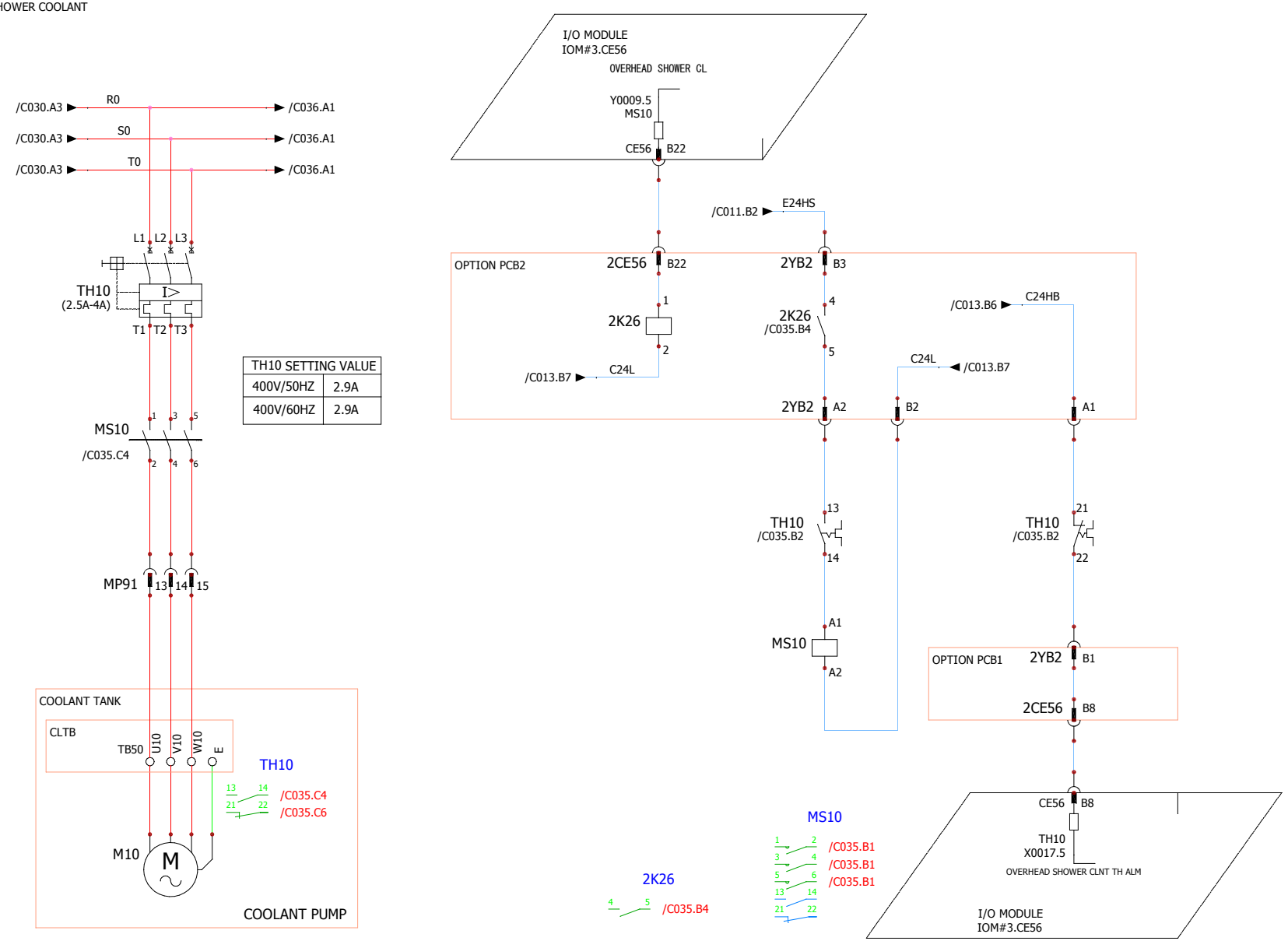
F22 - BASE COOLANT



TH16 SETTING VALUE	
400V/50HZ	2.9A
400V/60HZ	2.9A

REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	BASE / FIXTURE COOLANT (OPT)(REAR TANK)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C034
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

F23 - SHOWER COOLANT



TH10 SETTING VALUE	
400V/50HZ	2.9A
400V/60HZ	2.9A

REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	SHOWER COOLANT (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C035
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

F24 - TSC

/C035.A2 R0 /C041.A1
/C035.A2 S0 /C041.A1
/C035.A2 T0 /C041.A1

L1 L2 L3
TH8 (4.0-6.3A)
T1 T2 T3

TH8
13 14 /C036.C4
21 22 /C036.C5

TH8 SETTING VALUE	
400V/50HZ	5.3A
400V/60HZ	5.3A

MS8
/C036.C4
1 3 5
2 4 6

MP91 16 17 18

COOLANT TANK
CLTB
TB50 U8 V8 W8 E

F02 - OPTION
SECONDARY TANK
MP105 9 10 11 PE
SEC TB
TB90 U8 V8 W8 E

M8 M

TSC 1.5Mpa MOTOR (MAIN COOLANT TANK / SEC COOLANT TANK)
TSC 3.0Mpa MOTOR (SEC COOLANT TANK)

I/O MODULE
IOM#3.CE56
THROUGH SP CL
Y0009.4
MS8
CE56 A22

/C011.B2 E24HS

OPTION PCB2
2CE56 A22
2K25 1 2
/C013.B7 C24L

2YB1 B3
2K25 /C036.B4 4 5
C24L /C013.B7
/C013.B6 C24HB
2YB1 A2 B2 A1

TH8 /C036.B2 13 14
TH8 /C036.B2 21 22

MS8 A1 A2

OPTION PCB2
2YB1 B1
2CE56 A8

CE56 A8
TH8
X0017.4
TS CLNT PUMP MTR TH ALM
I/O MODULE
IOM#3.CE56

MS8
1 2 /C036.B1
3 4 /C036.B1
5 6 /C036.B1
13 14
21 22

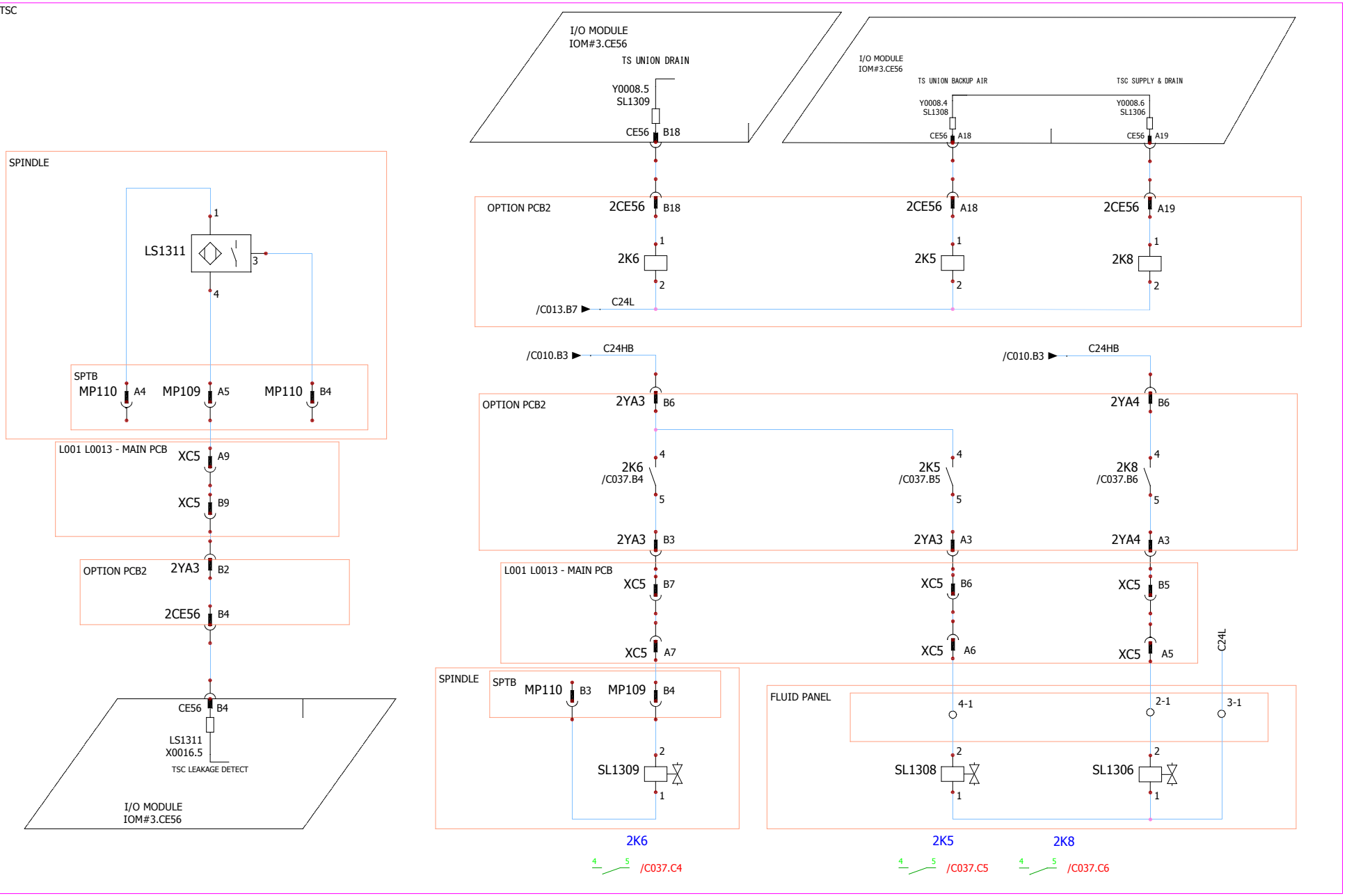
2K25
4 5 /C036.B4

REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

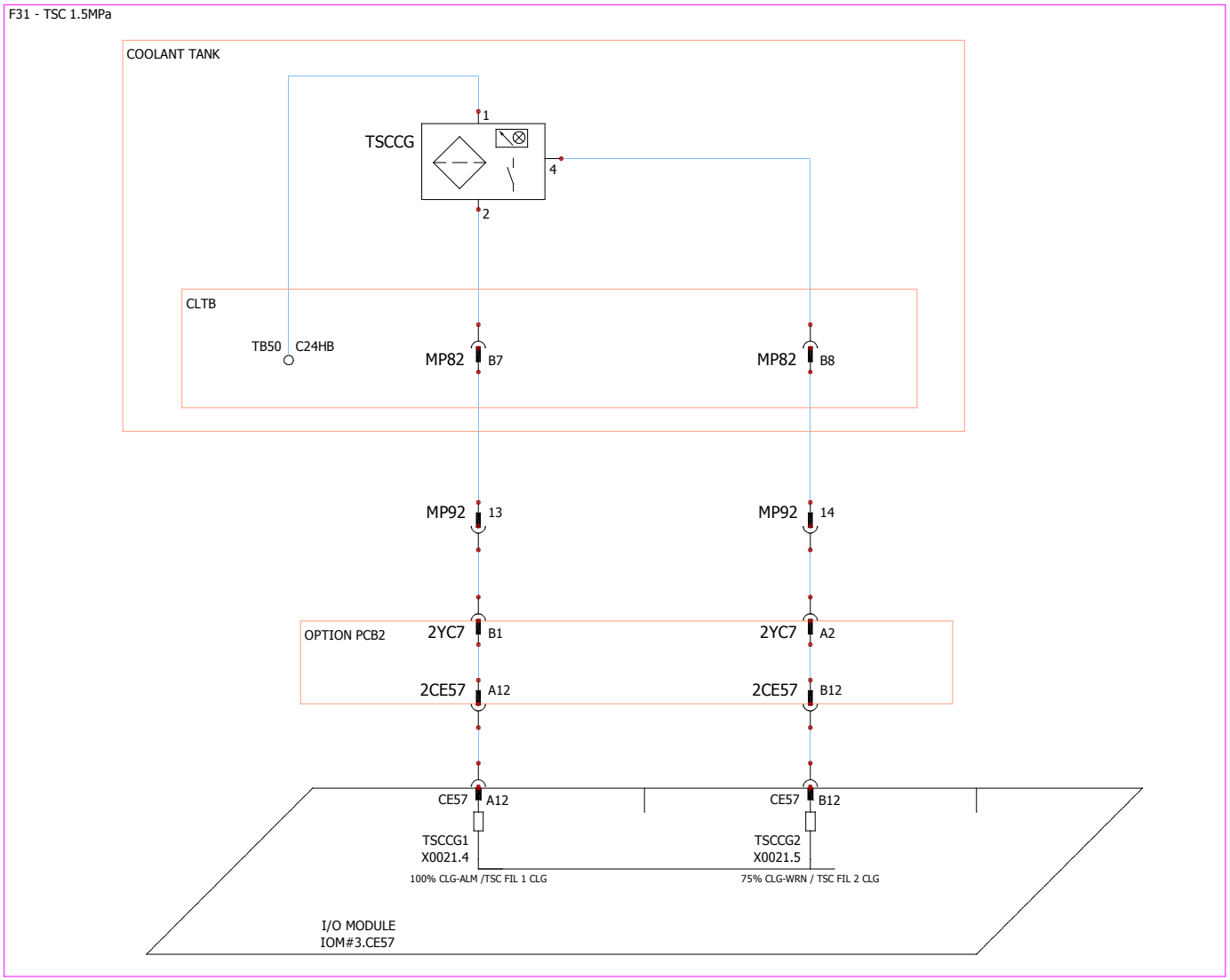
MAKINO
MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
TSC 1.5MPa & 3MPa PWR (OPT)	V2631I-C1E-Y	Ver.	B
		PAGE	C036

F24 - TSC

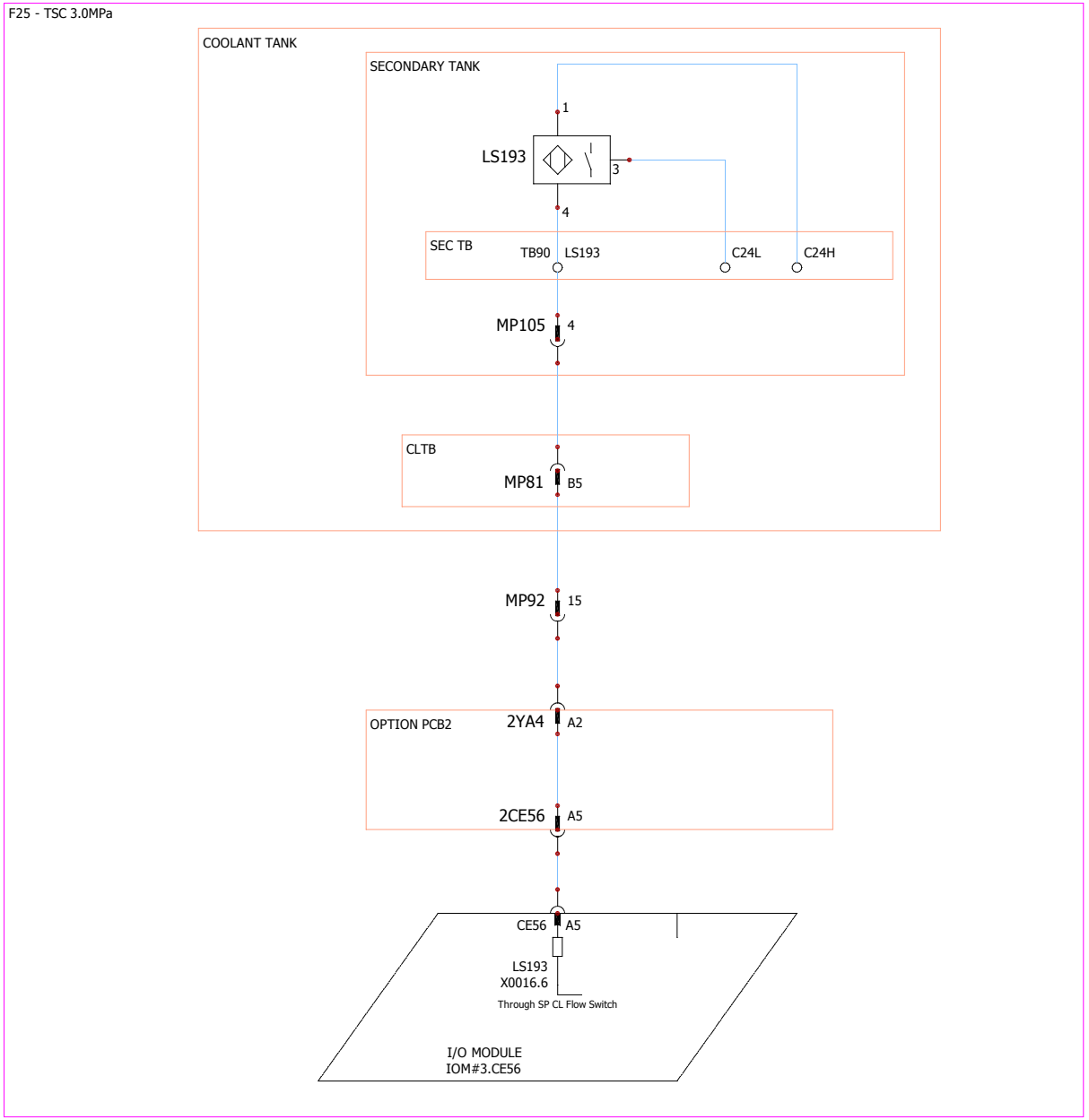


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TSC 1.5MPa & 3MPa CONTROL (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C037
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	TSC 1.5MPa CONTROL (OPT)	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C038
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

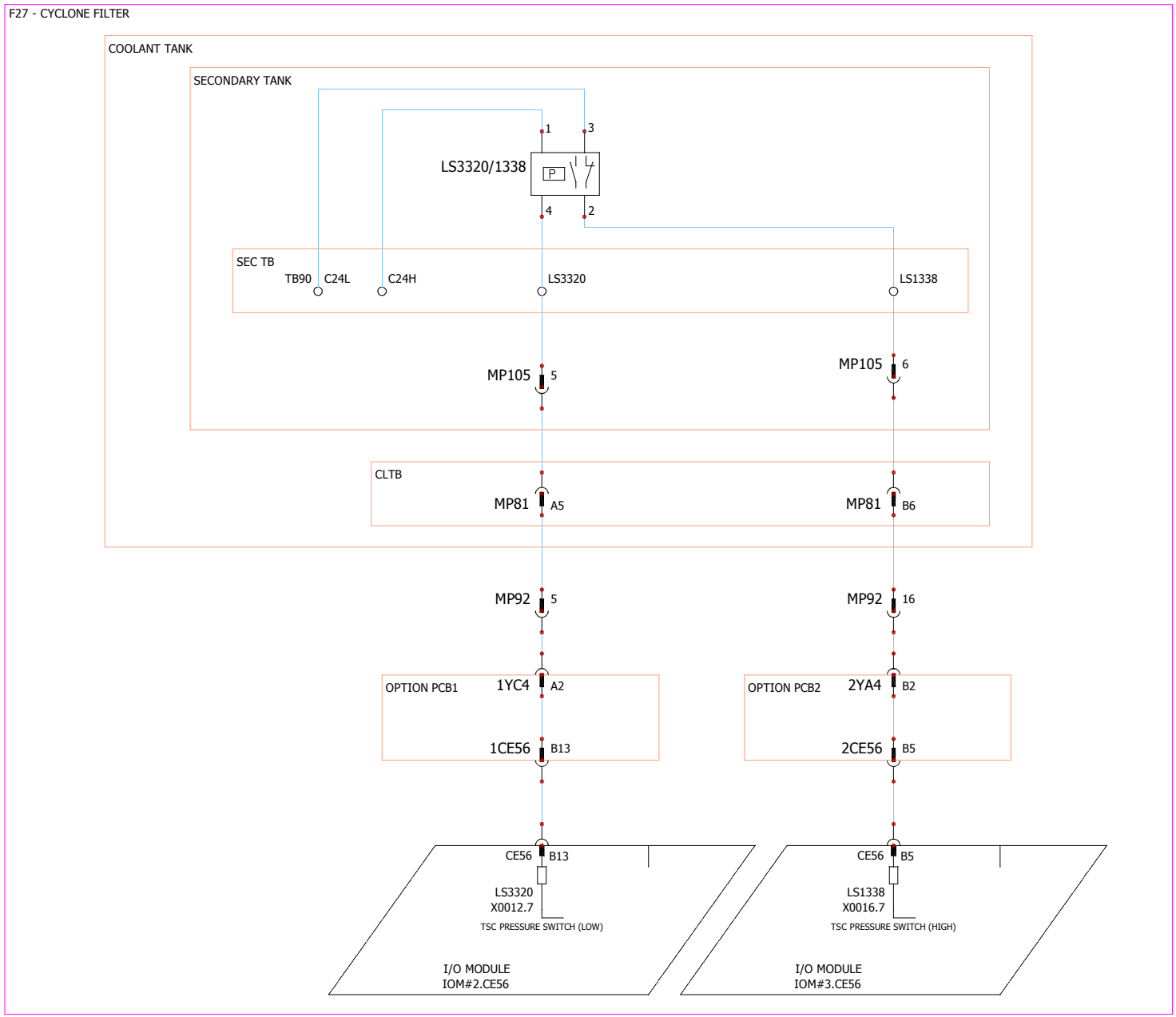
MAKINO
MAKINO ASIA PTE LTD



REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

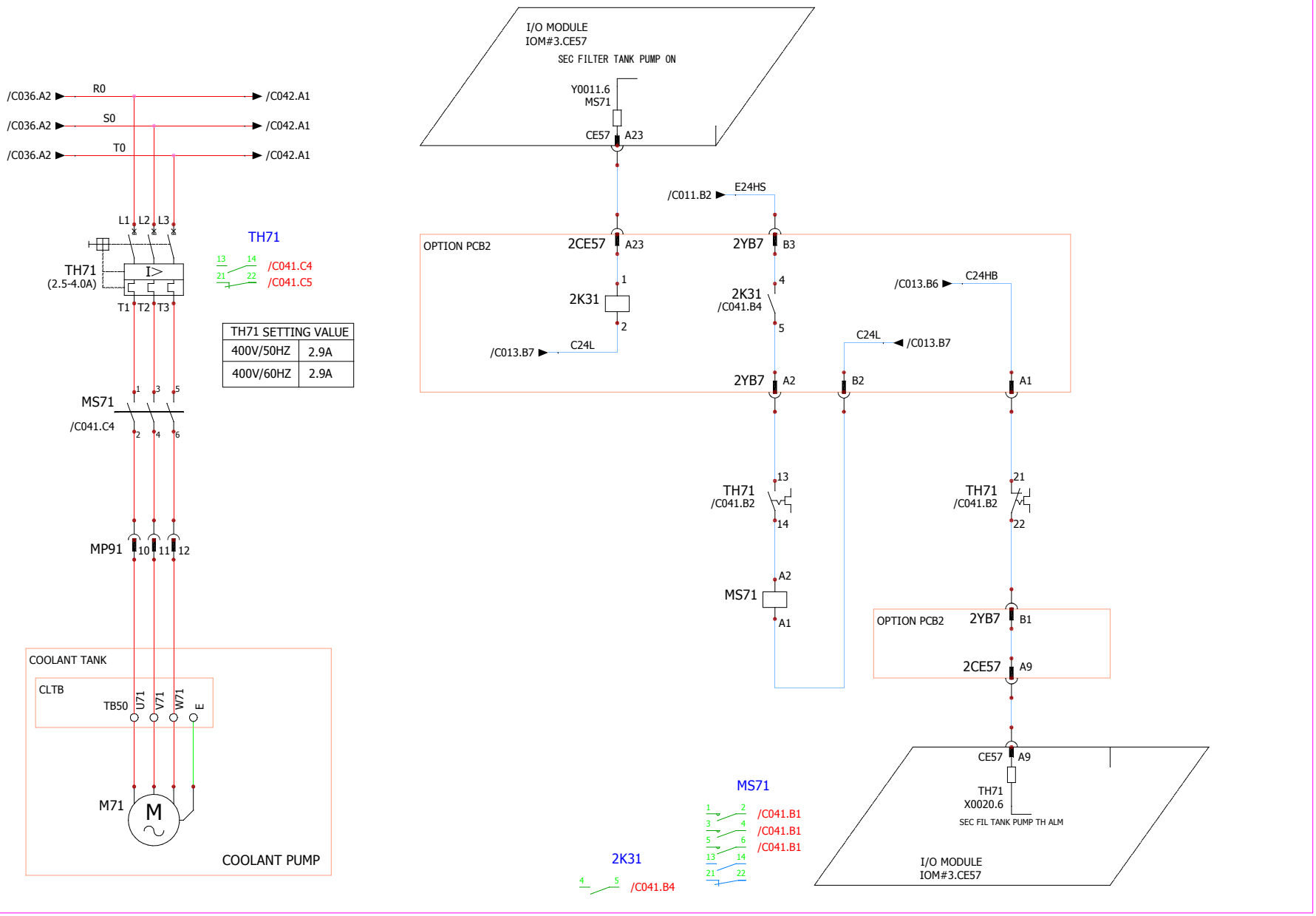
MAKINO
MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
TSC 3.0MPa CONTROL (OPT)	V2631I-C1E-Y	Ver.	B
		PAGE	C039



REVISION						
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TITLE: CYCLONE FILTER PRESSURE SW (OPT)	DRAWING No.: V2631I-C1E-Y
	----.--.--	CHECKED	AVINASH			MACHINE: F3/ F5 Pro 6
	----.--.--	DRAWN	ANISHA			Ver.: B
	----.--.--	DATE	28/6/2018			PAGE: C040

F26 - SECONDARY TANK



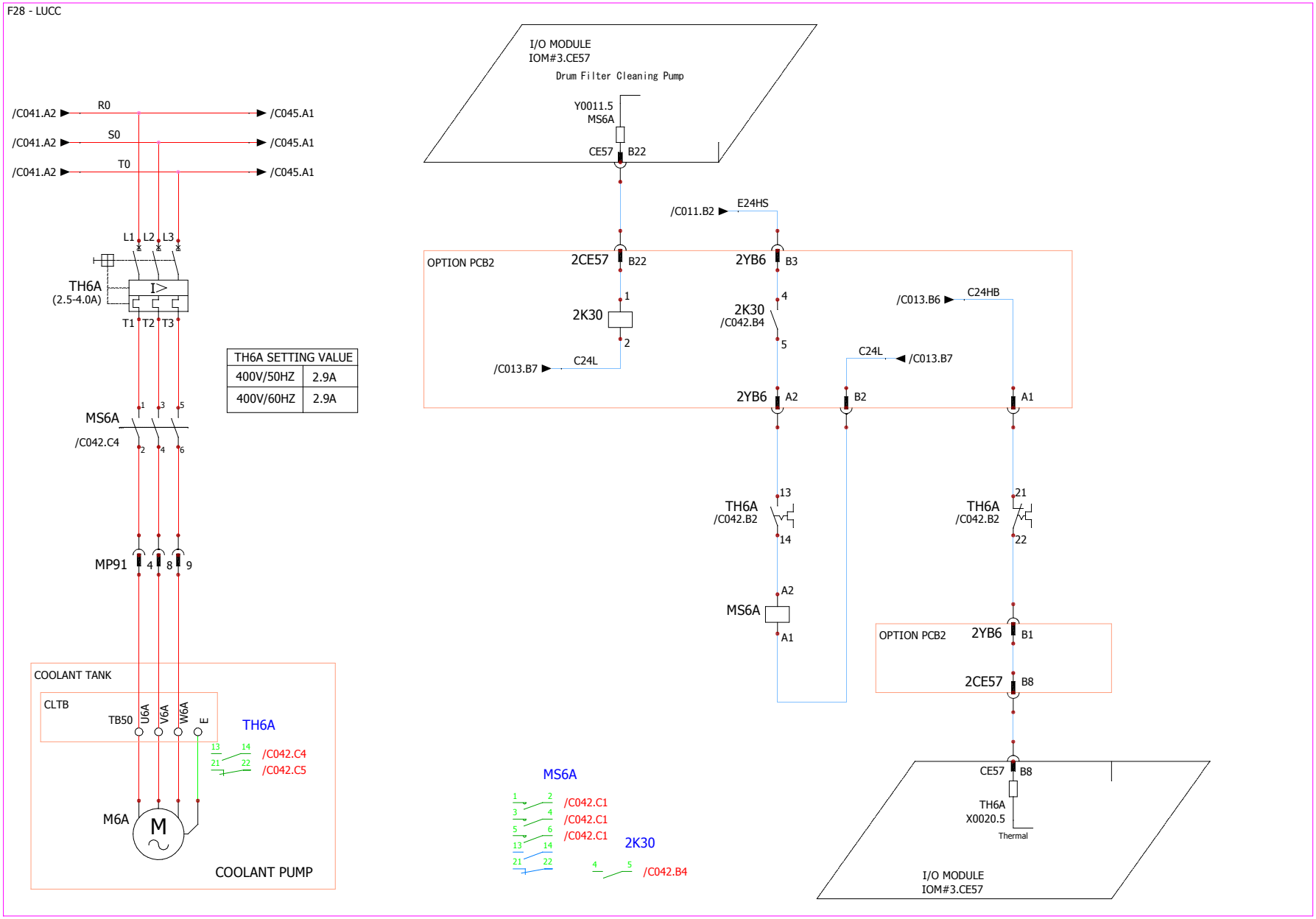
TH71 SETTING VALUE	
400V/50HZ	2.9A
400V/60HZ	2.9A

REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

MAKINO
MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
SECONDARY TRANSFER PUMP (OPT)	V2631I-C1E-Y	Ver.	B
		PAGE	C041

F28 - LUCC

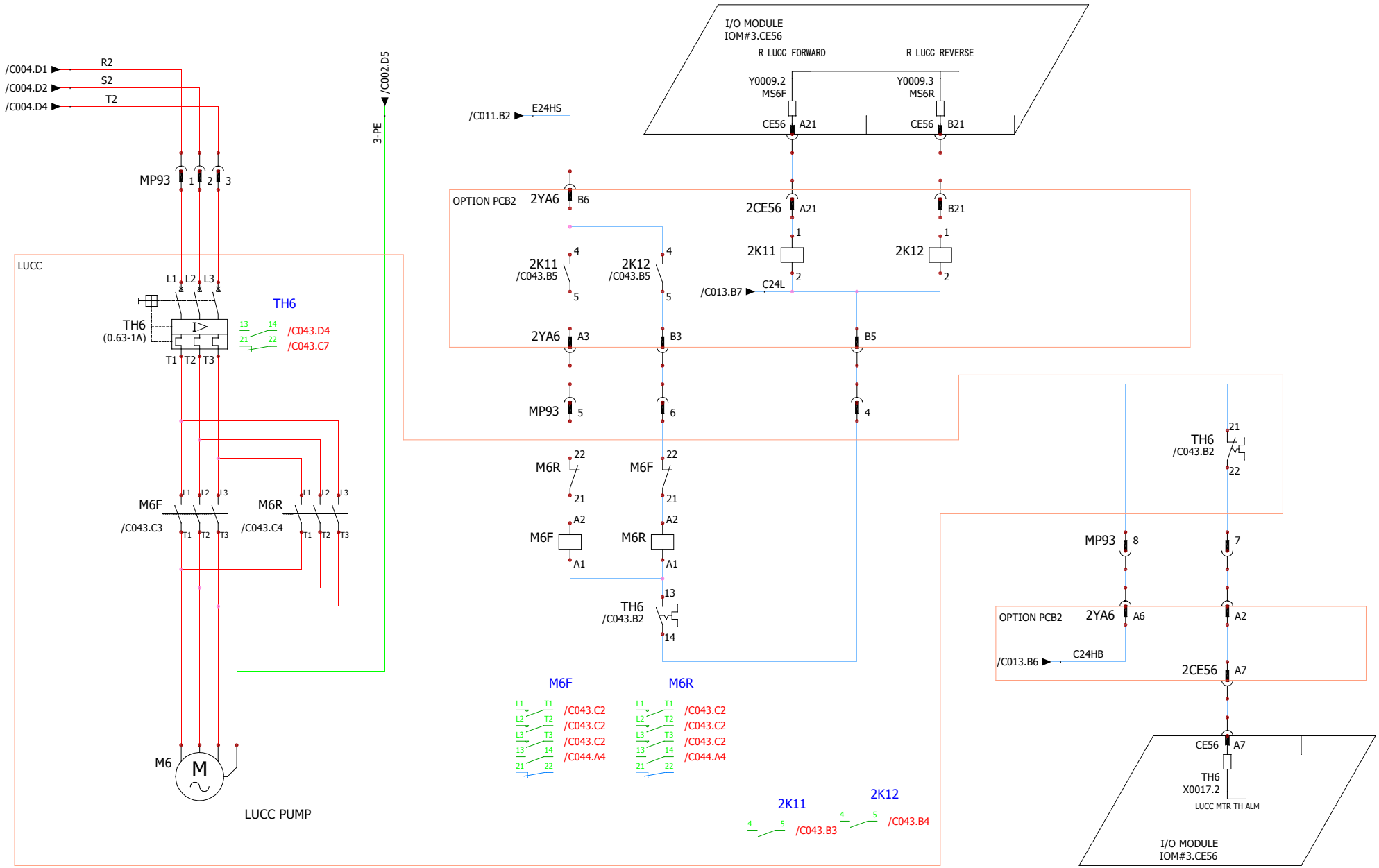


TH6A SETTING VALUE	
400V/50HZ	2.9A
400V/60HZ	2.9A

REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

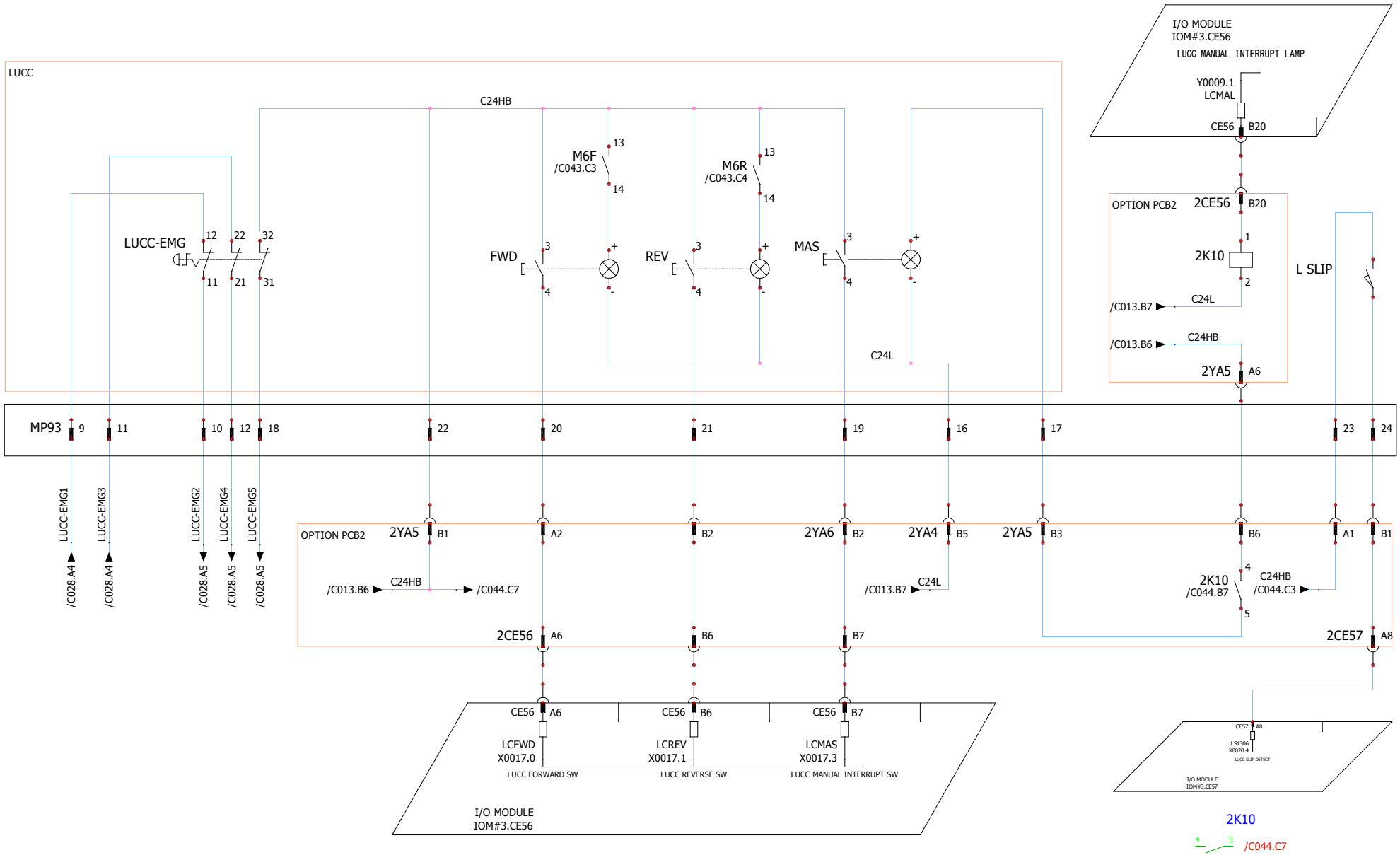
MAKINO
MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
LUCC DRUM (OPT)	V2631I-C1E-Y	Ver.	B
		PAGE	C042



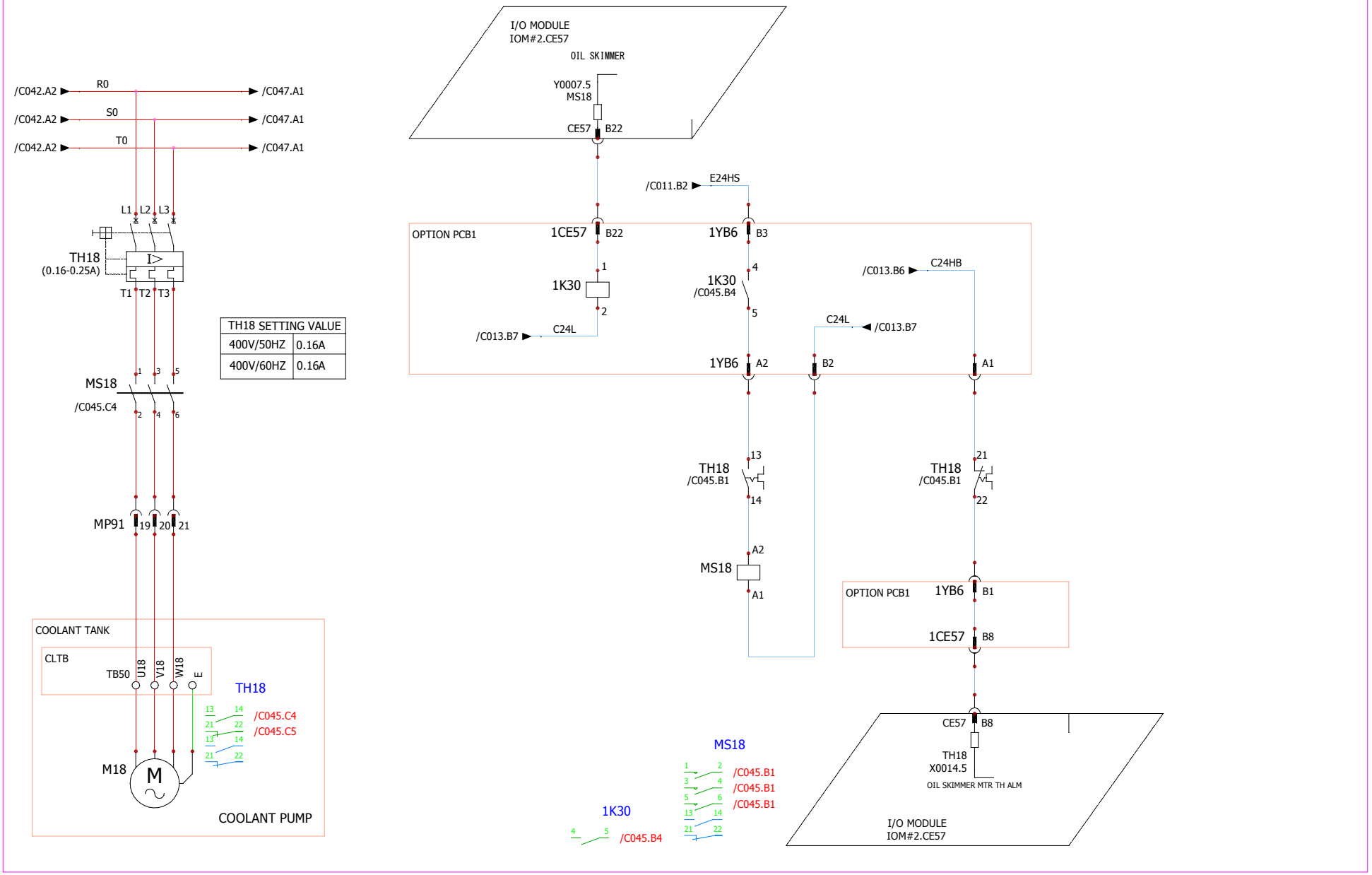
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	LIFT UP CHIP CYR PWR (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C043
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

F28 - LUCC



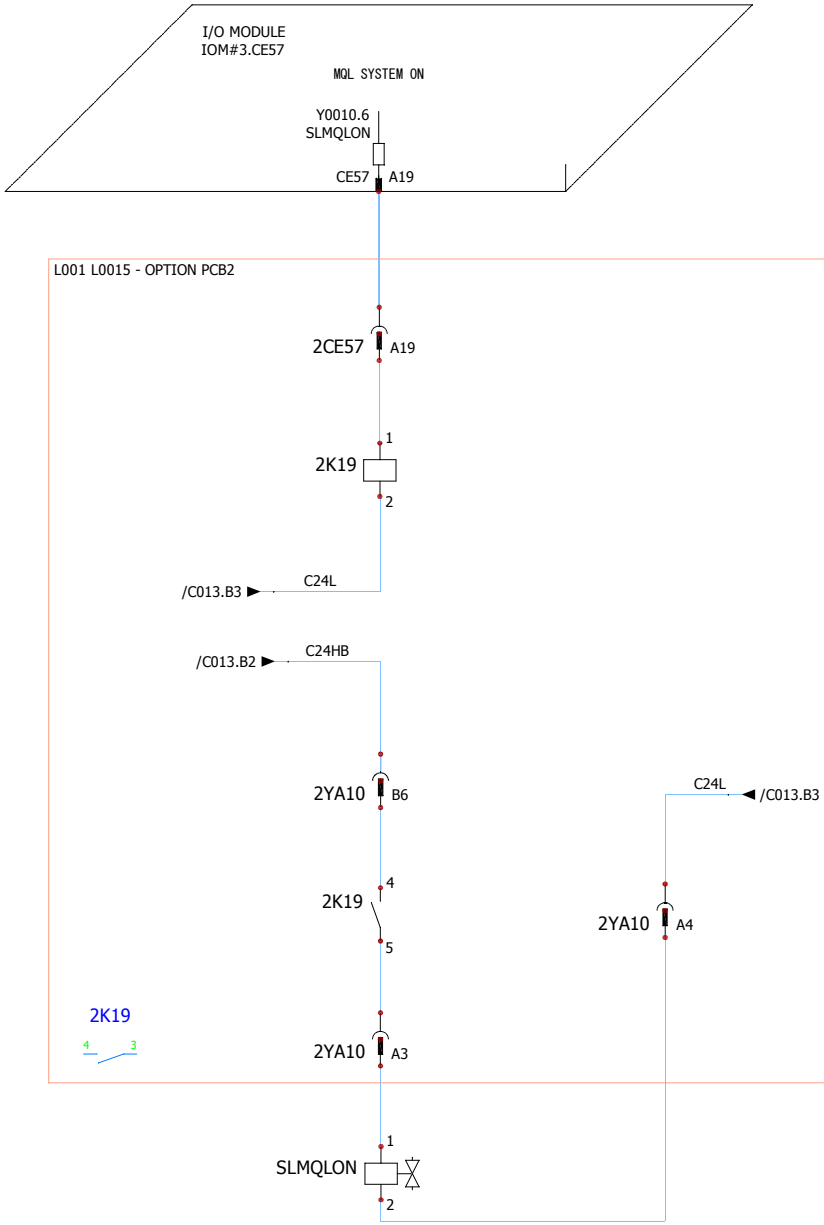
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	LIFT UP CHIP CYR CONTROL (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C044
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

F29 - OIL SKIMMER



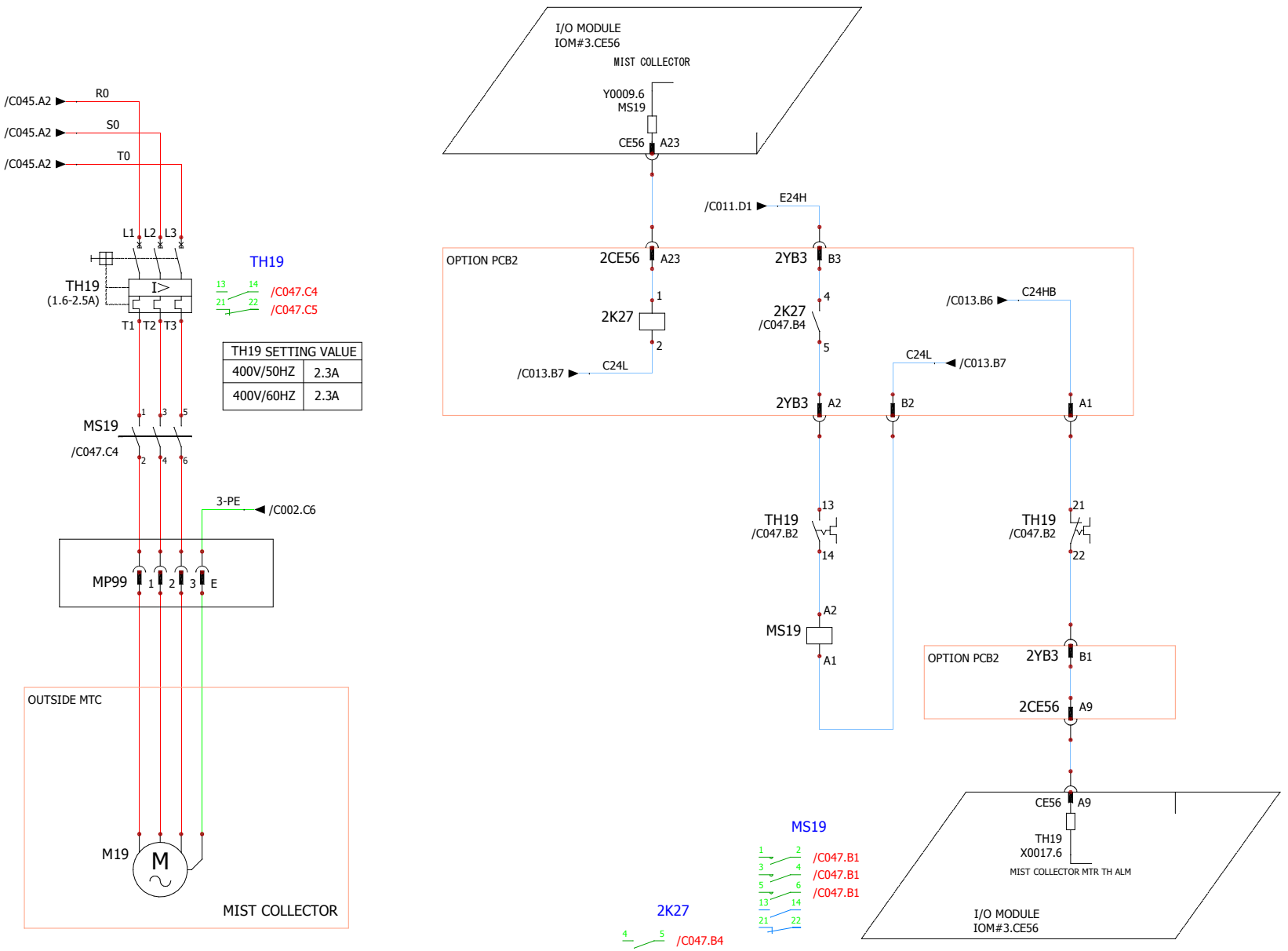
TH18 SETTING VALUE	
400V/50HZ	0.16A
400V/60HZ	0.16A

REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	OIL SKIMMER (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C045
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	OIL MIST (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C046
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

F54 - MIST COLLECTOR

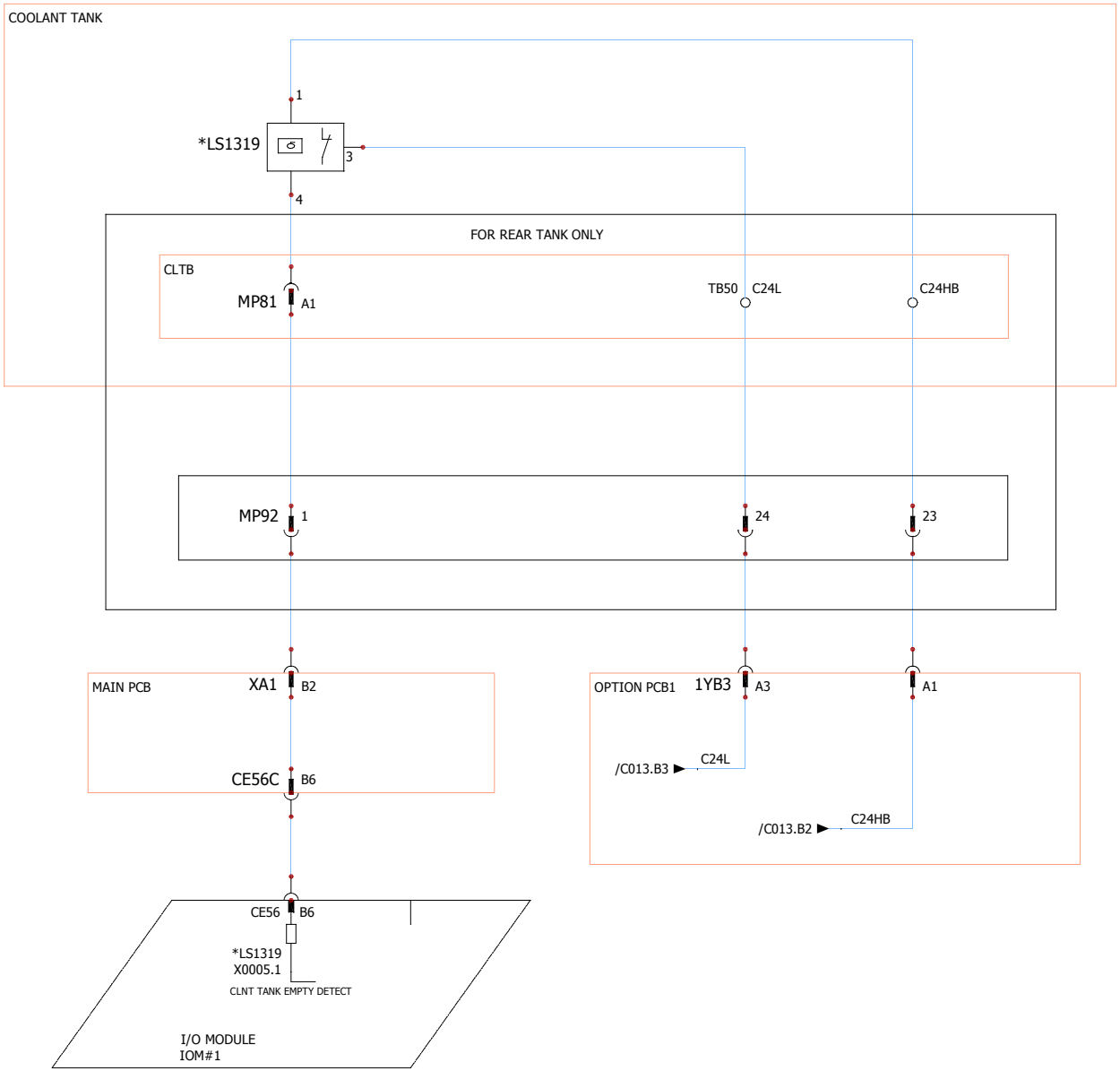


TH19 SETTING VALUE	
400V/50HZ	2.3A
400V/60HZ	2.3A

REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

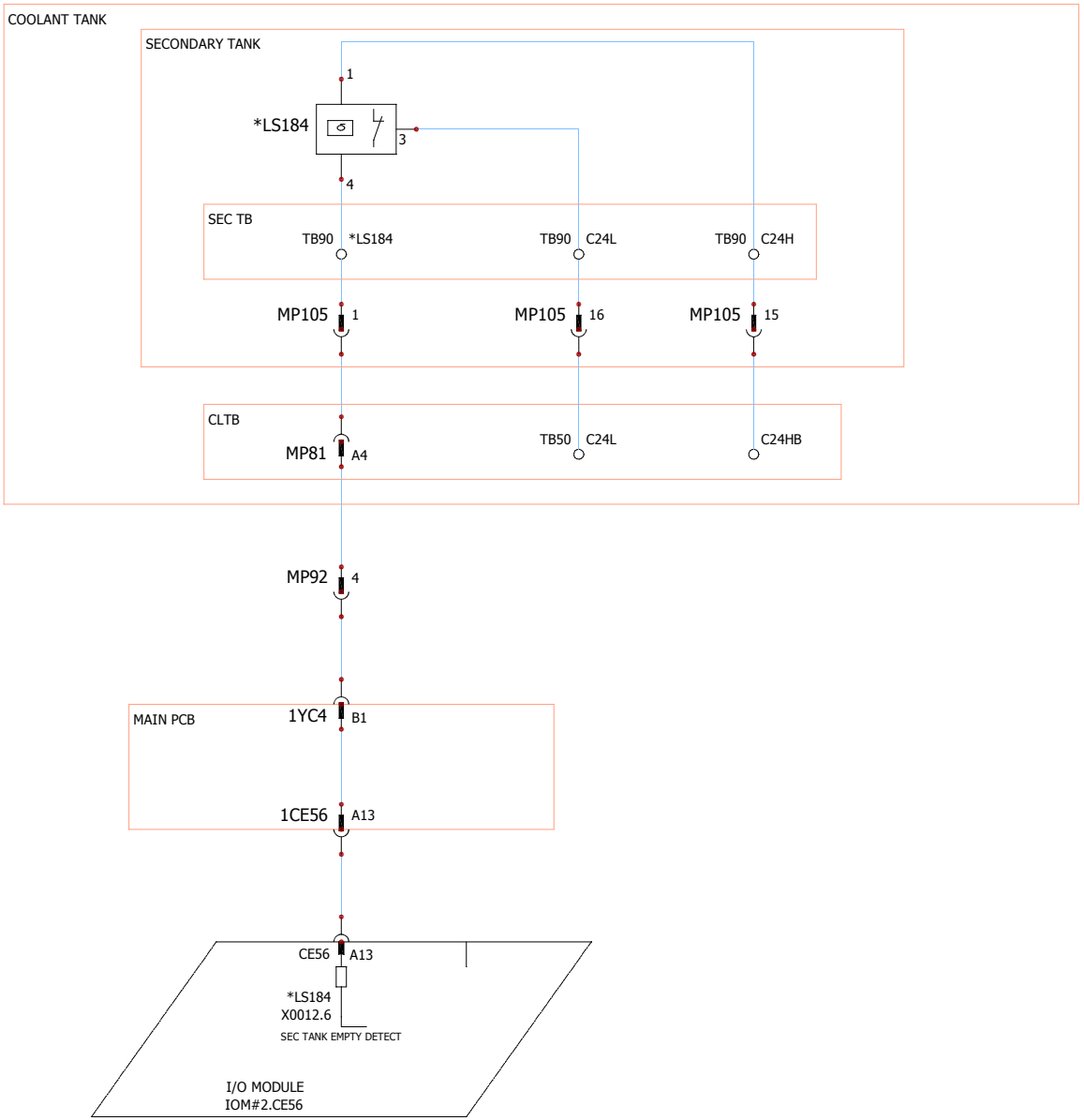
MAKINO
MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
MIST COLLECTOR (OPT)	V2631I-C1E-Y	Ver.	B
		PAGE	C047

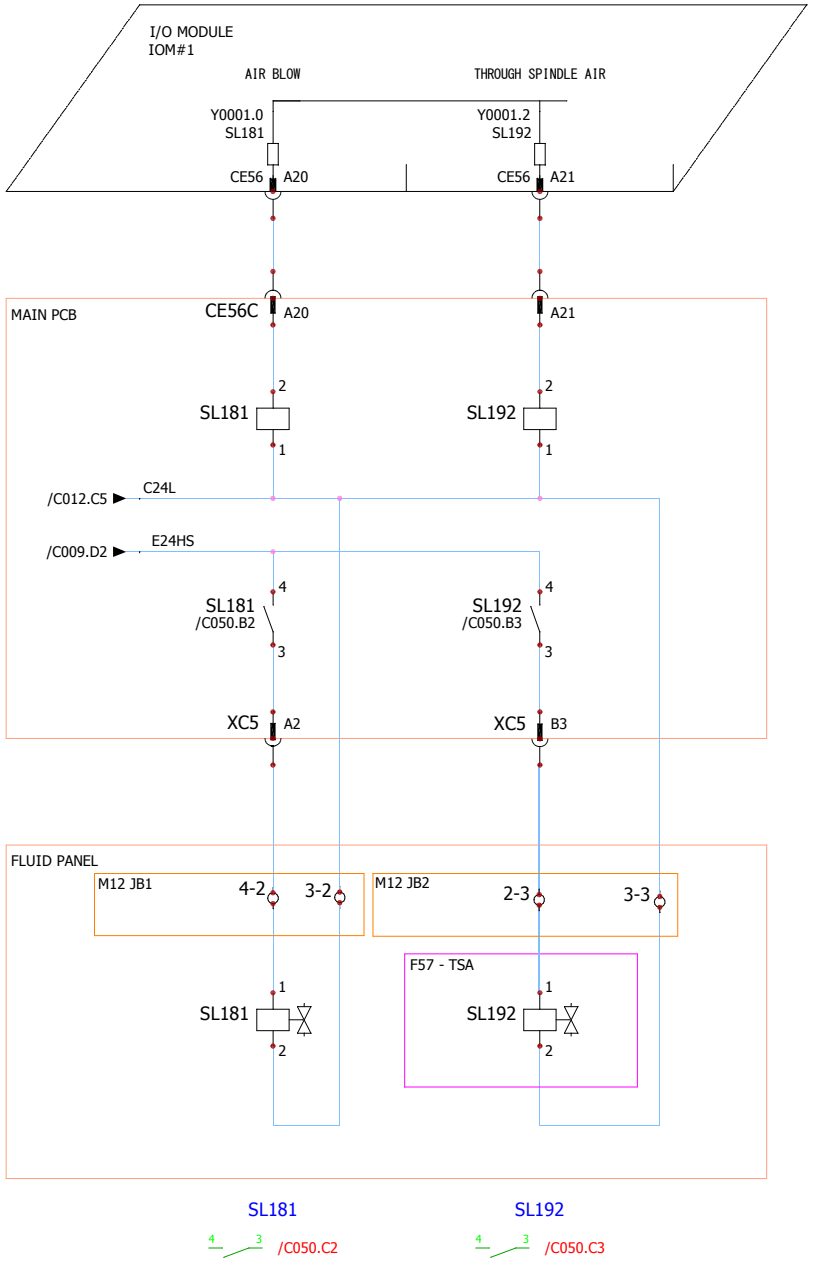


REVISION						
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TITLE: COOLANT LEVEL SENSOR - MAIN TANK	DRAWING No.: V2631I-C1E-Y
	----.--.--	CHECKED	AVINASH			MACHINE: F3/ F5 Pro 6
	----.--.--	DRAWN	ANISHA			Ver.: B
	----.--.--	DATE	28/6/2018			PAGE: C048

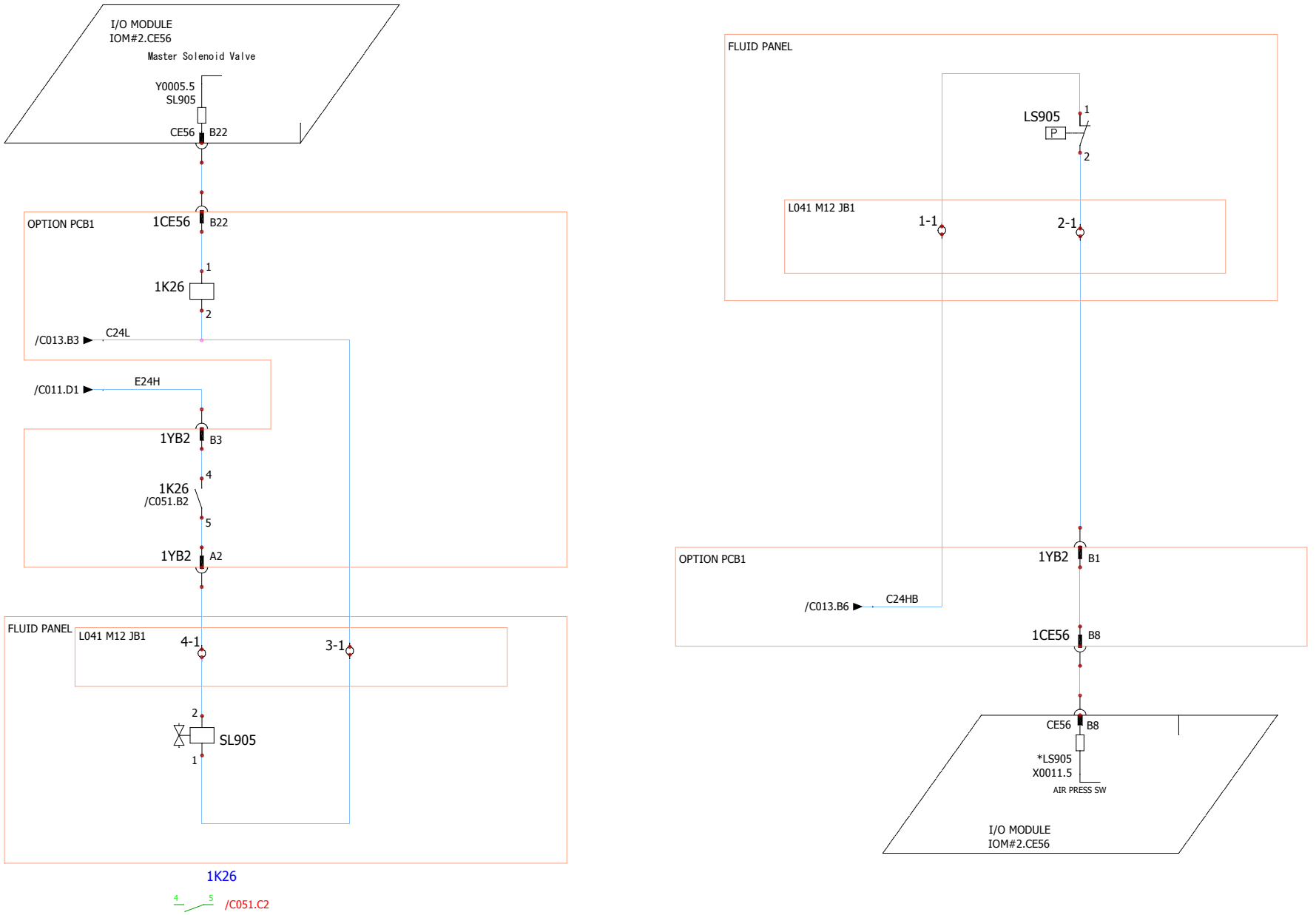
F26 - SECONDARY TANK



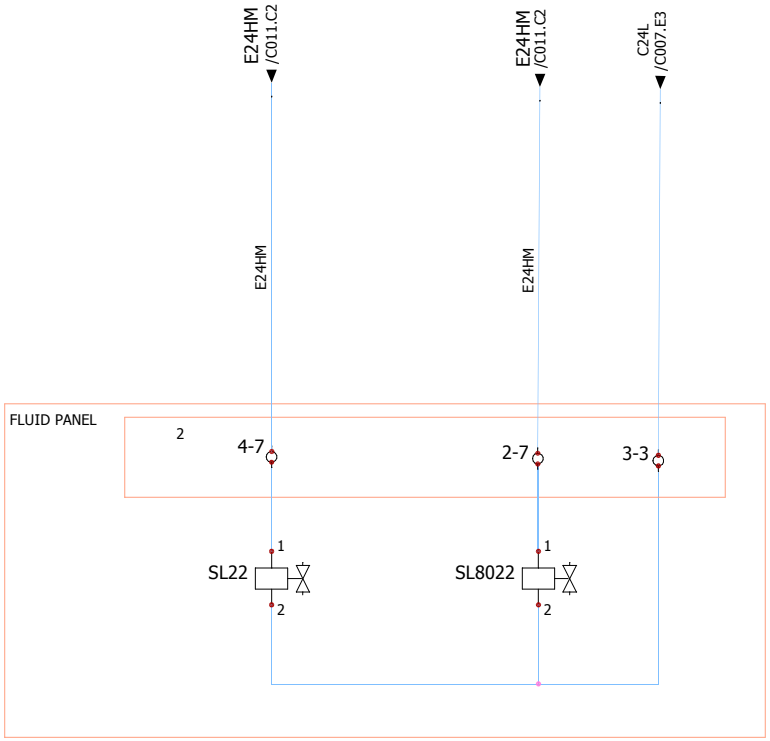
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	COOLANT LEVEL SENSOR - SEC TANK (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C049
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



REVISION							
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE F3/ F5 Pro 6
	----.--.--	CHECKED	AVINASH		AIR BLOW & TSA (OPT)	V2631I-C1E-Y	Ver. B
	----.--.--	DRAWN	ANISHA				PAGE C050
	----.--.--	DATE	28/6/2018				

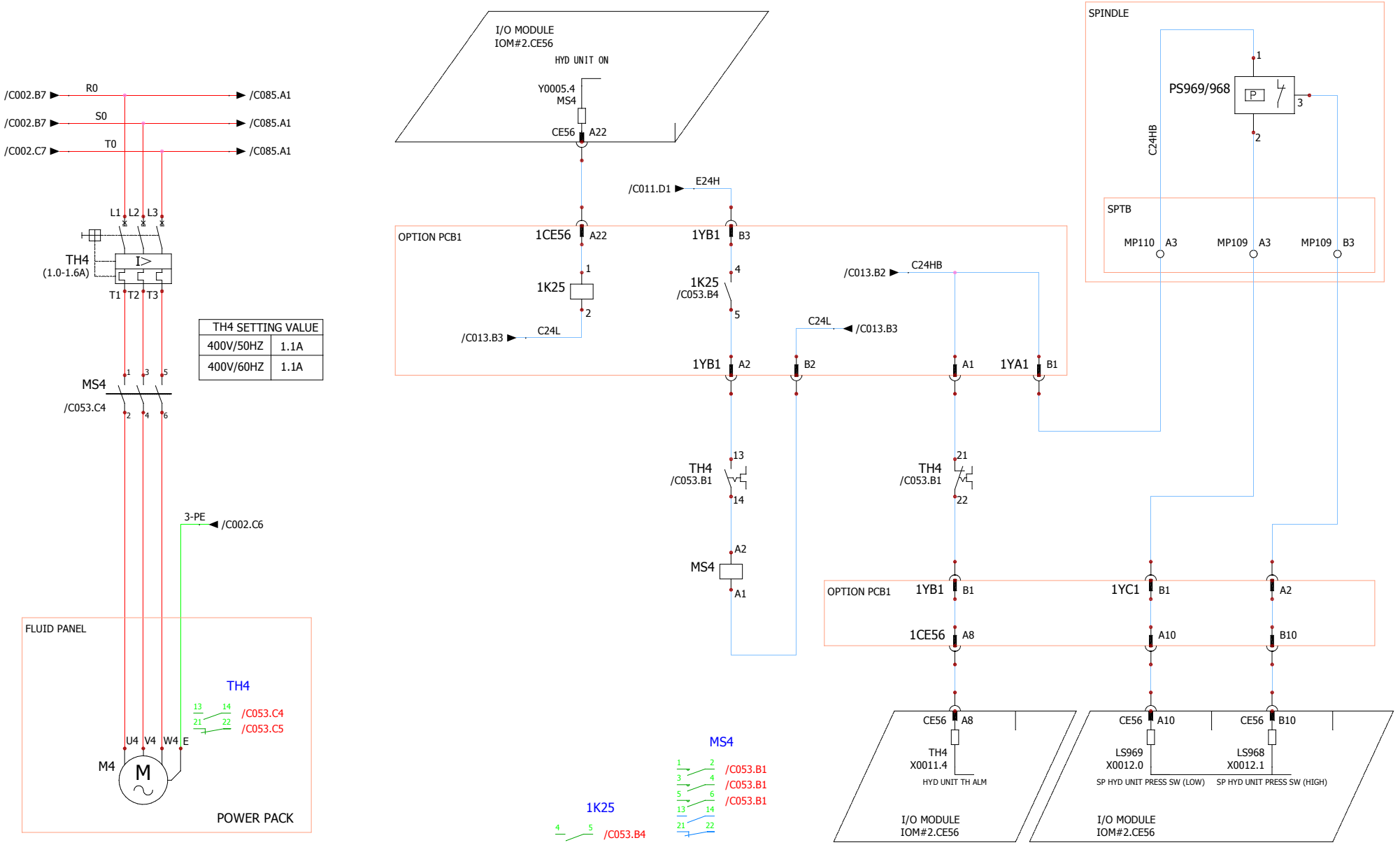


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	MASTER SOLENOID	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C051
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	ATC MASTER SOLENOID	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C052
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

MAKINO
MAKINO ASIA PTE LTD



REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

MAKINO
MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
POWER PACK (TOOL CL/UCL)	V2631I-C1E-Y	Ver.	B
		PAGE	C053

F32 - COOLANT CHILLER

I/O MODULE
IOM#1
COOLANT CHILLER ON
Y0001.6
CLOMON
CE56 A23

MAIN PCB
CE56C A23
XA2 B7

CLOMON 8 1

/C055.C2 ▶ C24L

/C004.D2 R2
/C004.D3 S2
/C004.D4 T2
3-PE

MP91 22 23 24 PE
FOR REAR TANK USE ONLY

CLTB
TB50 R2 S2 T2 PE

S2

COOLANT TEMP CONTROLLER
R S T PE

OUTSIDE MTC
THERMOSTAT
MTH1 MTH2

RE2 RE1

FOR REAR TANK ONLY
CLTB
MP81 B2 B1

MP92 12 11

CLOMON /C054.D2 2 4 3

12 11
A2
TB50 C24HB

2

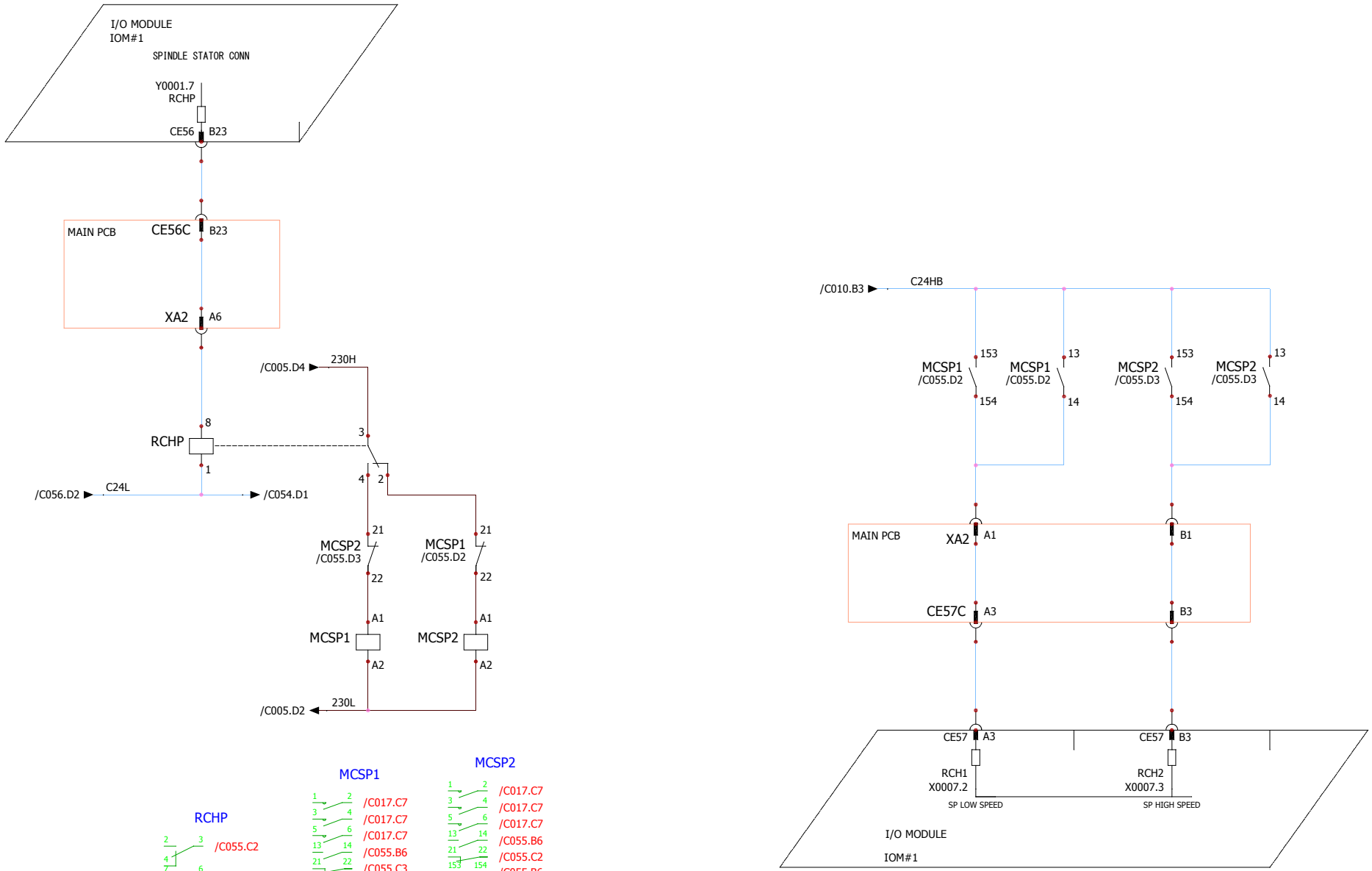
MAIN PCB
XA1 A3
CE56C A7

CE56 A7
*OMAL2
X0005.2
COOLANT CHILLER UNIT ALARM

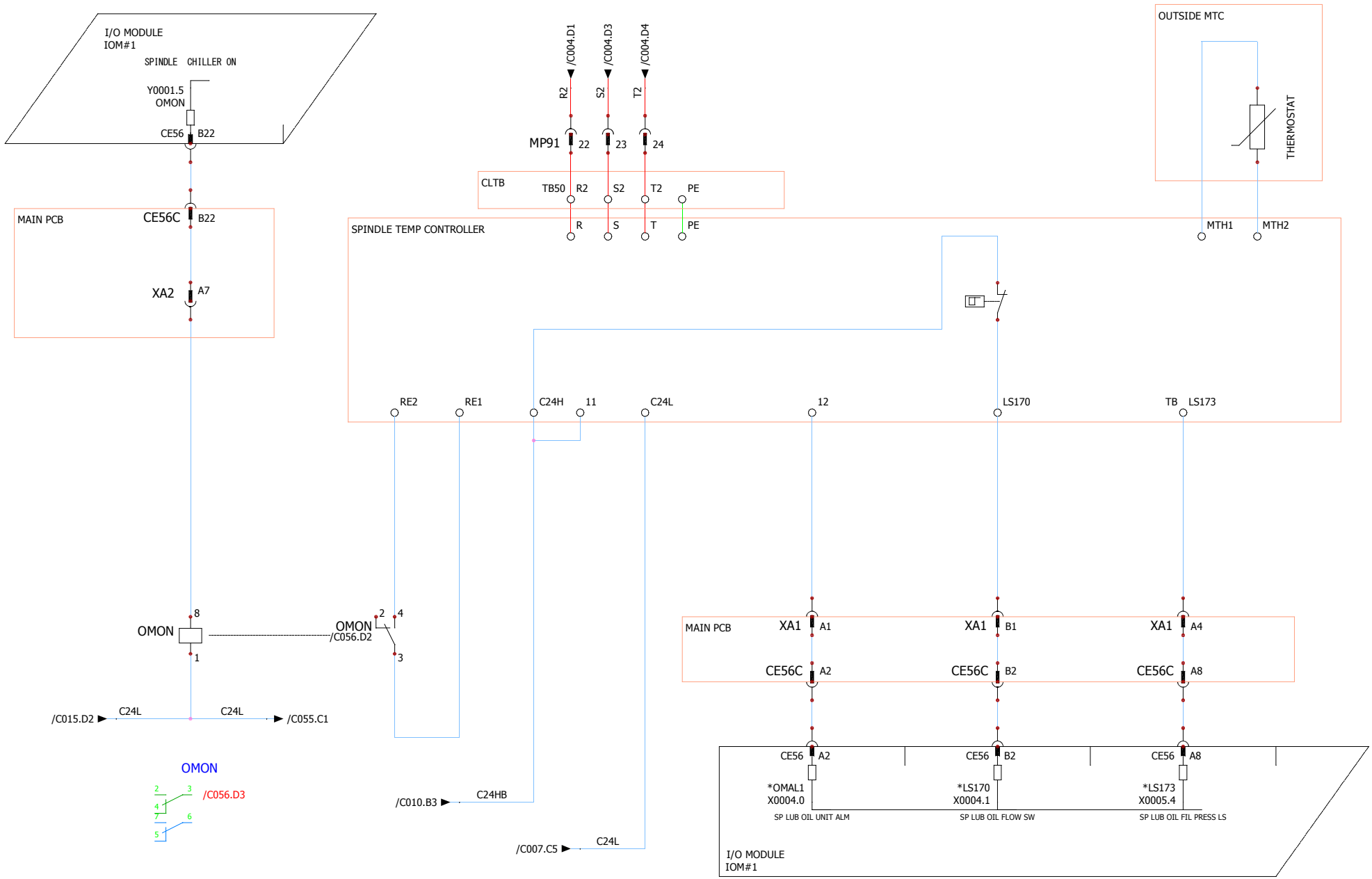
I/O MODULE
IOM#1

CLOMON
/C054.D3

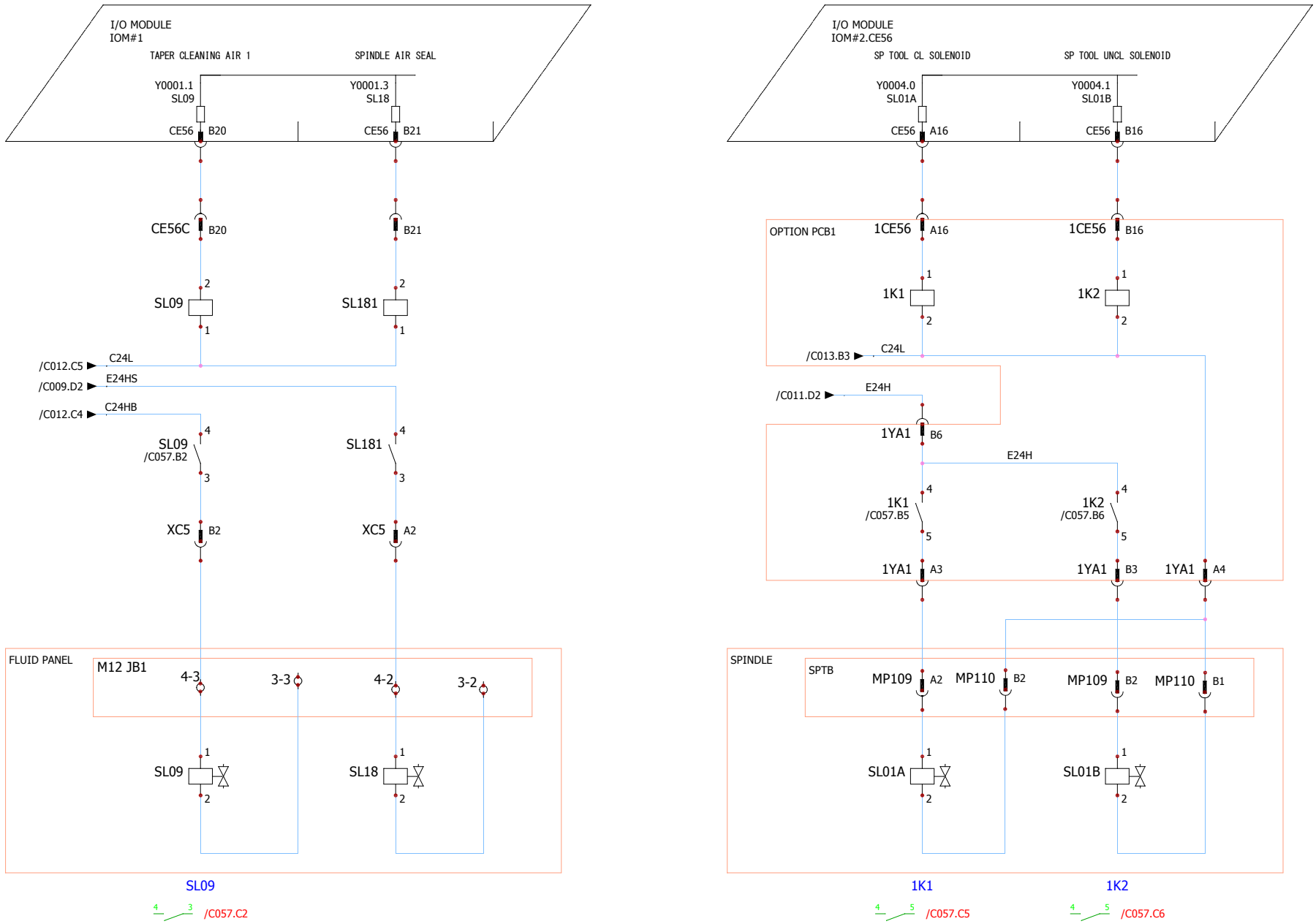
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	COOLANT CHILLER (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C054
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



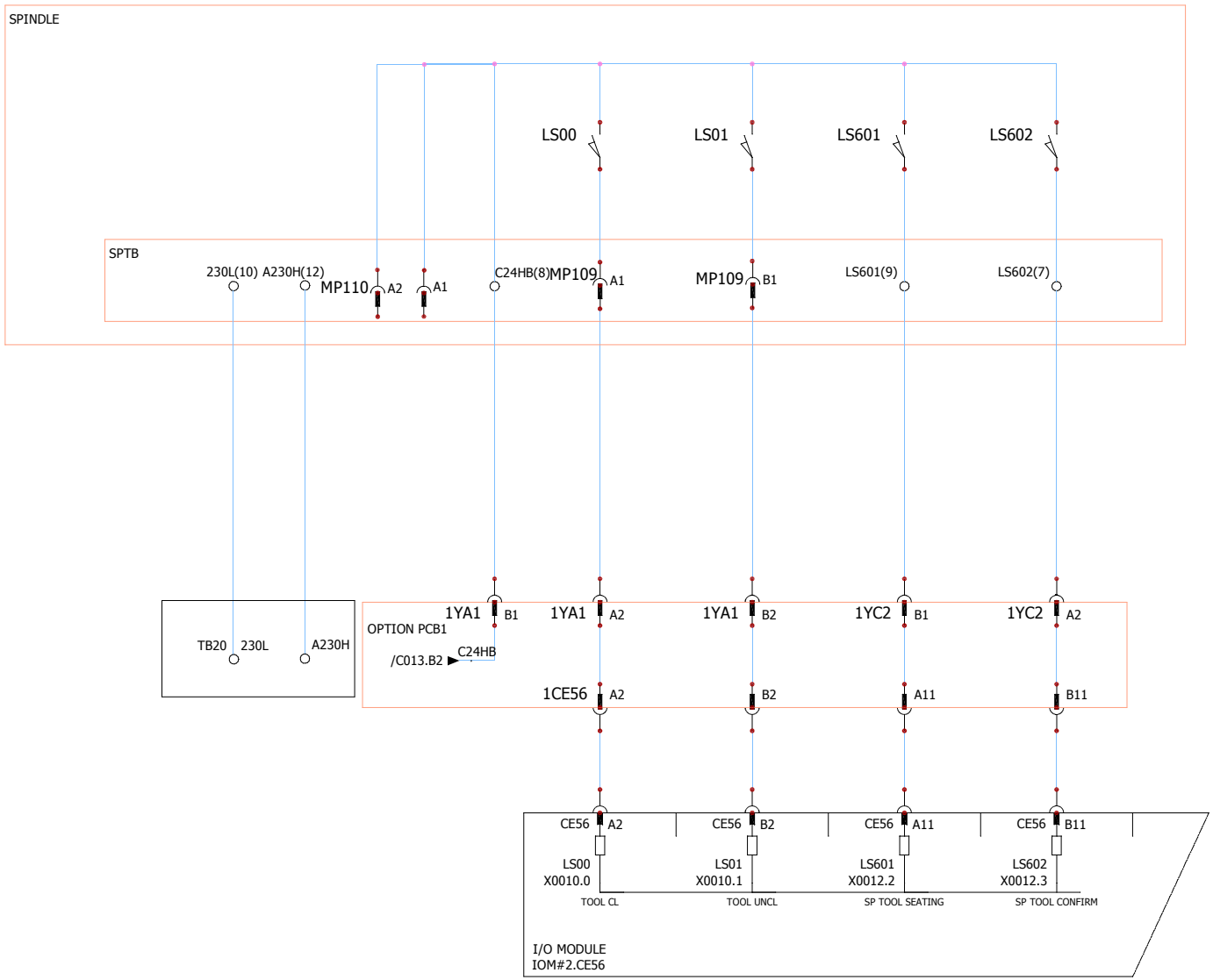
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	SPINDLE SWITCHING CONTROL	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C055
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	SPINDLE TEMPERATURE CONTROLLER	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C056
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

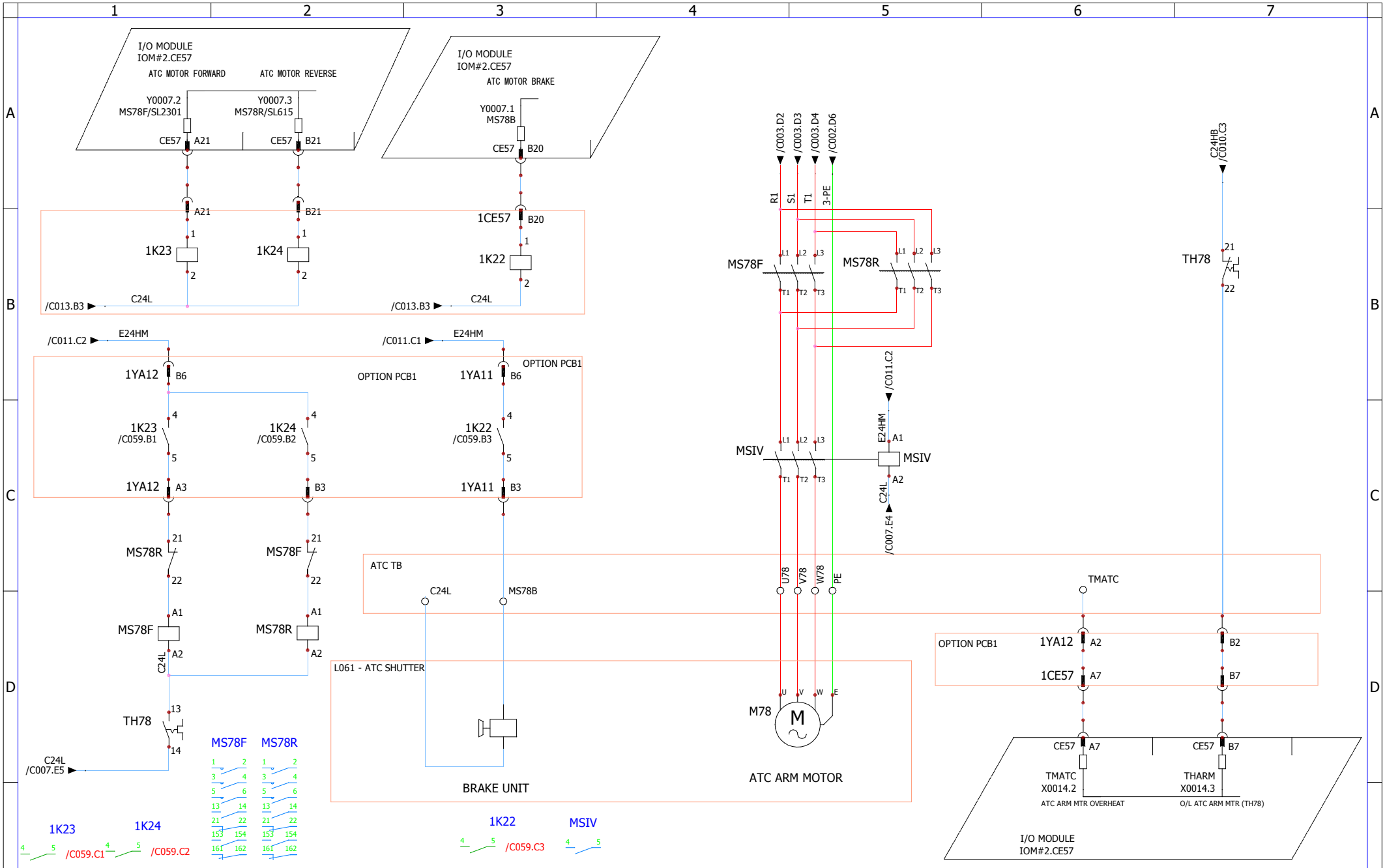


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TOOL CLAMP/UNCLAMP SOL	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C057
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

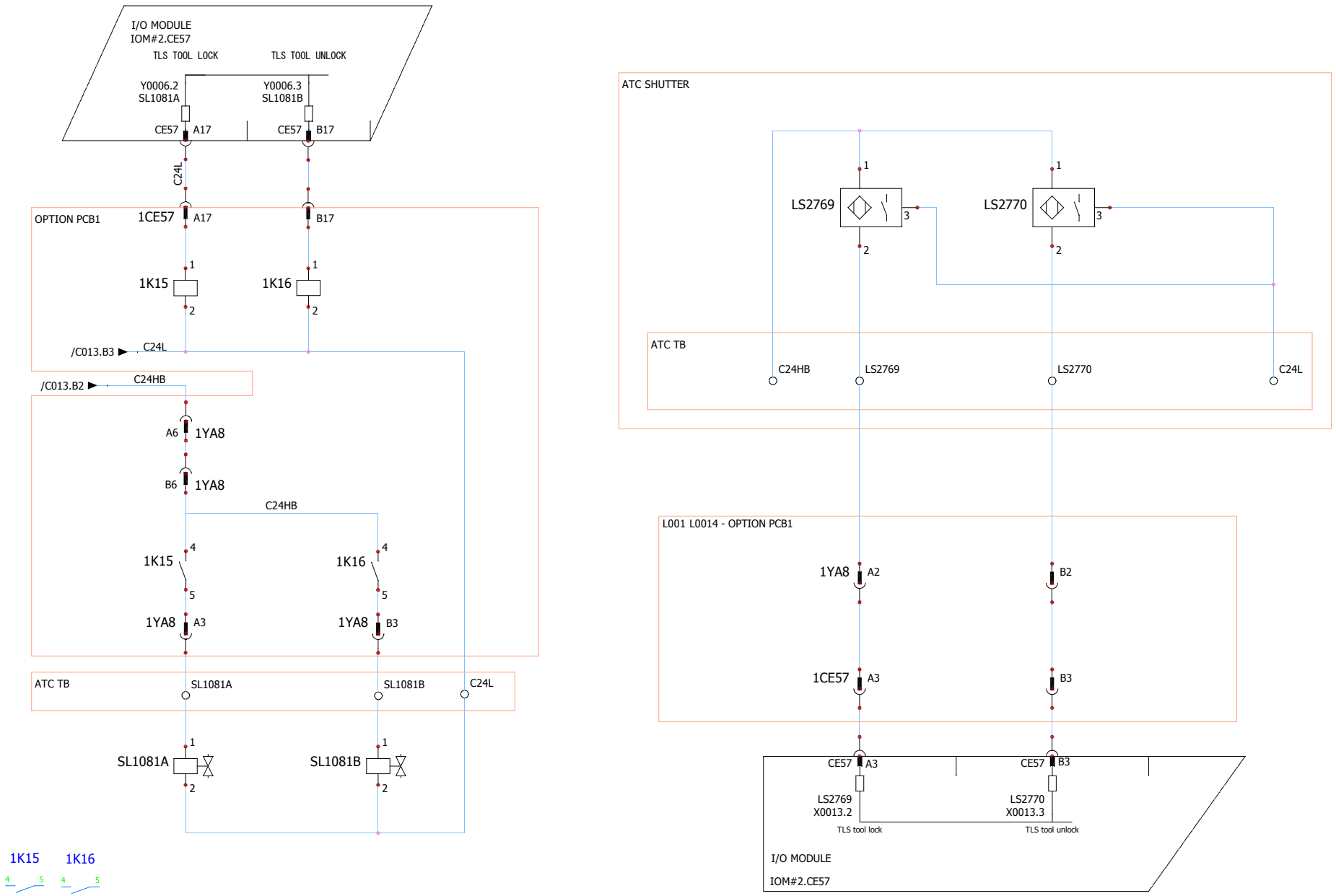


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	TOOL CLAMP/UNCLAMP LIMIT SW	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C058
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

MAKINO
MAKINO ASIA PTE LTD

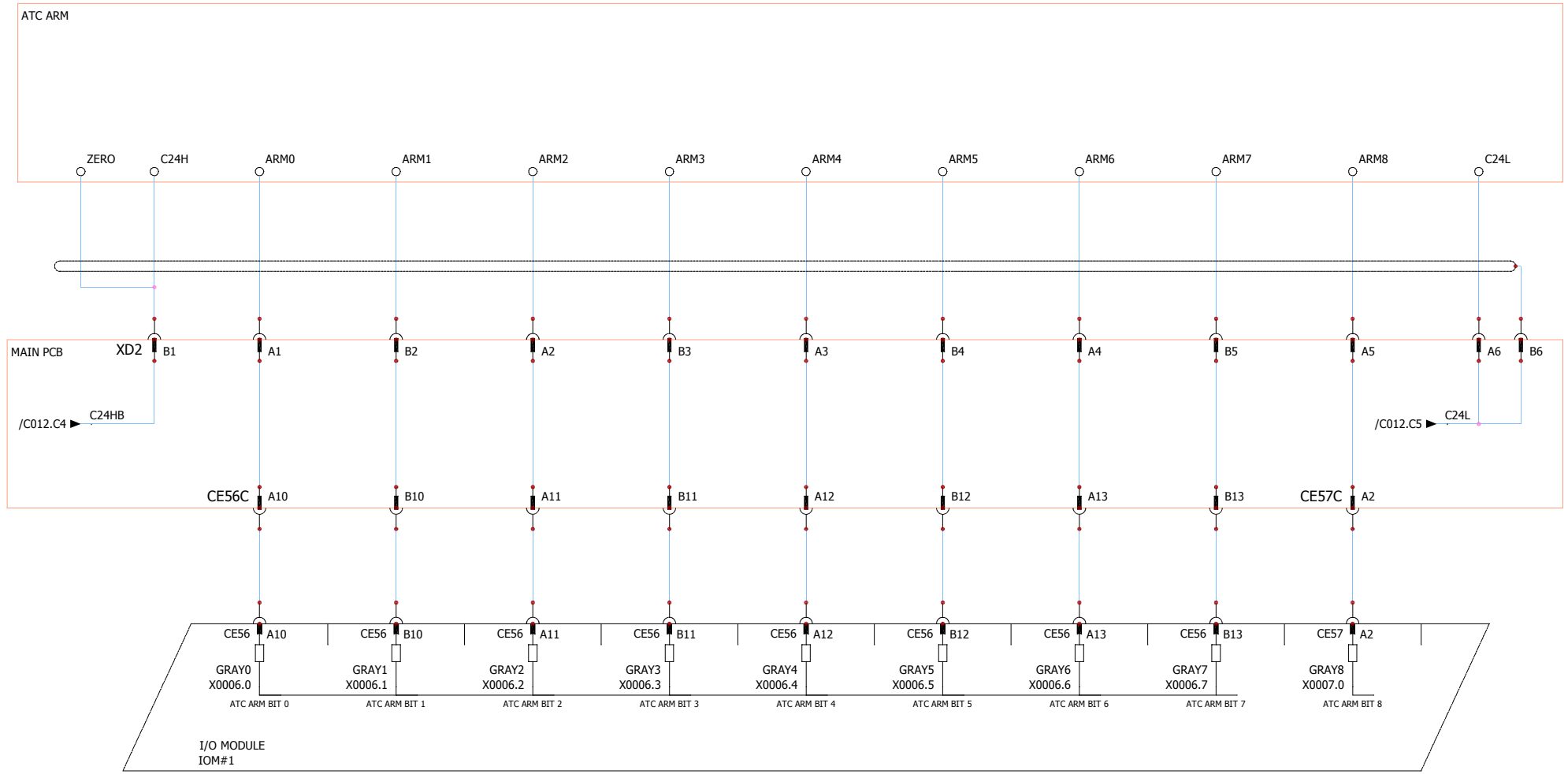


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	ATC ARM MOTOR	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C059
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

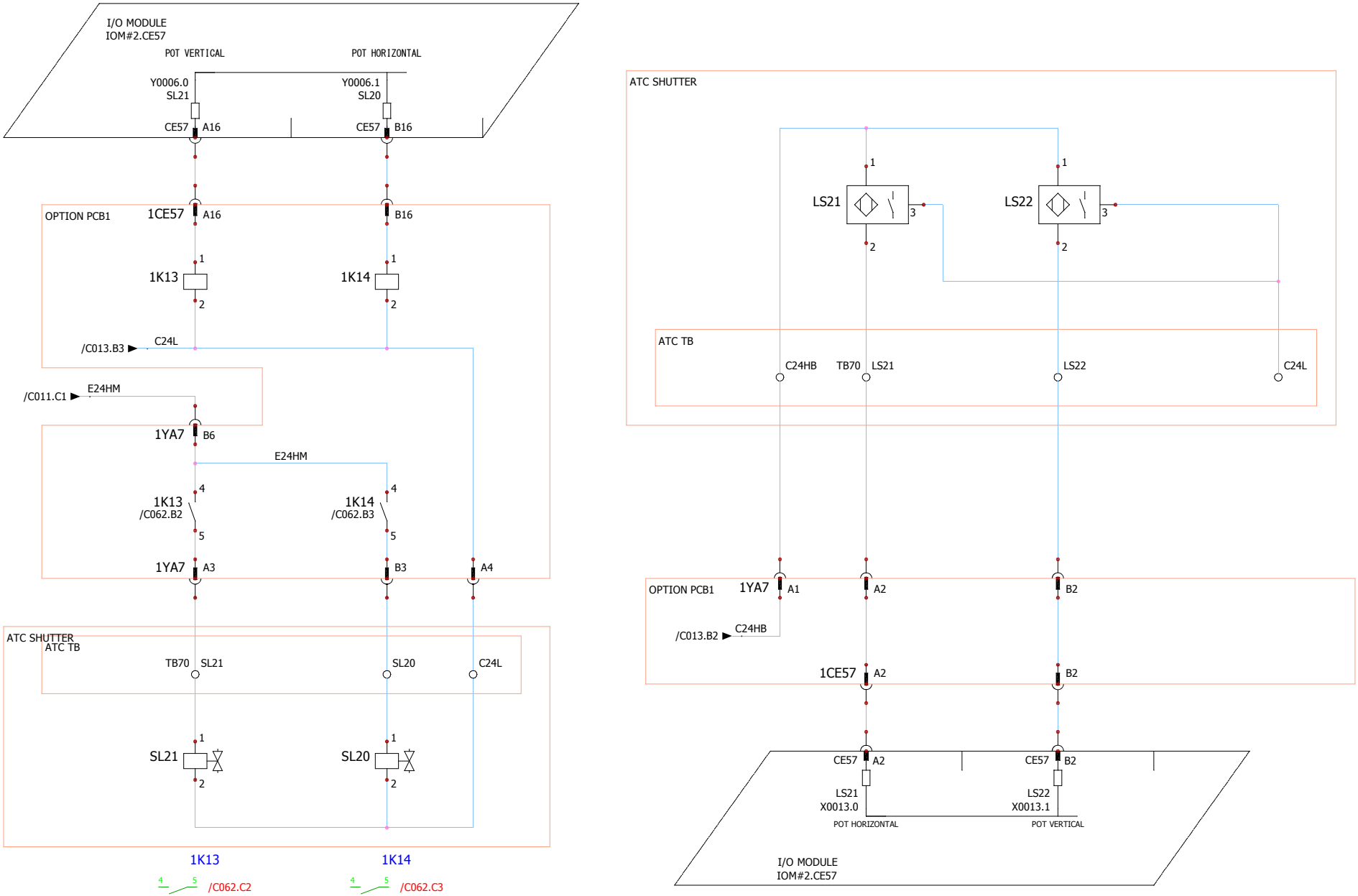


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	ATC TOOL EXTRACTOR	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C060
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

MAKINO
MAKINO ASIA PTE LTD

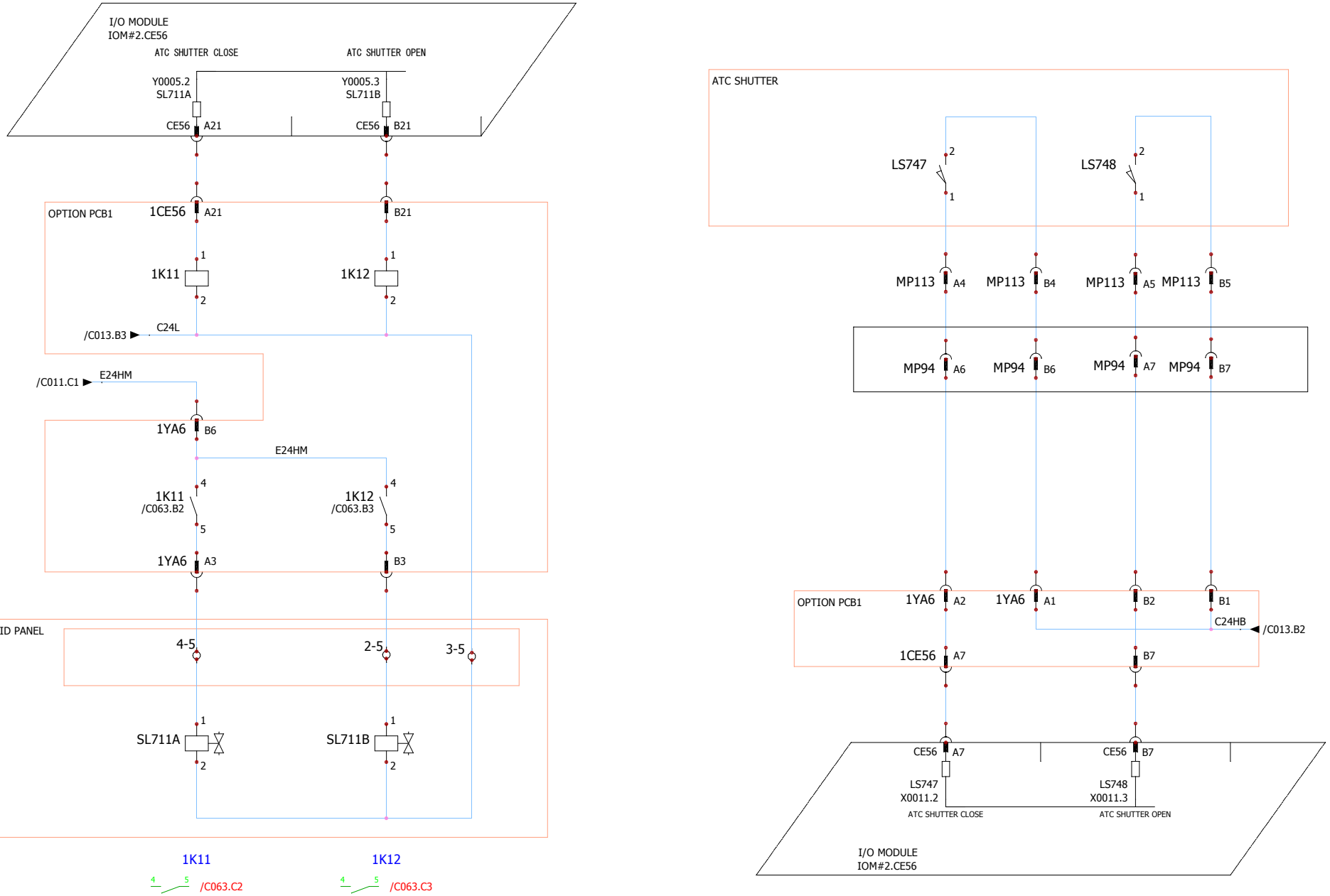


REVISION						
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TITLE: ATC ARM ENCODER	DRAWING No.: V2631I-C1E-Y
	----.--.--	CHECKED	AVINASH			MACHINE: F3/ F5 Pro 6
	----.--.--	DRAWN	ANISHA			Ver.: B
	----.--.--	DATE	28/6/2018			PAGE: C061

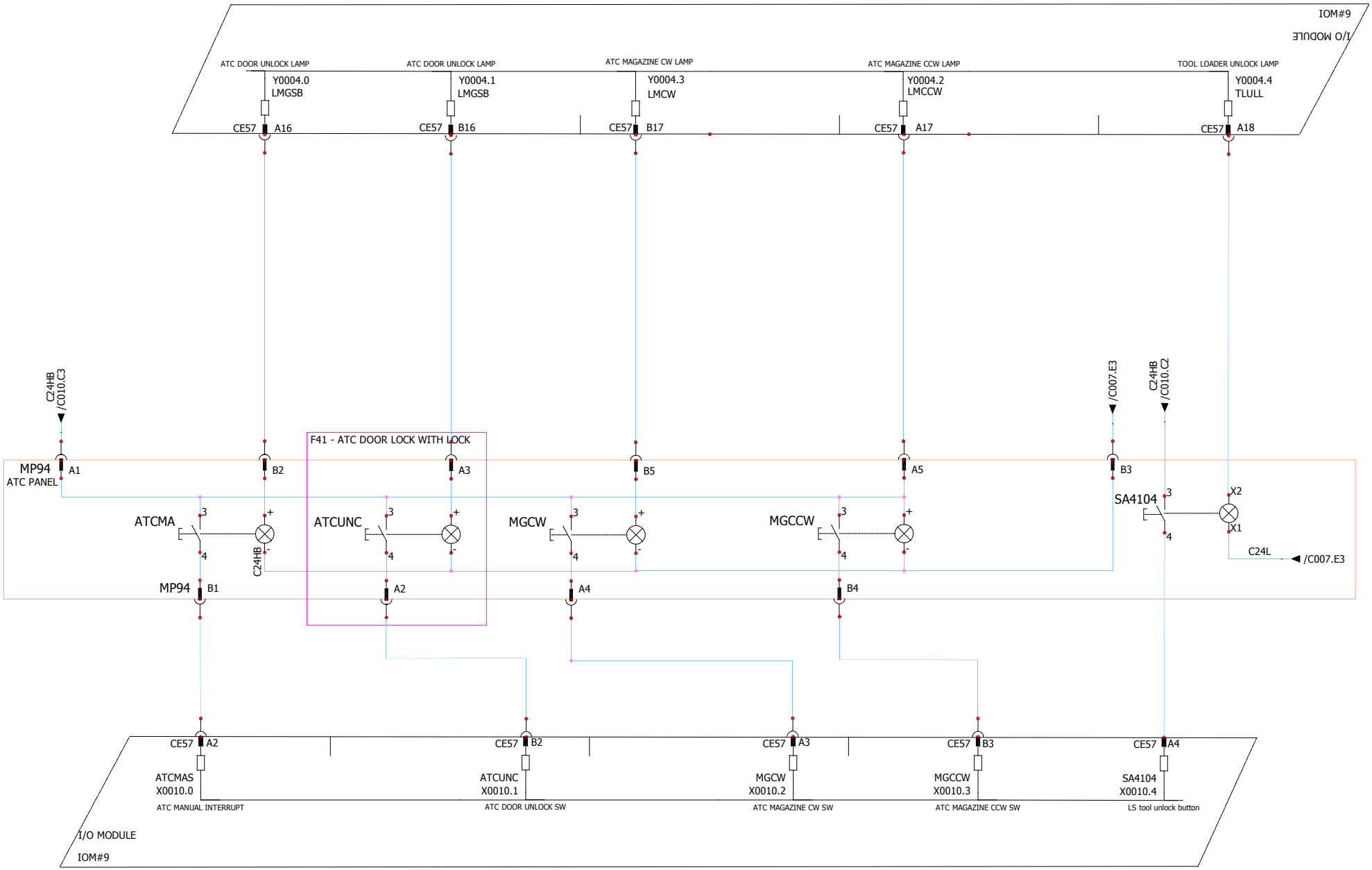


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	ATC POT CONTROL	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C062
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

MAKINO
MAKINO ASIA PTE LTD

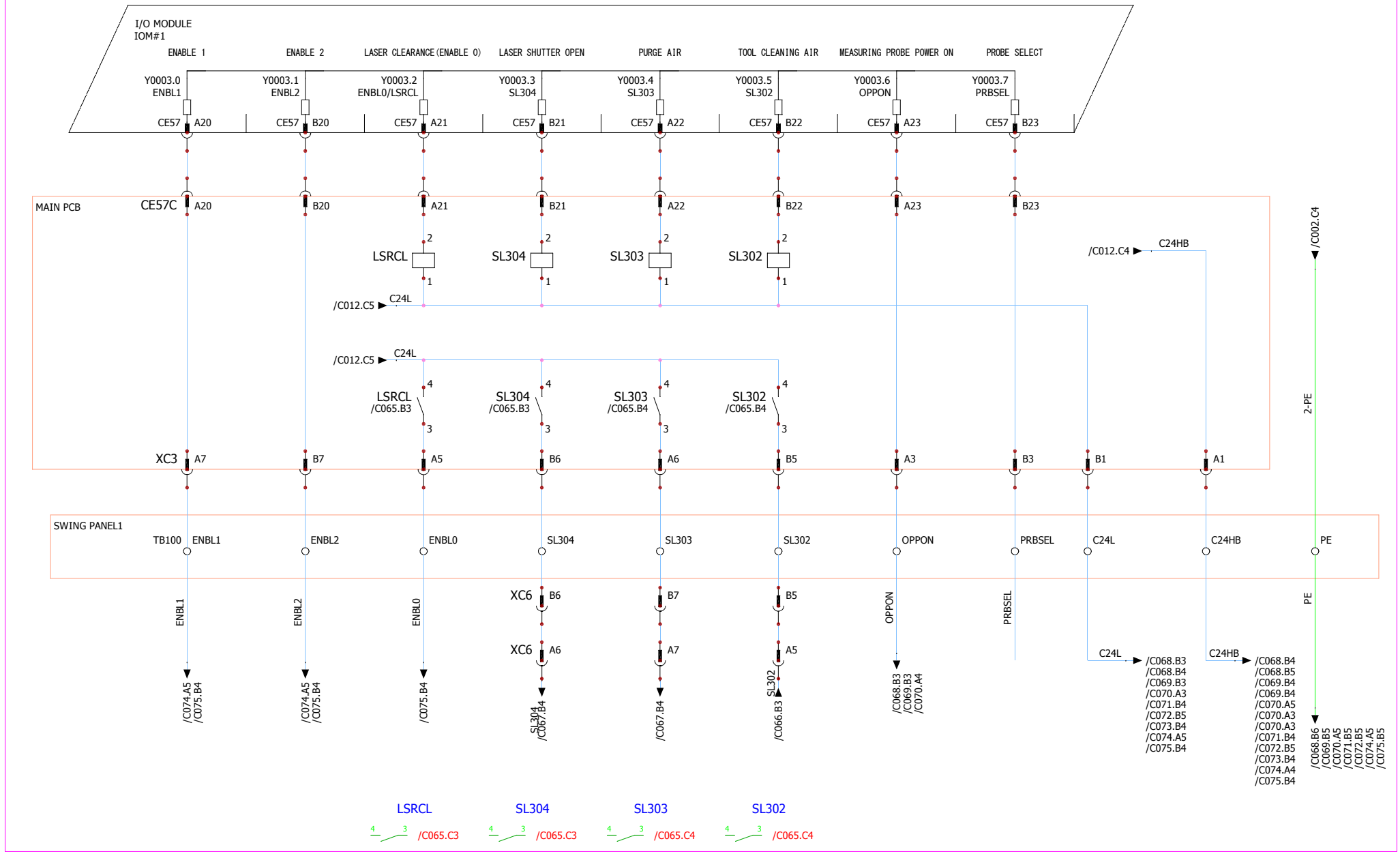


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	ATC SHUTTER	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C063
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



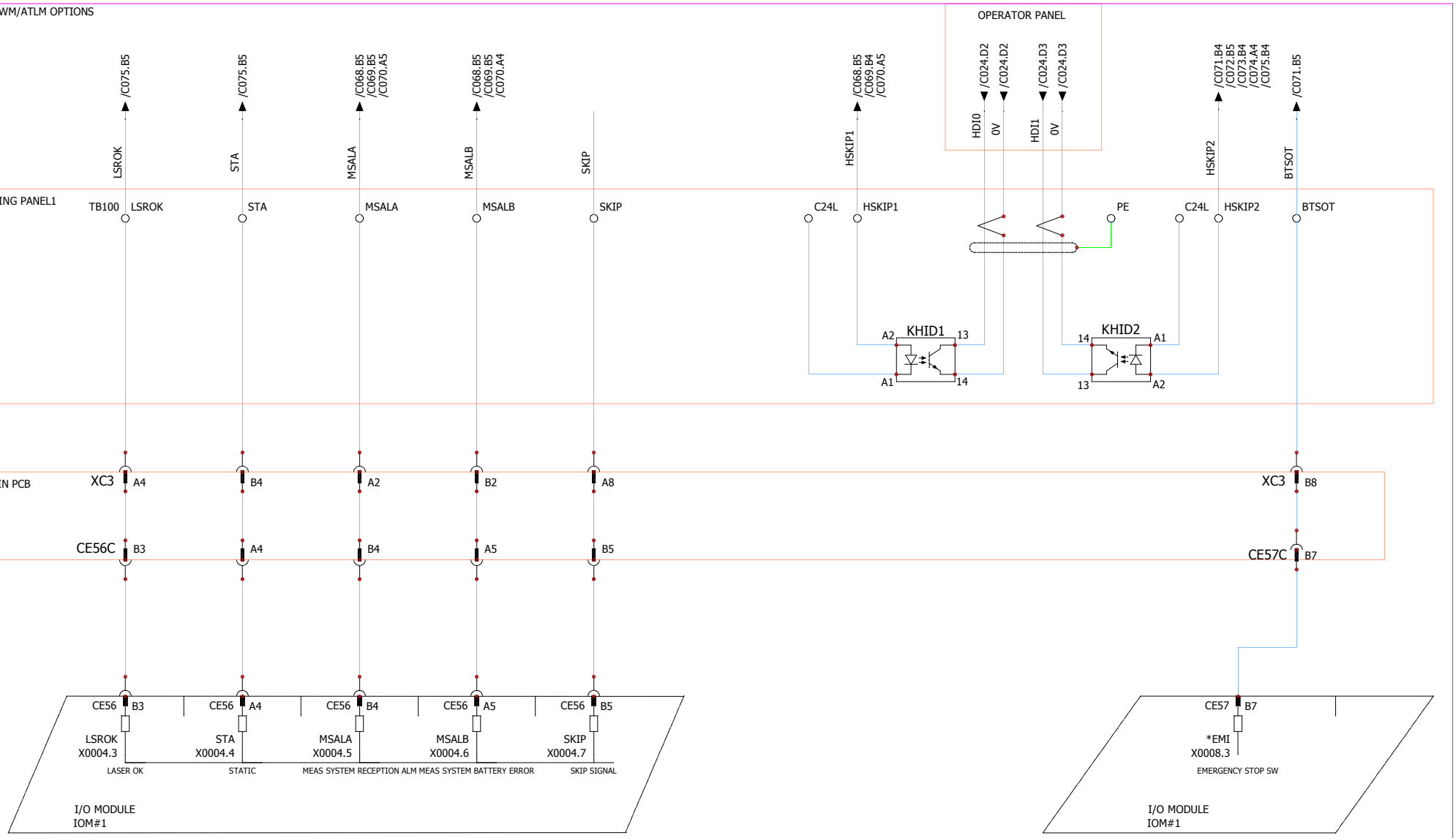
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	ATC PANEL	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C064
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

F42 - AWM/ATLM OPTIONS



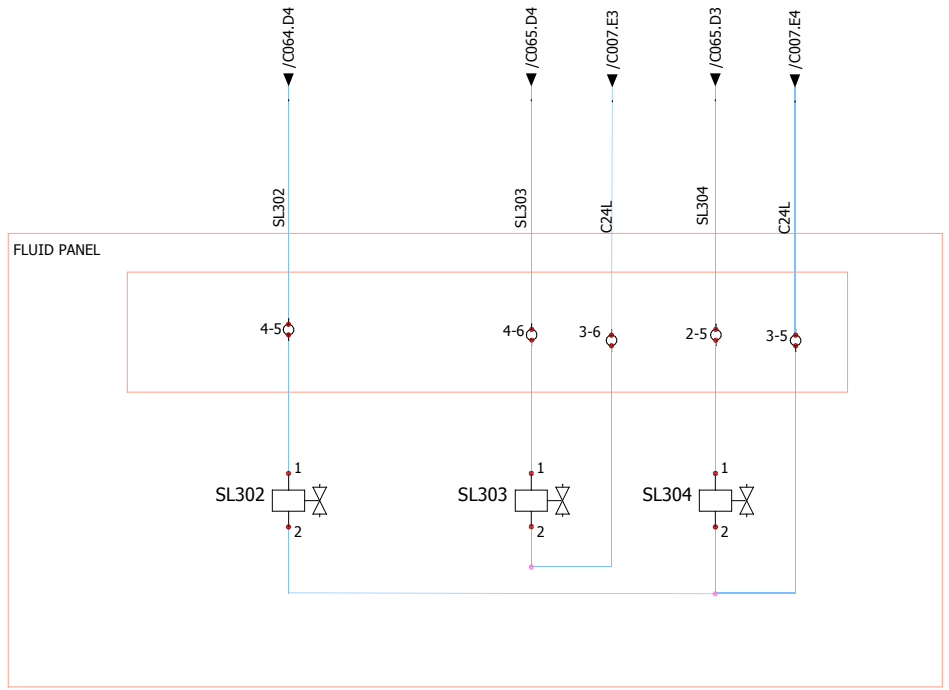
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	AWM & ATLM INTERFACE (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C065
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

F42 - AWM/ATLM OPTIONS



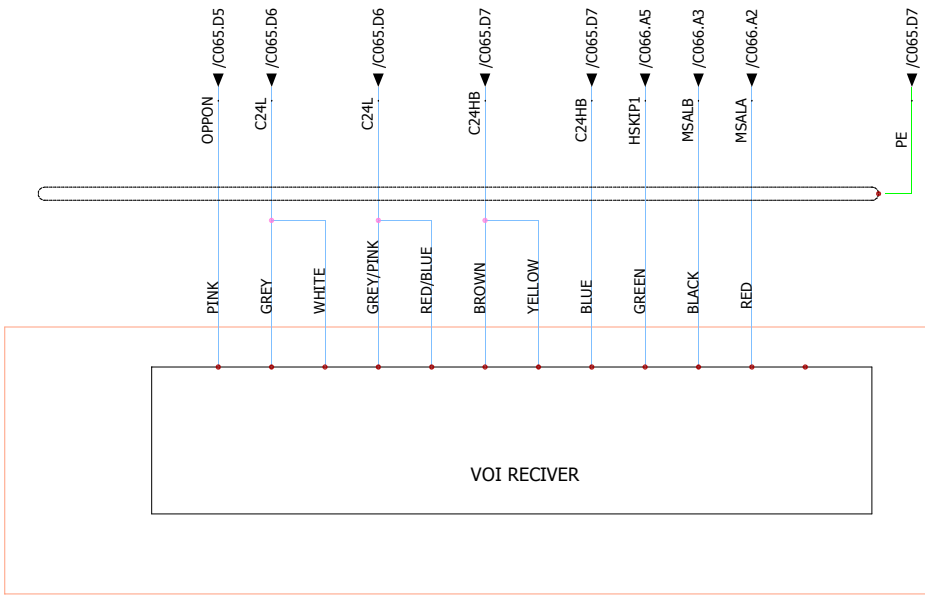
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	AWM & ATLM INTERFACE (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C066
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

F42 - AWM/ATLM OPTIONS



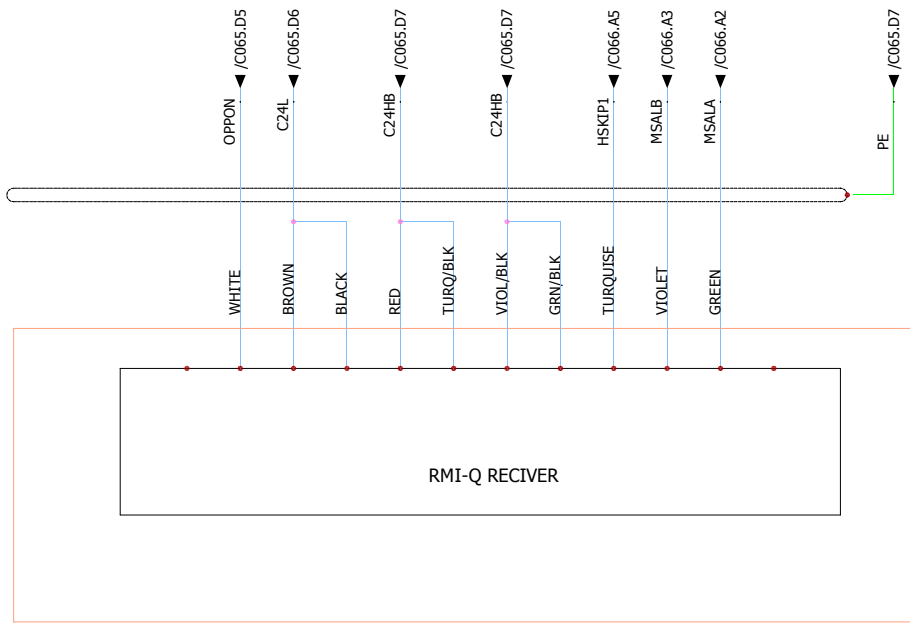
REVISION					MAKINO MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH		ATLM SOLENOIDS (OPT)	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	C067
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					

F42 - AWM/ATLM OPTIONS



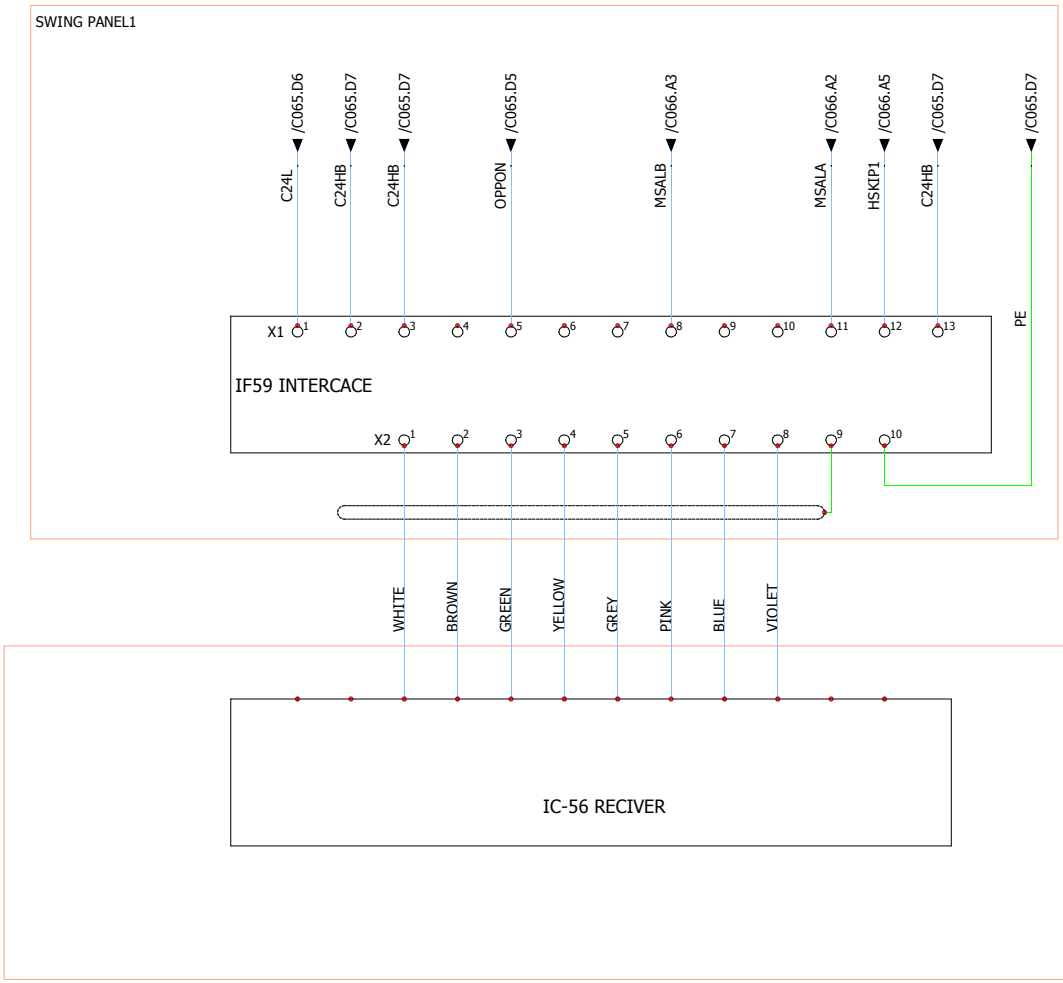
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	AWM MARPOSS (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C068
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

F42 - AWM/ATLM OPTIONS



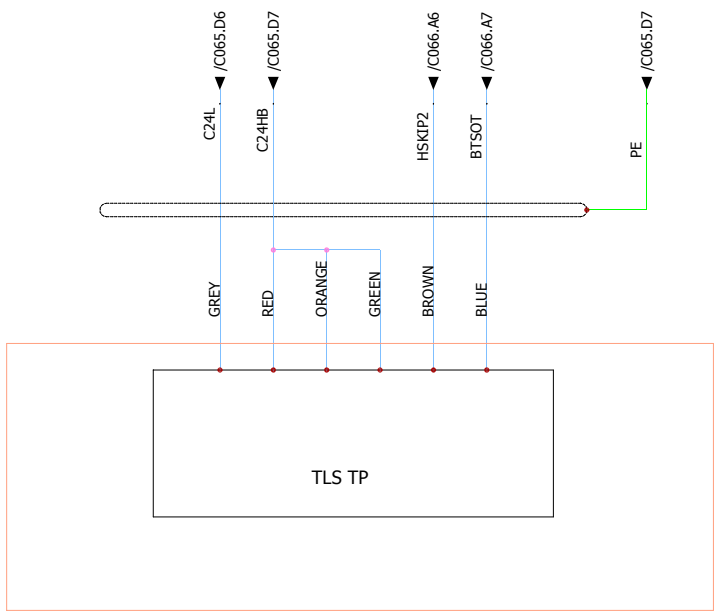
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	AWM RENISHAW (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C069
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

F42 - AWM/ATLM OPTIONS



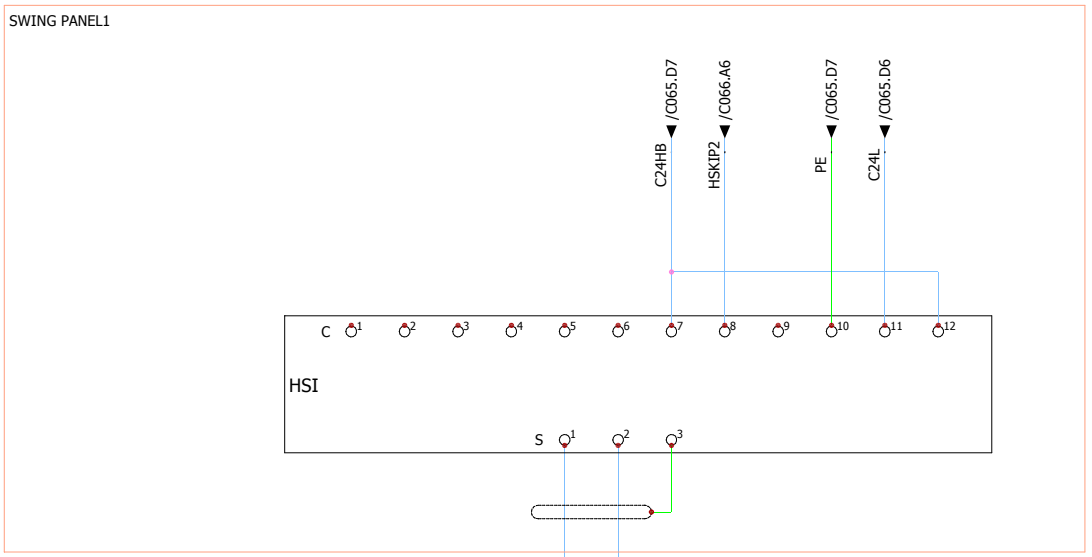
REVISION						
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TITLE: AWM BLUM (OPT)	DRAWING No.: V2631I-C1E-Y
	----.--.--	CHECKED	AVINASH			MACHINE: F3/ F5 Pro 6
	----.--.--	DRAWN	ANISHA			Ver.: B
	----.--.--	DATE	28/6/2018			PAGE: C070

F42 - AWM/ATLM OPTIONS



REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH			ATLM TP MARPOSS (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH					PAGE	C071
	----.--.--	DRAWN	ANISHA						
	----.--.--	DATE	28/6/2018						

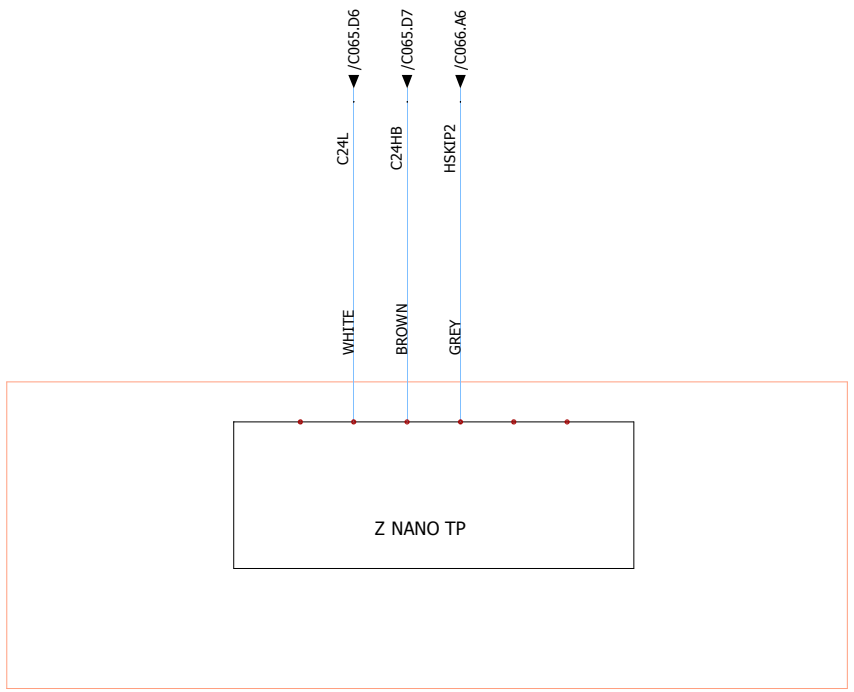
F42 - AWM/ATLM OPTIONS



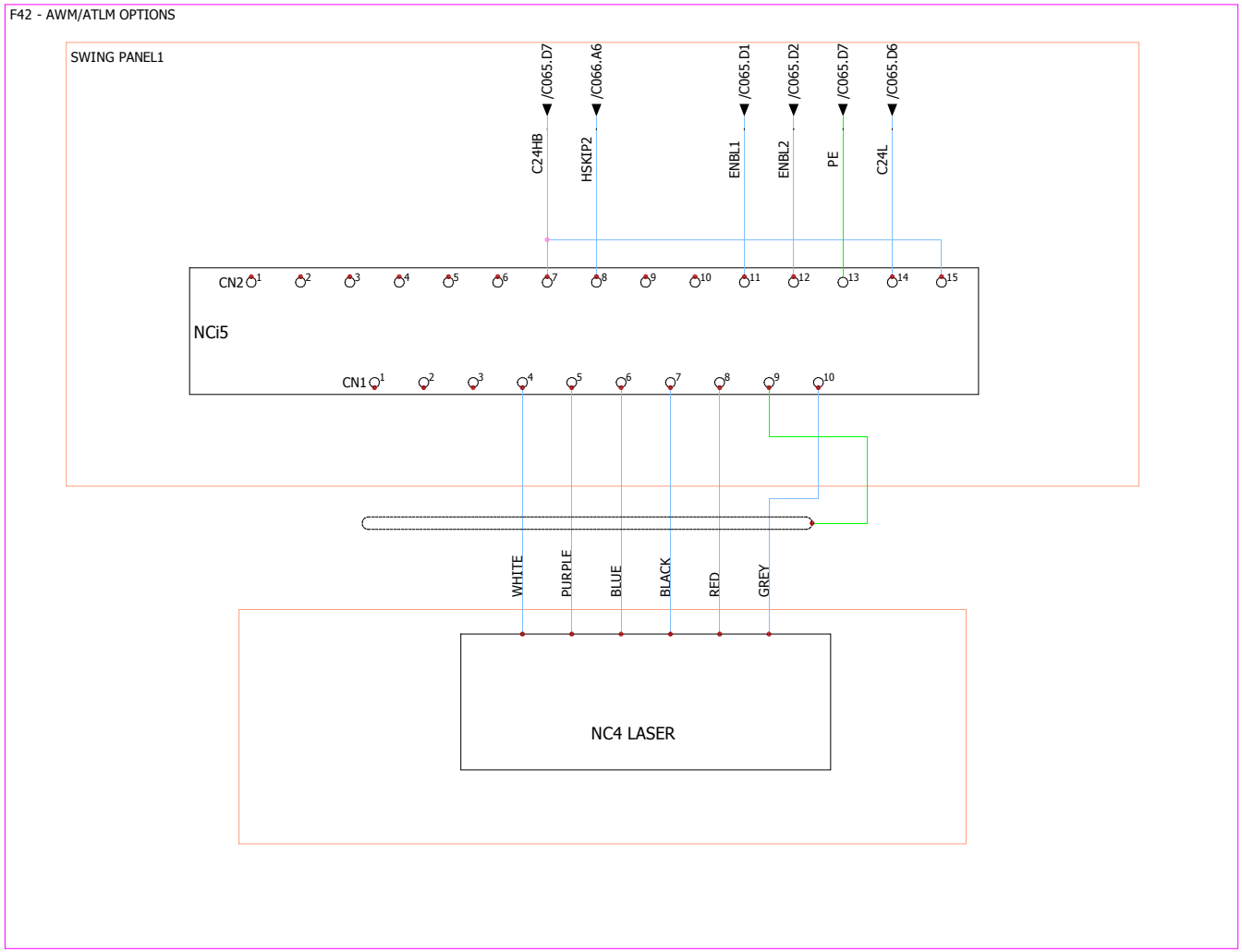
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	ATLM TP RENISHAW (OPT)	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C072
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

MAKINO
MAKINO ASIA PTE LTD

F42 - AWM/ATLM OPTIONS



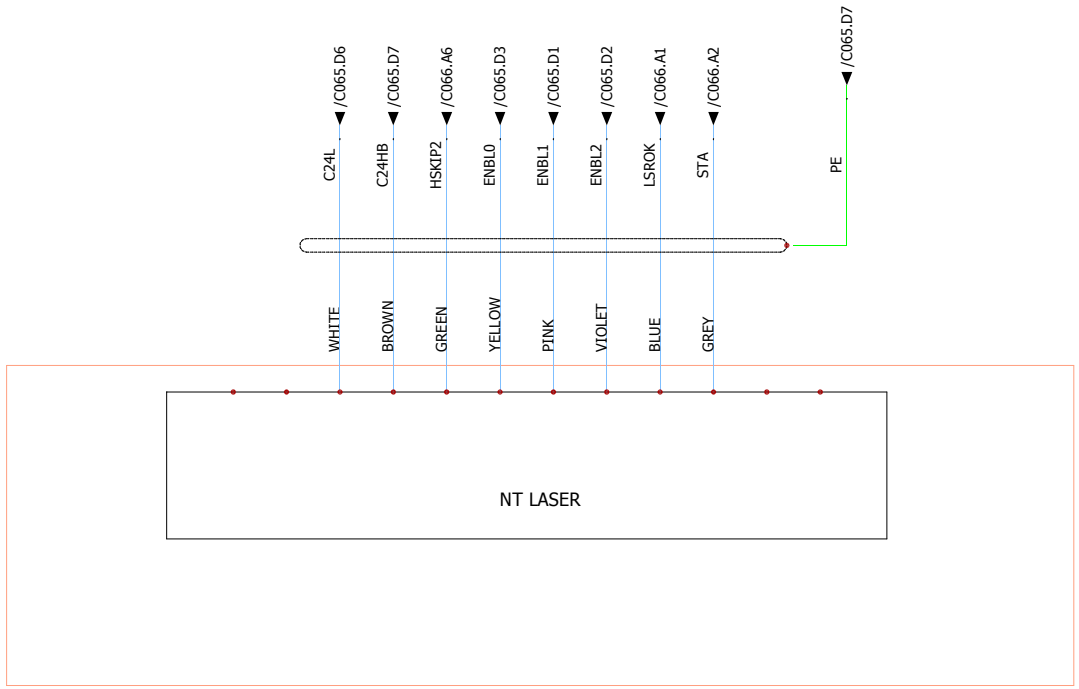
REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH		ATLM TP BLUM (OPT)	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	C073
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	ATLM LASER RENISHAW (OPT)	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C074
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

MAKINO
MAKINO ASIA PTE LTD

F42 - AWM/ATLM OPTIONS



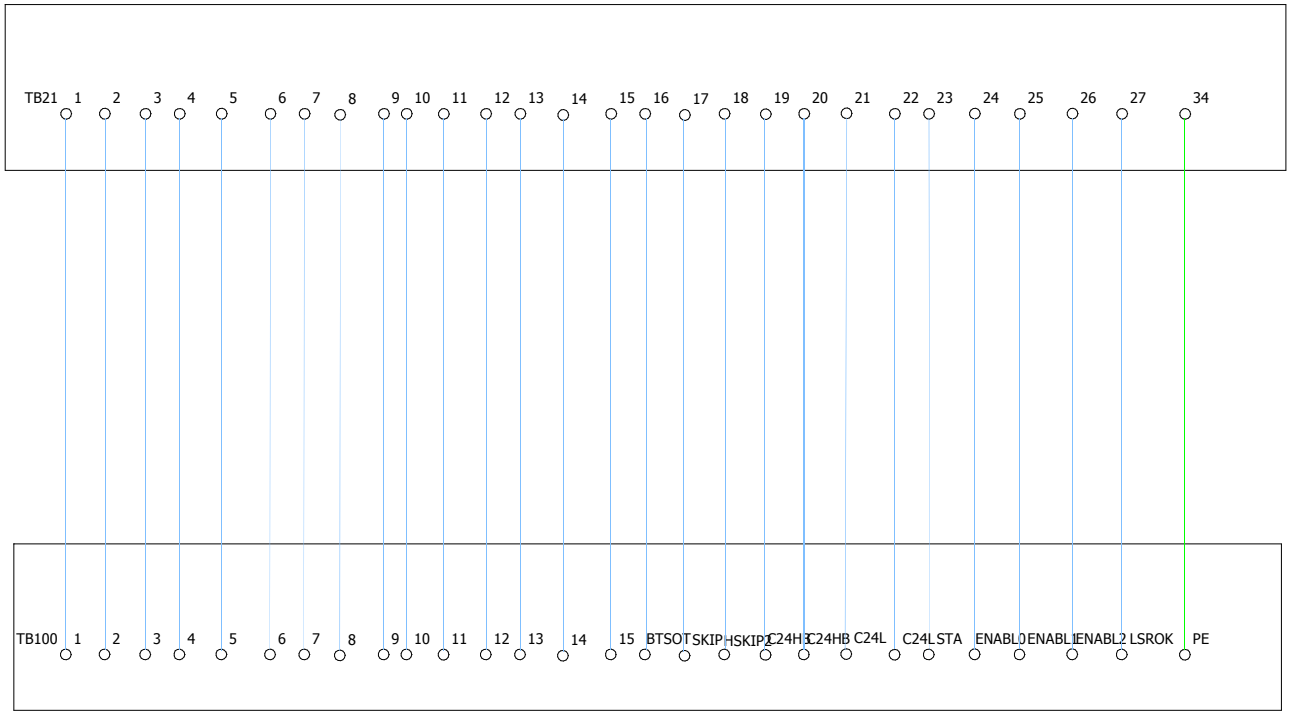
REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

MAKINO
MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
ATLM LASER BLUM (OPT)	V2631I-C1E-Y	Ver.	B
		PAGE	C075

TB21 CONNECTION

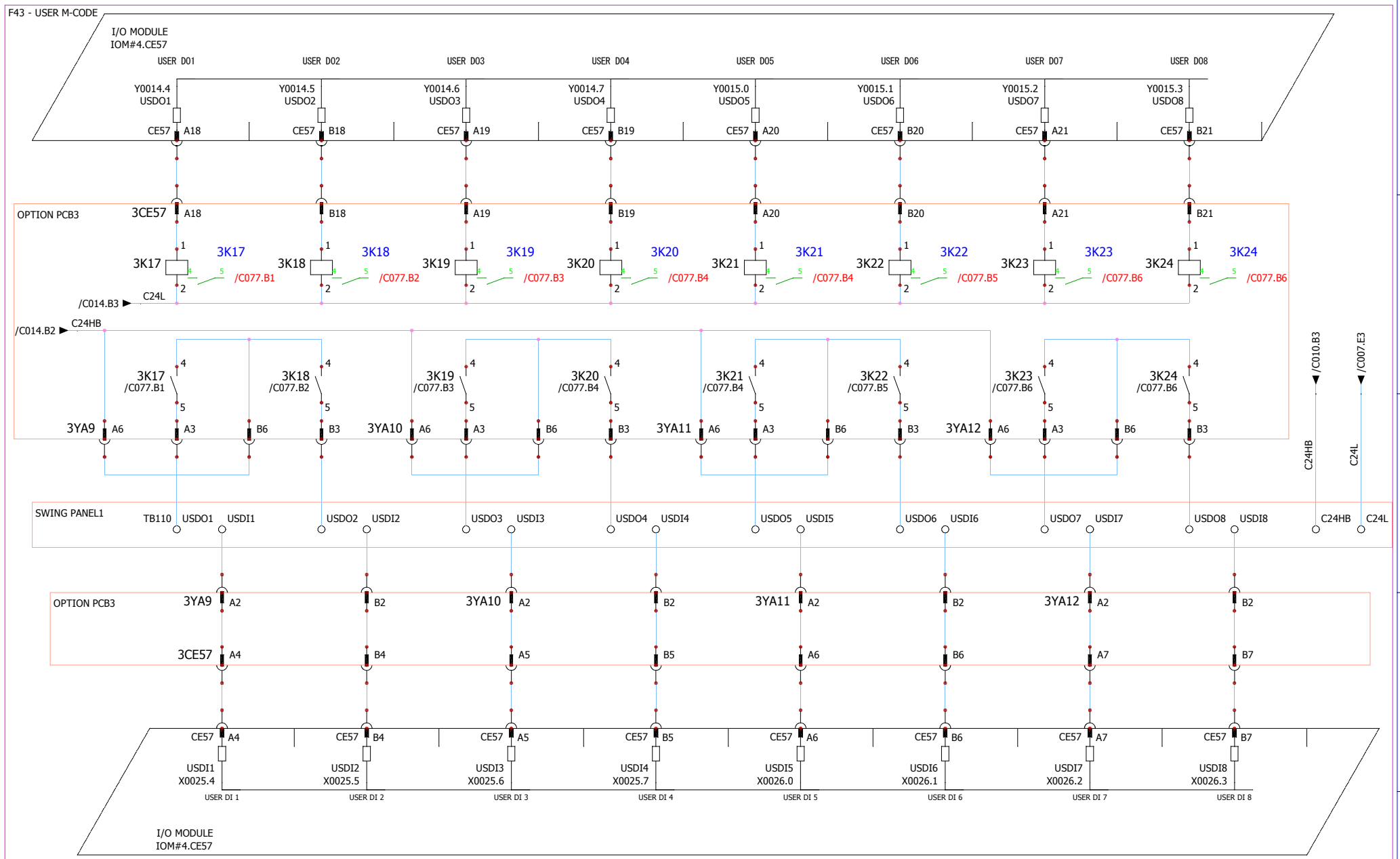
MARPOSS LASER	BLUM LASER	RENISHAW LASER	MARPOSS TP	BLUM TP	RENISHAW TP
1 (GREY / PINK)	19 (BROWN)	4 (WHITE)	16 (YELLOW)	18 (GREY)	1 (BLUE)
2 (WHITE / YELLOW)	21 (WHITE)	5 (PURPLE)	17 (GREEN)	20 (BROWN)	2 (RED)
3 (WHITE / GREEN)	23 (GREEN)	6 (BLUE)	18 (ORANGE)	22 (WHITE)	34 (SHEILD)
4 (--)	24 (YELLOW)	7 (BLACK)	20 (BLACK)		
5 (--)	25 (PINK)	8 (RED)	22 (SHEILD)		
6 (YELLOW)	26 (VIOLET)	9 (SCREEN)			
7 (BROWN)	27 (BLUE)	10 (GREY)			
8 (RED)					
9 (RED / BLUE)					
10 (YELLOW/BROWN)					
11 (BROWN / GREEN)					
12 (--)					
13 (GREEN)					
14 (BLUE)					
15 (WHITE)					



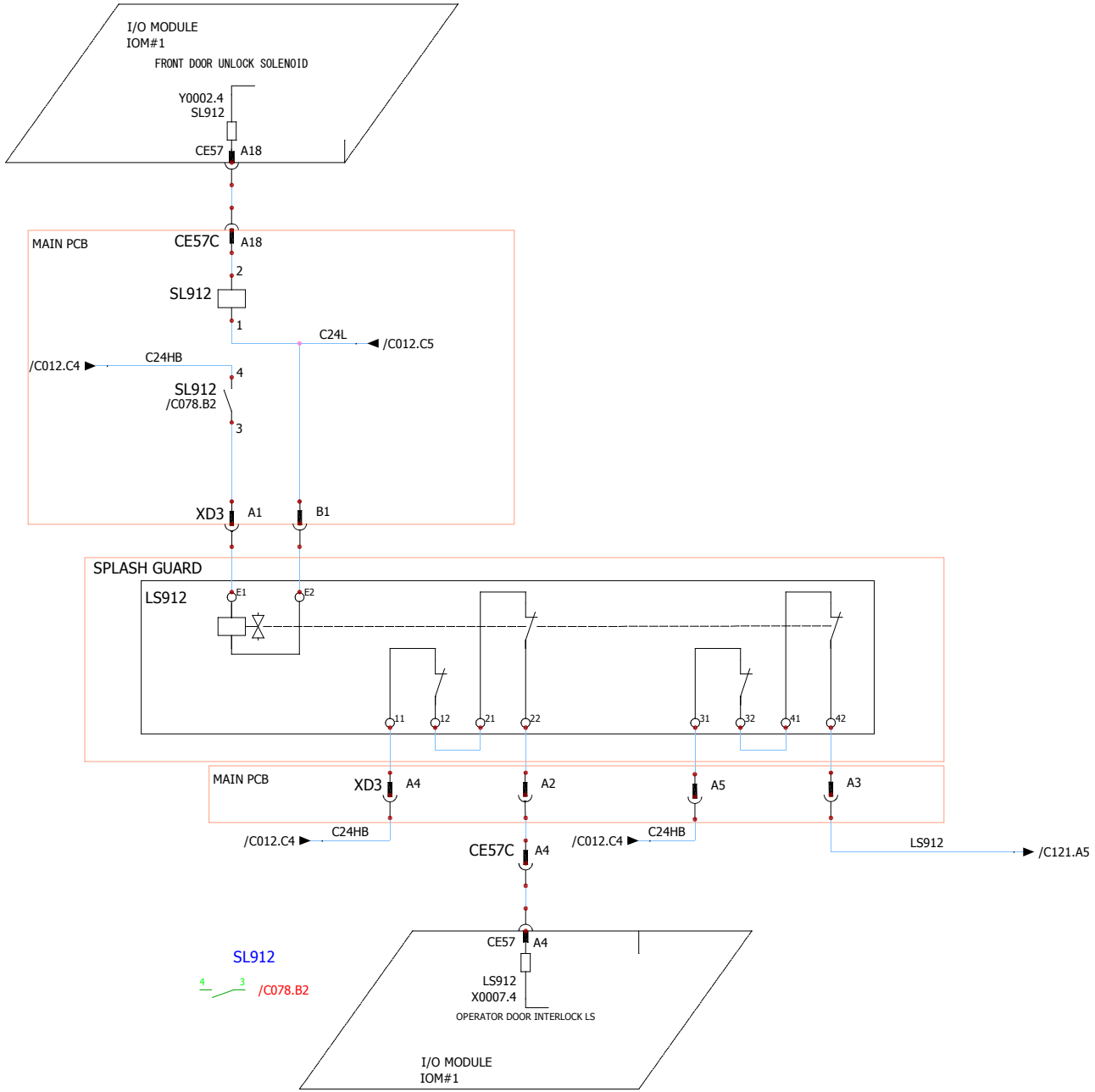
REVISION				
	----.--.--	APPROVED	AVINASH
	----.--.--	CHECKED	AVINASH
	----.--.--	DRAWN	ANISHA
	----.--.--	DATE	28/6/2018

MAKINO
MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
ATLM INTERFACE BOX (OPT)	V2631I-C1E-Y	Ver.	B
		PAGE	C076

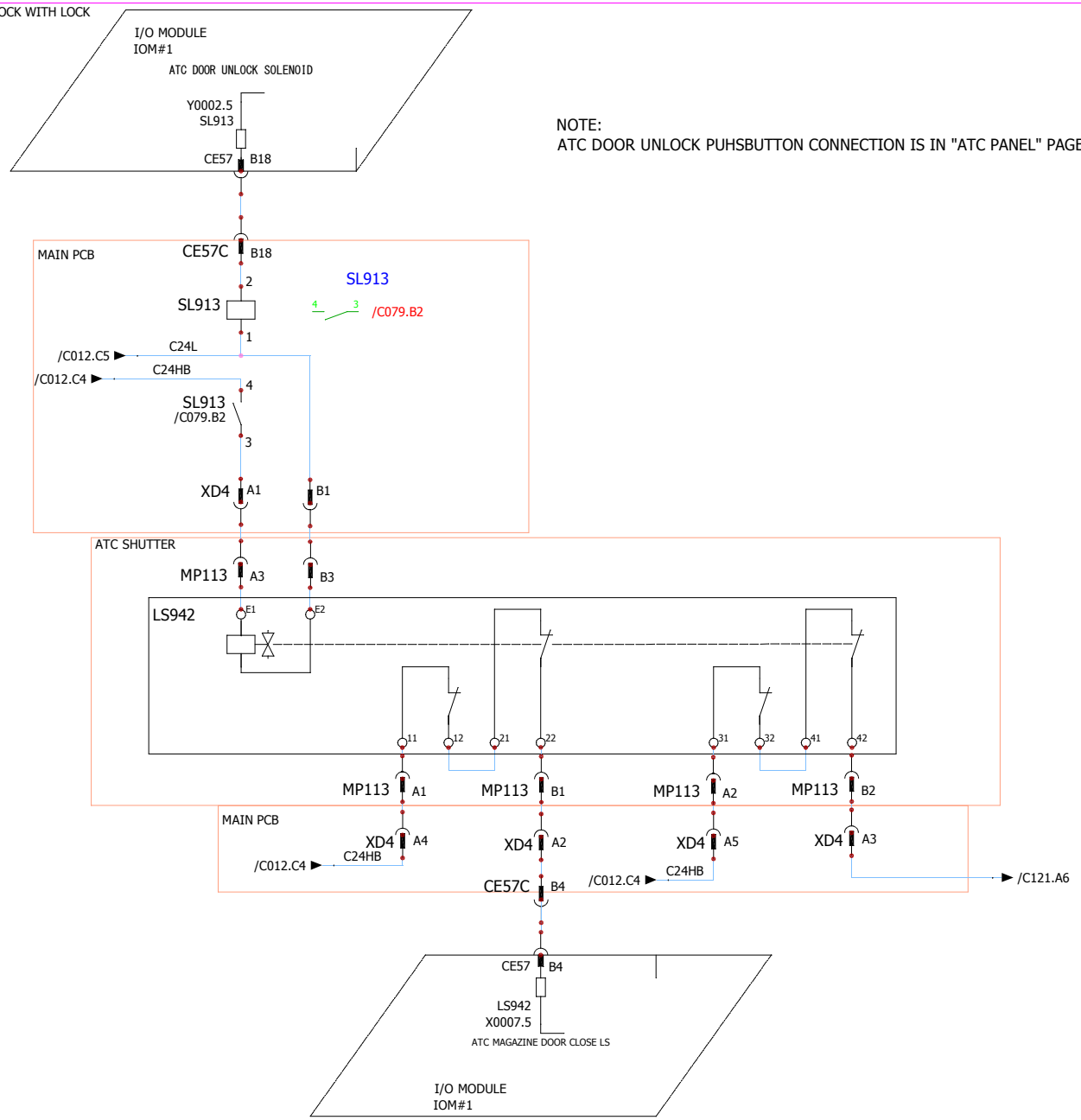


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	USER M-CODE (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C077
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



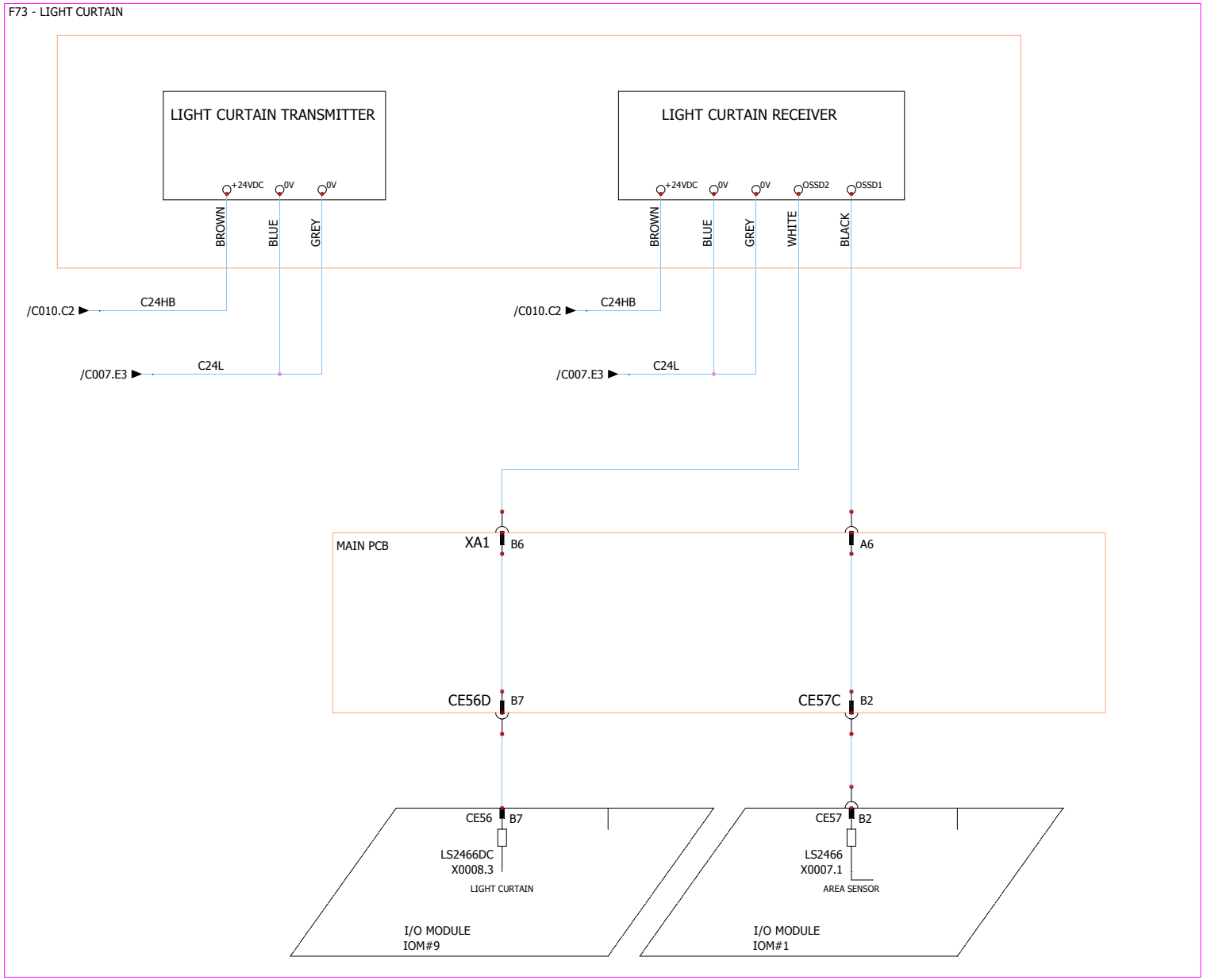
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	S/G DOOR LOCK W INTERLOCK	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C078
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

F41 - ATC DOOR LOCK WITH LOCK

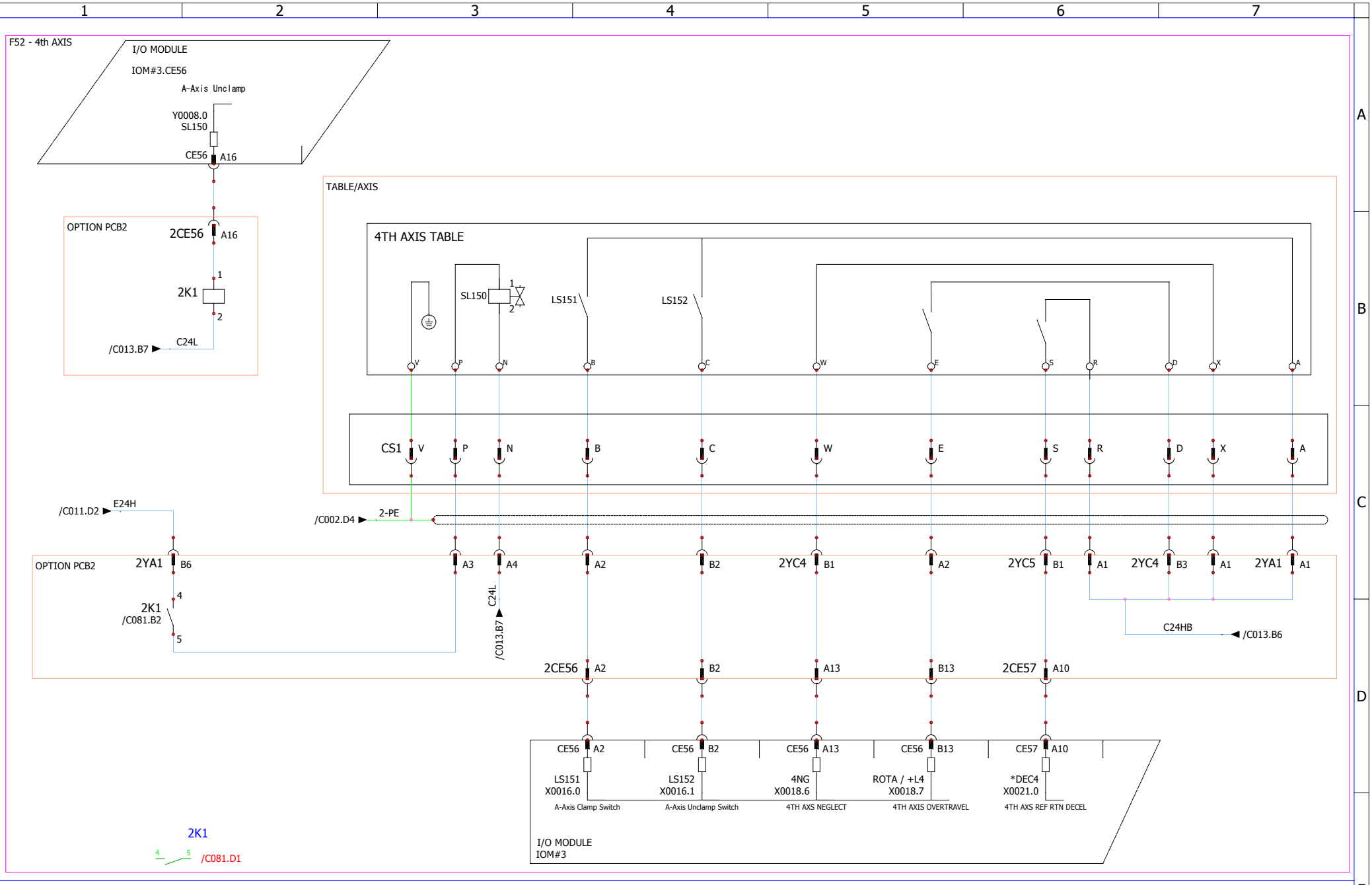


NOTE:
ATC DOOR UNLOCK PUHSBUTTON CONNECTION IS IN "ATC PANEL" PAGE

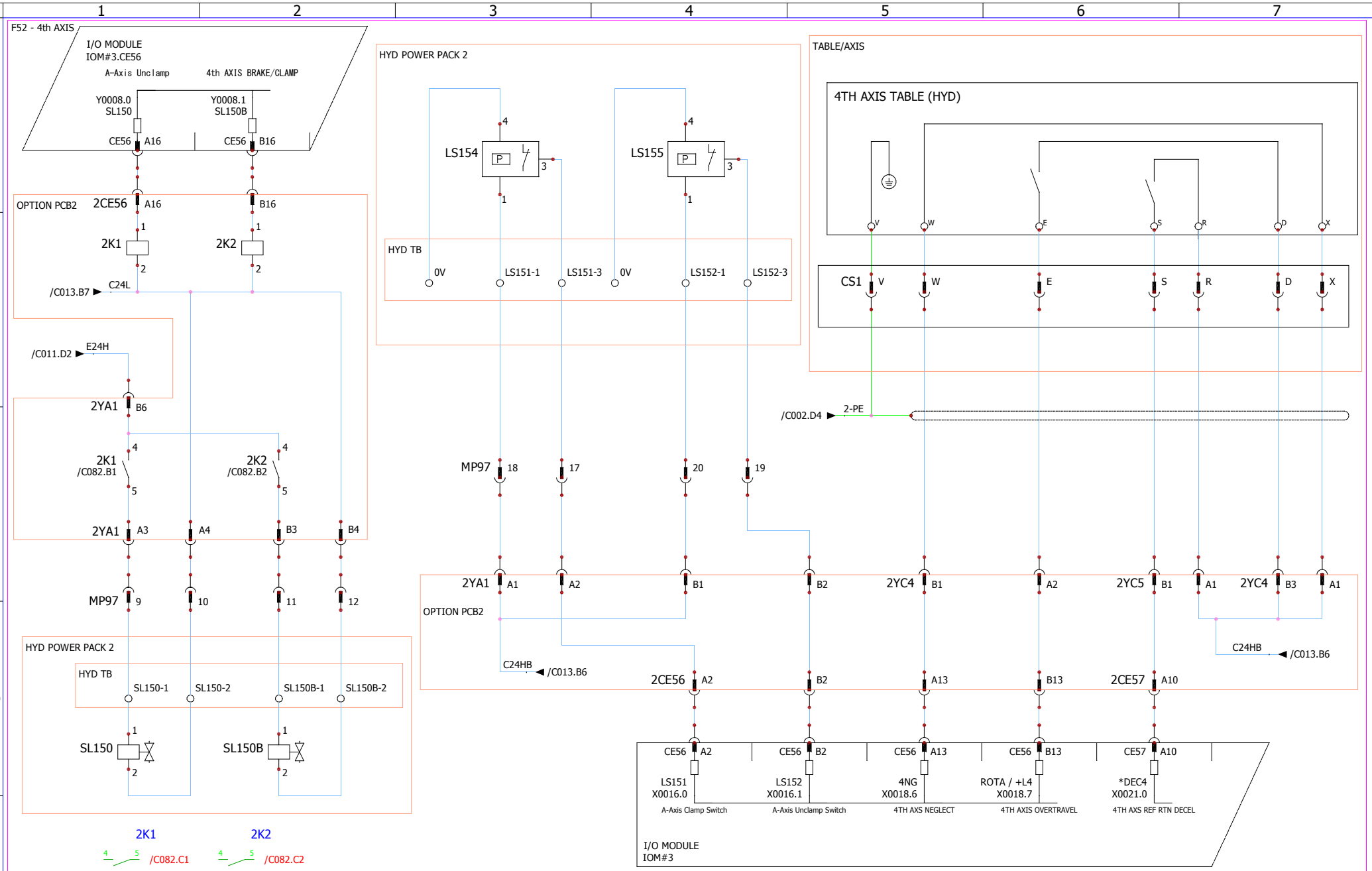
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	ATC DOOR LOCK (STD) W INTERLOCK (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C079
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	LIGHT CURTAIN (OPT)	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	C080
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					

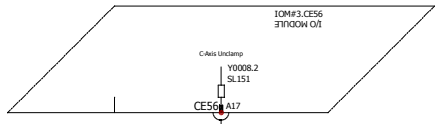


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	4TH AXIS CONTROL (PNEUMATIC)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C081
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	4TH AXIS CONTROL (HYDRAULIC)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C082
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

F53 - 5th AXIS



OPTION PCB2
2CE56 A17
1
2K3
2
/C013.B7 C24L

TABLE/AXIS

5TH AXIS TABLE

SL151 1 2
LS154
LS155

V P N B C W E S R D X A

CS3 V P N B C W E S R D X A

/C011.D2 E24H
/C002.D4 2-PE

OPTION PCB2
2YA2 B6
4
2K3
/C083.B1
5

A3 A4 A2 B2 2YC3 B1 A2 2YC5 A2 B3 2YC3 B3 A1 2YA2 A1

C24L /C013.B7

C24HB /C013.B6

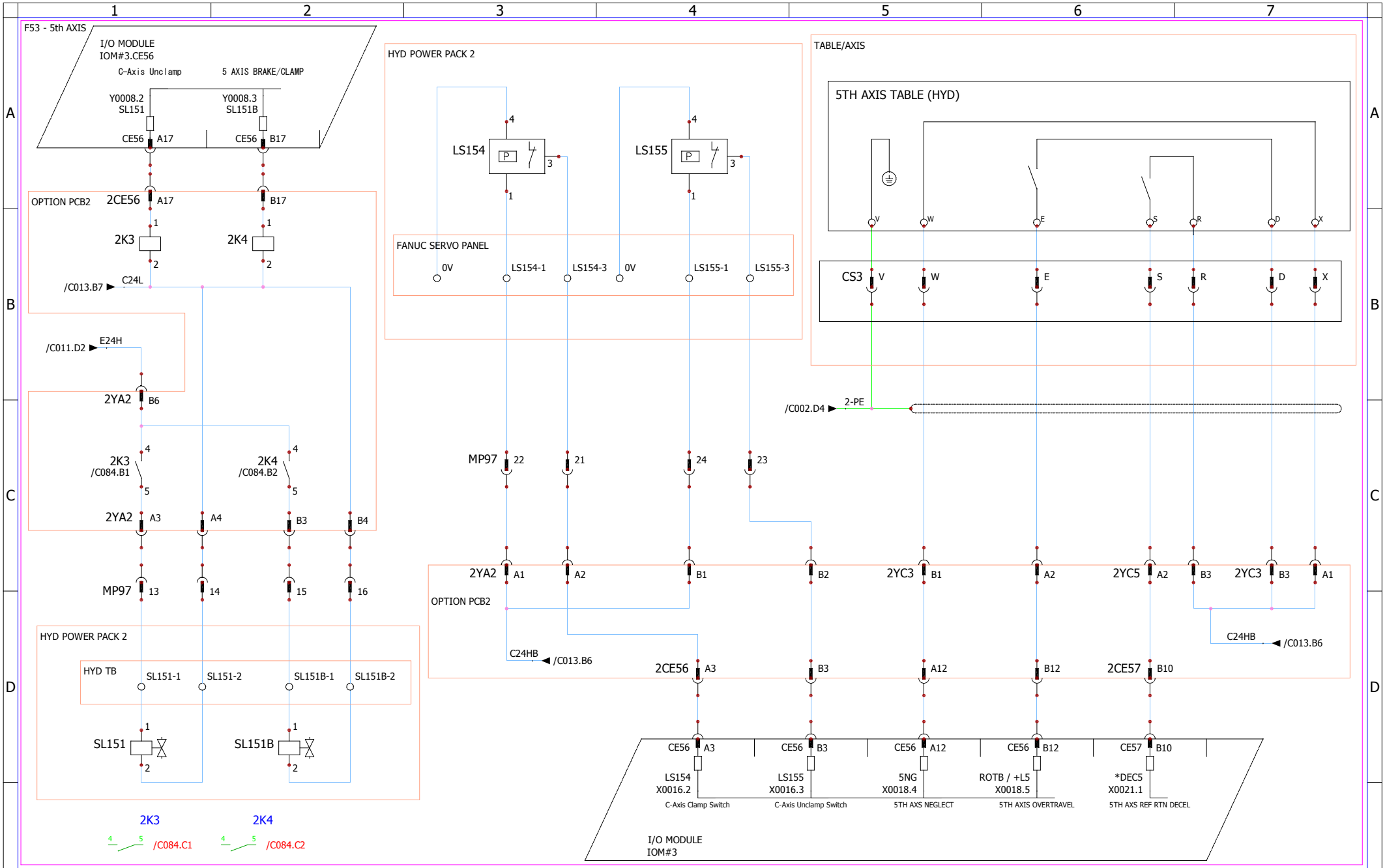
2CE56 A3 B3 A12 B12 2CE57 B10

CE56 A3 — LS154 X0016.2 — C-Axis Clamp Switch
CE56 B3 — LS155 X0016.3 — C-Axis Unclamp Switch
CE56 A12 — 5NG X0018.4 — 5TH AXS NEGLECT
CE56 B12 — ROTB / +L5 X0018.5 — 5TH AXIS OVERTRAVEL
CE57 B10 — *DEC5 X0021.1 — 5TH AXS REF RTN DECEL

I/O MODULE
IOM#3

2K3
4 5 /C083.C1

REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	5TH AXIS CONTROL (PNEUMATIC)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C083
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	5TH AXIS CONTROL (HYDRAULIC)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C084
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

F58 - HYD PP-2

/C053.A2 R0 /C097.A1
/C053.A2 S0 /C097.A1
/C053.A2 T0 /C097.A1

L1 L2 L3
TH260-2
(4.0-6.3A)
T1 T2 T3

TH260-2
13 14 /C085.C4
21 22 /C085.C5

TH260-2 SETTING VALUE	
400V/50HZ	5.8A
400V/60HZ	5.8A

MS260-2
/C085.C4
1 3 5
2 4 6

3-PE /C002.D6

MP97A 1 2 3 E

HYD POWER PACK 2
HYD TB
U16 V16 W16 E

M260-2
M
POWER PACK MOTOR

I/O MODULE
IOM#3.CE57
2nd HDY POWER PACK
Y0011.7
M260-2
CE57 B23

E24H /C011.D2

OPTION PCB2
2CE57 B23
2K32
2
1
/C013.B7 C24L

2YB8 B3
2K32
/C085.B3
4
5
C24L /C013.B7
2YB8 A2 B2 A1

/C013.B6 C24HB

TH260-2
/C085.B1
13
14

MS260-2
A1
A2

TH260-2
/C085.B1
21
22

HYD POWER PACK 2
F02 - OPTION
SYSPS-2
SYSPS-2-2
4
3
1
HYD TB
SYSPS2-1
SYSPS2-3
SYSPS2-2-3
0V

MP97 7 3 2 1

OPTION PCB2
2YB8 B1
2YC6 A1 B1 A2 B2
/C013.B6 C24HB
C24L /C013.B7
2CE57 B9 A11 B11

CE57 B9
TH260-2
X0020.7
2nd HYD PP MTR OVERLOAD

CE57 A11
SYSPS-2
X0021.2
2nd HYD PP SYSTEM PRESS (High)

CE57 B11
SYSPS2-2
X0021.3
2nd HYD PP SYSTEM PRESS (Low)

I/O MODULE
IOM#3.CE57

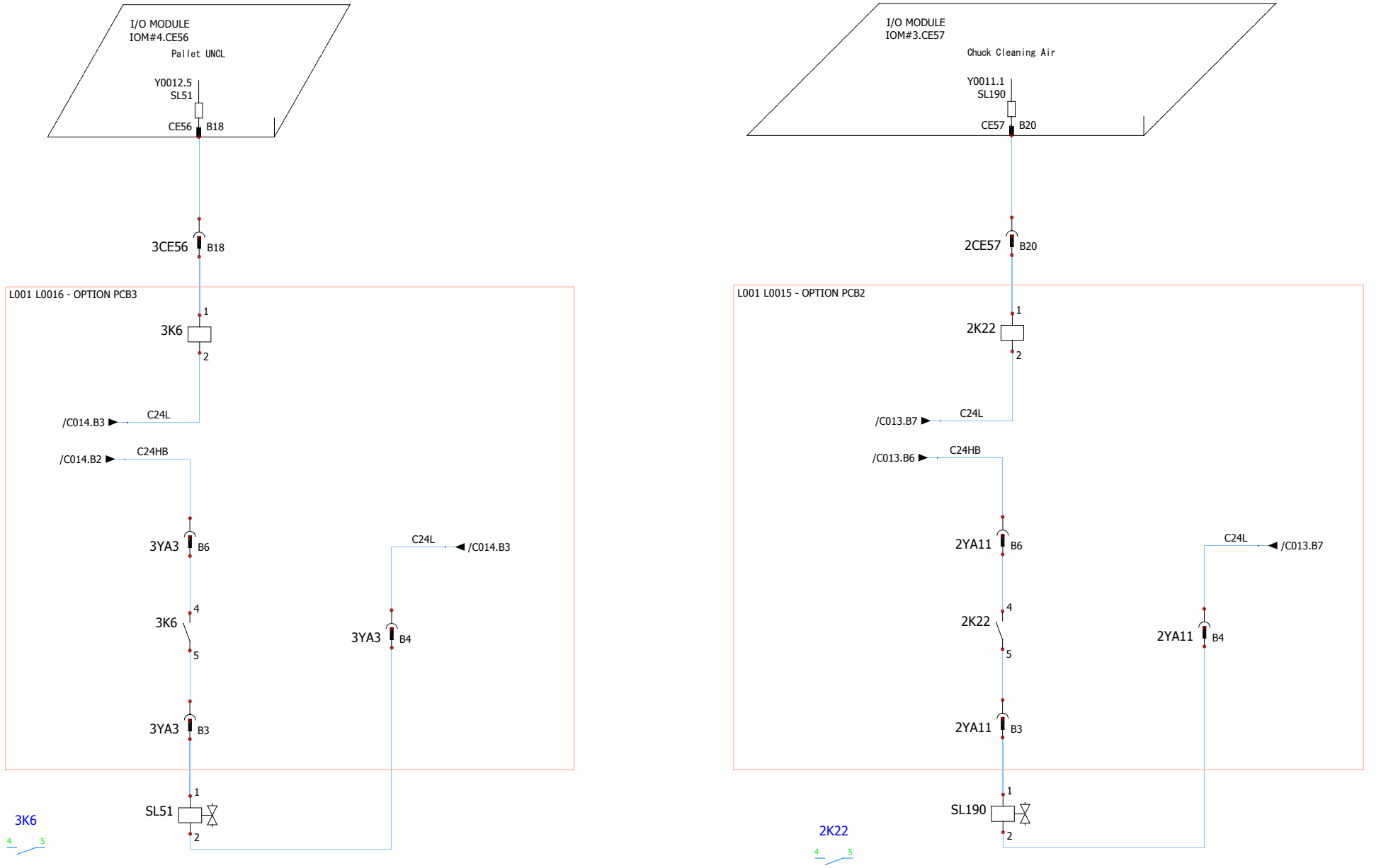
MS260-2
1 2 /C085.B1
3 4 /C085.B1
5 6 /C085.B1
13 14
21 22

2K32
4 5 /C085.B4

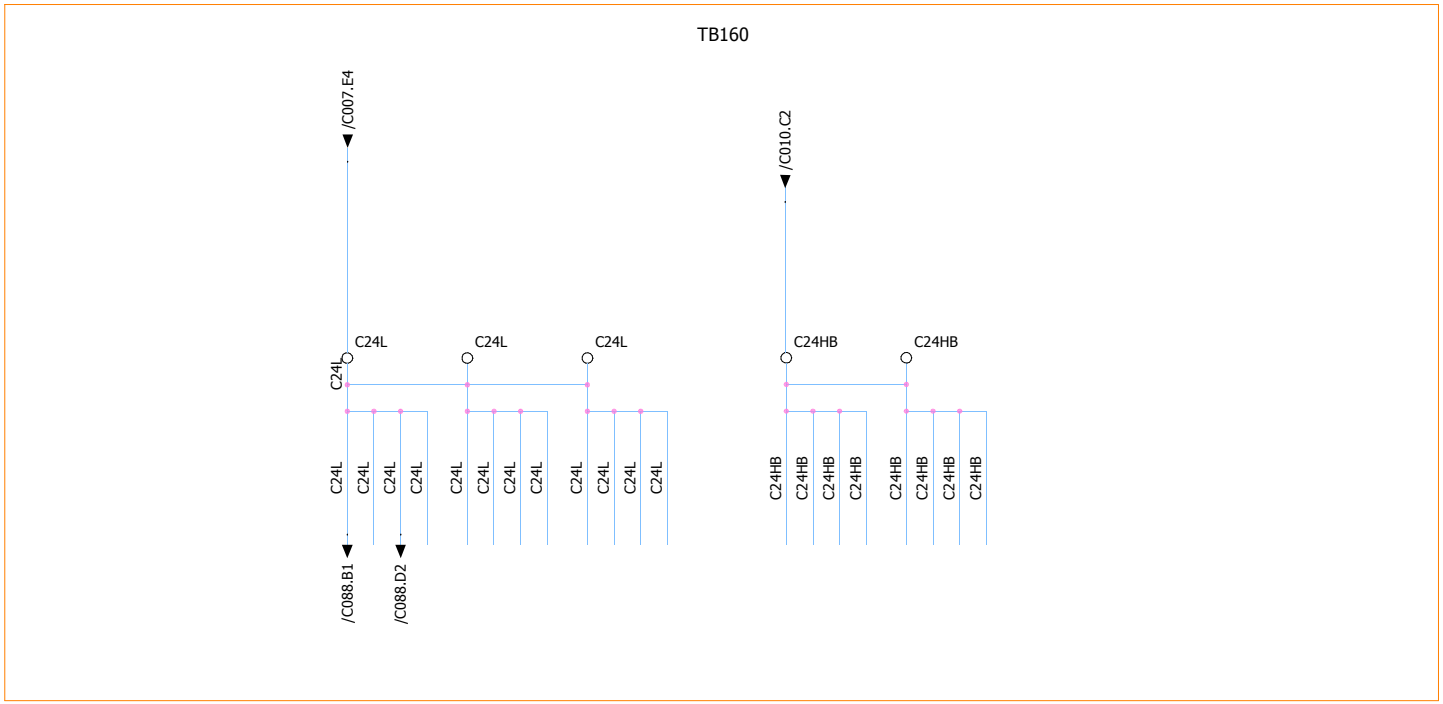
REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

MAKINO
MAKINO ASIA PTE LTD

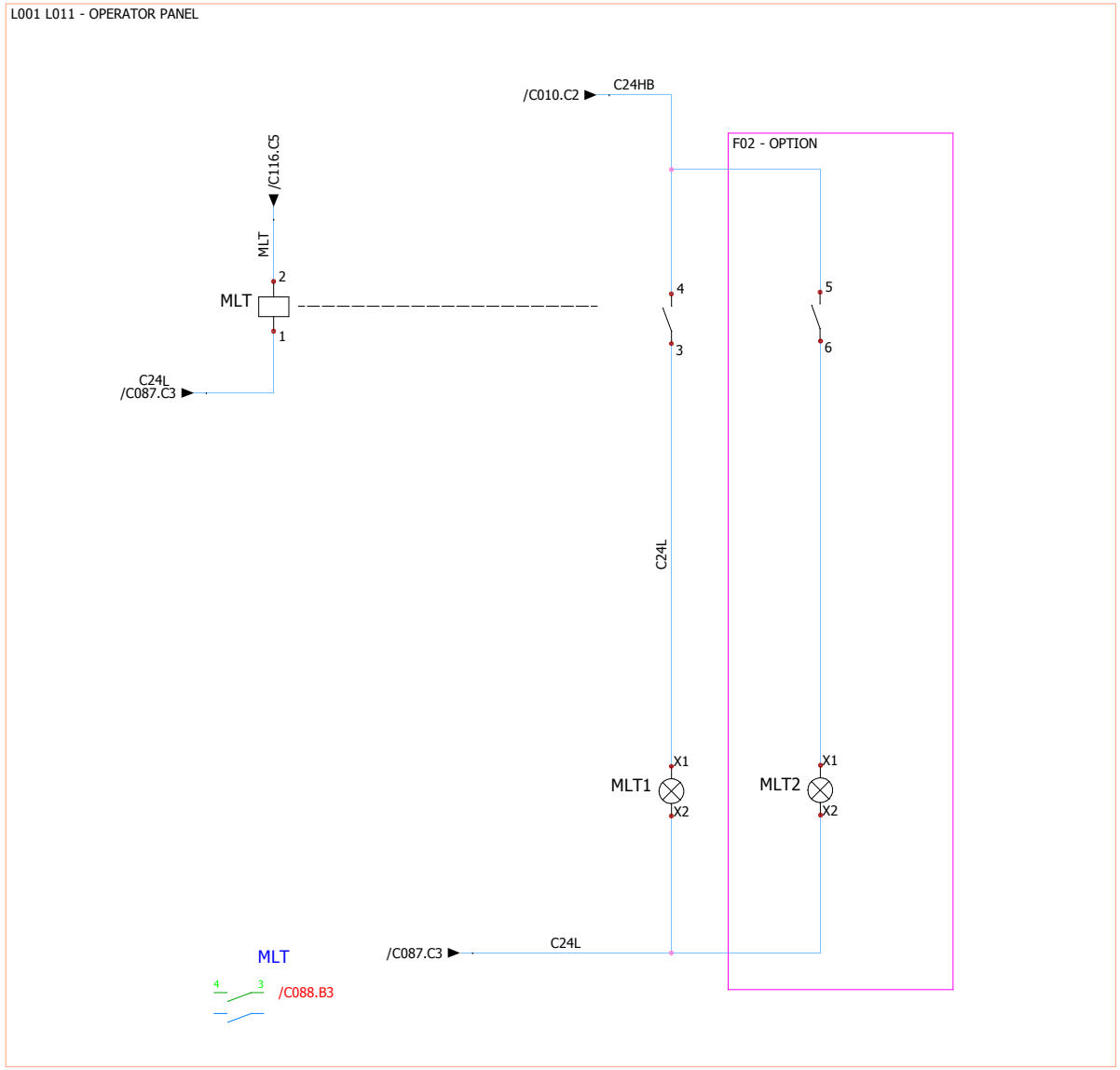
TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
RWH POWER PACK (OPT)	V2631I-C1E-Y	Ver.	B
		PAGE	C085



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TABLE CHUCK CL/UCL (OPT)	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	C086
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					



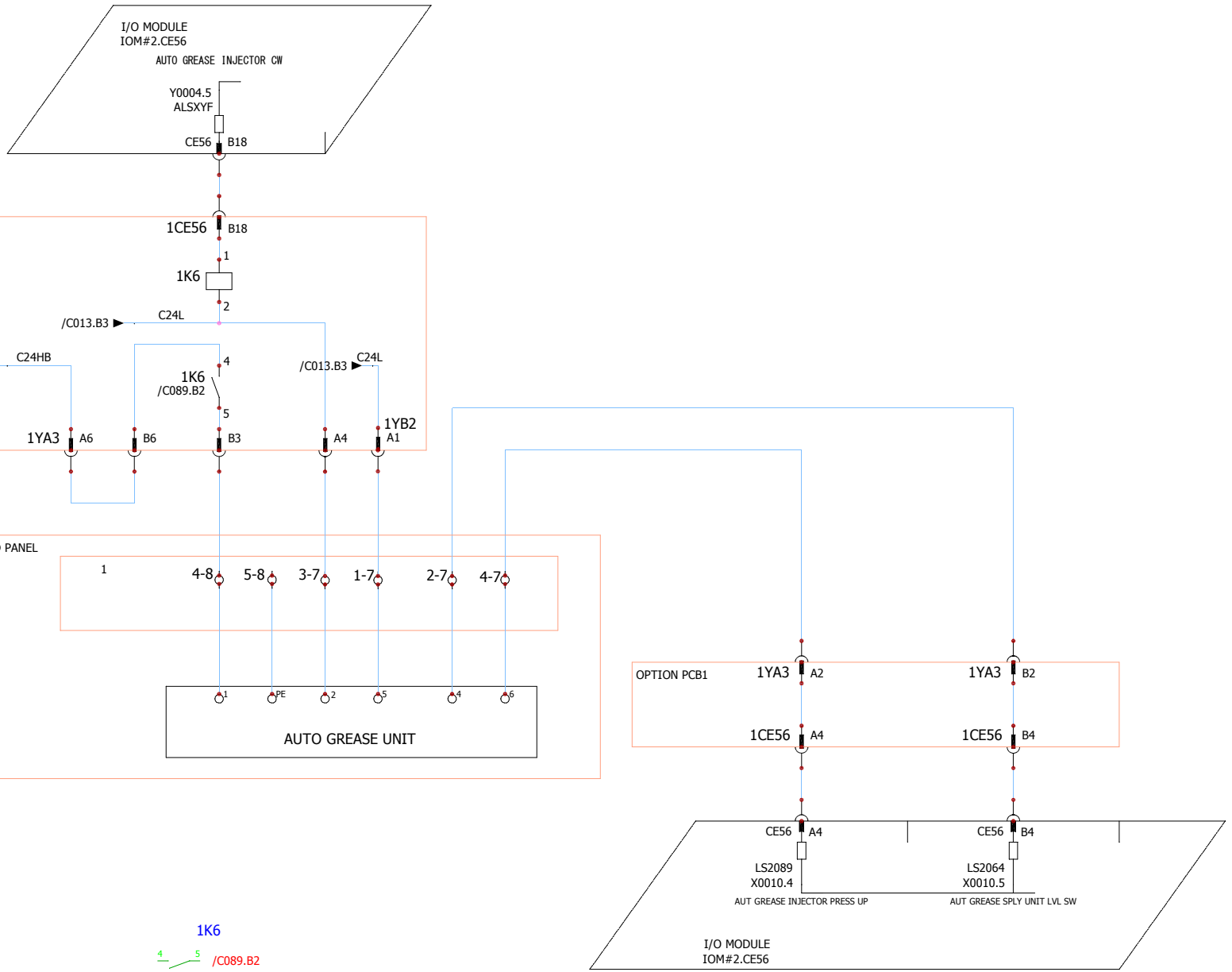
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	CONTROL VOLTAGE DISTRIBUTION OPE	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C087
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	MACHINE LIGHT	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C088
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

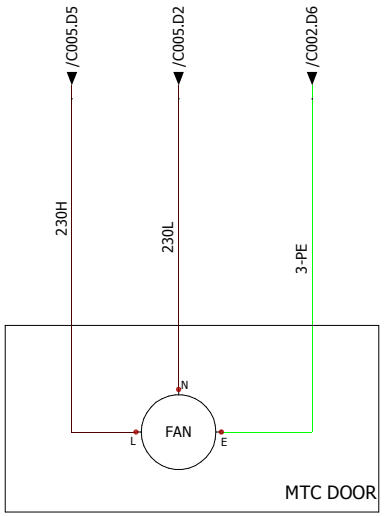
MAKINO
MAKINO ASIA PTE LTD

F56 - AUTO GREASE

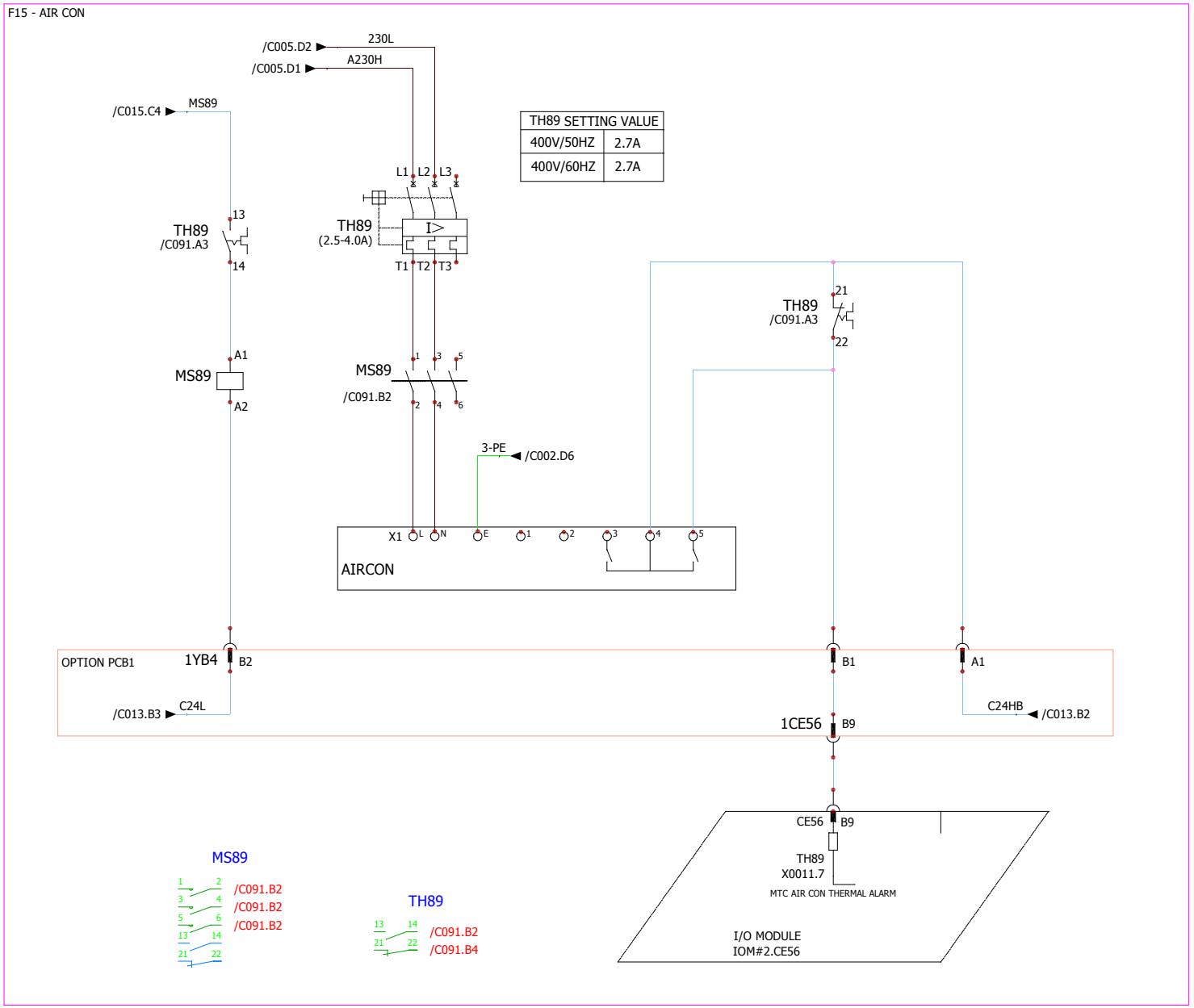


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	AUTO GREASE	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C089
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

MAKINO
MAKINO ASIA PTE LTD

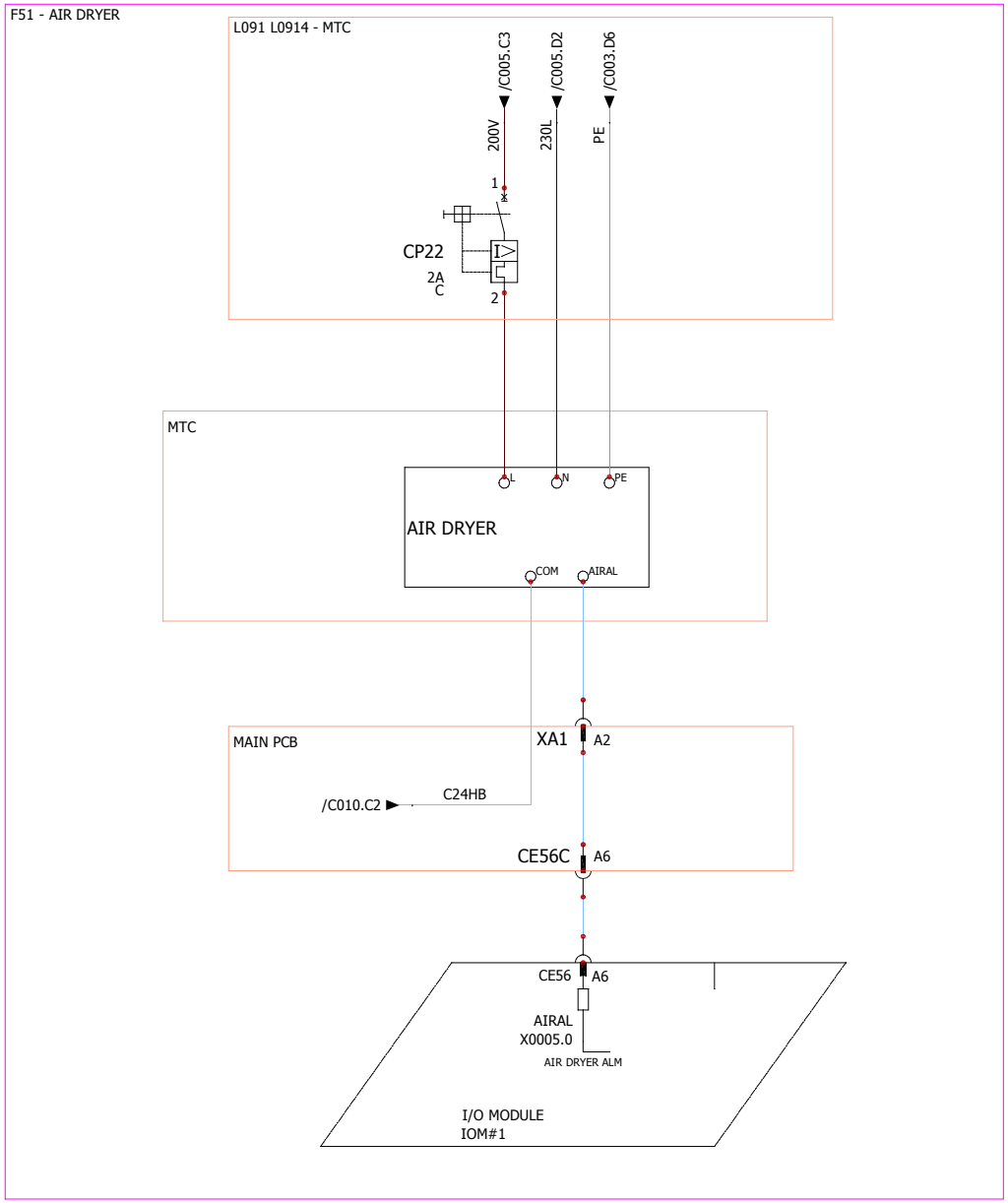


REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH		MTC FAN	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	C090
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					



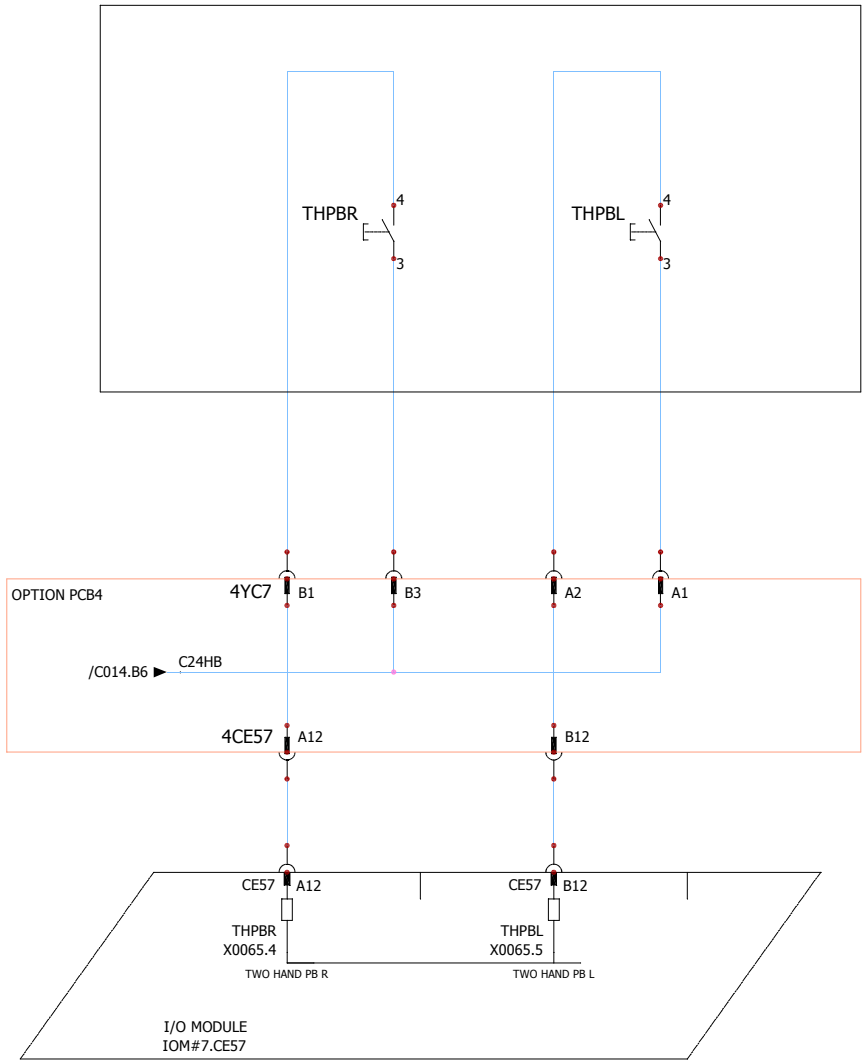
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	MTC AIR CONDITIONER (OPT)	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C091
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

MAKINO
MAKINO ASIA PTE LTD



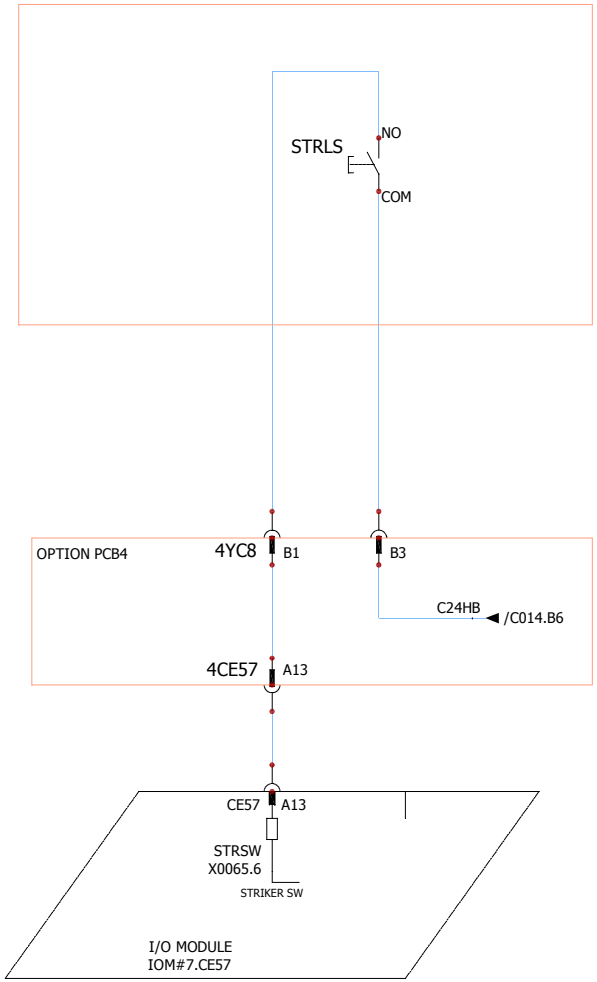
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	AIR DRYER (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C092
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

F72 - 2 HAND PB



REVISION					MAKINO	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TWO HAND PUSH BUTTON (OPT)	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	C093
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					

F71 - STRIKER SW

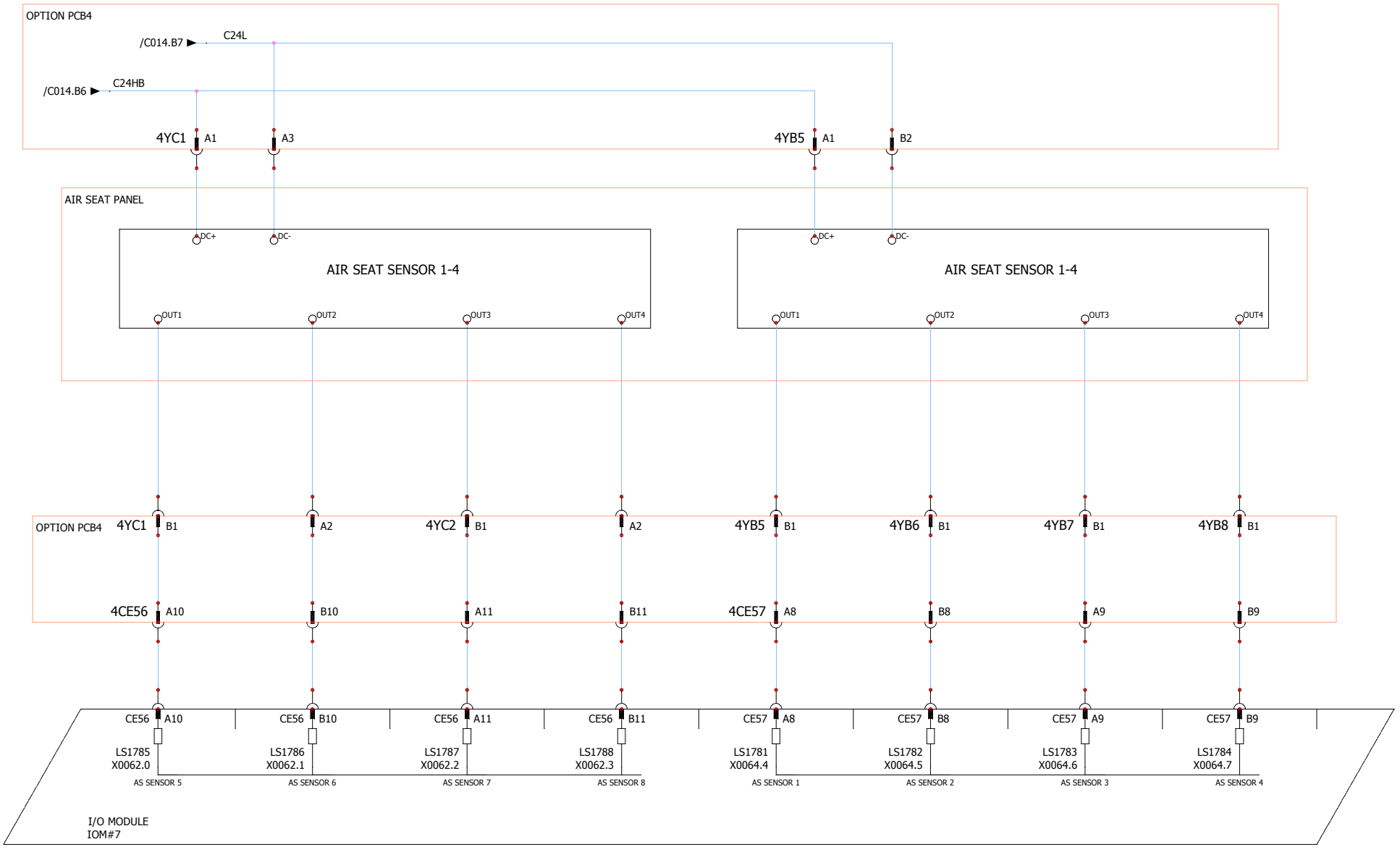


REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

MAKINO
MAKINO ASIA PTE LTD

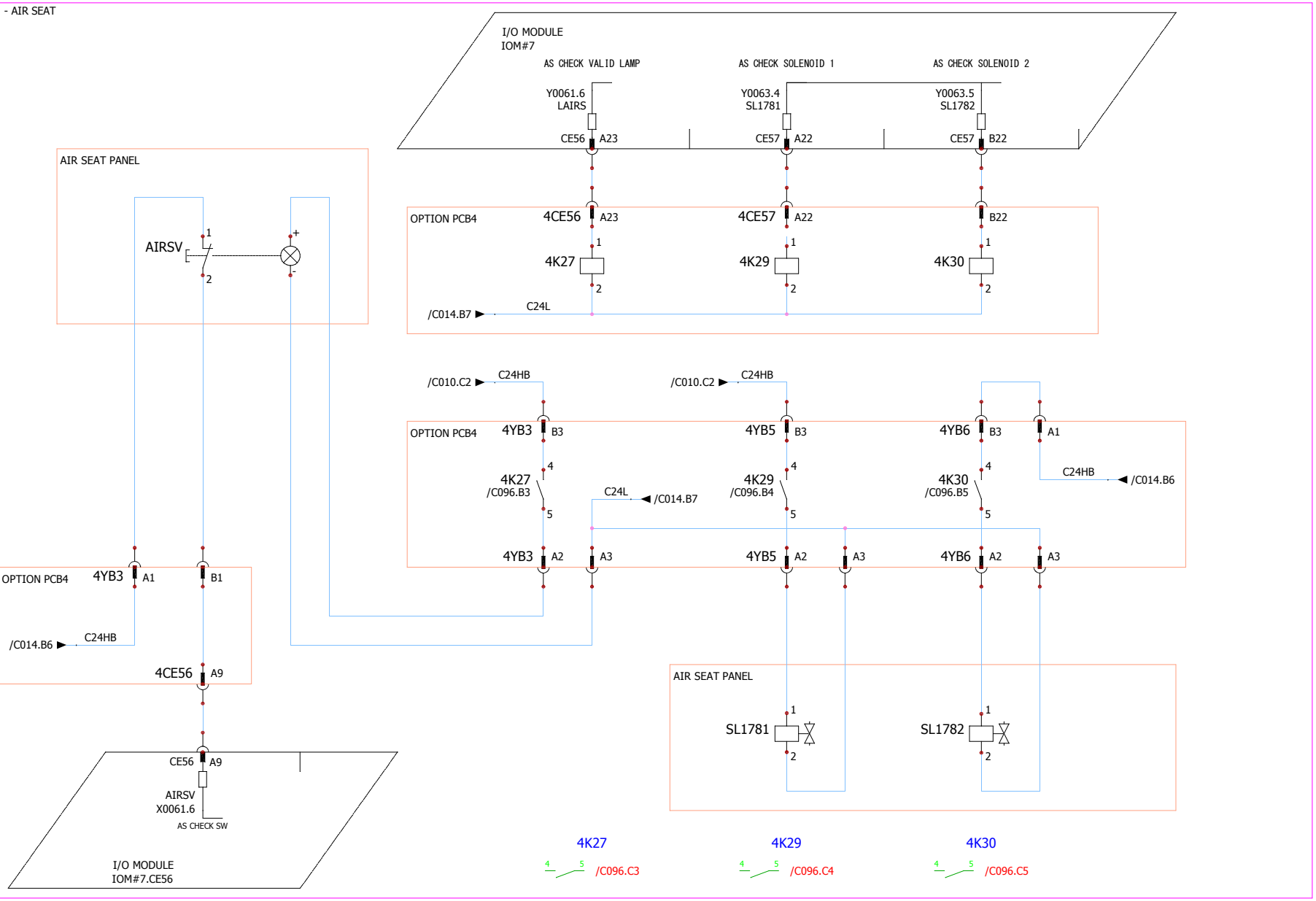
TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
STRIKER SWITCH (OPT)	V2631I-C1E-Y	Ver.	B
		PAGE	C094

F63 - AIR SEAT

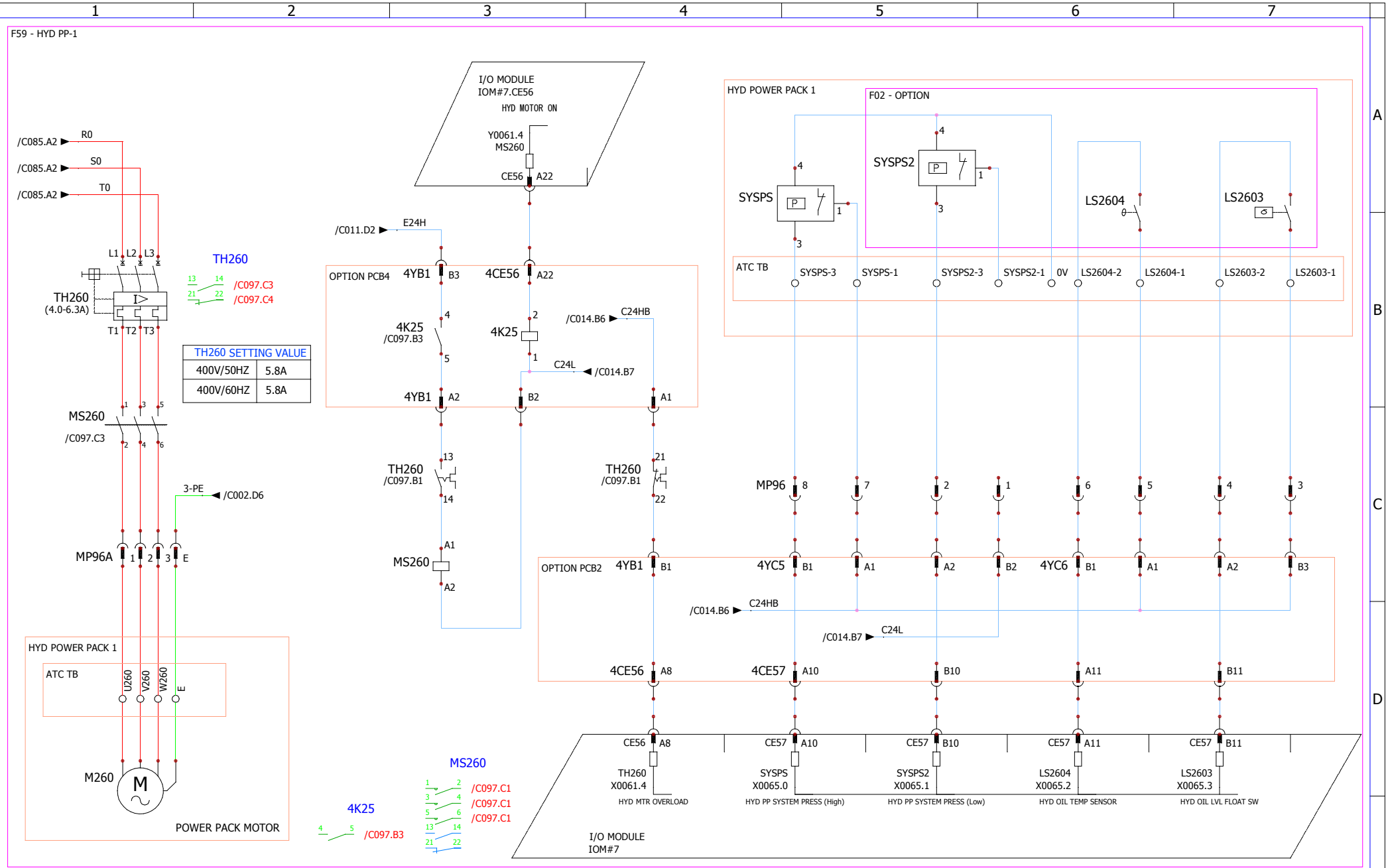


REVISION							
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	CHECKED	AVINASH		AIR SEAT CHECK (1-8 SENSOR) (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	DRAWN	ANISHA				PAGE	C095
	----.--.--	DATE	28/6/2018					

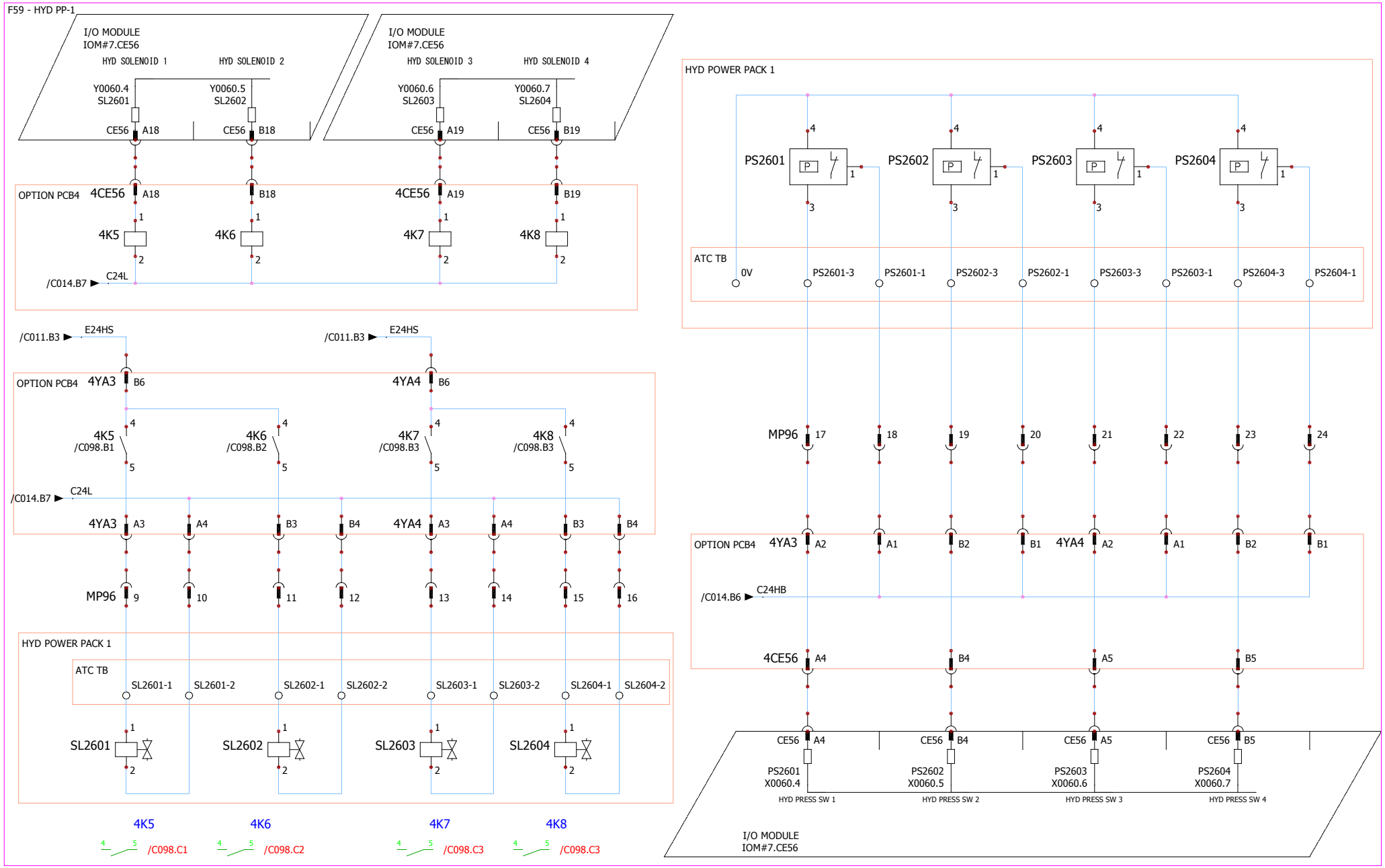
F63 - AIR SEAT



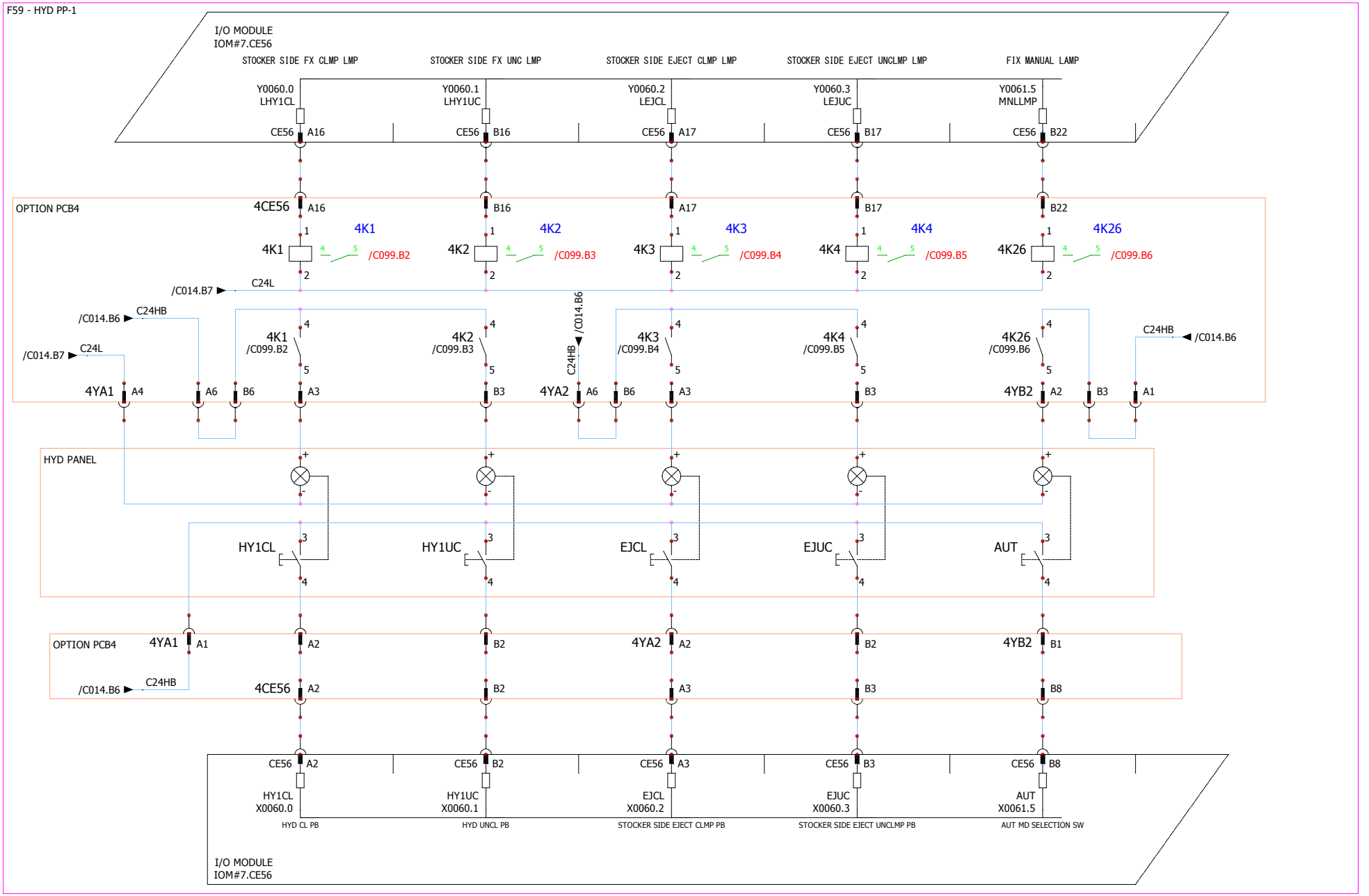
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	AIR SEAT CHECK (SOL & CHECK SW) (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C096
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	HYDRAULIC FIXTURE (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C097
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

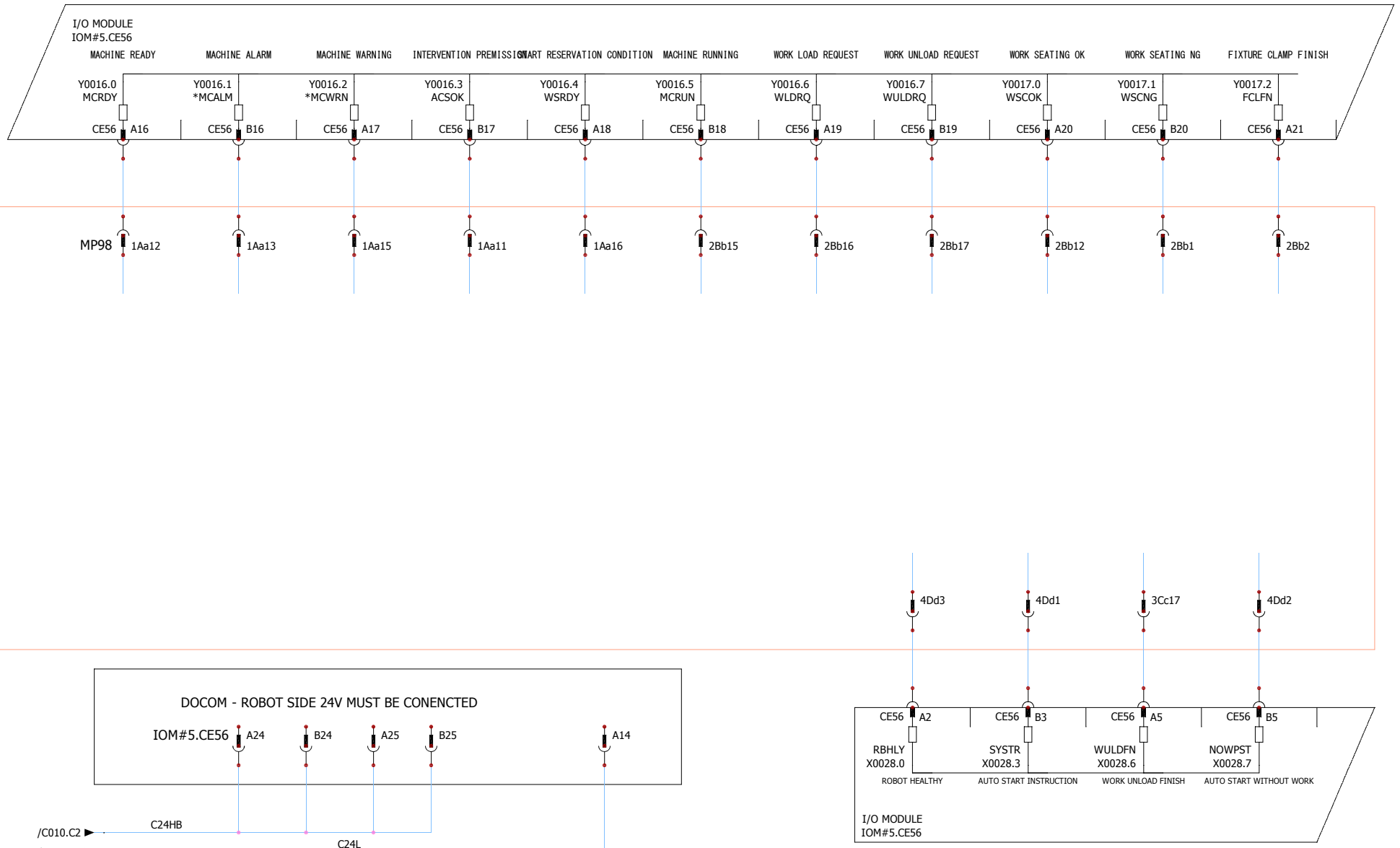


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	HYDRAULIC FIXTURE SOL/PS-1 (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C098
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



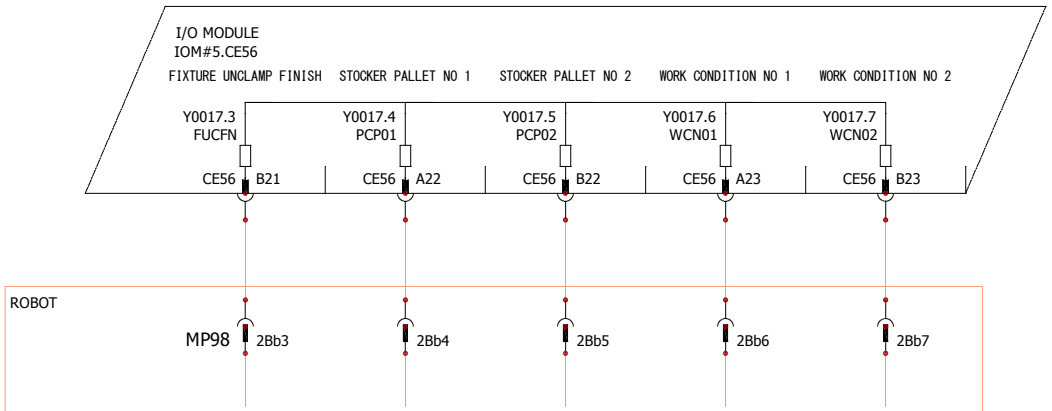
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	HYDRAULIC PANEL (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C099
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

F75 - ROBOT

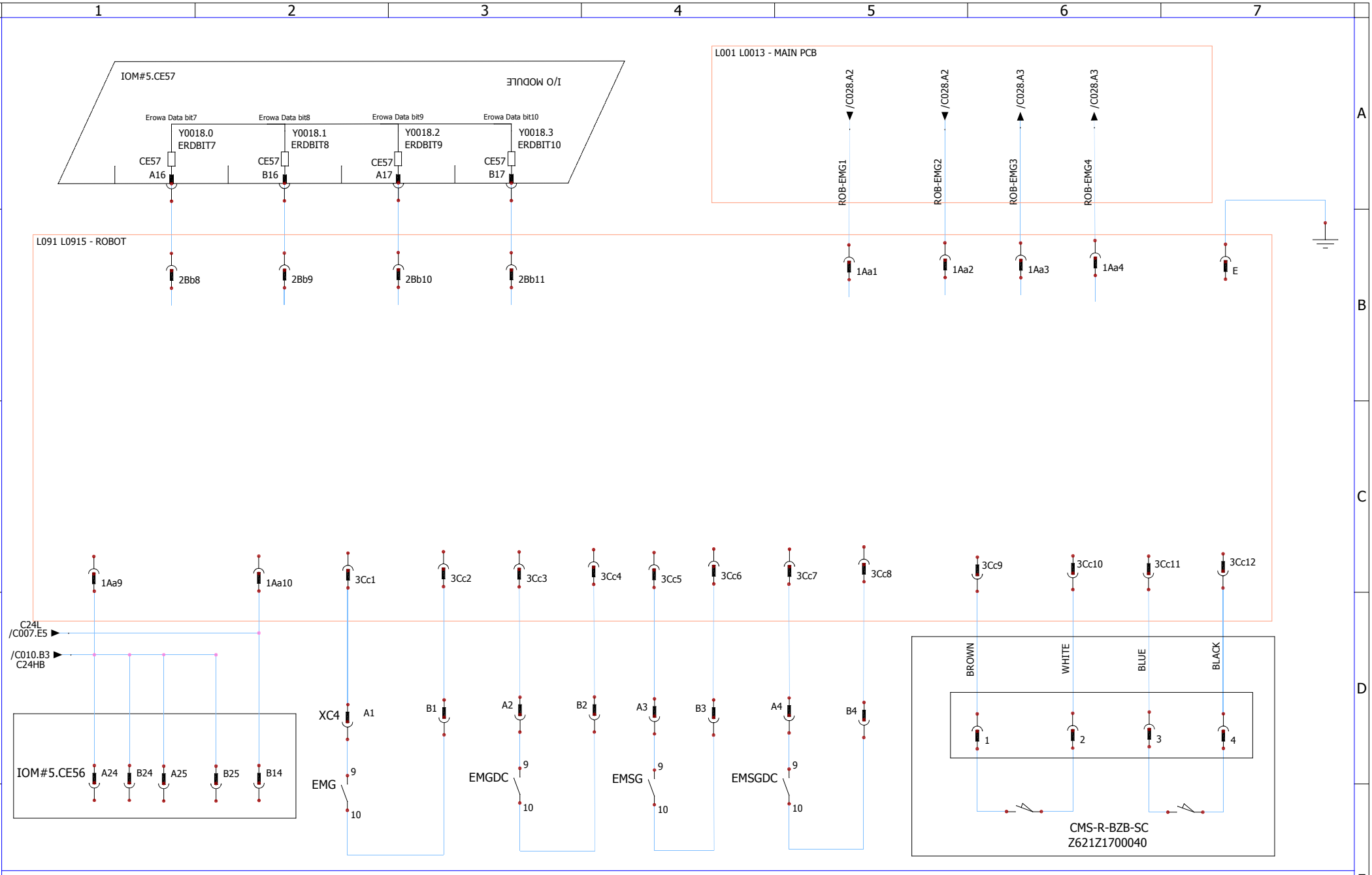


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	ROBOT INTERFACE-1 (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C100
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

F75 - ROBOT

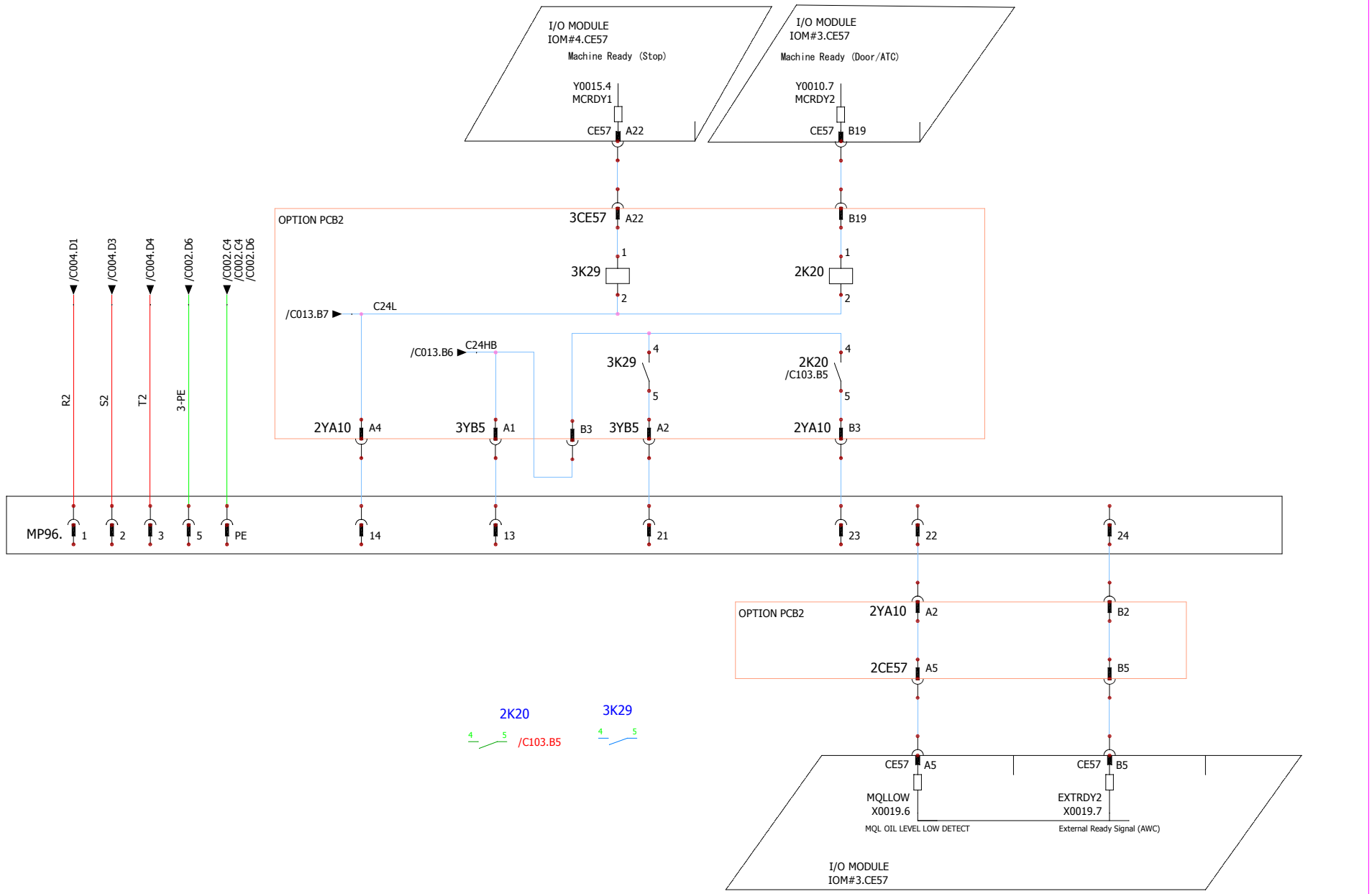


REVISION							
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	CHECKED	AVINASH		ROBOT INTERFACE-2 (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	DRAWN	ANISHA				PAGE	C101
	----.--.--	DATE	28/6/2018					

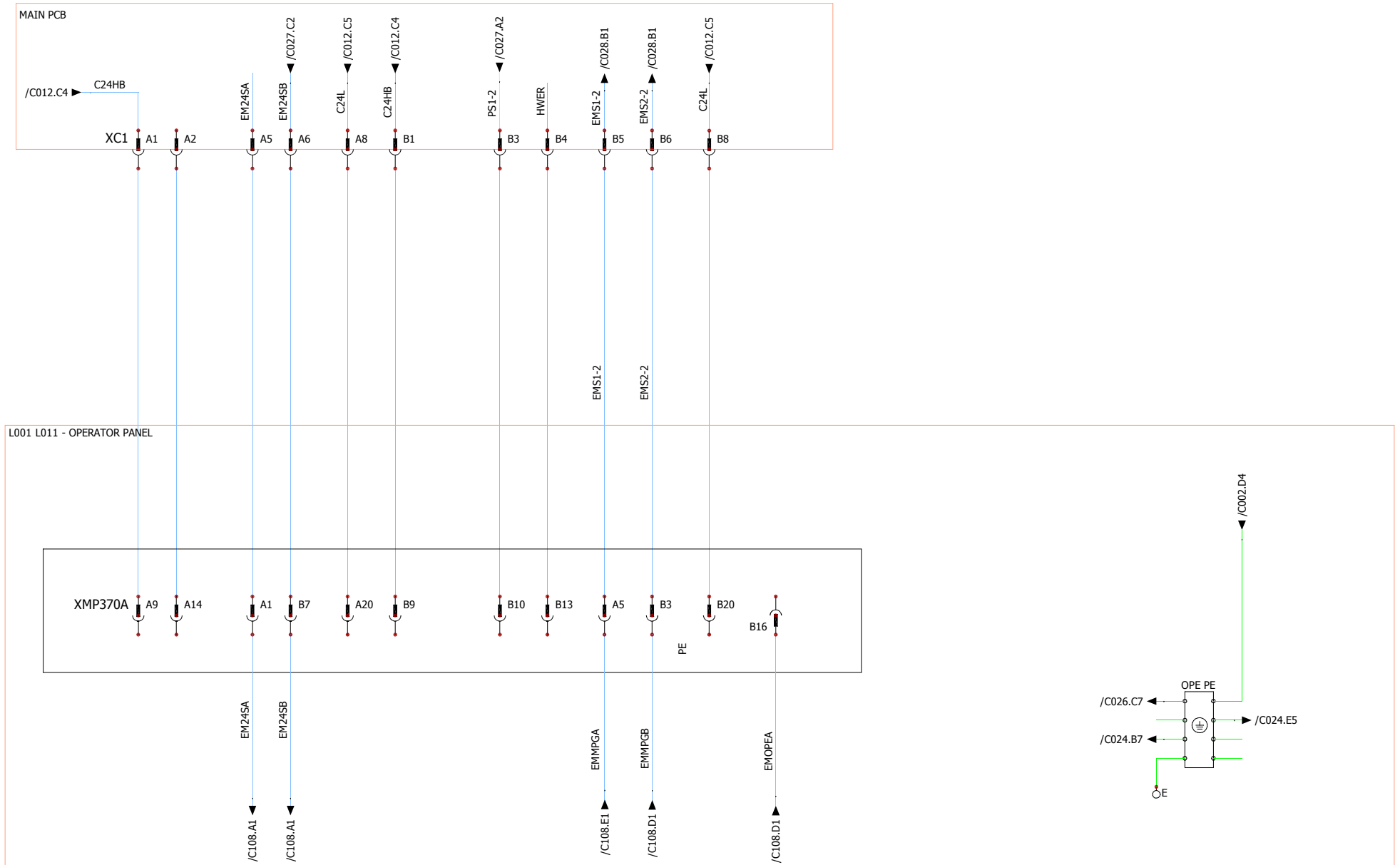


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	ROBOT INTERFACE-3 (OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C102
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

F61 - EXT HYD PUMP

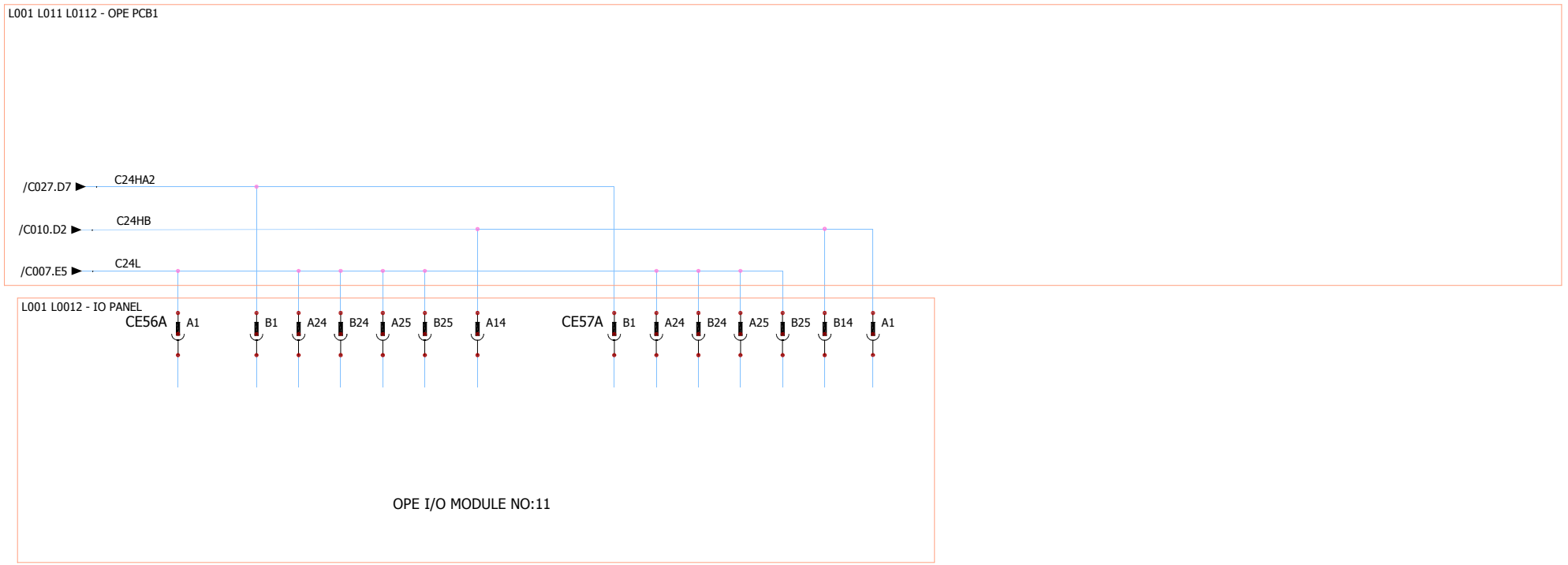


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	EXTERNAL HYDRAULIC PUMP I/F(OPT)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C103
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

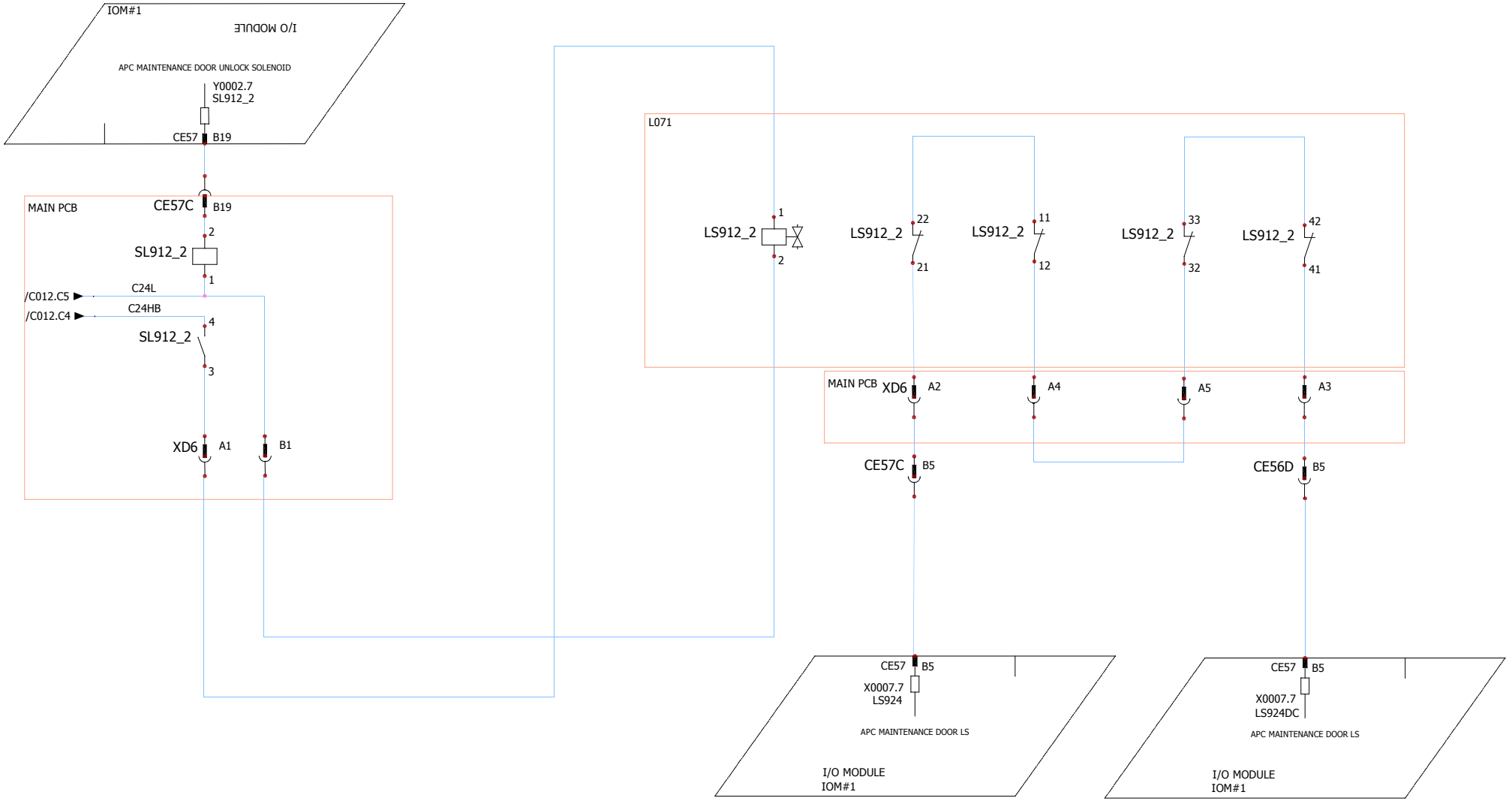


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	OPE PCB - POWER INPUT	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C104
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

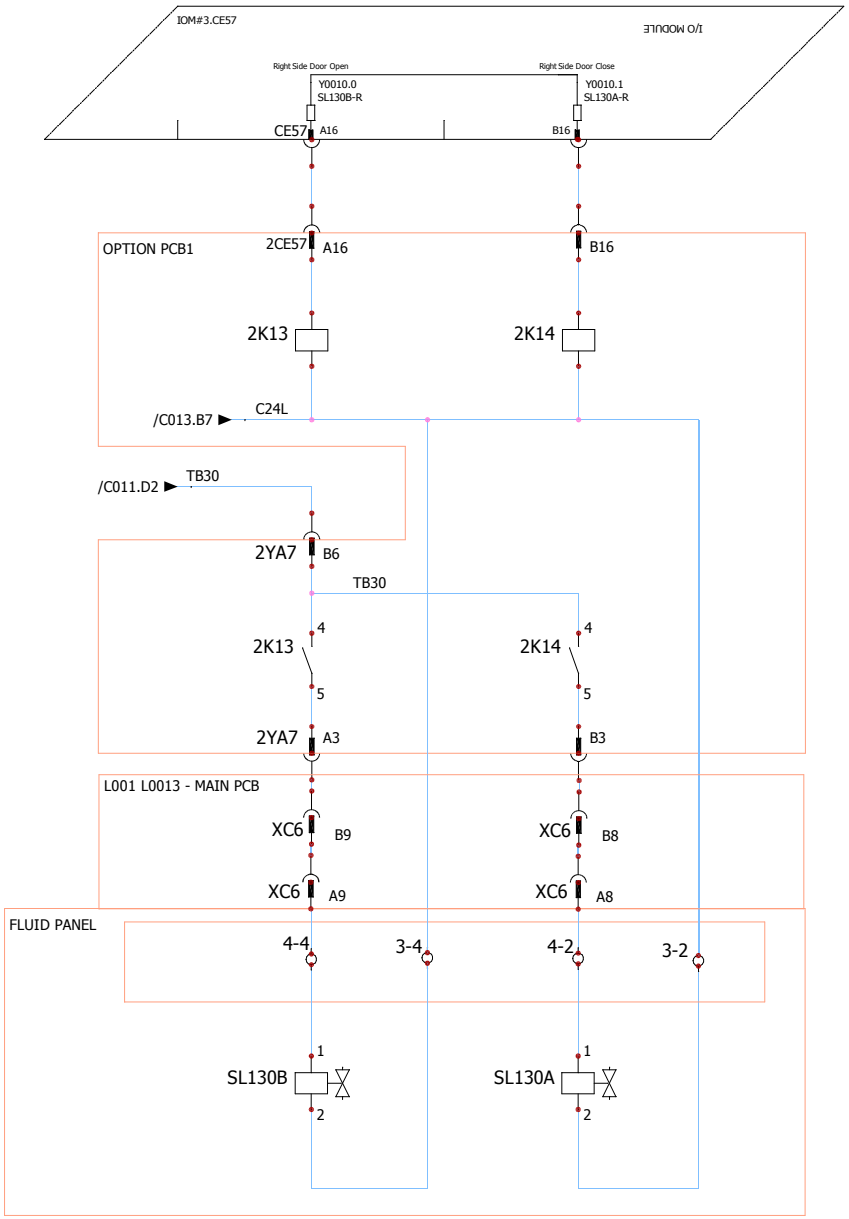
MAKINO
MAKINO ASIA PTE LTD

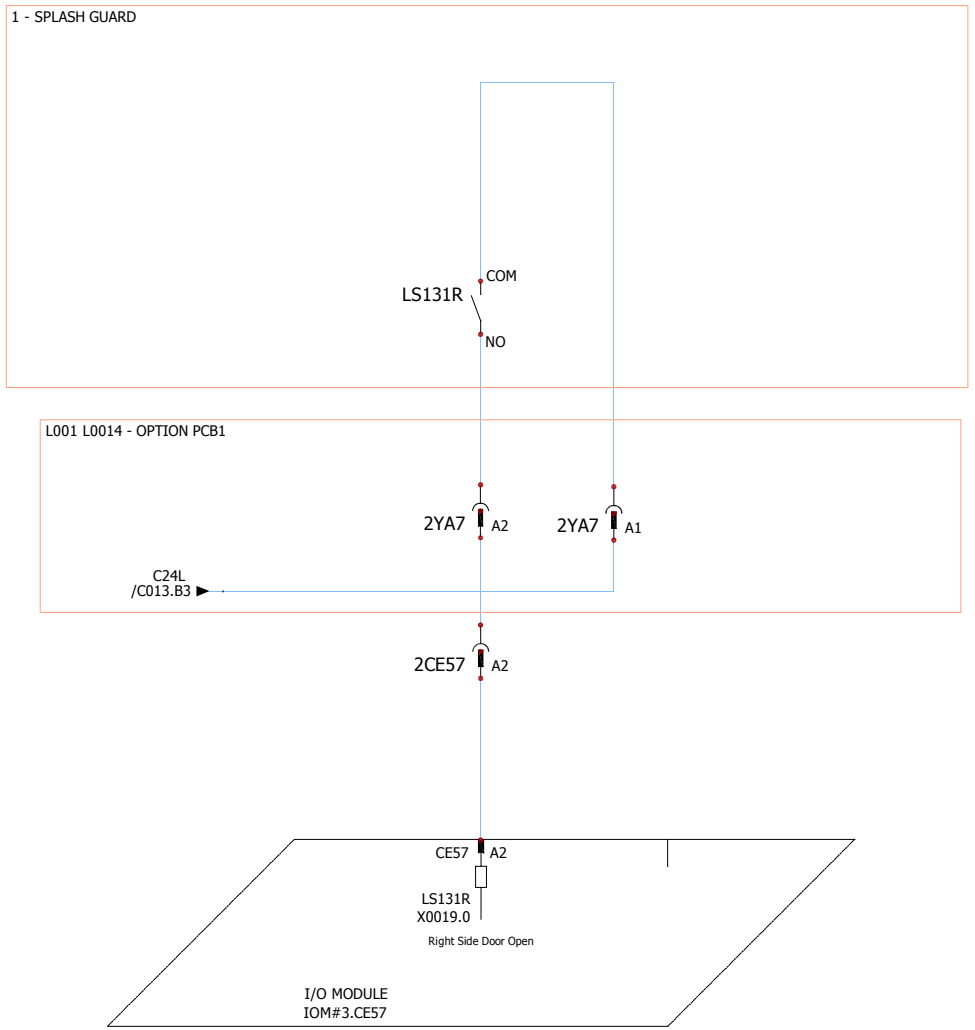


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	24V FOR OPE PCB & IO MODULE	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C105
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

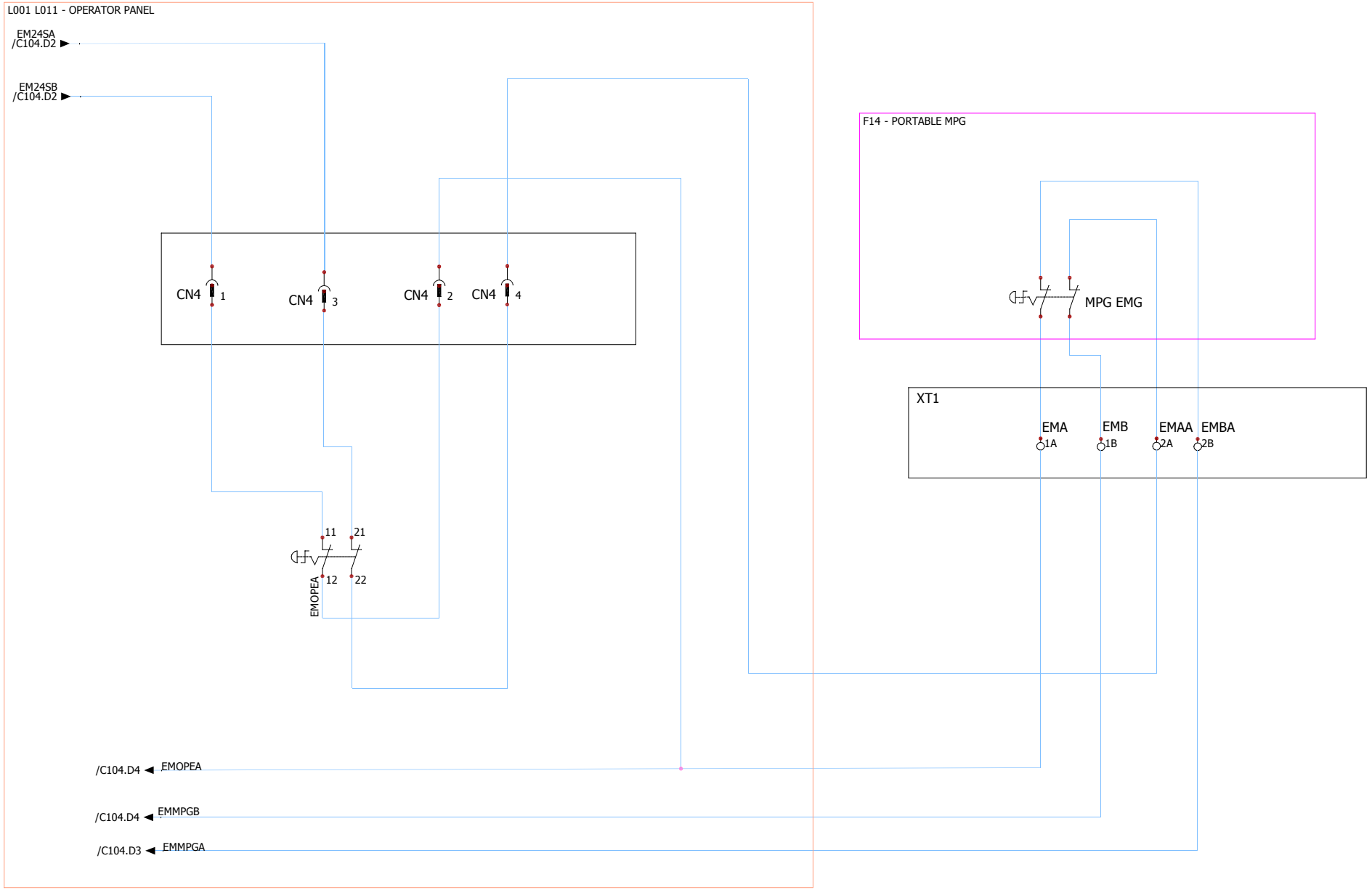


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	LOADER ACCESS DOOR	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C106
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					





REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	LOADER ACCESS DOOR-RH SIDE	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	C107
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					

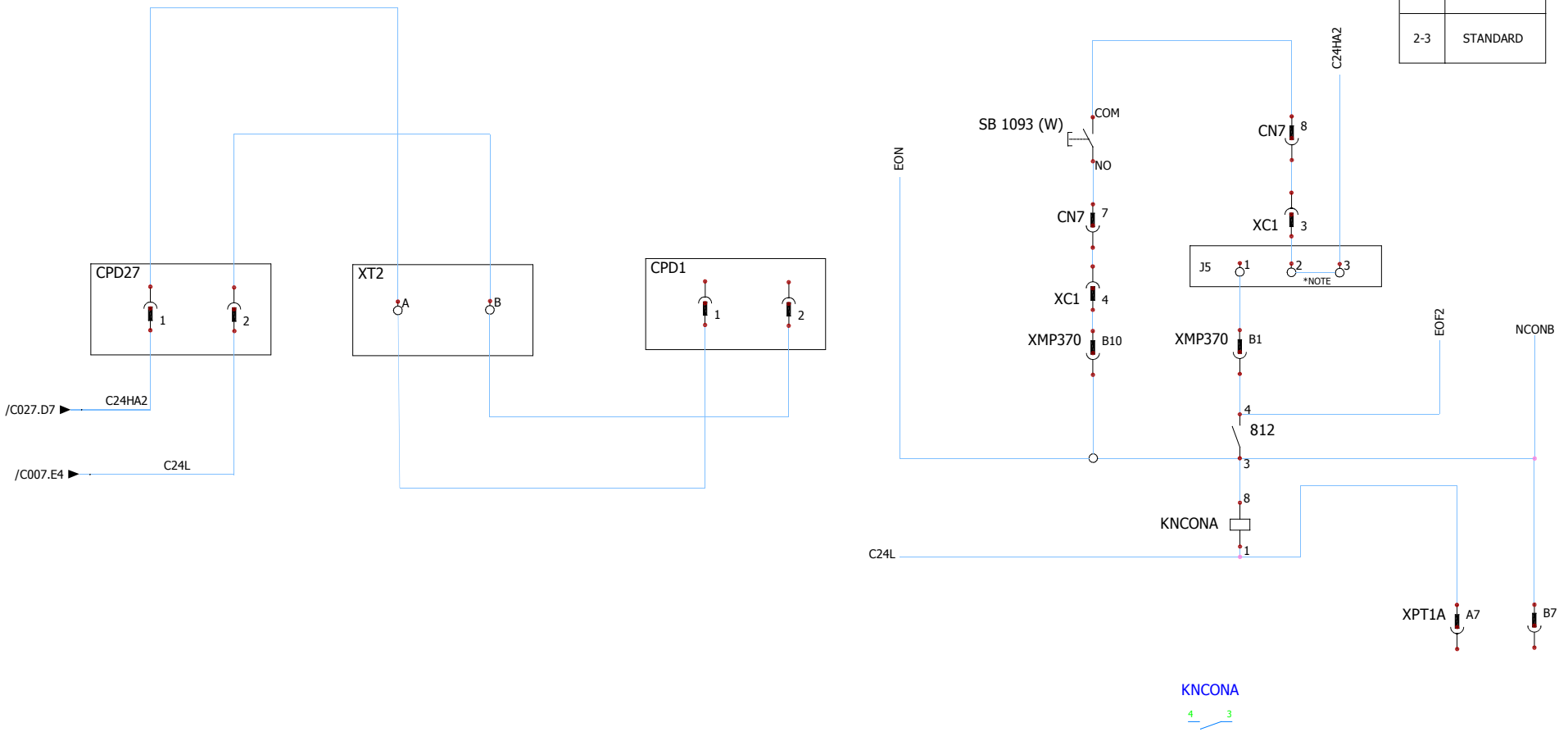


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	OPERATOR PANEL EMG CIRCUIT	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C108
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

L001 L011 L0112 - OPE PCB1

*NOTE: Jumper J5 default setting is 1-2.
jumper setting must change to 2-3

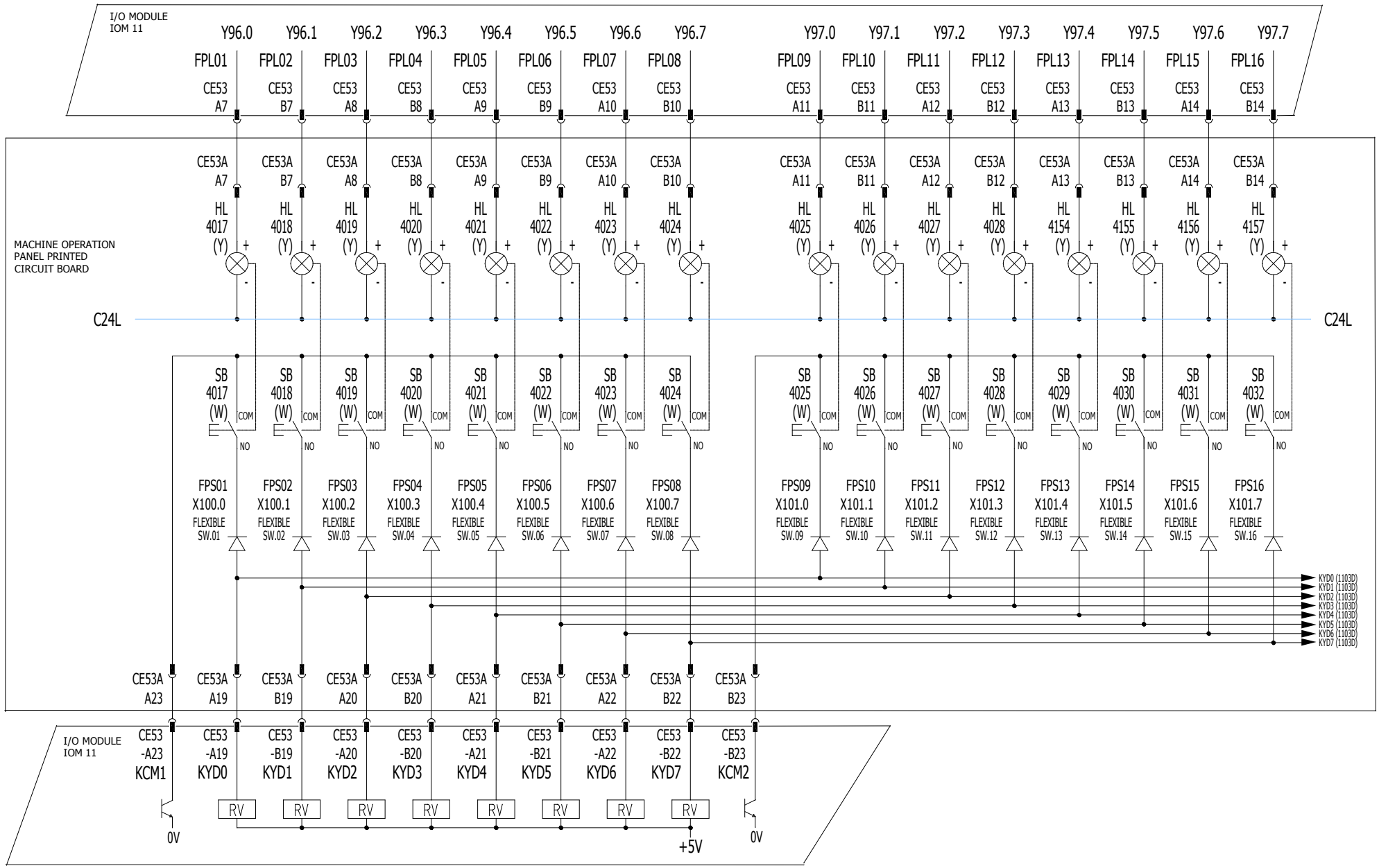
	J5
1-2	MFG SETTING
2-3	STANDARD



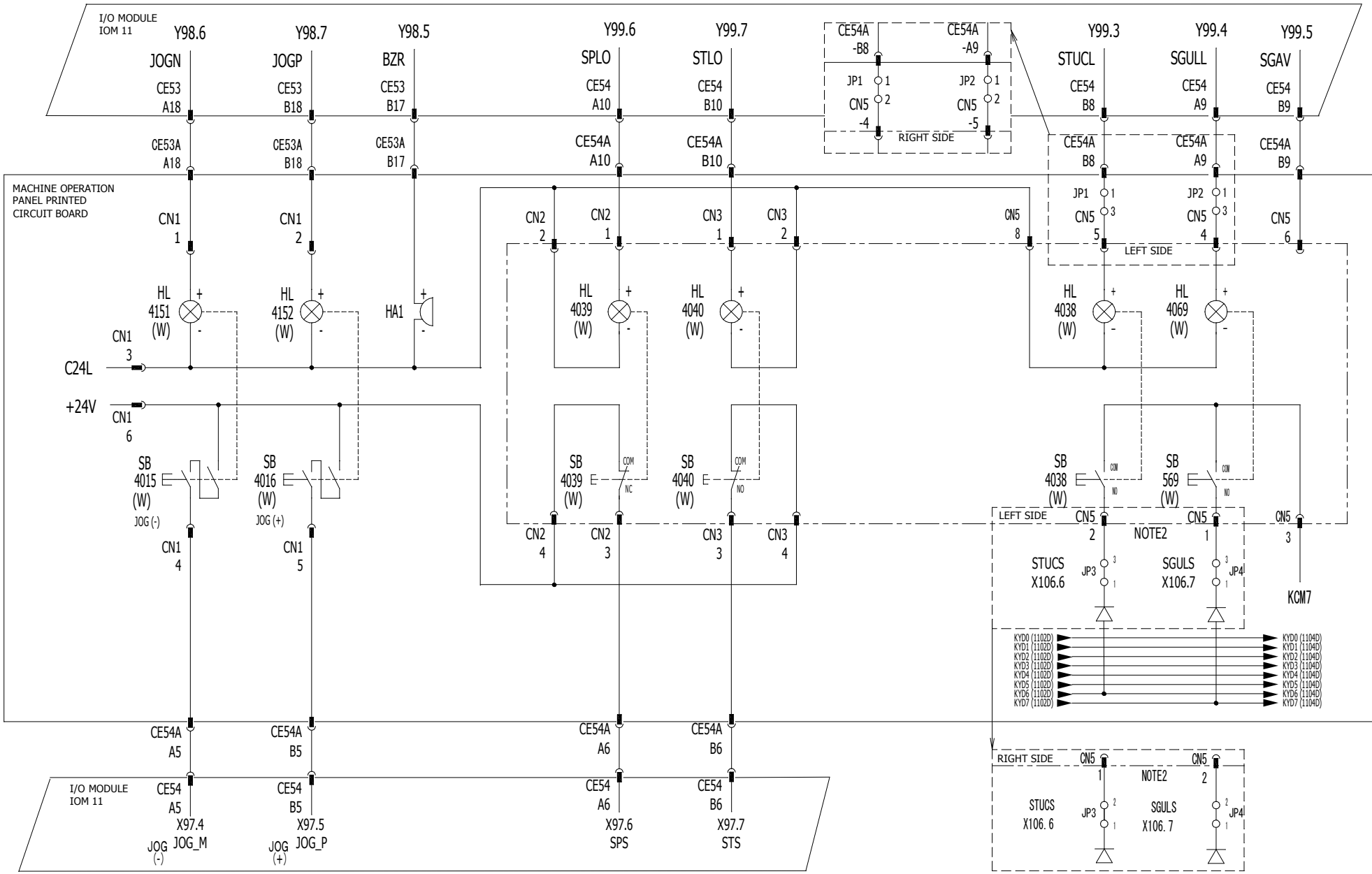
REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

MAKINO
MAKINO ASIA PTE LTD

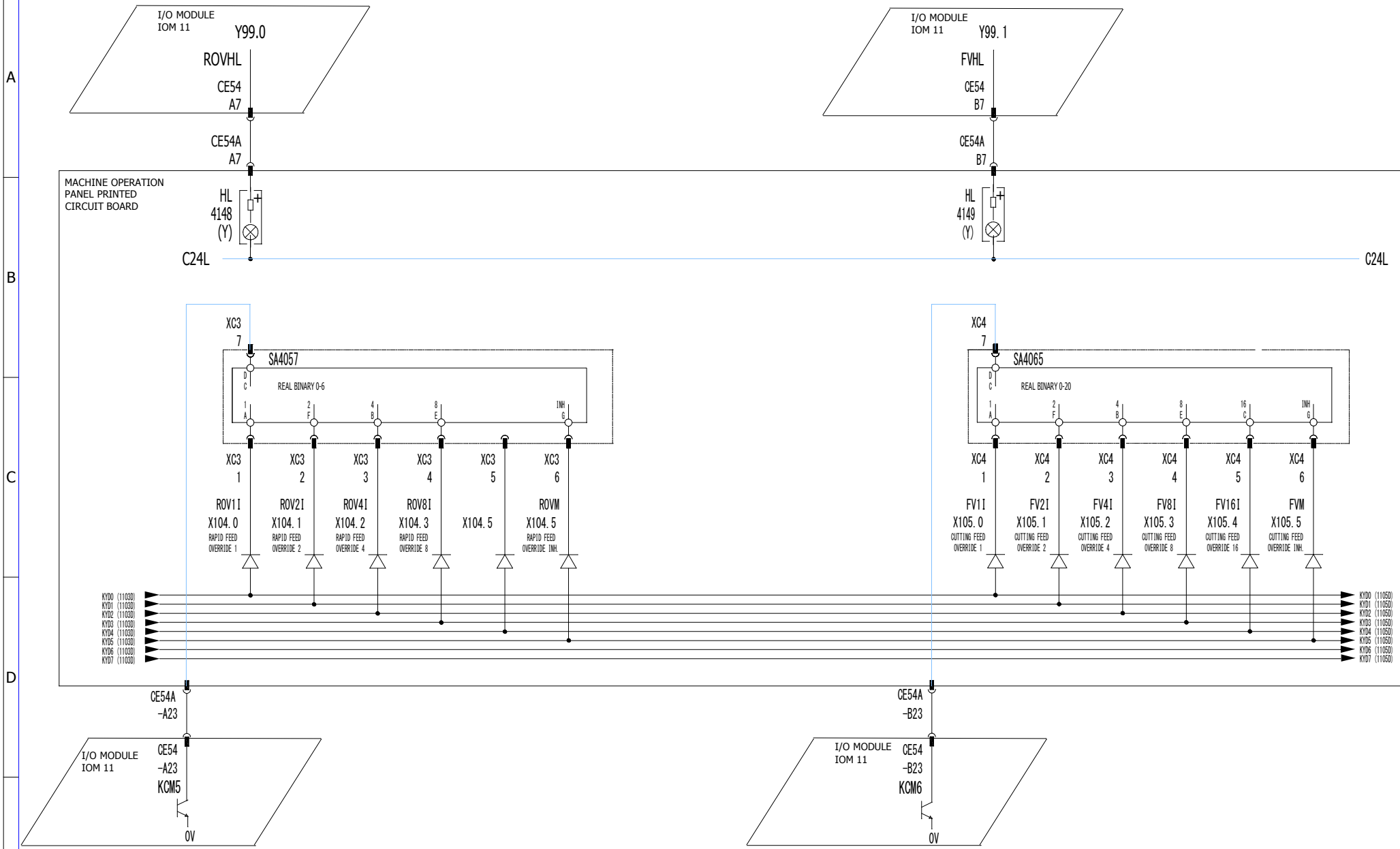
TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
OPERATOR PANEL CONTROL CIRCUIT	V2631I-C1E-Y	Ver.	B
		PAGE	C109



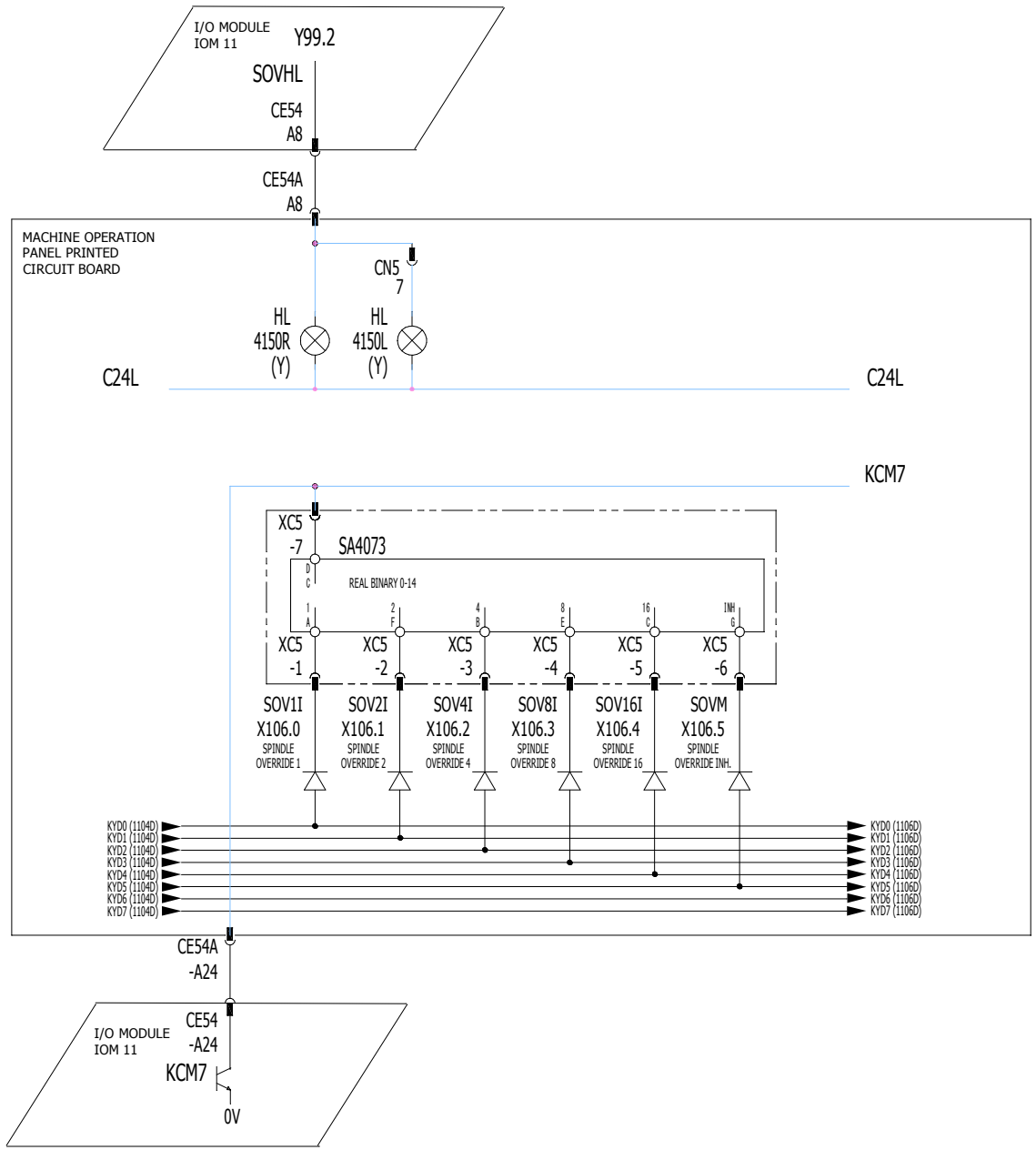
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	OPERATOR PANEL CONTROL CIRCUIT	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C110
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



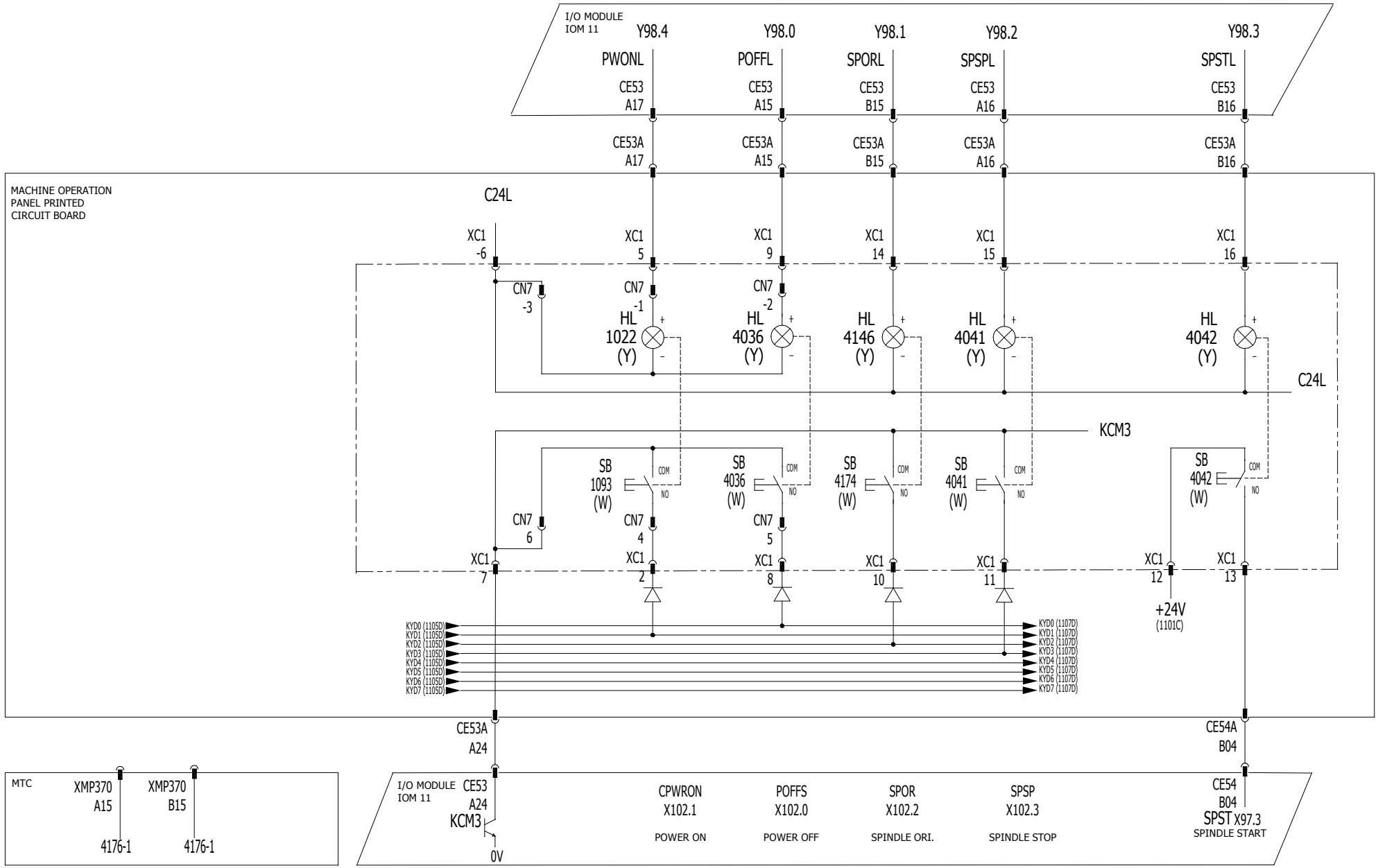
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	OPERATOR PANEL CONTROL CIRCUIT	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C111
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	OPERATOR PANEL CONTROL CIRCUIT	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C112
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



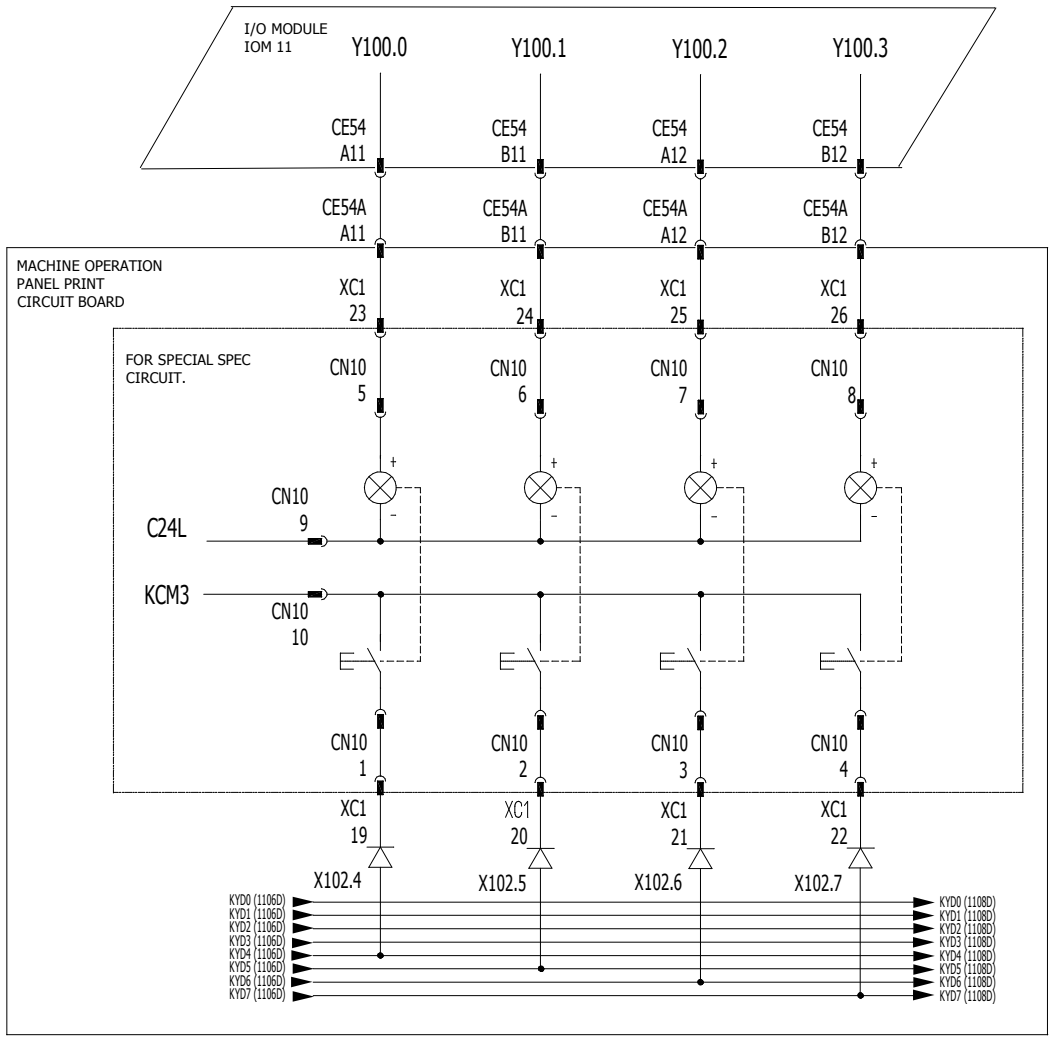
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	OPERATOR PANEL CONTROL CIRCUIT	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C113
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



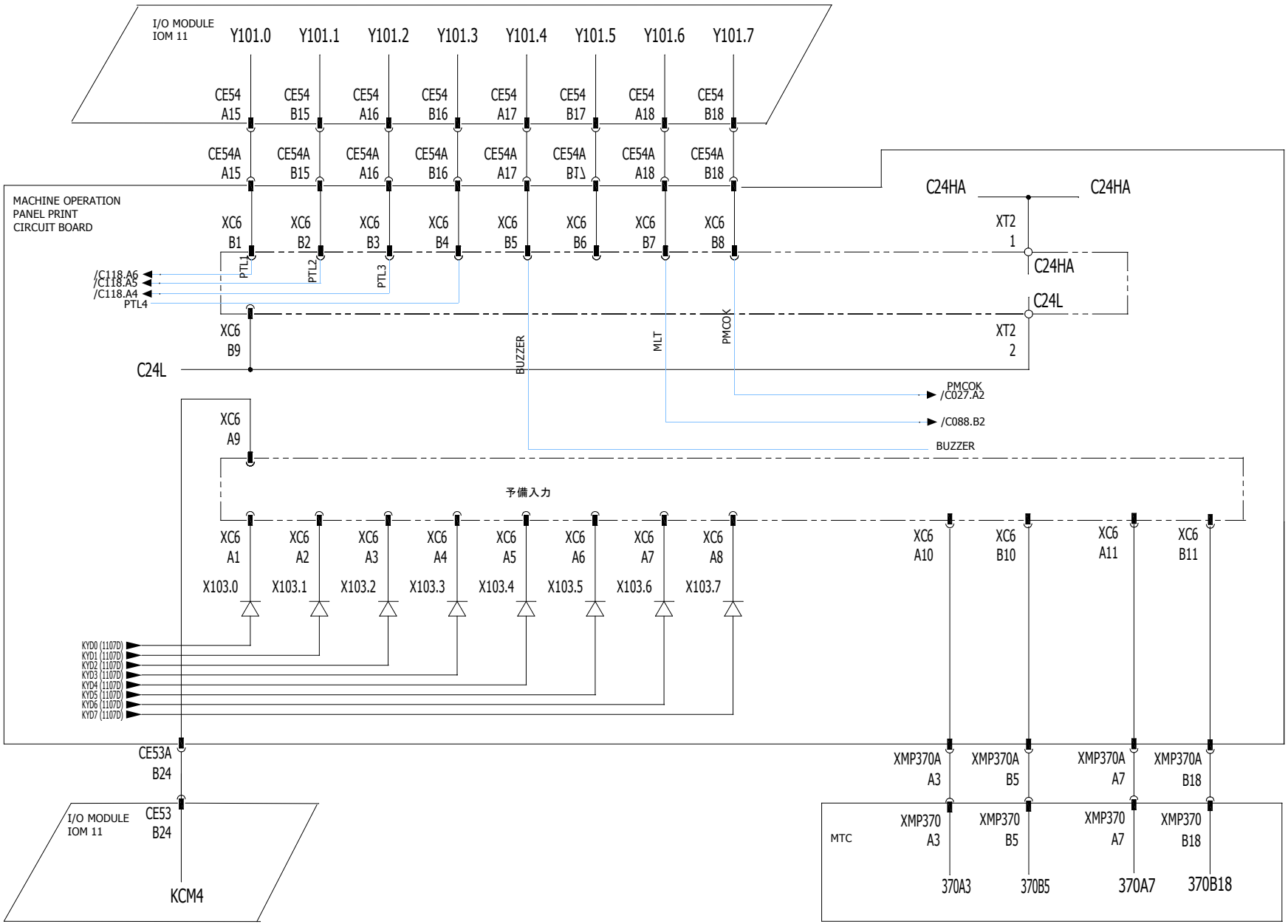
REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

MAKINO
MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
OPERATOR PANEL CONTROL CIRCUIT	V2631I-C1E-Y	Ver.	B
		PAGE	C114



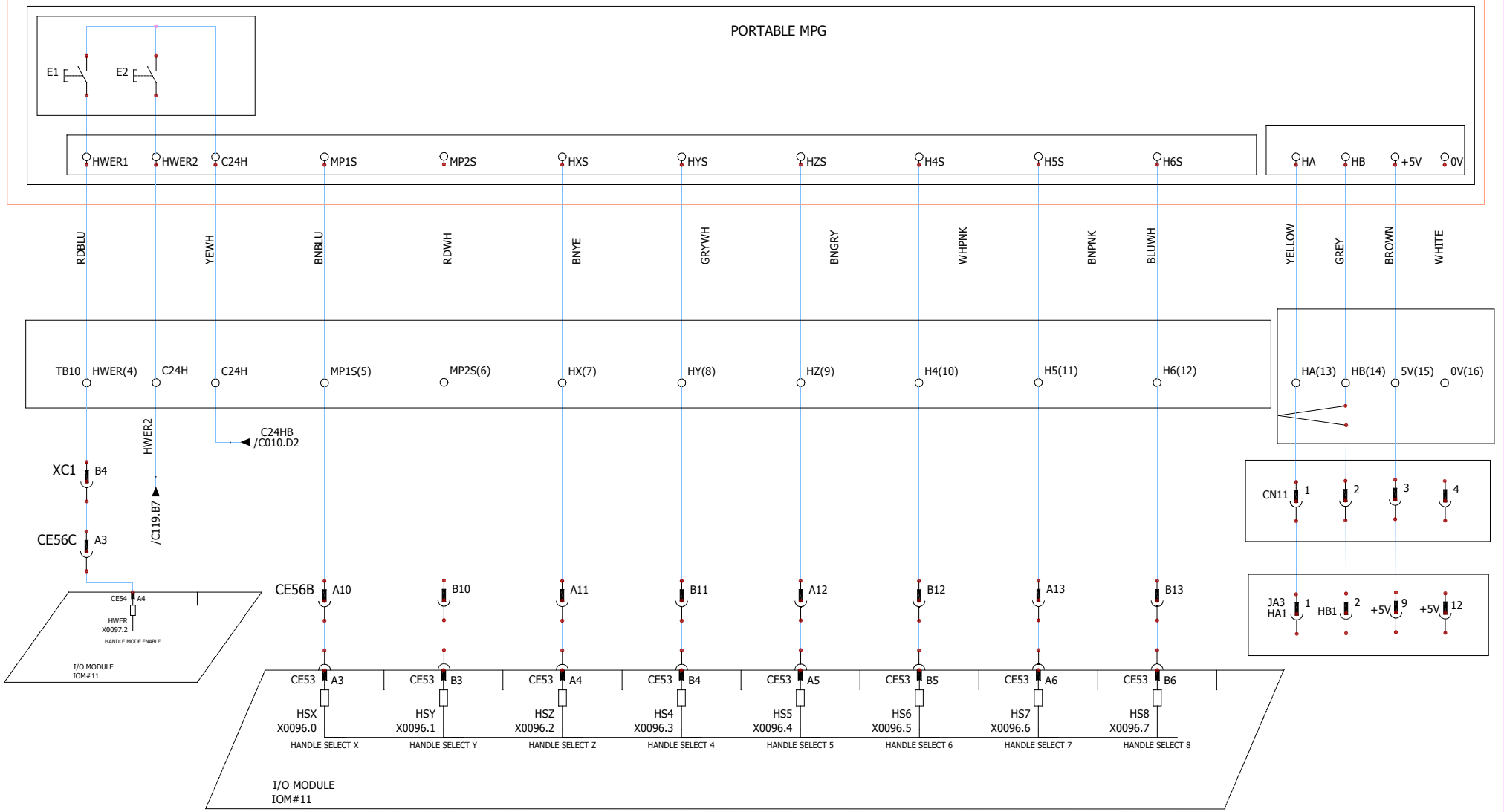
REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH		OPERATOR PANEL CONTROL CIRCUIT	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	C115
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					



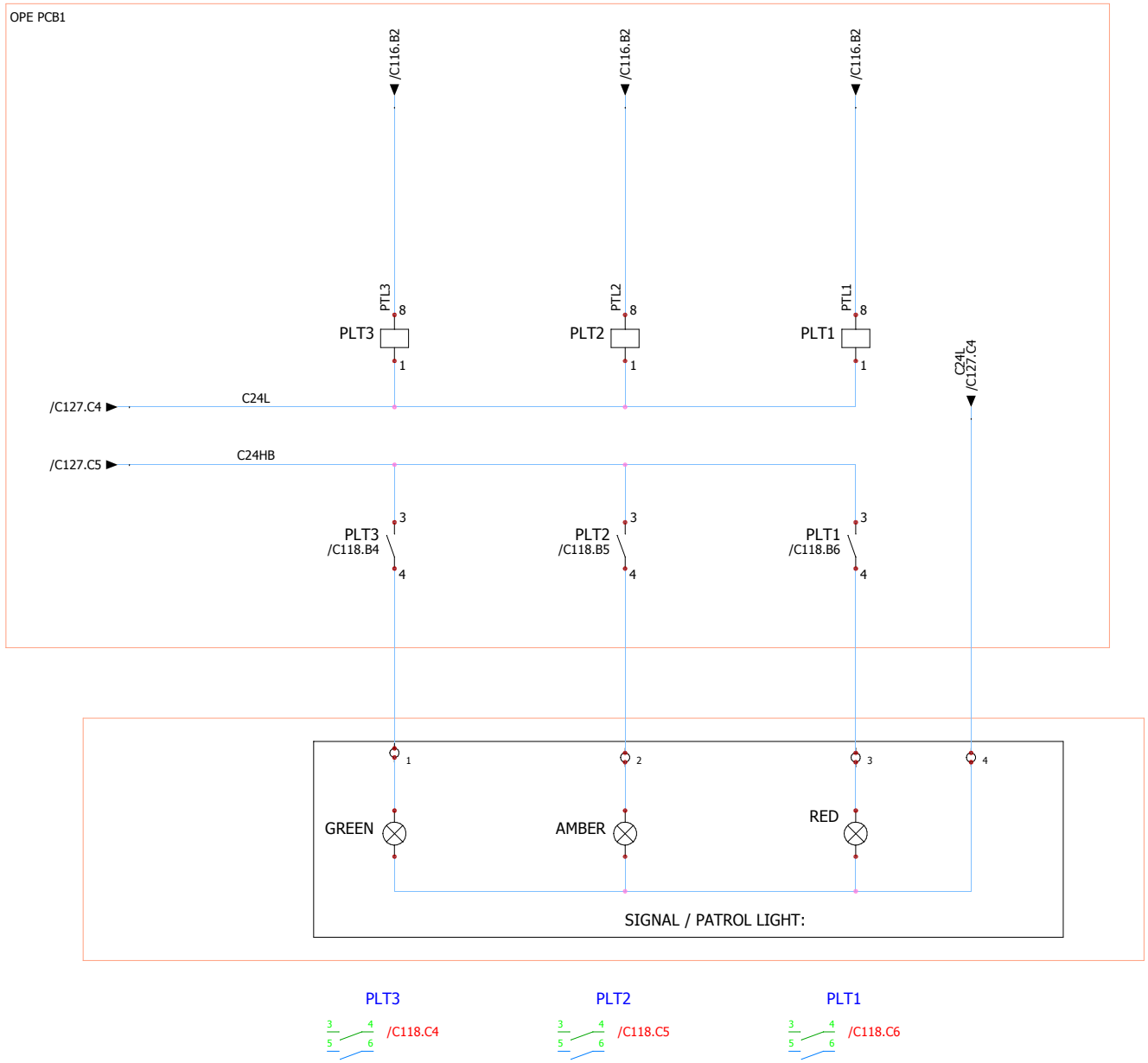
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	OPERATOR PANEL CONTROL CIRCUIT	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C116
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

F14 - PORTABLE MPG

L001 L011 L0115 - OPER BOTTOM PANEL (PORTABLE MPG)

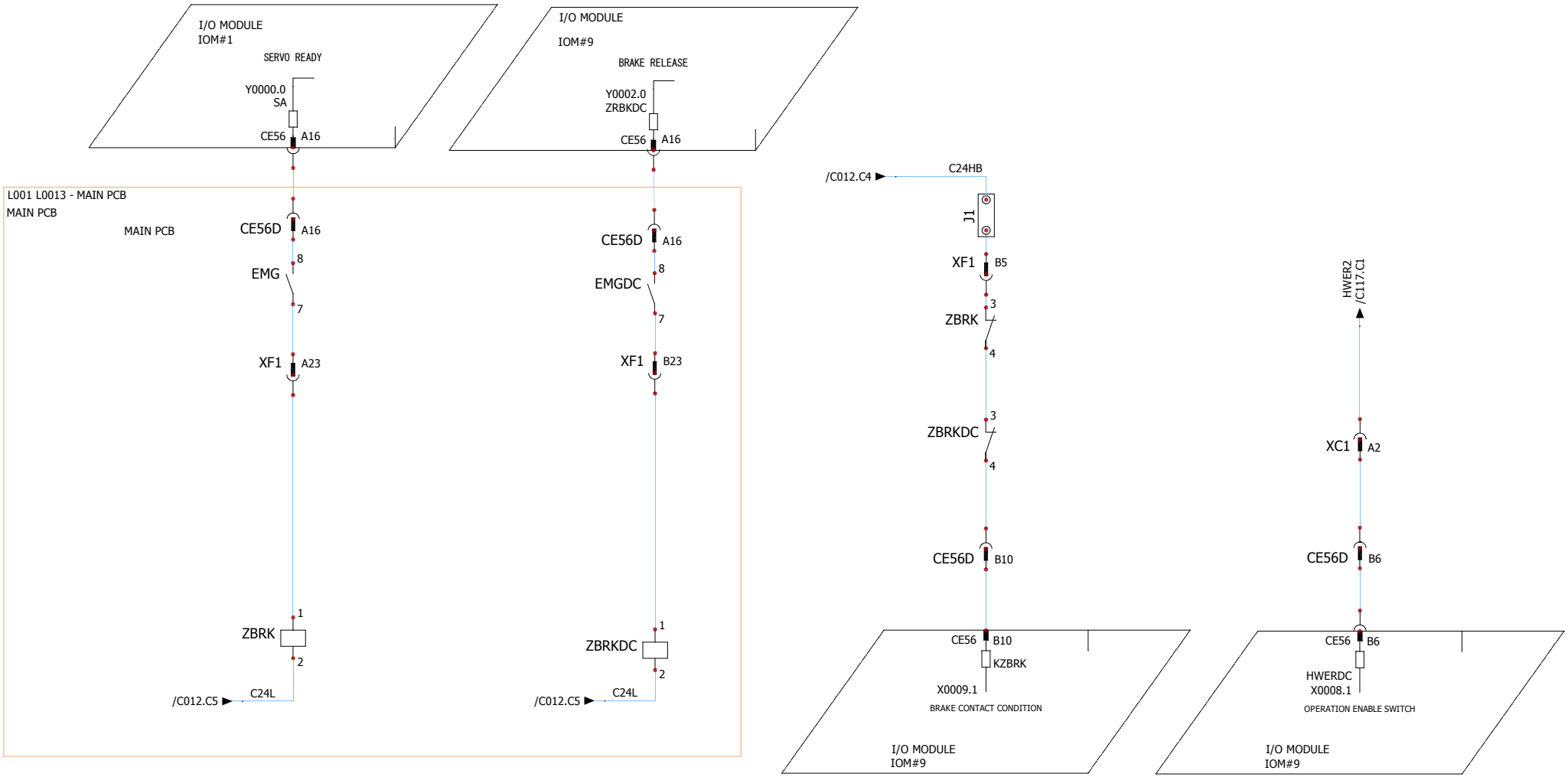


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	MANUAL PULSE GENERATOR (PORTABLE)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C117
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

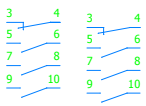


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	SIGNAL / PATROL LIGHT	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C118
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

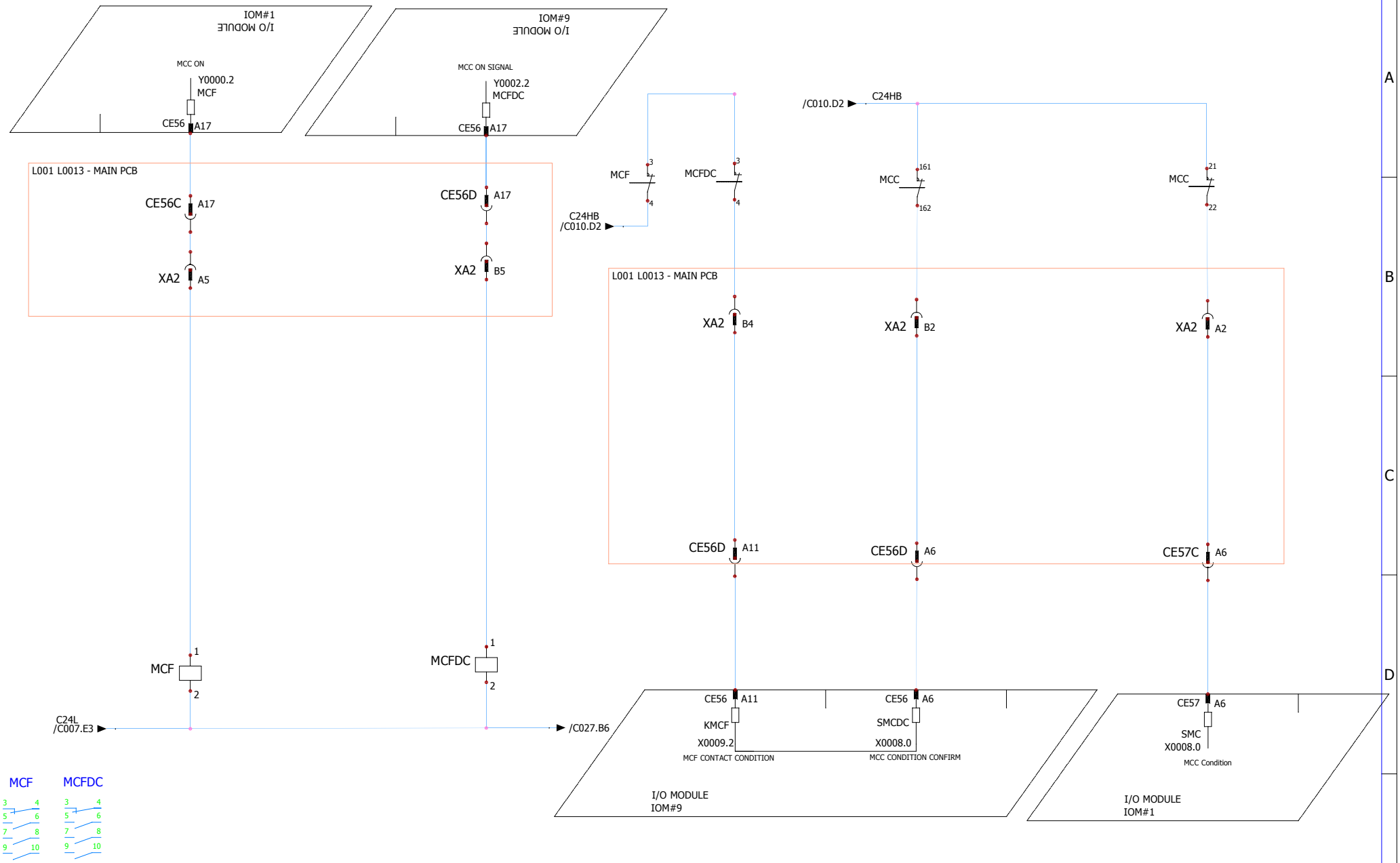
MAKINO
MAKINO ASIA PTE LTD



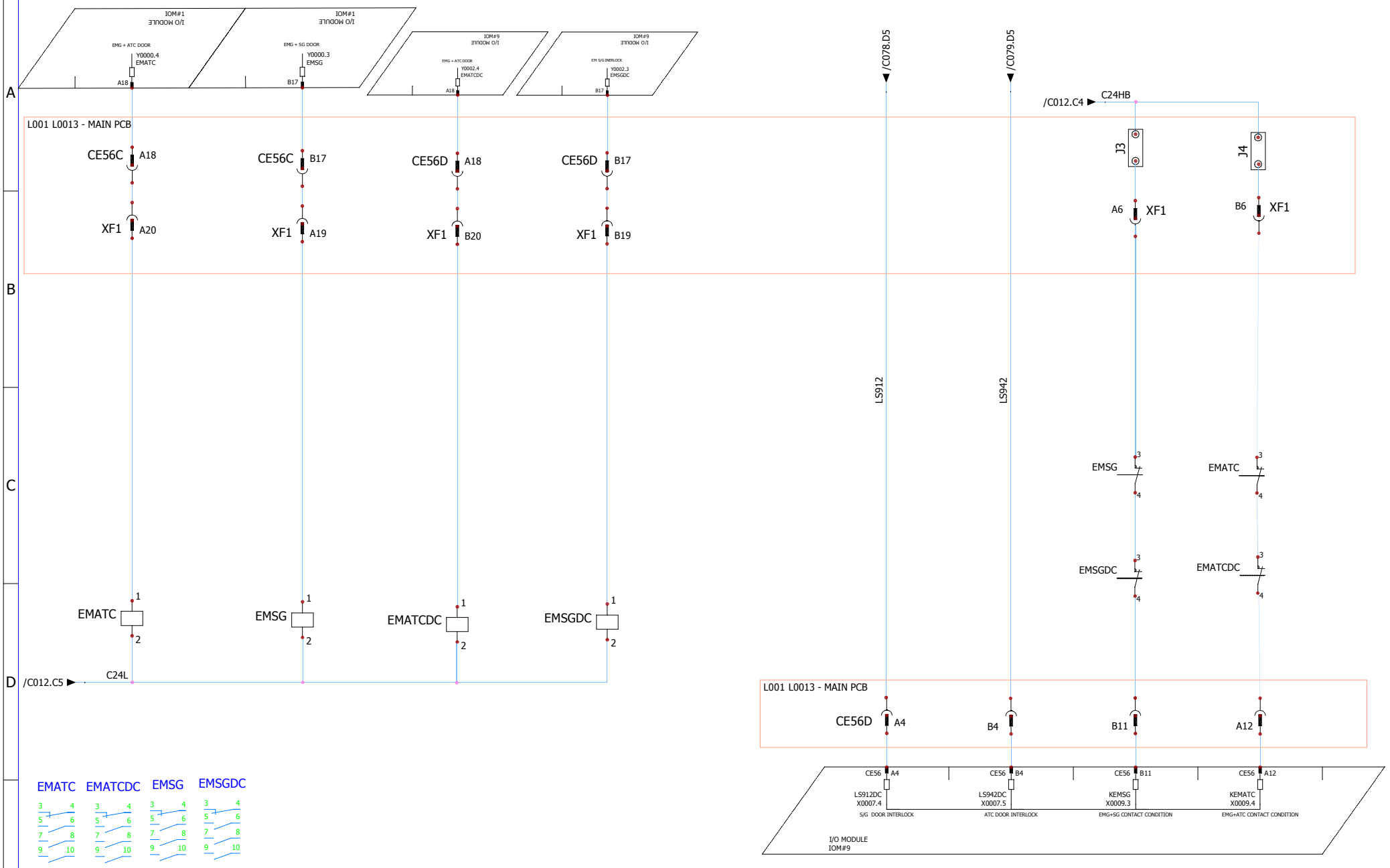
ZBRK ZBRKDC



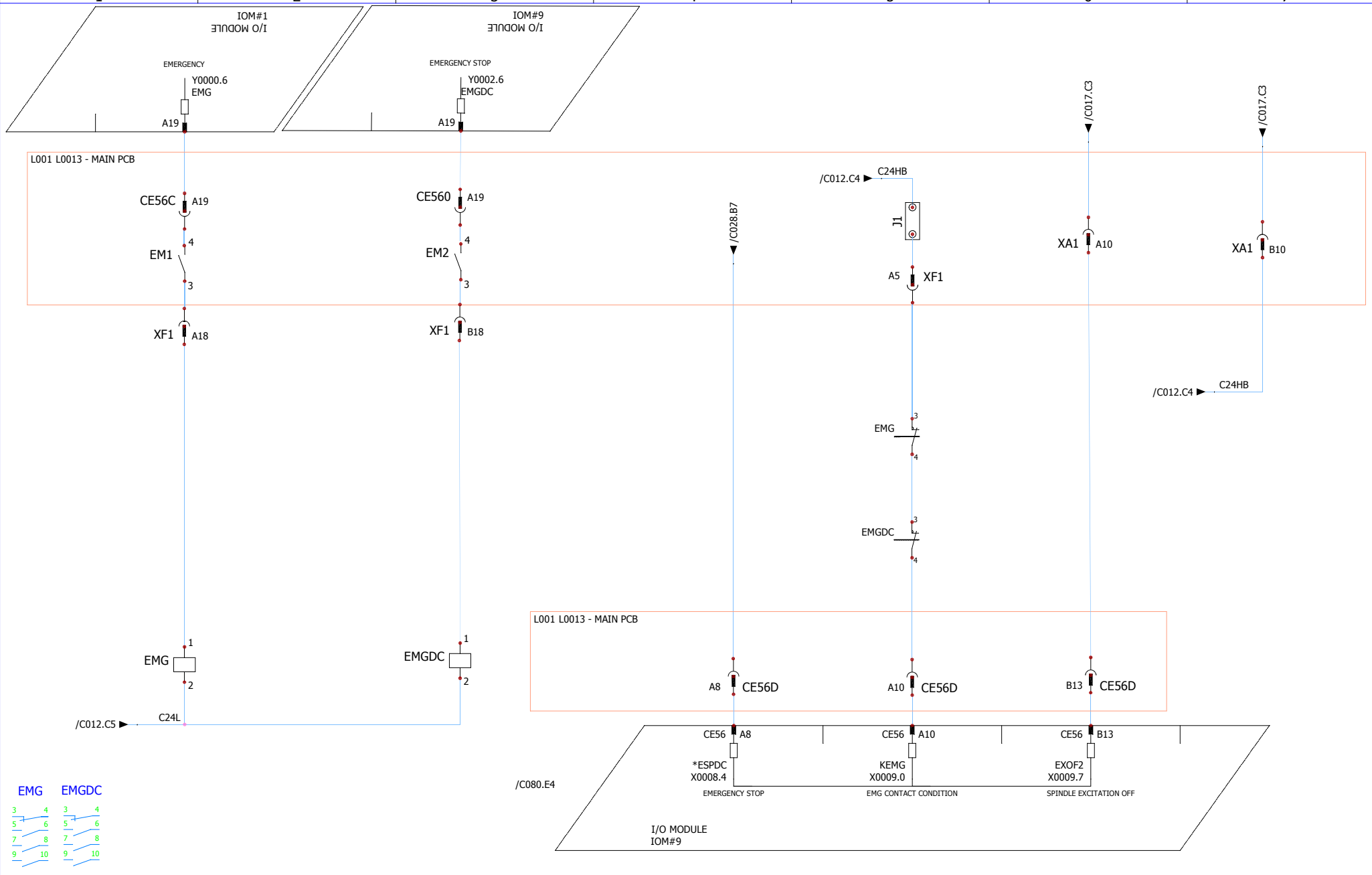
REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH		DCS ZBRK CONNECTION	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	C119
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	DCS SMC CONNECTION	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	C120
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					

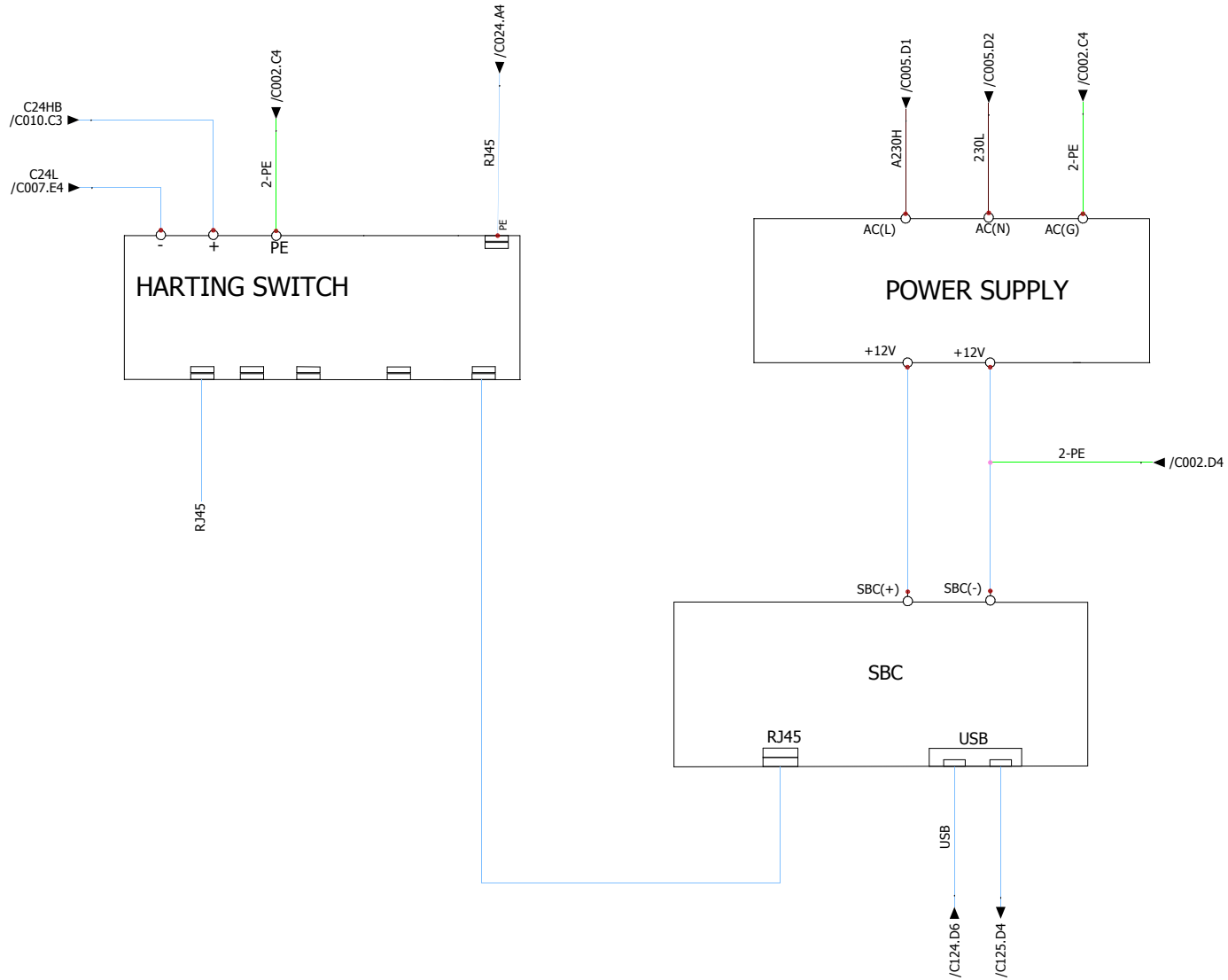


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	DCS SG/ATC CONNECTION	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C121
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					

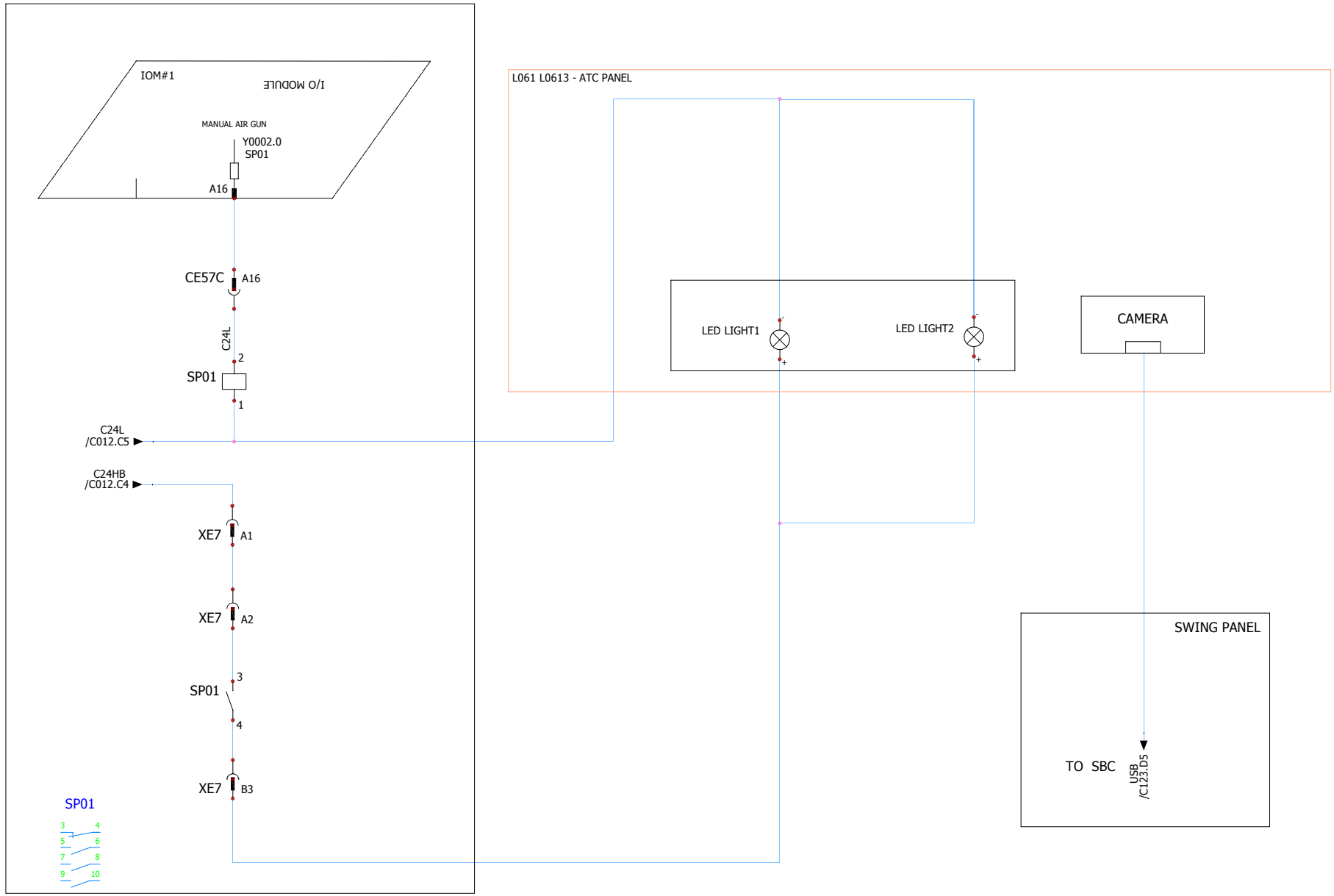


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	DCS EMG CONNECTION	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C122
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

MAKINO
MAKINO ASIA PTE LTD

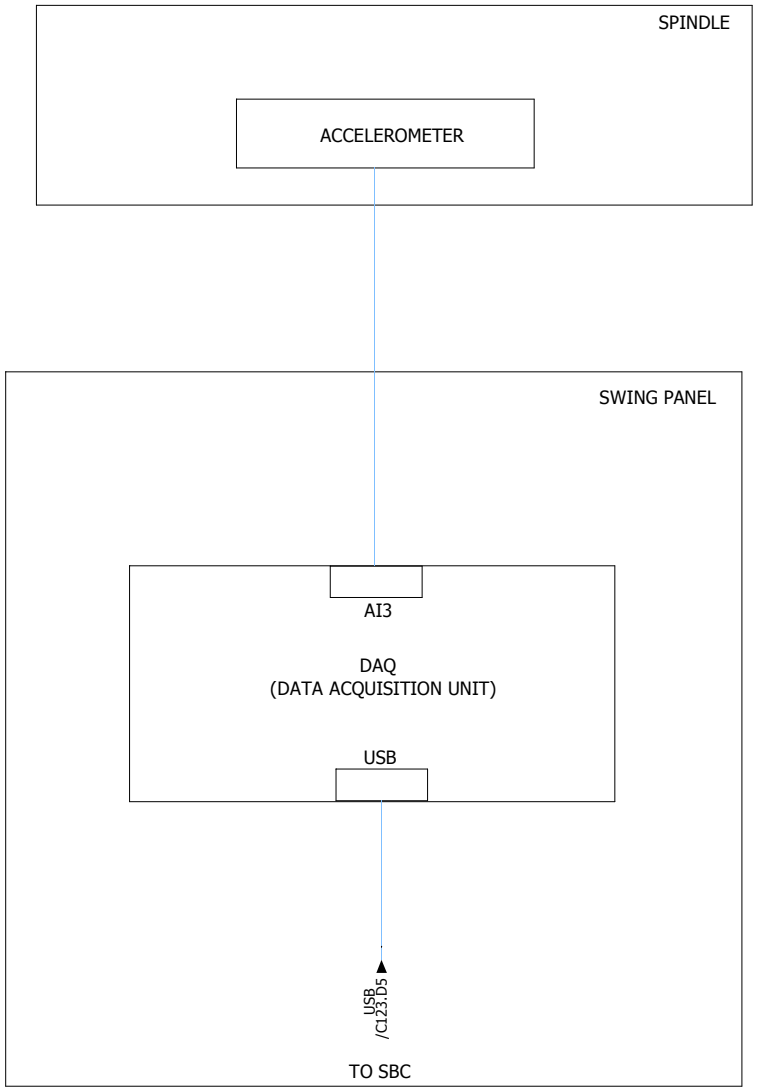


REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH		AUXILLARY COMPUTER SYSTEM(SBC)(OPT)	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	C123
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	VISION BTS (OPT)	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C124
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

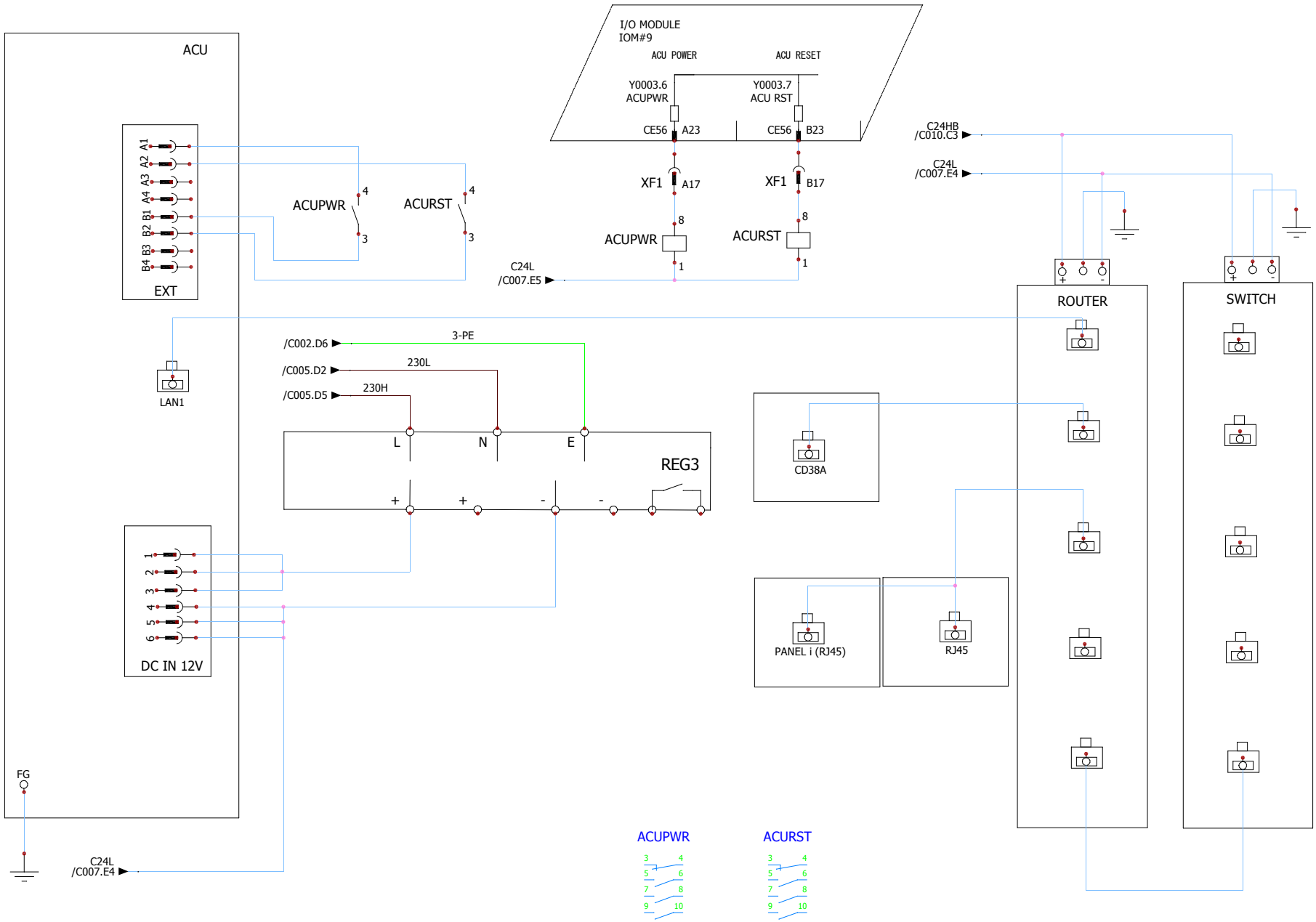
MAKINO
MAKINO ASIA PTE LTD



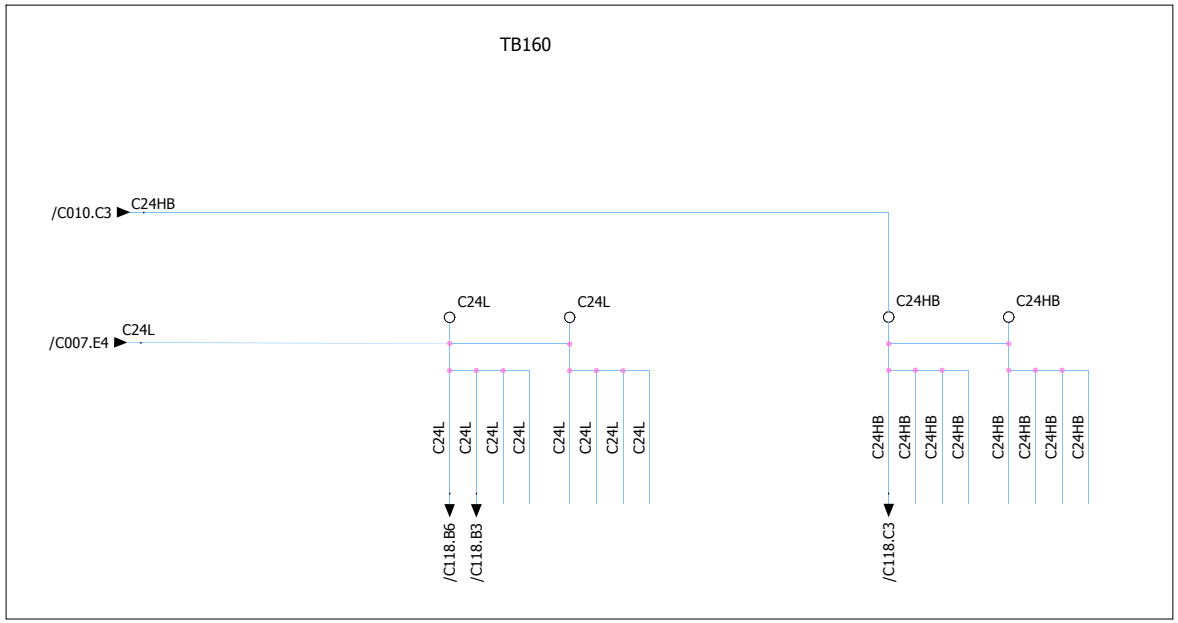
REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

MAKINO
MAKINO ASIA PTE LTD

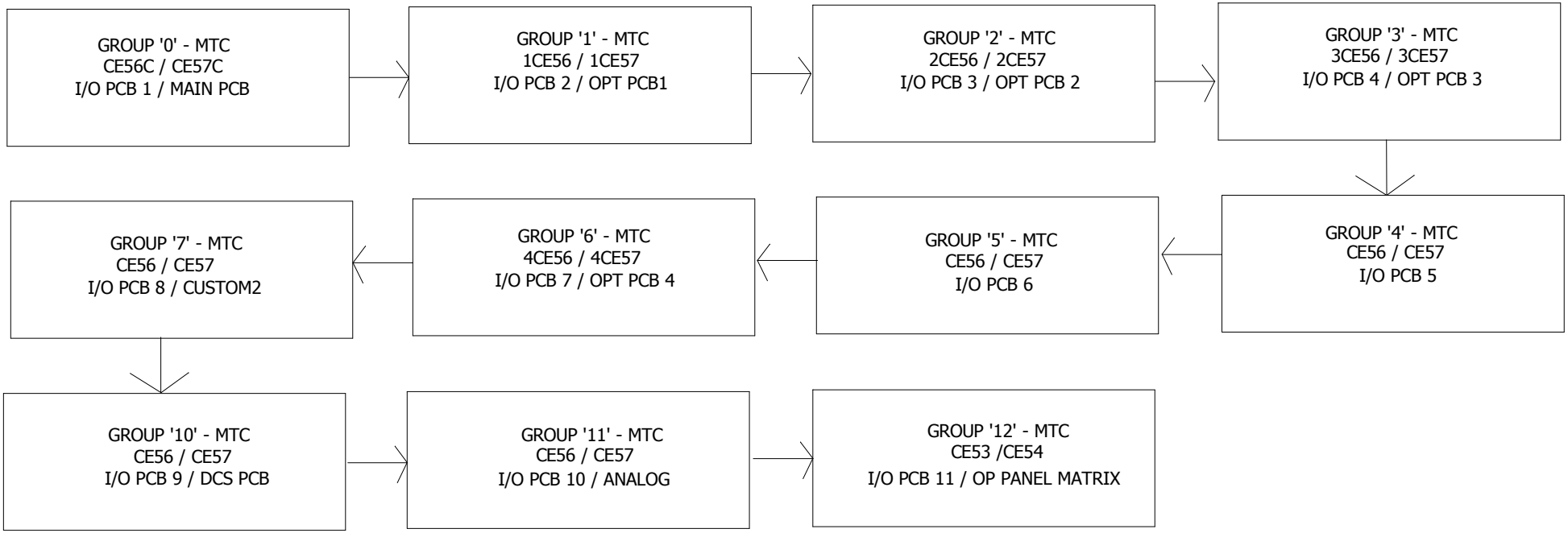
TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
SPINDLE ACTIVE CARE (OPT)	V2631I-C1E-Y	Ver.	B
		PAGE	C125



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	ACU & ETHERNET ROUTER (OPTION)	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH				PAGE	C126
	----.--.--	DRAWN	ANISHA					
	----.--.--	DATE	28/6/2018					



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD CONTROL VOLTAGE DISTRIBUTION OPE (C24HB)	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	C127
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				



GROUP '0' - -->	MTC (BASIC)
GROUP '1' - -->	MTC (BASIC)
GROUP '2' - -->	RWH,LUCC,TSC,MIST COLLECTOR,..(OPTION)
GROUP '3' - -->	APC,USER I/O,...(OPTION)
GROUP '4' - -->	ROBOT / GANTRY INTERFACE,...(OPTION)
GROUP '5' - -->	PRAGATI INDEXER/ CUSTOM /SPARE I/O (OPTION)
GROUP '6' - -->	FIXTURE,TWO HAND PB,AIR SEAT CHECK,STRIKER SWITCH ..(OPTION)
GROUP '7' - -->	CUSTOM2
GROUP '10' - -->	DCS (OPTION)
GROUP '11' - -->	ANALOG
GROUP '12' - -->	OPERATOR PANEL (BASIC)

REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	6/28/2018	

MAKINO
MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
I/O CONFIGURATION	V2631I-C1E-Y	Ver.	B
		PAGE	C128

I/O OPERATOR PNL, 48/32 A03B-0824-K203	IOM#1

IOM1-CE56

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0004.7	SKIP	SKIP SIGNAL	B5	/C066.D1
X0004.6	MSALB	MEAS SYSTEM BATTERY ERROR	A5	/C066.D1
X0004.5	MSALA	MEAS SYSTEM RECEPTION ALM	B4	/C066.D1
X0004.4	STA	STATIC	A4	/C066.D1
X0004.3	LSROK	LASER OK	B3	/C066.D1
X0004.2	HWER	HANDLE ENABLE	A3	
X0004.1	*LS170	SP LUB OIL FLOW SW	B2	/C056.E4
X0004.0	*OMAL1	SP LUB OIL UNIT ALM	A2	/C056.E4

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0005.7	LS2308	OIL AIR UNIT AIR PRESS DETECT	B9	
X0005.6	*LS2307	OIL AIR UNIT LOW LVL	A9	
X0005.5	TH154	OIL AIR UNIT TH ALM	B8	
X0005.4	*LS173	SP LUB OIL FIL PRESS LS	A8	/C056.E4
X0005.3	*LS1343	COOLANT CHILLER FLOW SWITCH	B7	/C066.D1
X0005.2	*OMAL2	COOLANT CHILLER UNIT ALARM	A7	/C054.E4
X0005.1	*LS1319	CLNT TANK EMPTY DETECT	B6	/C048.E3
X0005.0	AIRAL	AIR DRYER ALM	A6	/C092.E4

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0006.7	GRAY7	ATC ARM BIT 7	B13	/C061.E1
X0006.6	GRAY6	ATC ARM BIT 6	A13	/C061.E1
X0006.5	GRAY5	ATC ARM BIT 5	B12	/C061.E1
X0006.4	GRAY4	ATC ARM BIT 4	A12	/C061.E1
X0006.3	GRAY3	ATC ARM BIT 3	B11	/C061.E1
X0006.2	GRAY2	ATC ARM BIT 2	A11	/C061.E1
X0006.1	GRAY1	ATC ARM BIT 1	B10	/C061.E1
X0006.0	GRAY0	ATC ARM BIT 0	A10	/C061.E1

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
Y0000.7	POUT	AUTO POWER OFF	B19	/C015.A1
Y0000.6	EMG	EMERGENCY	A19	/C122.A2
Y0000.5	EMAPC	EMG + APC DOOR	B18	
Y0000.4	EMATC	EMG + ATC DOOR	A18	/C121.A1
Y0000.3	EMSG	EMG + SG DOOR	B17	/C121.A2
Y0000.2	MCF	MCC ON	A17	/C120.A2
Y0000.1	OTR	OVER TRAVEL RELEASE	B16	
Y0000.0	SA	SERVO READY	A16	/C119.A1

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
Y0001.7	RCHP	SPINDLE STATOR CONN	B23	/C055.A1
Y0001.6	CLOMON	COOLANT CHILLER ON	A23	/C054.A1
Y0001.5	OMON	SPINDLE CHILLER ON	B22	/C056.A1
Y0001.4	SL2300	SPINDLE LUBRICATION AIR (OILAIR UNIT)	A22	
Y0001.3	SL18	SPINDLE AIR SEAL	B21	/C057.A1
Y0001.2	SL192	THROUGH SPINDLE AIR	A21	/C050.A1
Y0001.1	SL09	TAPER CLEANING AIR 1	B20	/C057.A1
Y0001.0	SL181	AIR BLOW	A20	/C050.A1

REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

MAKINO
MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
DI/DO ADDRESS TABLE - IO MODULE 1	V2631I-C1E-Y	Ver.	B
		PAGE	D001

I/O OPERATOR PNL, 48/32 A03B-0824-K203	IOM#1

IOM1-CE57

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0007.7	LS924	APC MAINTENANCE DOOR LS	B5	/C066.D1
X0007.6	LS966	APC DOOR CLOSE LS	A5	/C066.D1
X0007.5	LS942	ATC MAGAZINE DOOR CLOSE LS	B4	/C079.E3
X0007.4	LS912	OPERATOR DOOR INTERLOCK LS	A4	/C078.E2
X0007.3	RCH2	SP HIGH SPEED	B3	/C055.E5
X0007.2	RCH1	SP LOW SPEED	A3	/C055.E5
X0007.1	LS2466	AREA SENSOR	B2	/C080.E5
X0007.0	GRAY8	ATC ARM BIT 8	A2	/C061.E1

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0008.7	OT2Z	OVER TRAVEL Z	B9	/C029.D5
X0008.6	OT2Y	OVER TRAVEL Y	A9	/C029.D3
X0008.5	OT2X	OVER TRAVEL X	B8	/C029.D2
X0008.4	*ESP	EMERGENCY STOP SIGNAL	A8	/C029.D7
X0008.3	*EMI	EMERGENCY STOP SW	B7	/C066.D6
X0008.2	FPTP	FUSE Blown	A7	/C008.E2
X0008.1	CPTP	CP Tripped	B6	/C006.E5
X0008.0	SMC	MCC Condition	A6	/C120.E6

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0009.7	24VOK	Power Supply OK	B13	
X0009.6	DPAL	DIFFERENCIAL PRESSURE ALARM	A13	
X0009.5	BLAL	FAN ABNORMAL ALARM	B12	
X0009.4	SPARE1	SPARE1	A12	
X0009.3	*DECM	MAG AXS REF RTN DECEL	B11	/C029.D6
X0009.2	*DECZ	3RD AXS REF RTN DECEL	A11	/C029.D4
X0009.1	*DECY	2ND AXS REF RTN DECEL	B10	/C029.D2
X0009.0	*DECX	1ST AXS REF RTN DECEL	A10	/C029.D1

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
Y0002.7	SL912_2	APC MAINTENANCE DOOR UNLOCK SOLENOID	B19	/C106.A2
Y0002.6	SL914	APC DOOR UNLOCK SOLENOID	A19	
Y0002.5	SL913	ATC DOOR UNLOCK SOLENOID	B18	/C079.A2
Y0002.4	SL912	FRONT DOOR UNLOCK SOLENOID	A18	/C078.A1
Y0002.3	MS154	OIL AIR UNIT PUMP ON	B17	
Y0002.2	EMSG_2	APC Maintenance Door	A17	
Y0002.1	BLCOM	SUCTION UNIT ENABLE	B16	
Y0002.0	SL---	MANUAL AIR GUN	A16	/C124.A2

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
Y0003.7	PRBSEL	PROBE SELECT	B23	/C065.A1
Y0003.6	OPPON	MEASURING PROBE POWER ON	A23	/C065.A1
Y0003.5	SL302	TOOL CLEANING AIR	B22	/C065.A1
Y0003.4	SL303	PURGE AIR	A22	/C065.A1
Y0003.3	SL304	LASER SHUTTER OPEN	B21	/C065.A1
Y0003.2	ENBL0/LSRCL	LASER CLEARANCE(ENABLE 0)	A21	/C065.A1
Y0003.1	ENBL2	ENABLE 2	B20	/C065.A1
Y0003.0	ENBL1	ENABLE 1	A20	/C065.A1

REVISION			
	----.--.--	APPROVED	AVINASH
	----.--.--	CHECKED	AVINASH
	----.--.--	DRAWN	ANISHA
	----.--.--	DATE	28/6/2018

MAKINO
MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
DI/DO ADDRESS TABLE - IO MODULE 1	V2631I-C1E-Y	Ver.	B
		PAGE	D002

I/O OPERATOR PNL, 48/32 A03B-0824-K203	IOM#2

IOM2-CE56

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0010.7	LS138L	AUTO FRONT DOOR OPEN LEFT	B5	
X0010.6	LS138R	AUTO FRONT DOOR OPEN RIGHT	A5	
X0010.5	LS2064	AUT GREASE SPLY UNIT LVL SW	B4	/C089.E4
X0010.4	LS2089	AUT GREASE INJECTOR PRESS UP	A4	/C089.E4
X0010.3	TH16	BASE CLNT TH ALM	B3	/C033.E5
X0010.2	TH2	NOZZLE CLNT MTR TH ALM	A3	/C031.E5
X0010.1	LS01	TOOL UNCL	B2	/C058.E4
X0010.0	LS00	TOOL CL	A2	/C058.E4

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0011.7	TH89	MTC AIR CON THERMAL ALARM	B9	/C091.E4
X0011.6	MS11C	WORK WASH GUN PUMP ON	A9	
X0011.5	*LS905	AIR PRESS SW	B8	/C051.E5
X0011.4	TH4	HYD UNIT TH ALM	A8	/C053.E5
X0011.3	LS748	ATC SHUTTER OPEN	B7	/C063.E5
X0011.2	LS747	ATC SHUTTER CLOSE	A7	/C063.E5
X0011.1	TH7A	SPIRAL CHIP CONV MTR TH ALM A	B6	/C030.E6
X0011.0	TH7	SPIRAL CHIP CONV MTR TH ALM	A6	/C030.E6

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0012.7	LS3320	TSC PRESSURE SWITCH (LOW)	B13	/C040.E3
X0012.6	*LS184	SEC TANK EMPTY DETECT	A13	/C049.E3
X0012.5	LS2323	OIL AIRUNIT FLOW UPPER CHECK	B12	
X0012.4	LS2312	OIL AIR UNIT FLOW LOWER CHECK	A12	
X0012.3	LS602	SP TOOL CONFIRM	B11	/C058.E4
X0012.2	LS601	SP TOOL SEATING	A11	/C058.E4
X0012.1	LS968	SP HYD UNIT PRESS SW (HIGH)	B10	/C053.E6
X0012.0	LS969	SP HYD UNIT PRESS SW (LOW)	A10	/C053.E6

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
Y0004.7	SL132A-1	AUTO FRONT DOOR CLOSE	B19	
Y0004.6	SL132B-1	AUTO FRONT DOOR OPEN	A19	
Y0004.5	ALSXYF	AUTO GREASE INJECTOR CW	B18	/C089.A2
Y0004.4	ALSXYR	AUTO GREASE INJECTOR CCW	A18	
Y0004.3	MS16	BASE CL	B17	/C033.A3
Y0004.2	MS2	NOZZLE CL	A17	/C031.A3
Y0004.1	SL01B	SP TOOL UNCL SOLENOID	B16	/C057.A4
Y0004.0	SL01A	SP TOOL CL SOLENOID	A16	/C057.A4

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
Y0005.7	SLATCF	ATC FLUSH	B23	
Y0005.6	SL750	Work Wash Gun	A23	
Y0005.5	SL905	Master Solenoid Valve	B22	/C051.A1
Y0005.4	MS4	HYD UNIT ON	A22	/C053.A3
Y0005.3	SL711B	ATC SHUTTER OPEN	B21	/C063.A1
Y0005.2	SL711A	ATC SHUTTER CLOSE	A21	/C063.A1
Y0005.1	MS7R	SPIRAL CONVEYOR REV	B20	/C030.A5
Y0005.0	MS7F	SPIRAL CONVEYOR FWD	A20	/C030.A5

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TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
DI/DO ADDRESS TABLE - IO MODULE 2	V2631I-C1E-Y	Ver.	B
		PAGE	D003

I/O OPERATOR PNL, 48/32 A03B-0824-K203	IOM#2

IOM2-CE57

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0013.7	LS80	ATC SIDE TOOL EXIST CHCK	B5	
X0013.6	LS1944	HSK Tool Orientation Confirmation†	A5	
X0013.5	MGCCW	ATC MAGAZINE CCW SW	B4	
X0013.4	MGCW	ATC MAGAZINE CW SW	A4	
X0013.3	LS2770	TLS tool unlock	B3	/C060.E4
X0013.2	LS2769	TLS tool lock	A3	/C060.E4
X0013.1	LS22	POT VERTICAL	B2	/C062.E4
X0013.0	LS21	POT HORIZONTAL	A2	/C062.E4

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0014.7	LS963	ATC MAINTENANCE COVER CONFIRM	B9	
X0014.6	LS181	NOZZLE CLNT FLOW SW	A9	
X0014.5	TH18	OIL SKIMMER MTR TH ALM	B8	/C045.E5
X0014.4	LS137R	AUTO DOOR CLOSE SLOW DOWN	A8	
X0014.3	THARM	O/L ATC ARM MTR (TH78)	B7	/C059.E6
X0014.2	TMATC	ATC ARM MTR OVERHEAT	A7	/C059.E6
X0014.1	ABRK/LS4104	ATC Arm Motor Brake	B6	
X0014.0	LS4103	Spindle Preload check 1	A6	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0015.7	LS1639	Tool Unclamp Power Pack Low Level Sensor	B13	
X0015.6	LS1641	Tool Unclamp Power Pack Temperature Sensor	A13	
X0015.5	LS176	Spindle Cooler Oil Level LOW	B12	
X0015.4			A12	
X0015.3	LS1337	CL Tank Lower Float for CL Supply	B11	
X0015.2	LS1325	CL Tank Upper Float for CL Supply	A11	
X0015.1	LS192	SEC FIL TANk LOWER FLOAT	B10	
X0015.0	LS191	SEC FIL TANK UPPER FLOAT	A10	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
Y0006.7	PIVRM/SL613	INVERTER SPEED 2(MID SPEED)	B19	
Y0006.6	PIVRH/SL612	INVERTER SPEED 1(HIGH SPEED)	A19	
Y0006.5			B18	
Y0006.4			A18	
Y0006.3	SL1081B	TLS TOOL UNLOCK	B17	/C060.A1
Y0006.2	SL1081A	TLS TOOL LOCK	A17	/C060.A1
Y0006.1	SL20	POT HORIZONTAL	B16	/C062.A1
Y0006.0	SL21	POT VERTICAL	A16	/C062.A1

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
Y0007.7	SL16	TAPER CLEANNING AIR 2	B23	
Y0007.6	NZSLVLV	NOZZLE SOL VALVE	A23	/C031.A3
Y0007.5	MS18	OIL SKIMMER	B22	/C045.A3
Y0007.4	SL132A-2	AUTO DOOR CLOSE SLOW DOWN	A22	
Y0007.3	MS78R/SL615	ATC MOTOR REVERSE	B21	/C059.A1
Y0007.2	MS78F/SL2301	ATC MOTOR FORWARD	A21	/C059.A1
Y0007.1	MS78B	ATC MOTOR BRAKE	B20	/C059.A3
Y0007.0	PIVRL/SL614	INVERTER SPEED 3(LOW SPEED)	A20	

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MAKINO
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TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
DI/DO ADDRESS TABLE - IO MODULE 2	V2631I-C1E-Y	Ver.	B
		PAGE	D004

I/O OPERATOR PNL, 48/32 A03B-0824-K203	IOM#3

IOM3-CE56

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0016.7	LS1338	TSC PRESSURE SWITCH (HIGH)	B5	/C040.E4
X0016.6	LS193	Through SP CL Flow Switch	A5	/C039.E3
X0016.5	LS1311	TSC LEAKAGE DETECT	B4	/C037.E1
X0016.4	*LS1340	TS UNION ABNORMAL DETECT	A4	
X0016.3	LS155	C-Axis Unclamp Switch	B3	/C083.E3
X0016.2	LS154	C-Axis Clamp Switch	A3	/C083.E3
X0016.1	LS152	A-Axis Unclamp Switch	B2	/C081.E3
X0016.0	LS151	A-Axis Clamp Switch	A2	/C081.E3

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0017.7	OMAL3	ROT AXS CHILLER UNIT ALM	B9	
X0017.6	TH19	MIST COLLECTOR MTR TH ALM	A9	/C047.E5
X0017.5	TH10	OVERHEAD SHOWER CLNT TH ALM	B8	/C035.E5
X0017.4	TH8	TS CLNT PUMP MTR TH ALM	A8	/C036.E5
X0017.3	LCMAS	LUCC MANUAL INTERRUPT SW	B7	/C044.E3
X0017.2	TH6	LUCC MTR TH ALM	A7	/C043.E6
X0017.1	LCREV	LUCC REVERSE SW	B6	/C044.E3
X0017.0	LCFWD	LUCC FORWARD SW	A6	/C044.E3

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0018.7	ROTA / +L4	4TH AXIS OVERTRAVEL	B13	/C081.E3
X0018.6	4NG	4TH AXS NEGLECT	A13	/C081.E3
X0018.5	ROTB / +L5	5TH AXIS OVERTRAVEL	B12	/C083.E3
X0018.4	5NG	5TH AXS NEGLECT	A12	/C083.E3
X0018.3	-L5	5TH AXS MINUS SIDE STROKE END	B11	
X0018.2	-L4	4TH AXS MINUS SIDE STROKE END	A11	
X0018.1			B10	
X0018.0			A10	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
Y0008.7	SL1310	TS SUCTION SOLENOID	B19	
Y0008.6	SL1306	TSC SUPPLY & DRAIN	A19	/C037.A5
Y0008.5	SL1309	TS UNION DRAIN	B18	/C037.A4
Y0008.4	SL1308	TS UNION BACKUP AIR	A18	/C037.A5
Y0008.3	SL151B	5 AXIS BRAKE/CLAMP	B17	
Y0008.2	SL151	C-Axis Unclamp	A17	/C083.A2
Y0008.1	SL150B	4th AXIS BRAKE/CLAMP	B16	
Y0008.0	SL150	A-Axis Unclamp	A16	/C081.A1

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
Y0009.7	OMON3	Feed Axis Chiller	B23	
Y0009.6	MS19	MIST COLLECTOR	A23	/C047.A3
Y0009.5	MS10	OVERHEAD SHOWER CL	B22	/C035.A3
Y0009.4	MS8	THROUGH SP CL	A22	/C036.A3
Y0009.3	MS6R	R LUCC REVERSE	B21	/C043.A4
Y0009.2	MS6F	R LUCC FORWARD	A21	/C043.A4
Y0009.1	LCMAL	LUCC MANUAL INTERRUPT LAMP	B20	/C044.A6
Y0009.0	LCALML	LUCC ALARM LAMP	A20	

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TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
DI/DO ADDRESS TABLE - IO MODULE 3	V2631I-C1E-Y	Ver.	B
		PAGE	D005

I/O OPERATOR PNL, 48/32 A03B-0824-K203	IOM#3

IOM3-CE57

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0019.7	EXTRDY2	External Ready Signal (AWC)	B5	/C103.E5
X0019.6	MQLLOW	MQL OIL LEVEL LOW DETECT	A5	/C103.E5
X0019.5	LSWCCL2B	WORK CHUCK 2B CL CONFIRM	B4	
X0019.4	LSWCCL2A	WORK CHUCK 2A CL CONFIRM	A4	
X0019.3	LSWCCL1B	WORK CHUCK 1B CL CONFIRM	B3	
X0019.2	LSWCCL1A	WORK CHUCK 1A CL CONFIRM	A3	
X0019.1	LS130R	Right Side Door Close	B2	
X0019.0	LS131R	Right Side Door Open	A2	/C107.D5

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0020.7	TH260-2	2nd HYD PP MTR OVERLOAD	B9	/C085.E4
X0020.6	TH71	SEC FIL TANK PUMP TH ALM	A9	/C041.E5
X0020.5	TH6A	Thermal	B8	/C042.E5
X0020.4	LS1306	LUCC SLIP DETECT	A8	/C044.D6
X0020.3	LS130L	Left Side Door Close	B7	
X0020.2	LS131L	Left Side Door Open	A7	
X0020.1	0	0	B6	
X0020.0	TH23	Chuck Cleaning CL Thermal Alarm	A6	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0021.7	*LS2340	ROT AXS CHILLER UNIT FLOW SW	B13	
X0021.6	LS1339	TSC SUCTION FILTER CLOG DETECT (Before Pump)	A13	
X0021.5	TSCCG2	75% CLG-WRN / TSC FIL 2 CLG	B12	/C038.E3
X0021.4	TSCCG1	100% CLG-ALM /TSC FIL 1 CLG	A12	/C038.E3
X0021.3	SYSPS2-2	2nd HYD PP SYSTEM PRESS (Low)	B11	/C085.E4
X0021.2	SYSPS-2	2nd HYD PP SYSTEM PRESS (High)	A11	/C085.E4
X0021.1	*DEC5	5TH AXS REF RTN DECEL	B10	/C083.E3
X0021.0	*DEC4	4TH AXS REF RTN DECEL	A10	/C081.E3

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
Y0010.7	MCRDY2	Machine Ready (Door/ATC)	B19	/C103.A4
Y0010.6	SLMQLON	MQL SYSTEM ON	A19	/C046.A1
Y0010.5	SLWCUC2B	WORK CHUCK 2B UNCL	B18	
Y0010.4	SLWCUC2A	WORK CHUCK 2A UNCL	A18	
Y0010.3	SLWCUC1B	WORK CHUCK 1B UNCL	B17	
Y0010.2	SLWCUC1A	WORK CHUCK 1A UNCL	A17	
Y0010.1	SL130A-R	Right Side Door Close	B16	/C107.A3
Y0010.0	SL130B-R	Right Side Door Open	A16	/C107.A3

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
Y0011.7	M260-2	2nd HDY POWER PACK	B23	/C085.A3
Y0011.6	MS71	SEC FILTER TANK PUMP ON	A23	/C041.A3
Y0011.5	MS6A	Drum Filter Cleaning Pump	B22	/C042.A3
Y0011.4	SL1301	Coolant Supply Start	A22	
Y0011.3	SL130A-L	Left Side Door Close	B21	
Y0011.2	SL130B-L	Left Sode Door Open	A21	
Y0011.1	SL190	Chuck Cleaning Air	B20	/C086.A5
Y0011.0	MS23	Chuck Cleaning Coolant	A20	

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TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
DI/DO ADDRESS TABLE - IO MODULE 3	V2631I-C1E-Y	Ver.	B
		PAGE	D006

I/O OPERATOR PNL, 48/32 A03B-0824-K203	IOM#4

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DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0022.7	L152-2/LS680	ADDITIONAL 3RD AXS UNCL/Shifter first section Adv	B5	
X0022.6	L151-2/LS679	ADDITIONAL 3RD AXS CL/Shifter first section ret	A5	
X0022.5	LS56	Pallet CL (Low)	B4	
X0022.4	LS55	Pallet CL (High)	A4	
X0022.3	LS102	Pallet CCW	B3	
X0022.2	LS103	Pallet CW	A3	
X0022.1	LS100	APC Arm Down	B2	
X0022.0	LS101	APC Arm Lift	A2	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0024.7	LS1648L	APC STOCKER DOOR LEFT OPRN	B13	
X0024.6	LS1648R	APC STOCKER DOOR RIGHT OPEN	A13	
X0024.5	7NG/LS2715	7TH AXS NEGLECT/Tool Confirmation Holder Side	B12	
X0024.4	6NG / LS99	6TH AXS NEGLECT/Tool Confirmation Mag. Side	A12	
X0024.3	LS2603-2	2nd HYD OIL LVL FLOAT SW	B11	
X0024.2	LS2604-2	2nd HYD OIL TEMP SENSOR	A11	
X0024.1			B10	
X0024.0			A10	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
Y0012.7	BRK6	6 AXIS BRAKE/CLAMP	B19	
Y0012.6	SL150-2	ADDITIONAL 3RD AXS UNCL	A19	
Y0012.5	SL51	Pallet UNCL	B18	/C086.A1
Y0012.4	SL51A	Pallet CL	A18	
Y0012.3	SL101A	APC Arm CCW	B17	
Y0012.2	SL101B	APC Arm CW	A17	
Y0012.1	SL100A	APC Arm Down	B16	
Y0012.0	SL100B	APC Arm Lift	A16	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0023.7	LS674	Tool Loader Retract	B9	
X0023.6	LS2761	POSITION OF TOOL CONFIRMATION	A9	
X0023.5			B8	
X0023.4	TH40	Drawing Pump Thermal Alarm	A8	
X0023.3	LS90/LS76	Carrier Pot UNCL / Tool Advance	B7	
X0023.2	LS89/LS77	Carrier Pot CL/Tool Retract	A7	
X0023.1	L155-2/LS750	ADDITIONAL 4TH AXS UNCL/Shifter second secton Adv	B6	
X0023.0	L154-2/LS751	ADDITIONAL 4TH AXS CL/Shifter second section Ret	A6	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
Y0013.7	CR23	PRESET SPEED # 4	B23	
Y0013.6	SL52	Locate Pin Cleaning Air	A23	
Y0013.5	SL101C	APC Arm CW/CCW Slow Down	B22	
Y0013.4	MS40	CL Drawing Pump	A22	
Y0013.3	SL89B	Carrier Pot UNCL	B21	
Y0013.2	0	0	A21	
Y0013.1	BRK7	7 AXIS BRAKE/CLAMP	B20	
Y0013.0	SL151-2	ADDITIONAL 4TH AXS UNCL	A20	

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TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
DI/DO ADDRESS TABLE - IO MODULE 4	V2631I-C1E-Y	Ver.	B
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I/O OPERATOR PNL, 48/32 A03B-0824-K203	IOM#4

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DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0025.7	USDI4	USER DI 4	B5	/C077.E1
X0025.6	USDI3	USER DI 3	A5	/C077.E1
X0025.5	USDI2	USER DI 2	B4	/C077.E1
X0025.4	USDI1	USER DI 1	A4	/C077.E1
X0025.3	LS1973/LS1971	Pot HV Lock / ATC Shifter pot Lock	B3	
X0025.2	LS1974/LS1972	Pot HV Unlock /ATC Shifter Pot Unlock	A3	
X0025.1			B2	
X0025.0			A2	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0027.7	LS711	ATC Magazine Pot Exist Confirm	B13	
X0027.6	LS3720	ATC Pot Hrz/Vrt Angle Position	A13	
X0027.5	*DEC7	7TH AXS REF RTN DECEL	B12	
X0027.4	*DEC6	6TH AXS REF RTN DECEL	A12	
X0027.3	MR3	RETURN PUMP SIGNAL	B11	
X0027.2	FLC	BAG FILTER CLOG	A11	
X0027.1	PFALM/*ES	READY SIGNAL PBF/ERROR SIGNAL	B10	
X0027.0	PFPEND/AP	END OF PAPER ALARM PBF/ATTEND PAPER	A10	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
Y0014.7	USDO4	USER DO4	B19	/C077.A1
Y0014.6	USDO3	USER DO3	A19	/C077.A1
Y0014.5	USDO2	USER DO2	B18	/C077.A1
Y0014.4	USDO1	USER DO1	A18	/C077.A1
Y0014.3	SL776	TLS Tool Unload	B17	
Y0014.2	SL771B	POT HV UNLOCK	A17	
Y0014.1	CR3	INDEX#1	B16	
Y0014.0	CR1	COOLANT#1 HP PUMP	A16	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0026.7	THPF	MOTOR FAULT ALARM PBF	B9	
X0026.6			A9	
X0026.5	EXTDVI1	EXTERNAL DEVICE INPUT 1	B8	
X0026.4	EXTRDY1	External Ready Signal (Work)	A8	
X0026.3	USDI8	USER DI 8	B7	/C077.E1
X0026.2	USDI7	USER DI 7	A7	/C077.E1
X0026.1	USDI6	USER DI 6	B6	/C077.E1
X0026.0	USDI5	USER DI 5	A6	/C077.E1

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
Y0015.7	CR22	PRESET SPEED # 3	B23	
Y0015.6	CR21	PRESET SPEED # 2	A23	
Y0015.5	EXTDVO1	EXTERNAL DEVICE OUTPUT 1	B22	
Y0015.4	MCRDY1	Machine Ready (Stop)	A22	/C103.A3
Y0015.3	USDO8	USER DO8	B21	/C077.A1
Y0015.2	USDO7	USER DO7	A21	/C077.A1
Y0015.1	USDO6	USER DO6	B20	/C077.A1
Y0015.0	USDO5	USER DO5	A20	/C077.A1

REVISION			
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	----.--.--	DATE	28/6/2018

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TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
DI/DO ADDRESS TABLE - IO MODULE 4	V2631I-C1E-Y	Ver.	B
		PAGE	D008

I/O OPERATOR PNL, 48/32 A03B-0824-K203	IOM#5

IOM5-CE56

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0028.7	NOWPST	AUTO START WITHOUT WORK	B5	/C100.E5
X0028.6	WULDFN	WORK UNLOAD FINISH	A5	/C100.E5
X0028.5	WLDFN	WORK LOAD FINISH	B4	
X0028.4	SYSTP	AUTO STOP INSTRUCTION	A4	
X0028.3	SYSTR	AUTO START INSTRUCTION	B3	/C100.E5
X0028.2	*RBINT	ROBOT IN INTERFERENCE ZONE	A3	
X0028.1	*RBALM	ROBOT ALARM	B2	
X0028.0	RBHLY	ROBOT HEALTHY	A2	/C100.E5

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0029.7	WPNPRT	WORK NUMBER PARITY	B9	
X0029.6	WPN08	WORK NUMBER 08	A9	
X0029.5	WPN04	WORK NUMBER 04	B8	
X0029.4	WPN02	WORK NUMBER 02	A8	
X0029.3	WPN01	WORK NUMBER 01	B7	
X0029.2	FUCRQ	FIXTURE UNCLAMP REQUEST	A7	
X0029.1	FCLRQ	FIXTURE CLAMP REQUEST	B6	
X0029.0	FLSON	FIXTURE WASH INSTRUCTION	A6	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0030.7	RBI24	ROBOT INTERFACE INPUT 24	B13	
X0030.6	RBI23	ROBOT INTERFACE INPUT 23	A13	
X0030.5	RBI22	ROBOT INTERFACE INPUT 22	B12	
X0030.4	RBI21	ROBOT INTERFACE INPUT 21	A12	
X0030.3	RBI20	ROBOT INTERFACE INPUT 20	B11	
X0030.2	RBI19	ROBOT INTERFACE INPUT 19	A11	
X0030.1	RBI18	ROBOT INTERFACE INPUT 18	B10	
X0030.0	RBI17	ROBOT INTERFACE INPUT 17	A10	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
Y0016.7	WULDRQ	WORK UNLOAD REQUEST	B19	/C100.A1
Y0016.6	WLDRQ	WORK LOAD REQUEST	A19	/C100.A1
Y0016.5	MCRUN	MACHINE RUNNING	B18	/C100.A1
Y0016.4	WSRDY	START RESERVATION CONDITION	A18	/C100.A1
Y0016.3	ACSOK	INTERVENTION PREMISSION	B17	/C100.A1
Y0016.2	*MCWRN	MACHINE WARNING	A17	/C100.A1
Y0016.1	*MCALM	MACHINE ALARM	B16	/C100.A1
Y0016.0	MCRDY	MACHINE READY	A16	/C100.A1

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
Y0017.7	WCN02	WORK CONDITION NO 2	B23	/C101.A1
Y0017.6	WCN01	WORK CONDITION NO 1	A23	/C101.A1
Y0017.5	PCP02	STOCKER PALLET NO 2	B22	/C101.A1
Y0017.4	PCP01	STOCKER PALLET NO 1	A22	/C101.A1
Y0017.3	FUCFN	FIXTURE UNCLAMP FINISH	B21	/C101.A1
Y0017.2	FCLFN	FIXTURE CLAMP FINISH	A21	/C100.A1
Y0017.1	WSCNG	WORK SEATING NG	B20	/C100.A1
Y0017.0	WSCOK	WORK SEATING OK	A20	/C100.A1

REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

MAKINO
MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
DI/DO ADDRESS TABLE - IO MODULE 5	V2631I-C1E-Y	Ver.	B
		PAGE	D009

I/O OPERATOR PNL, 48/32 A03B-0824-K203	IOM#5

IOM5-CE57

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0031.7			B5	
X0031.6			A5	
X0031.5			B4	
X0031.4			A4	
X0031.3			B3	
X0031.2			A3	
X0031.1			B2	
X0031.0			A2	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0033.7	PDIG28	EXTERNAL PROGRAM SELECTOR SW 28	B13	
X0033.6	PDIG24	EXTERNAL PROGRAM SELECTOR SW 24	A13	
X0033.5	PDIG22	EXTERNAL PROGRAM SELECTOR SW 22	B12	
X0033.4	PDIG21	EXTERNAL PROGRAM SELECTOR SW 21	A12	
X0033.3	PDIG18	EXTERNAL PROGRAM SELECTOR SW 18	B11	
X0033.2	PDIG14	EXTERNAL PROGRAM SELECTOR SW 14	A11	
X0033.1	PDIG12	EXTERNAL PROGRAM SELECTOR SW 12	B10	
X0033.0	PDIG11	EXTERNAL PROGRAM SELECTOR SW 11	A10	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
Y0018.7			B19	
Y0018.6			A19	
Y0018.5	3R_COP4		B18	
Y0018.4	3R_TCP4		A18	
Y0018.3	ERDBIT10	Erowa Data bit10	B17	/C102.A3
Y0018.2	ERDBIT9	Erowa Data bit9	A17	/C102.A3
Y0018.1	ERDBIT8	Erowa Data bit8	B16	/C102.A3
Y0018.0	ERDBIT7	Erowa Data bit7	A16	/C102.A3

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0032.7			B9	
X0032.6			A9	
X0032.5			B8	
X0032.4			A8	
X0032.3			B7	
X0032.2			A7	
X0032.1			B6	
X0032.0			A6	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
Y0019.7			B23	
Y0019.6	3RUCLCFM	SPINDLE TOOL UNCLAMP	A23	
Y0019.5	ERSPCHPOS	SPINDLE CHANGE POSITION	B22	
Y0019.4	ERP2CHPOS	PALLET2 CHANGE POSITION	A22	
Y0019.3	ERP1CHPOS	PALLET1 CHANGE POSITION	B21	
Y0019.2	EROPCHK1	CHUCK 1 OPEN	A21	
Y0019.1	UNCLR2	CHUCK2 UNCLAMP REQUEST	B20	
Y0019.0	UNCLR1	CHUCK1 UNCLAMP REQUEST	A20	

REVISION			
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	----.--.--	DRAWN	ANISHA
	----.--.--	DATE	28/6/2018

MAKINO
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TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
DI/DO ADDRESS TABLE - IO MODULE 5	V2631I-C1E-Y	Ver.	B
		PAGE	D010

I/O OPERATOR PANELI 48/32	IOM 6

IOM 6- CE56

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE56		PAGE ページ
X0034.7	CUSDI8	CUSTOMER DESIGN INPUT 8	B5		
X0034.6	CUSDI7	CUSTOMER DESIGN INPUT 7	A5		
X0034.5	CUSDI6	CUSTOMER DESIGN INPUT 6	B4		
X0034.4	CUSDI5	CUSTOMER DESIGN INPUT 5	A4		
X0034.3	CUSDI4	CUSTOMER DESIGN INPUT 4	B3		
X0034.2	CUSDI3	CUSTOMER DESIGN INPUT 3	A3		
X0034.1	CUSDI2	CUSTOMER DESIGN INPUT 2	B2		
X0034.0	CUSDI1	CUSTOMER DESIGN INPUT 1	A2		

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE56		PAGE ページ
X0036.7	CUSDI16	CUSTOMER DESIGN INPUT 24	B13		
X0036.6	CUSDI15	CUSTOMER DESIGN INPUT 23	A13		
X0036.5	CUSDI14	CUSTOMER DESIGN INPUT 22	B12		
X0036.4	CUSDI13	CUSTOMER DESIGN INPUT 21	A12		
X0036.3	CUSDI12	CUSTOMER DESIGN INPUT 20	B11		
X0036.2	CUSDI11	CUSTOMER DESIGN INPUT 19	A11		
X0036.1	CUSDI10	CUSTOMER DESIGN INPUT 18	B10		
X0036.0	CUSDI9	CUSTOMER DESIGN INPUT 17	A10		

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE56		PAGE ページ
Y0020.7	CUSDO8	CUSTOMER DESIGN OUTPUT 8	B19		
Y0020.6	CUSDO7	CUSTOMER DESIGN OUTPUT 7	A19		
Y0020.5	CUSDO6	CUSTOMER DESIGN OUTPUT 6	B18		
Y0020.4	CUSDO5	CUSTOMER DESIGN OUTPUT 5	A18		
Y0020.3	CUSDO4	CUSTOMER DESIGN OUTPUT 4	B17		
Y0020.2	CUSDO3	CUSTOMER DESIGN OUTPUT 3	A17		
Y0020.1	CUSDO2	CUSTOMER DESIGN OUTPUT 2	B16		
Y0020.0	CUSDO1	CUSTOMER DESIGN OUTPUT 1	A16		

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE56		PAGE ページ
X0035.7	CUSDI16	CUSTOMER DESIGN INPUT 16	B9		
X0035.6	CUSDI15	CUSTOMER DESIGN INPUT 15	A9		
X0035.5	CUSDI14	CUSTOMER DESIGN INPUT 14	B8		
X0035.4	CUSDI13	CUSTOMER DESIGN INPUT 13	A8		
X0035.3	CUSDI12	CUSTOMER DESIGN INPUT 12	B7		
X0035.2	CUSDI11	CUSTOMER DESIGN INPUT 11	A7		
X0035.1	CUSDI10	CUSTOMER DESIGN INPUT 10	B6		
X0035.0	CUSDI9	CUSTOMER DESIGN INPUT 9	A6		

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE56		PAGE ページ
Y0021.7	CUSDO16	CUSTOMER DESIGN OUTPUT 16	B23		
Y0021.6	CUSDO15	CUSTOMER DESIGN OUTPUT 15	A23		
Y0021.5	CUSDO14	CUSTOMER DESIGN OUTPUT 14	B22		
Y0021.4	CUSDO13	CUSTOMER DESIGN OUTPUT 13	A22		
Y0021.3	CUSDO12	CUSTOMER DESIGN OUTPUT 12	B21		
Y0021.2	CUSDO11	CUSTOMER DESIGN OUTPUT 11	A21		
Y0021.1	CUSDO10	CUSTOMER DESIGN OUTPUT 10	B20		
Y0021.0	CUSDO9	CUSTOMER DESIGN OUTPUT 9	A20		

REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

MAKINO
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TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
DI/DO ADDRESS TABLE - IO MODULE 6	V2631I-C1E-Y	Ver.	B
		PAGE	D011

I/O OPERATOR PANELI 48/32	IOM 6

IOM 6- CE57

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE57	PAGE ページ
X0037.7			B5	
X0037.6			A5	
X0037.5			B4	
X0037.4			A4	
X0034.3			B3	
X0037.2			A3	
X0037.1			B2	
X0037.0			A2	

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE57	PAGE ページ
X0039.7			B13	
X0039.6			A13	
X0039.5			B12	
X0039.4			A12	
X0039.3			B11	
X0039.2			A11	
X0039.1			B10	
X0039.0			A10	

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE57	PAGE ページ
Y0022.7			B19	
Y0022.6			A19	
Y0022.5			B18	
Y0022.4			A18	
Y0022.3			B17	
Y0022.2			A17	
Y0022.1			B16	
Y0022.0			A16	

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE57	PAGE ページ
X0038.7			B9	
X0038.6			A9	
X0038.5			B8	
X0038.4			A8	
X0038.3			B7	
X0038.2			A7	
X0038.1			B6	
X0038.0			A6	

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE57	PAGE ページ
Y0023.7			B23	
Y0023.6			A23	
Y0023.5			B22	
Y0023.4			A22	
Y0023.3			B21	
Y0023.2			A21	
Y0023.1			B20	
Y0023.0			A20	

REVISION			
	----.--.--	APPROVED	AVINASH
	----.--.--	CHECKED	AVINASH
	----.--.--	DRAWN	ANISHA
	----.--.--	DATE	28/6/2018

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TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
DI/DO ADDRESS TABLE - IO MODULE 6	V2631I-C1E-Y	Ver.	B
		PAGE	D012

I/O OPERATOR PNL, 48/32 A03B-0824-K203	IOM#7

IOM7-CE56

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0060.7	PS2604	HYD PRESS SW 4	B5	/C098.E4
X0060.6	PS2603	HYD PRESS SW 3	A5	/C098.E4
X0060.5	PS2602	HYD PRESS SW 2	B4	/C098.E4
X0060.4	PS2601	HYD PRESS SW 1	A4	/C098.E4
X0060.3	EJUC	STOCKER SIDE EJECT UNCLMP PB	B3	/C099.E2
X0060.2	EJCL	STOCKER SIDE EJECT CLMP PB	A3	/C099.E2
X0060.1	HY1UC	HYD UNCL PB	B2	/C099.E2
X0060.0	HY1CL	HYD CL PB	A2	/C099.E2

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0062.7	LS1884	POSITION SENSOR 4	B13	
X0062.6	LS1883	POSITION SENSOR 3	A13	
X0062.5	LS1882	POSITION SENSOR 2	B12	
X0062.4	LS1881	POSITION SENSOR 1	A12	
X0062.3	LS1788	AS SENSOR 8	B11	/C095.E1
X0062.2	LS1787	AS SENSOR 7	A11	/C095.E1
X0062.1	LS1786	AS SENSOR 6	B10	/C095.E1
X0062.0	LS1785	AS SENSOR 5	A10	/C095.E1

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
Y0060.7	SL2604	HYD SOLENOID 4	B19	/C098.A2
Y0060.6	SL2603	HYD SOLENOID 3	A19	/C098.A2
Y0060.5	SL2602	HYD SOLENOID 2	B18	/C098.A1
Y0060.4	SL2601	HYD SOLENOID 1	A18	/C098.A1
Y0060.3	LEJUC	STOCKER SIDE EJECT UNCLMP LMP	B17	/C099.A2
Y0060.2	LEJCL	STOCKER SIDE EJECT CLMP LMP	A17	/C099.A2
Y0060.1	LHY1UC	STOCKER SIDE FX UNC LMP	B16	/C099.A2
Y0060.0	LHY1CL	STOCKER SIDE FX CLMP LMP	A16	/C099.A2

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0061.7	TH106	Table CL PUMP TH ALM	B9	
X0061.6	AIRSV	AS CHECK SW	A9	/C096.E1
X0061.5	AUT	AUT MD SELECTION SW	B8	/C099.E2
X0061.4	TH260	HYD MTR OVERLOAD	A8	/C097.E3
X0061.3	PS2608	HYD PRESS SW 8	B7	
X0061.2	PS2607	HYD PRESS SW 7	A7	
X0061.1	PS2606	HYD PRESS SW 6	B6	
X0061.0	PS2605	HYD PRESS SW 5	A6	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
Y0061.7	MS106	Table Coolant	B23	
Y0061.6	LAIRS	AS CHECK VALID LAMP	A23	/C096.A3
Y0061.5	MNLLMP	FIX MANUAL LAMP	B22	/C099.A2
Y0061.4	MS260	HYD MOTOR ON	A22	/C097.A3
Y0061.3	SL2608	HYD SOLENOID 8	B21	
Y0061.2	SL2607	HYD SOLENOID 7	A21	
Y0061.1	SL2606	HYD SOLENOID 6	B20	
Y0061.0	SL2605	HYD SOLENOID 5	A20	

REVISION				
	----.--.--	APPROVED	AVINASH	
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	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

MAKINO
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TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
DI/DO ADDRESS TABLE - IO MODULE 7	V2631I-C1E-Y	Ver.	B
		PAGE	D013

I/O OPERATOR PNL, 48/32 A03B-0824-K203	IOM#7

IOM7-CE57

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0063.7	PS2708	HYD PRESS SW 108	B5	
X0063.6	PS2707	HYD PRESS SW 107	A5	
X0063.5	PS2706	HYD PRESS SW 106	B4	
X0063.4	PS2705	HYD PRESS SW 105	A4	
X0063.3	PS2704	HYD PRESS SW 104	B3	
X0063.2	PS2703	HYD PRESS SW 103	A3	
X0063.1	PS2702	HYD PRESS SW 102	B2	
X0063.0	PS2701	HYD PRESS SW 101	A2	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0065.7	LDRMD	LOADER MODE OFF/ON	B13	
X0065.6	STRSW	STRIKER SW	A13	/C094.E4
X0065.5	THPBL	TWO HAND PB L	B12	/C093.E3
X0065.4	THPBR	TWO HAND PB R	A12	/C093.E3
X0065.3	LS2603	HYD OIL LVL FLOAT SW	B11	/C097.E3
X0065.2	LS2604	HYD OIL TEMP SENSOR	A11	/C097.E3
X0065.1	SYSPS2	HYD PP SYSTEM PRESS (Low)	B10	/C097.E3
X0065.0	SYSPS	HYD PP SYSTEM PRESS (High)	A10	/C097.E3

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
Y0062.7	SL2708	HYD SOLENOID 108	B19	
Y0062.6	SL2707	HYD SOLENOID 107	A19	
Y0062.5	SL2706	HYD SOLENOID 106	B18	
Y0062.4	SL2705	HYD SOLENOID 105	A18	
Y0062.3	SL2704	HYD SOLENOID 104	B17	
Y0062.2	SL2703	HYD SOLENOID 103	A17	
Y0062.1	SL2702	HYD SOLENOID 102	B16	
Y0062.0	SL2701	HYD SOLENOID 101	A16	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0064.7	LS1784	AS SENSOR 4	B9	/C095.E1
X0064.6	LS1783	AS SENSOR 3	A9	/C095.E1
X0064.5	LS1782	AS SENSOR 2	B8	/C095.E1
X0064.4	LS1781	AS SENSOR 1	A8	/C095.E1
X0064.3	LS1792	AS SESNROR 12	B7	
X0064.2	LS1791	AS SESNROR 11	A7	
X0064.1	LS1790	AS SESNROR 10	B6	
X0064.0	LS1789	AS SESNROR 9	A6	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
Y0063.7	SL1784	AS CHECK SOLENOID 4	B23	
Y0063.6	SL1783	AS CHECK SOLENOID 3	A23	
Y0063.5	SL1782	AS CHECK SOLENOID 2	B22	/C096.A3
Y0063.4	SL1781	AS CHECK SOLENOID 1	A22	/C096.A3
Y0063.3	SL1804	FIX. CLEAN SOLENOID 4	B21	
Y0063.2	SL1803	FIX. CLEAN SOLENOID 3	A21	
Y0063.1	SL1802	FIX. CLEAN SOLENOID 2	B20	
Y0063.0	SL1801	FIX. CLEAN SOLENOID 1	A20	

REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

MAKINO
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TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
DI/DO ADDRESS TABLE - IO MODULE 7	V2631I-C1E-Y	Ver.	B
		PAGE	D014

I/O OPERATOR PNL, 48/32 A03B-0824-K203	IOM#9

IOM9-CE56

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0007.7	LS912DC_2	APC MAINTENANCE DOOR INTERLOCK	B5	
X0007.6	LS966DC	APC DOOR INTERLOCK	A5	
X0007.5	LS942DC	ATC DOOR INTERLOCK	B4	/C121.E5
X0007.4	LS912DC	S/G DOOR INTERLOCK	A4	/C121.E5
X0007.3	SPI4		B3	
X0007.2	KAPCR	STO APC1	A3	
X0007.1	KATCA	STO ATC2	B2	
X0007.0	KATCM	STO ATC1	A2	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0009.7	EXOF2	SPINDLE EXCITATION OFF	B13	/C122.E4
X0009.6	KEMROB	EMG+APC MAINTENANCE CONTACT CONDITION	A13	
X0009.5	KEMAPC	EMG+APC CONTACT CONDITION	B12	
X0009.4	KEMATC	EMG+ATC CONTACT CONDITION	A12	/C121.E5
X0009.3	KEMSG	EMG+SG CONTACT CONDITION	B11	/C121.E5
X0009.2	KMCF	MCF CONTACT CONDITION	A6	/C120.E4
X0009.1	KBRK	BRAKE CONTACT CONDITION	B10	/C119.D4
X0009.0	KEMG	EMG CONTACT CONDITION	A10	/C122.E4

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
Y0002.7	EMROBDC	EMG + APC MAINTENANCE DOOR	B19	
Y0002.6	EMGDC	EMERGENCY STOP	A19	/C122.A3
Y0002.5	EMAPCDC	EMG + APC DOOR	B18	
Y0002.4	EMATCDC	EMG + ATC DOOR	A18	/C121.A3
Y0002.3	EMSGDC	EM S/G INERLOCK	B17	/C121.A4
Y0002.2	MCFDC	MCC ON SIGNAL	A17	/C120.A3
Y0002.1	SPO1		B16	
Y0002.0	ZRBKDC	BRAKE RELEASE	A16	/C119.A3

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
X0008.7	KMS8	MS8 CONTACT CONDITION	B9	
X0008.6	KMS2	MS2 CONTACT CONDITION	A9	
X0008.5	KMSIV	ATC DRM CONTACT CONDITION	B8	
X0008.4	*ESPDC	EMERGENCY STOP	A8	/C122.E4
X0008.3	LS2466DC	LIGHT CURTAIN	B7	/C080.E4
X0008.2	SPI5		A7	
X0008.1	HWERDC	OPERATION ENABLE SWITCH	B6	/C119.D6
X0008.0	SMCDC	MCC CONDITION CONFIRM	A11	/C120.E4

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE56	PAGE ページ
Y0003.7	ACU RST	ACU RESET	B23	/C126.A4
Y0003.6	ACUPWR	ACU POWER	A23	/C126.A4
Y0003.5	SPO5		B22	
Y0003.4	SPO4		A22	
Y0003.3	SPO3		B21	
Y0003.2	SPO2		A21	
Y0003.1	KCCW1	PALLET CCW	B20	
Y0003.0	KCW1	PALLET CW	A20	

REVISION						
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TITLE	DRAWING No.
	----.--.--	CHECKED	AVINASH		DI/DO ADDRESS TABLE - IO MODULE 9	V2631I-C1E-Y
	----.--.--	DRAWN	ANISHA			
	----.--.--	DATE	28/6/2018			

MACHINE	F3/ F5 Pro 6
Ver.	B
PAGE	D015

I/O OPERATOR PNL, 48/32 A03B-0824-K203	IOM#9

IOM9-CE57

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0010.7			A11	
X0010.6			B5	
X0010.5			A5	
X0010.4	SA4104	LS tool unlock button	B4	/C064.E1
X0010.3	MGCCW	ATC MAGAZINE CCW SW	A4	/C064.E1
X0010.2	MGCW	ATC MAGAZINE CW SW	B3	/C064.E1
X0010.1	ATCUNC	ATC DOOR UNLOCK SW	A2	/C064.E1
X0010.0	ATCMAS	ATC MANUAL INTERRUPT	A2	/C064.E1

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0012.7			A16	
X0012.6			B13	
X0012.5			A13	
X0012.4			B12	
X0012.3			A12	
X0012.2			B6	
X0012.1			A6	
X0012.0			B10	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
Y0004.7			A20	
Y0004.6			B19	
Y0004.5			A19	/C064.A7
Y0004.4	TLULL	TOOL LOADER UNLOCK LAMP	B18	/C064.A7
Y0004.3	LMCW	ATC MAGAZINE CW LAMP	A18	/C064.A7
Y0004.2	LMCCW	ATC MAGAZINE CCW LAMP	B17	/C064.A7
Y0004.1	LMGSB	ATC DOOR UNLOCK LAMP	A17	/C064.A7
Y0004.0	ATCMAL	ATC MANUAL INTERRUPT LAMP	B16	/C064.A7

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
X0011.7			A10	
X0011.6			B9	
X0011.5			A9	
X0011.4			B8	
X0011.3			A8	
X0011.2			B7	
X0011.1			A7	
X0011.0			B11	

DGN No. 診断番号	SYMBOL 記号	SIGNAL NAME 信号名称	CONN No. コネ No. CE57	PAGE ページ
Y0005.7			B23	
Y0006.6			B23	
Y0005.5			A23	
Y0005.4			B22	
Y0005.3			A22	
Y0005.2			B21	
Y0005.1			A21	
Y0005.0			B20	

REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

MAKINO
MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
DI/DO ADDRESS TABLE - IO MODULE 9	V2631I-C1E-Y	Ver.	B
		PAGE	D016

I/O OPERATOR PANELI 48/32	IOM 11

IOM 11- CE53

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE53		PAGE ページ
X0096.7	H8S/HZ2S	HANDLE SELECT 8	B6	+24V	/C111.D6
X0096.6	H7S	HANDLE SELECT 7	A6		/C111.D5
X0096.5	H6S	HANDLE SELECT 6	B5		/C111.D5
X0096.4	H5S	HANDLE SELECT 5	A5	CE53 -B2	/C111.D4
X0096.3	H4S	HANDLE SELECT S	B4		/C111.D4
X0096.2	HZS	HANDLE SELECT Z	A4		/C111.D3
X0096.1	HYS	HANDLE SELECT Y	B3		/C111.D3
X0096.0	HXS	HANDLE SELECT X	A3		/C111.D2

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE53		PAGE ページ
X0100.7	FPS8	FLEXIBLE SWITCH 08	B22	KCM1	/C104.C4
X0100.6	FPS7	FLEXIBLE SWITCH 07	A22		/C104.C4
X0100.5	FPS6	FLEXIBLE SWITCH 06	B21		/C104.C3
X0100.4	FPS5	FLEXIBLE SWITCH 05	A21		/C104.C3
X0100.3	FPS4	FLEXIBLE SWITCH 04	B20	A23	/C104.C3
X0100.2	FPS3	FLEXIBLE SWITCH 03	A20		/C104.C2
X0100.1	FPS2	FLEXIBLE SWITCH 02	B19		/C104.C2
X0100.0	FPS1	FLEXIBLE SWITCH 01	A19		/C104.C2

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE53		PAGE ページ
X0102.7	RSVS4		B22	KCM3	/C109.D4
X0102.6	RSVS3		A22		/C109.D3
X0102.5	ADCLS	FRONT AUTO DOOR CLOSE	B21		/C109.D3
X0102.4	ADOPN	FRONT AUTO DOOR OPEN	A21		/C109.D2
X0102.3	SPSP	SPINDLE STOP	B20	A24	/C108.E5
X0102.2	SPOR	SPINDLE ORIENTION	A20		/C108.E5
X0102.1	CPWRON	CONTROL POWER ON	B19		/C108.E4
X0102.0	POFFS	POWER OFF	A19		/C108.E4

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE53		PAGE ページ
X0101.7	FPS16	FLEXIBLE SWITCH 16	B22	KCM2	/C104.C7
X0101.6	FPS15	FLEXIBLE SWITCH 15	A22		/C104.C7
X0101.5	FPS14	FLEXIBLE SWITCH 14	B21		/C104.C6
X0101.4	FPS13	FLEXIBLE SWITCH 13	A21		/C104.C6
X0101.3	FPS12	FLEXIBLE SWITCH 12	B20	B23	/C104.C6
X0101.2	FPS11	FLEXIBLE SWITCH 11	A20		/C104.C5
X0101.1	FPS10	FLEXIBLE SWITCH 10	B19		/C104.C5
X0101.0	FPS9	FLEXIBLE SWITCH 09	A19		/C104.C5

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE53		PAGE ページ
X0103.7			B22	KCM4	/C110.C4
X0103.6			A22		/C110.C4
X0103.5			B21		/C110.C4
X0103.4			A21		/C110.C3
X0103.3			B20	B24	/C110.C3
X0103.2			A20		/C110.C3
X0103.1			B19		/C110.C2
X0103.0			A19		/C110.C2

REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

MAKINO
MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
DI/DO ADDRESS TABLE - IO MODULE 11	V2631I-C1E-Y	Ver.	B
		PAGE	D017

I/O OPERATOR PANELI 48/32	IOM 11

IOM 11- CE54

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE54		PAGE ページ
X0097.7	STS	CYCLE START	B6	+24V	/C105.E4
X0097.6	SPS	FEED HOLD	A6	CE54 -B2	/C105.E4
X0097.5	JOG_P	JOG(+)	B5		/C105.E2
X0097.4	JOG_M	JOG(-)	A5		/C105.E2
X0097.3	SPST	SPINDLE START	B4		
X0097.2	HWER	HANDLE MODE ENABLE	A4		
X0097.1	MP2S	HANDLE PULSE RATE×100	B3		
X0097.0	MP1S	HANDLE PULSE RATE×10	A3		

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE54		PAGE ページ
X0104.7			B22	KCM5	
X0104.6			A22		
X0104.5	ROVM	RAPID TRAVERSE OVERRIDE SWITCH (INHIBIT)	B21		/C106.C3
X0104.4			A21		/C106.C3
X0104.3	ROV8I	RAPID TRAVERSE OVERRIDE SWITCH (8)	B20	A23	/C106.C3
X0104.2	ROV4I	RAPID TRAVERSE OVERRIDE SWITCH (4)	A20		/C106.C2
X0104.1	ROV2I	RAPID TRAVERSE OVERRIDE SWITCH (2)	B19		/C106.C2
X0104.0	ROV1I	RAPID TRAVERSE OVERRIDE SWITCH (1)	A19		/C106.C2

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE54		PAGE ページ
X0106.7	SGULS	OPERATOR DOOR LOCK RELEASE	B22	KCM7	
X0106.6	STUCS	SPINDLE TOOL UNCLAMP	A22		
X0106.5	SOVM	SPINDLE OVERRIDE SWITCH (INHIBIT)	B21		/C107.D5
X0106.4	SOV16I	SPINDLE OVERRIDE SWITCH (16)	A21		/C107.D5
X0106.3	SOV8I	SPINDLE OVERRIDE SWITCH (8)	B20	A24	/C107.D4
X0106.2	SOV4I	SPINDLE OVERRIDE SWITCH (4)	A20		/C107.D4
X0106.1	SOV2I	SPINDLE OVERRIDE SWITCH (2)	B19		/C107.D4
X0106.0	SOV1I	SPINDLE OVERRIDE SWITCH (1)	A19		/C107.D3

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE54		PAGE ページ
X0105.7			B22	KCM6	
X0105.6			A22		
X0105.5	FVM	FEEDRATE OVERRIDE SWITCH (INHIBIT)	B21		/C106.C7
X0105.4	FV16I	FEEDRATE OVERRIDE SWITCH (16)	A21		/C106.C7
X0105.3	FV8I	FEEDRATE OVERRIDE SWITCH (8)	B20	B23	/C106.C6
X0105.2	FV4I	FEEDRATE OVERRIDE SWITCH (4)	A20		/C106.C6
X0105.1	FV2I	FEEDRATE OVERRIDE SWITCH (2)	B19		/C106.C6
X0105.0	FV1I	FEEDRATE OVERRIDE SWITCH (1)	A19		/C106.C5

REVISION				
	----.--.--	APPROVED	AVINASH	
	----.--.--	CHECKED	AVINASH	
	----.--.--	DRAWN	ANISHA	
	----.--.--	DATE	28/6/2018	

MAKINO
MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
DI/DO ADDRESS TABLE - IO MODULE 11	V2631I-C1E-Y	Ver.	B
		PAGE	D018

I/O OPERATOR PANELI 48/32	IOM 11

IOM 11- CE53

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE53		PAGE ページ
Y0096.7	FPL8	FLEXIBLE SWITCH 08	B10	C24HA	/C104.A4
Y0096.6	FPL7	FLEXIBLE SWITCH 07	A10		/C104.A4
Y0096.5	FPL6	FLEXIBLE SWITCH 06	B9	A25 B25	/C104.A3
Y0096.4	FPL5	FLEXIBLE SWITCH 05	A9		/C104.A3
Y0096.3	FPL4	FLEXIBLE SWITCH 04	B8		/C104.A3
Y0096.2	FPL3	FLEXIBLE SWITCH 03	A8		/C104.A2
Y0096.1	FPL2	FLEXIBLE SWITCH 02	B7		/C104.A2
Y0096.0	FPL1	FLEXIBLE SWITCH 01	A7		/C104.A2

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE53		PAGE ページ
Y0098.7	JOGP	JOG(+)	B18	C24HA	/C105.A2
Y0098.6	JOGN	JOG(-)	A18		/C105.A2
Y0098.5	BZR	BUZZER	B17	A25 B25	/C105.A3
Y0098.4	PWONL	CONTROL POWER ON	A17		/C108.A4
Y0098.3	SPSTL	SPINDLE START	B16		/C108.A7
Y0098.2	SPSPL	SPINDLE STOP	A16		/C108.A6
Y0098.1	SPORL	SPINDLE START	B15		/C108.A5
Y0098.0	POFFL	POWER OFF	A15		/C108.A4

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE54		PAGE ページ
Y0100.7			B14	C24HA	
Y0100.6			A14		
Y0100.5			B13	B24 A25 B25	
Y0100.4			A13		
Y0100.3			B12		/C109.A4
Y0100.2			A12		/C109.A3
Y0100.1	LADCLS	LAMP AUTO DOOR CLOSE	B11		/C109.A3
Y0100.0	LADOPN	LAMP AUTO DOOR OPEN	A11		/C109.A2

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE53		PAGE ページ
Y0097.7	FPL16	FLEXIBLE SWITCH 16	B14	C24HA	/C104.A7
Y0097.6	FPL15	FLEXIBLE SWITCH 15	A14		/C104.A7
Y0097.5	FPL14	FLEXIBLE SWITCH 14	B13	A25 B25	/C104.A6
Y0097.4	FPL13	FLEXIBLE SWITCH 13	A13		/C104.A6
Y0097.3	FPL12	FLEXIBLE SWITCH 12	B12		/C104.A6
Y0097.2	FPL11	FLEXIBLE SWITCH 11	A12		/C104.A5
Y0097.1	FPL10	FLEXIBLE SWITCH 10	B11		/C104.A5
Y0097.0	FPL9	FLEXIBLE SWITCH 09	A11		/C104.A5

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE54		PAGE ページ
Y0099.7	STLO	CYCLE START	B10	C24HA	/C105.A4
Y0099.6	SPLO	FEED HOLD	A10		/C105.A4
Y0099.5	SGAV	OPERATOR DOOR LOCK RELEASE	B9	B24 A25 B25	/C105.A7
Y0099.4	SGULL	OPERATOR DOOR UNLOCK	A9		/C105.A7
Y0099.3	STUCL	SPINDLE TOOL UNCLAMP	B8		/C105.A6
Y0099.2	SOVHL	SPINDLE OVERRIDE LAMP	A8		/C106.A3
Y0099.1	FOVHL	CUTTING FEED OVERRIDE LAMP	B7		/C106.A5
Y0099.0	ROVHL	RAPID FEED OVERRIDE LAMP	A7		/C106.A2

	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE54		PAGE ページ
Y101.7	bWCOK	PMC RUNNING	B18	C24HA	/C110.A4
Y101.6	MLT	MC LIGHT IN SPLASH GUARD	A18		/C110.A4
Y101.5			B17	B24 A25 B25	/C110.A4
Y101.4	PBUZ	SIGNAL LIGHT BUZZER	A17		/C110.A3
Y101.3	PTL4	SIGNAL LIGHT 4	B16		/C110.A3
Y101.2	PTL3	SIGNAL LIGHT 3	A16		/C110.A3
Y101.1	PTL2	SIGNAL LIGHT 2	B15		/C110.A2
Y101.0	PTL1	SIGNAL LIGHT 1	A15		/C110.A2

REVISION			
	----.--.--	APPROVED	AVINASH
	----.--.--	CHECKED	AVINASH
	----.--.--	DRAWN	ANISHA
	----.--.--	DATE	28/6/2018

MAKINO
MAKINO ASIA PTE LTD

TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
DI/DO ADDRESS TABLE - IO MODULE 11	V2631I-C1E-Y	Ver.	B
		PAGE	D019

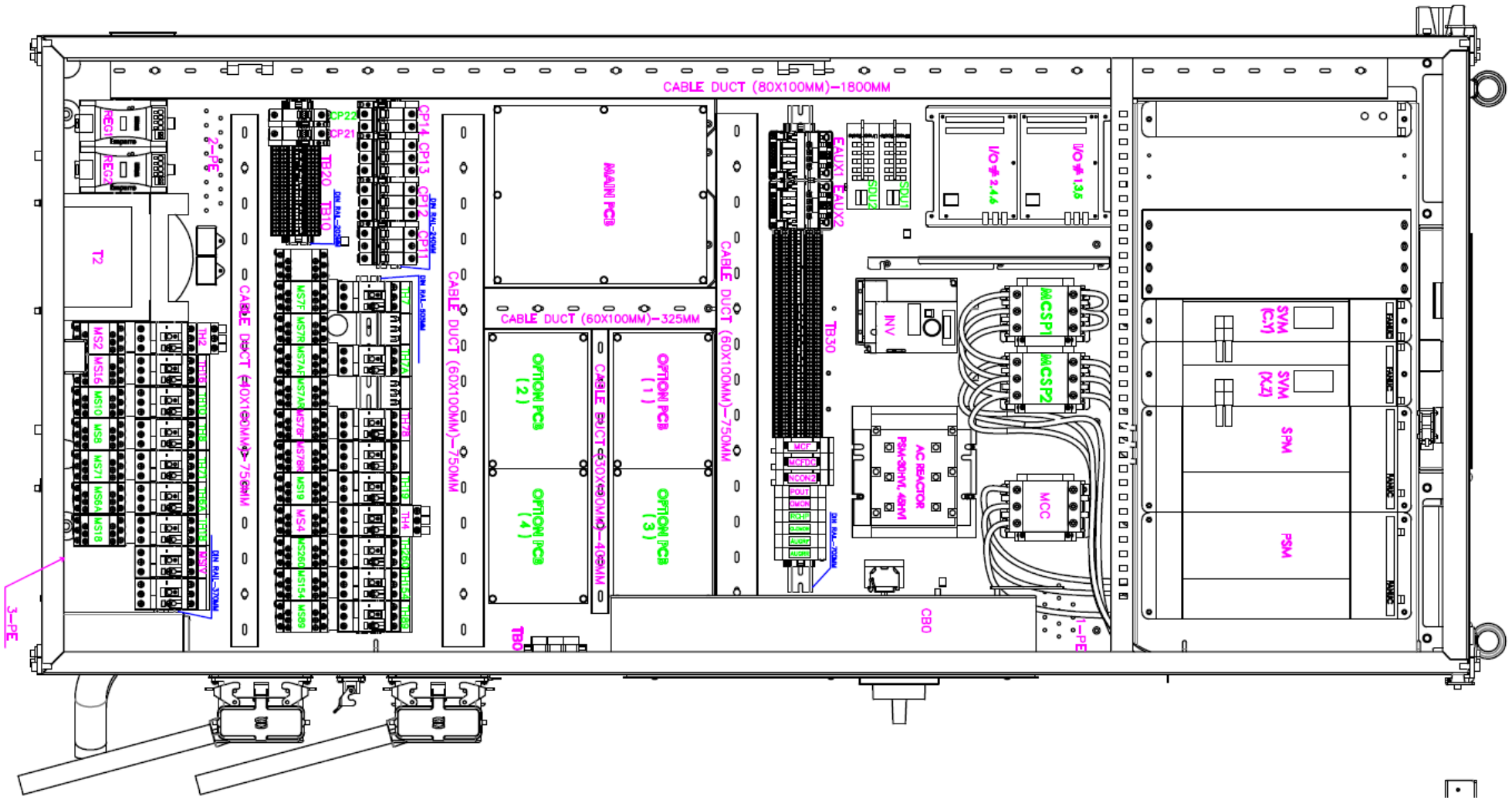
I/O OPERATOR PANELI 48/32	IOM 11

IOM 11- CE54

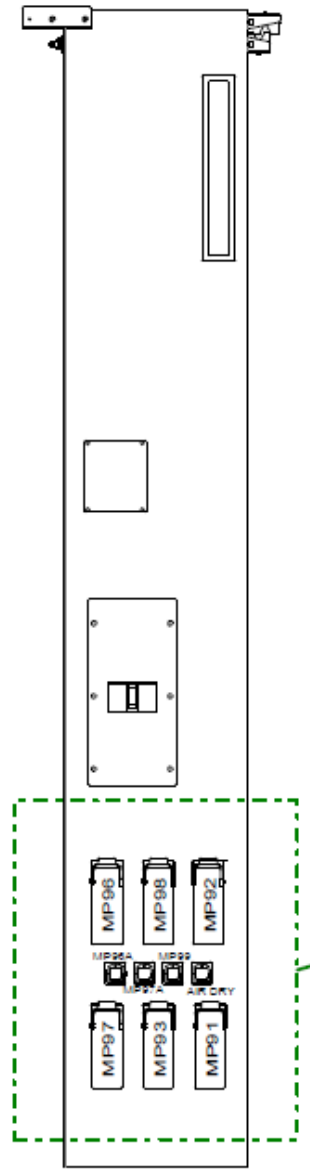
	SYMBOL 記号	SIGNAL NAME 信号名称	CONNECTOR No. コネクタ No. CE54		PAGE ページ
Y0102.7			B22	C24HA	
Y0102.6			A22		
Y0102.5			B21	B24	
Y0102.4			A21	A25	
Y0102.3			B20	B25	
Y0102.2			A20		
Y0102.1			B19		
Y0102.0			A19		

REVISION						
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TITLE	DRAWING No.
	----.--.--	CHECKED	AVINASH		DI/DO ADDRESS TABLE - IO MODULE 11	V2631I-C1E-Y
	----.--.--	DRAWN	ANISHA			
	----.--.--	DATE	28/6/2018			

MACHINE	F3/ F5 Pro 6
Ver.	B
PAGE	D020



REVISION					MAKINO MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH		MTC LAYOUT DETAILS	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	E001
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	6/28/2018					



OUTSIDE (RIGHT WALL) MTC VIEW

CONNECTOR No.	OPTION NAME	REMARKS
MP91	COOLANT POWER	24 WAY
MP92	COOLANT SIGNAL	46 WAY
MP93	LUCC	24 WAY
MP94	ATC PANEL	
MP95	FLUID OPTIONS	MTC INTERNAL CONNECTOR
MP96	FIXTURE SOL/PS	24 WAY
MP96A	FIXTURE PP1	4 WAY
MP97	FIXTURE SOL/PS	24 WAY
MP97A	FIXTURE PP2	4 WAY
MP98	ROBOT INTERFACE	72 WAY
MP99	MIST COLLECTOR	4 WAY

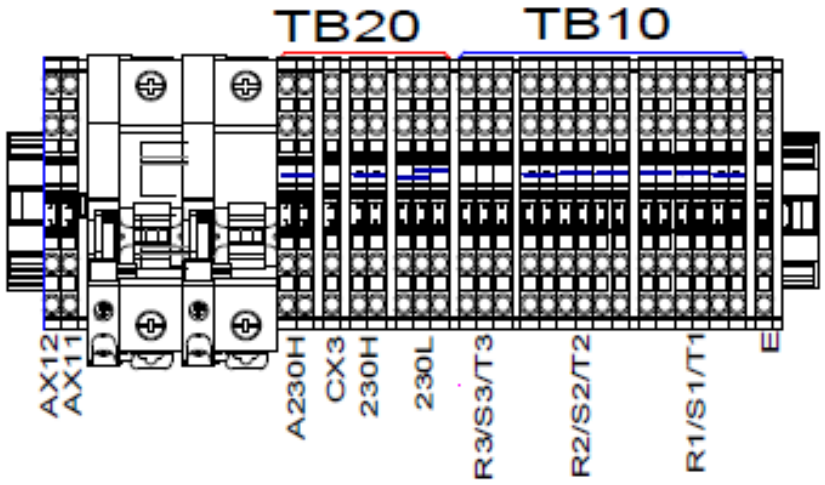
REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH			MTC LAYOUT DETAILS	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH					PAGE	E002
	----.--.--	DRAWN	ANISHA						
	----.--.--	DATE	28/6/2018						

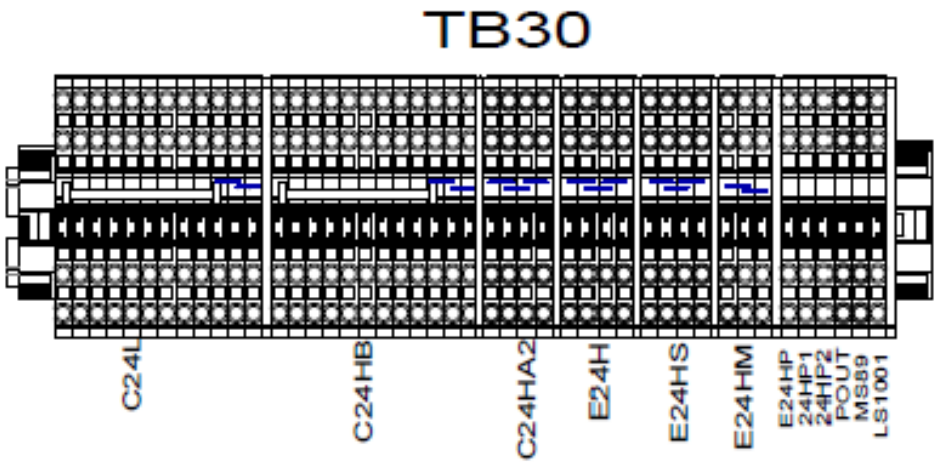
DESIGN DESIGNATION	PART DESCRIPTION	MANUFACTURER	MFR PART NO.	MAKINO PART NO.
PSM	αiPS 45HV-B	FANUC	A06B-6252-H030	Z513A0302510
SPM	SPM ALPHA SP 30HVi,30i-B SERIES (20K)	FANUC	A06B-6272-H030#H610	Z513A0302350
	SPM αiSP 22HV-B (30K)	FANUC	A06B-6272-H022#610	Z513A0302680
SVM X/Y/Z	SVM3 ALPHA SV 40/40/40HVI,30I-B SERIES	FANUC	A06B-6290-H328	Z513A0302360
SVM ATC	SVM ALPHA SV 10HVI,30I-B SERIES	FANUC	A06B-6290-H122	Z513A0302370
SVM 4/5	SVM ALPHAi SV10HV	FANUC	A06B-6290-H102	Z513A0301360
	SVM1 ALPHA I-20HVi, 30i-B SERIES	FANUC	A06B-6290-H103	Z513A0301370
	SERVO AMP AI SV 40HV	FANUC	A06B-6290-H104	A06B-6290-H104
	SVM ALPHAi SV80HV	FANUC	A06B-6290-H105	Z513A0301390
ACR	AC REACTOR FOR ALPHA I PS 30/45HVI 30IB	FANUC	A81L-0001-0191	A81L-0001-0191
I/O	I/O MODULE 48/32	FANUC	A03B-0824-K203	A03B-0824-K203
SDU	SDU UNIT FOR FEEDBACK SCALE	FANUC	A02B-0323-C205	A02B-0323-C205
LINE FILTER	LINE FILTER 100A	SOSHIN	NF3100C-YXZ	Z721A0010180
MCC	CONTACTOR	SCHNEIDER	LC1D95P7	L16A100
	SUPPRESSOR BLOCK	SCHNEIDER	LA4DA2U	L16A102
	CONTACT BLOCK	SCHNEIDER	LAD8N11	L16A103
MCSP1,MCSP2	CONTACTOR 80A	SCHNEIDER	LC1D80P7	L16A101
	SUPRESSOR BLOCK	SCHNEIDER	LA4DA2U	L16A102
	CONTACT BLOCK	SCHNEIDER	LAD8N11	L16A103
TB10,TB20,TB30	TB Grey 2.5mm 4W	WAGO	2002-1401	Z697I0140003
	TB Y/G 2.5mm 4W	WAGO	2002-1407	Z697I0140016
	TB Jumper, 10W	WAGO	2002-410	Z697I0140023
	TB Jumper, 2W	WAGO	2002-402	Z697I0130004
	TERM END PLATE, 4 CONN	WAGO	2002~1492	L15C232
	TERM END STOPPER, 6MM	WAGO	249~116	L15C218
RELAY PCB	CX12 Ver.B MAIN RELAY PCB	ST CIRCUITS	E21VS690A110-XX	E21VS690A110-XX
OPTION PCB	CX12 OPTION PCB	ST CIRCUITS	E21VS690A101-XX	E21VS690A101-XX
TB0	M.I.E TERMINAL BLOCK 1P 90A	SCL	IN60BK-C	L15C252
CB0	MCCB TERM COVER	FUJI	BW9BTCA-L3	L14A060
	MCCB 100A	FUJI	BW125JAG-3P100FK 200V	Z643A2160186
	MCCB 100A 200V(FOR MM ONLY)	FUJI	BW125JAGU-3P100FK 200V	Z643A2160187
CB0 ROTARY	MCCB HANDLE	FUJI	BE9N0CA	L14A143
RCHP,POUT,OMON, CLOMON,NCON2	RELAY	PANASONIC	AHN22324	Z643L1300201
	RELAY SOCKET	PANASONIC	AHNA21	Z643L1310201
REG1	EMPARRO PSUPPLY 3PH	MURR	85692	L35A022
REG2	EMPARRO PSUPPLY 3PH	MURR	85692	L35A022

REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH			MTC COMPONENT DETAILS	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH					PAGE	E003
	----.--.--	DRAWN	ANISHA						
	----.--.--	DATE	28/6/2018						

DESIGN DESIGNATION	PART DESCRIPTION	MANUFACTURER	MFR PART NO.	MAKINO PART NO.
	CB,IC60N 3P 10A C	SCHNEIDER	A9F74310	L14A112
CP11,CP13	AUX, SWITCH IC60N	SCHNEIDER	A9A26924	L14A097
	CP,3POLE,25A,C,IC60H	SCHNEIDER	A9F84325	Z643Y8020003
CP12	AUX, SWITCH IC60N	SCHNEIDER	A9A26924	L14A097
	CP ACTI9 D60H ,2P,6A	SCHNEIDER	A9F85206	L14A164
CP14	AUX, SWITCH IC60N	SCHNEIDER	A9A26924	L14A097
	CB,IC60N 1P 6A C	SCHNEIDER	A9F74106	L14A098
CP21	AUX, SWITCH IC60N	SCHNEIDER	A9A26924	L14A097
	CB,IC60N 1P 6A C	SCHNEIDER	A9F74106	L14A098
CP22	AUX, SWITCH IC60N	SCHNEIDER	A9A26924	L14A097
	CONT.3 POLE 24VDC	SCHNEIDER	LC1D09BD	L16A107
	COMBINATION BLOCK	SCHNEIDER	GV2-AF3	Z643Y2130020
	MOTOR STARTER THERMAL CONTACTOR 2.5A~4A	SCHNEIDER	GV2~ME08	L16A134
MS2/TH2	AUX.CONT SW,1NO1NC	SCHNEIDER	GV AE11	L16A092
	CONT.3 POLE 24VDC	SCHNEIDER	LC1D09BD	L16A107
	COMBINATION BLOCK	SCHNEIDER	GV2-AF3	Z643Y2130020
	MOTOR STARTER THERMAL CONTACTOR 1A~1.6A	SCHNEIDER	GV2~ME06	L16A133
MS4/TH4	AUX.CONT SW,1NO1NC	SCHNEIDER	GV AE11	L16A092
MS7F/MS7R/TH7, MS7AF/MS7AR/TH7A	MOTOR STARTER THERMAL CONTACTOR 0.63~1A	SCHNEIDER	GV2DM205BD	L16A144
	CONT.3 POLE 24VDC	SCHNEIDER	LC1D09BD	L16A107
MS6A/TH6A,MS10/TH10,MS16/TH16, MS71/TH71,MS89/TH89	MOTOR STARTER THERMAL CONTACTOR 2.5A~4A	SCHNEIDER	GV2~ME08	L16A134
	MOTOR STARTER THERMAL CONTACTOR 4~6.3A	SCHNEIDER	GV2DM110BD	L16A130
MS8/TH8,MS260/TH260	AUX.CONT SW,1NO1NC	SCHNEIDER	GV AE11	L16A092
	MOTOR STARTER THERMAL CONTACTOR 0.16~0.25A	SCHNEIDER	GV2DM102BD	L16A140
MS18/TH18	AUX.CONT SW,1NO1NC	SCHNEIDER	GV AE11	L16A092
	MOTOR STARTER THERMAL CONTACTOR 1.6~2.5A	SCHNEIDER	GV2DM107BD	L16A119
MS19/TH19,MS154/TH154	AUX.CONT SW,1NO1NC	SCHNEIDER	GV AE11	L16A092
SURGE P.L~L	SURGE P.L~L	OKAYA	RAV-152BYZ-2A	L13C014
SURGE P.L~G	SURGE P.L~G	OKAYA	RAV-801BXZ-4	L13C015
T2	TRANSFORMER 2KVA(CE,UL)380/400/415V-200/230V	SHIN NIPPON	L29A201	L29A201
SW296	KEY SWITCH	DECA	ADA16K6-AA6-DF-03	L11B198
LS1001	DOOR LIMIT SWITCH	AZBIL	SL1-PK	Z621A0180110
FAN	FAN 230VAC 150D	SUNON	A2175-HBT.T.TGN+FG-17B	L19A022
AIR CON	AIR CON	RITTAL	SK3303500	L19A020

REVISION							
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE / F3/ F5 Pro 6
	----.--.--	CHECKED	AVINASH		MTC COMPONENT DETAILS	V2631I-C1E-Y	Ver. / B
	----.--.--	DRAWN	ANISHA				PAGE / E004
	----.--.--	DATE	6/28/2018				





REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH		TB10/TB20/TB30 LAYOUT	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	E005
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					

MP81			
A1	*LS1319	B1	OM2-RE1
A2	*OMAL2	B2	OM2-RE2
A3	*LS1343	B3	
A4	*LS184	B4	
A5	LS3320	B5	LS193
A6	LS181	B6	LS1338
A7	LS191	B7	LS1339
A8	LS192	B8	SPARE1
A9	LS1325	B9	SPARE2
A10	LS1337	B10	SLATCF

MP82			
A1	SL750	B1	*OMAL3
A2	SPARE3	B2	OM3-RE1
A3	OM1-RE1	B3	OM3-RE2
A4	OM1-RE2	B4	*LS2340
A5	LS170	B5	
A6	LS173	B6	
A7	*OMAL1	B7	TSCCG1
A8	C24HB	B8	TSCCG2
A9	C24L	B9	
A10		B10	

MP113			
A1	LS942-11	B1	LS942-22
A2	LS942-31	B2	LS942-42
A3	LS942-E(1)	B3	LS942-E(2)
A4	LS747	B4	C24HB
A5	LS748	B5	C24HB

MP94			
A1	C24HB	B1	ATCMA-(4)
A2	ATCUNC-(4)	B2	ATCMA-(+)
A3	ATCUNC-(+)	B3	C24L
A4	MGCW-(4)	B4	MGCCW-4
A5	MGCCW-(+)	B5	MGCW-(+)
A6	LS747	B6	C24HB
A7	LS748	B7	C24HB
A8	EMG-11	B8	EMG-12
A9	EMG-21	B9	EMG-22
A10	EMG-31	B10	EMG-32

REVISION							
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE: F3/ F5 Pro 6
	----.--.--	CHECKED	AVINASH		CONNECTOR PIN DETAILS	V2631I-C1E-Y	Ver.: B
	----.--.--	DRAWN	ANISHA				PAGE: E006
	----.--.--	DATE	28/6/2018				

MP91			
1	U2	13	U10
2	V2	14	V10
3	W2	15	W10
4	U16	16	U8
5	V16	17	V8
6	W16	18	W8
7	U6A	19	U18
8	V6A	20	V18
9	W6A	21	W18
10	U71	22	R2
11	V71	23	S2
12	W71	24	T2

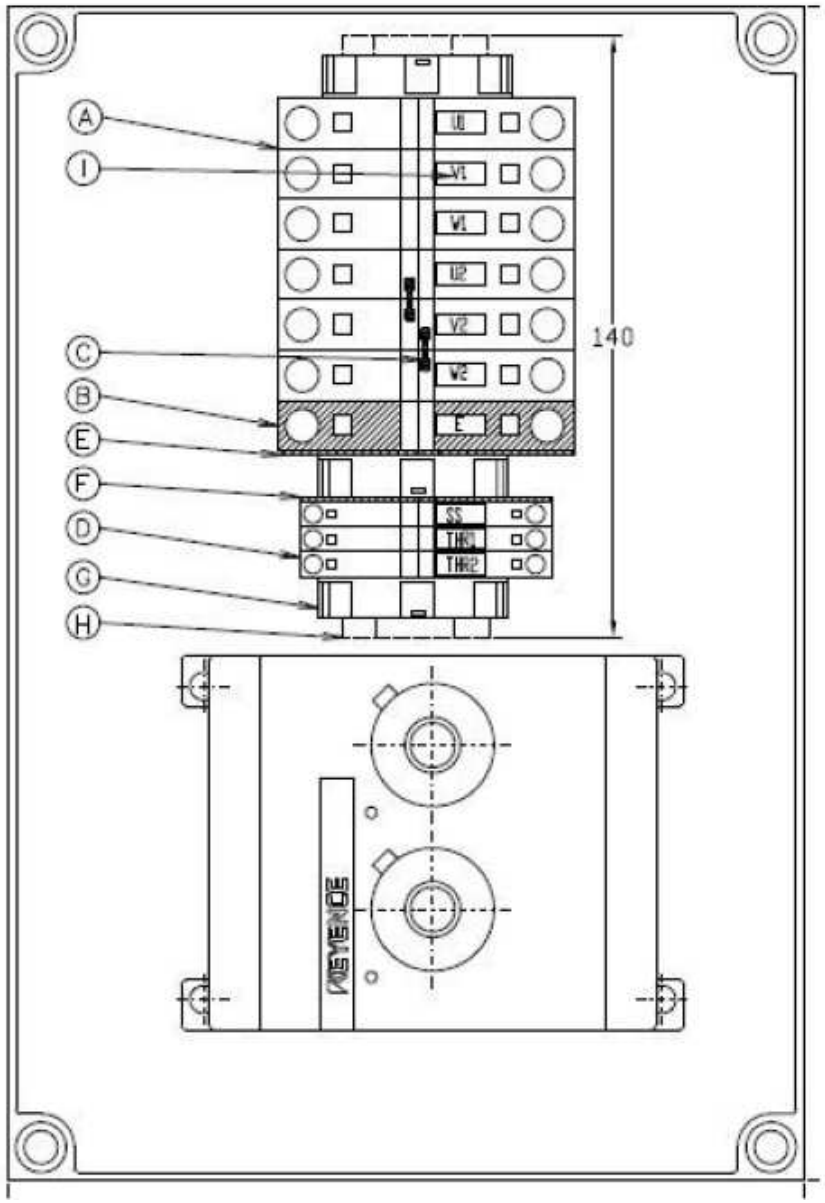
MP92							
1	*LS1319	14	TSCCG2	24	C24L	34	OM3-RE2
2	*OMAL2	15	LS193	25	OM1-RE1	35	*LS2340
3	*LS1343	16	LS1338	26	OM1-RE2	36	
4	*LS184	17	LS1339	27	LS170	37	
5	LS3320	18	SPARE1	28	LS173	38	
6	LS181	19	SPARE2	29	*OMAL1	39	
7	LS191	20	SLATCF	30	C24HB	40	
8	LS192	21	SL750	31	C24L	41	
9	LS1325	22	SPARE3	32	*OMAL3	42	
10	LS1337	23	C24HB	33	OM3-RE1	43	
11	OM2-RE1					44	
12	OM2-RE2					45	
13	TSCCG1					46	

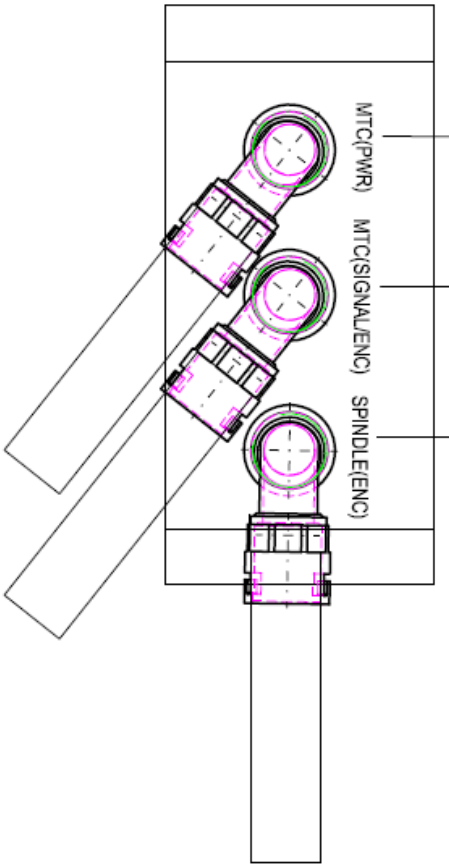
MP93			
1	TH6-L1	13	
2	TH6-L2	14	
3	TH6-L3	15	
4	TH6-14	16	C24L
5	M6R-22	17	MAS-(+)
6	M6F-22	18	EMG31
7	TH6-22	19	MAS-(4)
8	TH6-21	20	FWD-(4)
9	EMG-12	21	REV-(4)
10	EMG-11	22	C24HB
11	EMG-22	23	
12	EMG-21	24	

MP96			
1	C24L	13	SL2603-1
2	SYSPS2-3	14	SL2603-2
3	LS2603-1	15	SL2604-1
4	LS2603-2	16	SL2604-2
5	LS2604-1	17	PS2601-3
6	LS2604-2	18	PS260-1
7	SYSPS-1	19	PS2602-3
8	SYSPS-3	20	PS2602-1
9	SL2601-1	21	PS2603-3
10	SL2601-2	22	PS2603-1
11	SL2602-1	23	PS2604-3
12	SL2602-2	24	PS2604-1

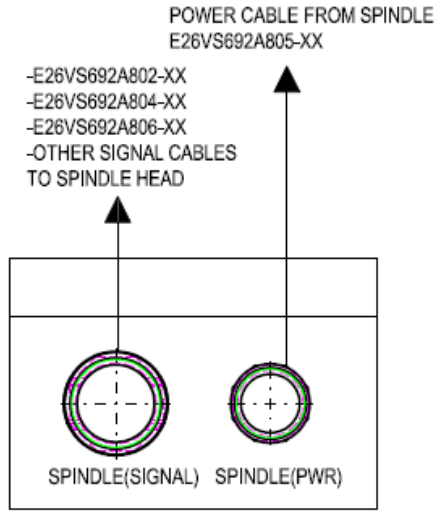
MP98											
1	ROB-EMG1	13		25	RBHLY	37	WPN02	49	MCRDY	61	PCP01
2	ROB-EMG2	14		26	*RBALM	38	WPN04	50	*MCALM	62	PCP02
3	ROB-EMG3	15		27	*RBINT	39	WPN08	51	*MCWRN	63	WCN01
4	ROB-EMG4	16		28	SYSTR	40	WPNPRT	52	ACSOK	64	WCN02
5	EM1-1	17	ROB-EMG5	29	SYSTP	41	RBI17	53	WSRDY	65	C24HB
6	EM1-2	18	ROB-EMG6	30	WLDFN	42	RBI18	54	MCRUN	66	C24HB
7	EM2-1	19		31	WULDFN	43	RBI19	55	WLDRQ	67	C24L
8	EM2-2	20		32	NOWPST	44	RBI20	56	WULDRQ	68	C24L
9		21		33	FLSON	45	RBI21	57	WSCOK	69	DOCOM
10		22		34	FCLRQ	46	RBI22	58	WSCNG	70	DOCOM
11		23		35	FUCRQ	47	RBI23	59	FCLFN	71	DOCOM
12		24		36	WPN01	48	RBI24	60	FUCFN	72	DOCOM

REVISION						
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	CHECKED	AVINASH		CONNECTION PIN DETAILS	V2631I-C1E-Y	Ver.	B
	----.--.--	DRAWN	ANISHA				PAGE	E007
	----.--.--	DATE	28/6/2018					



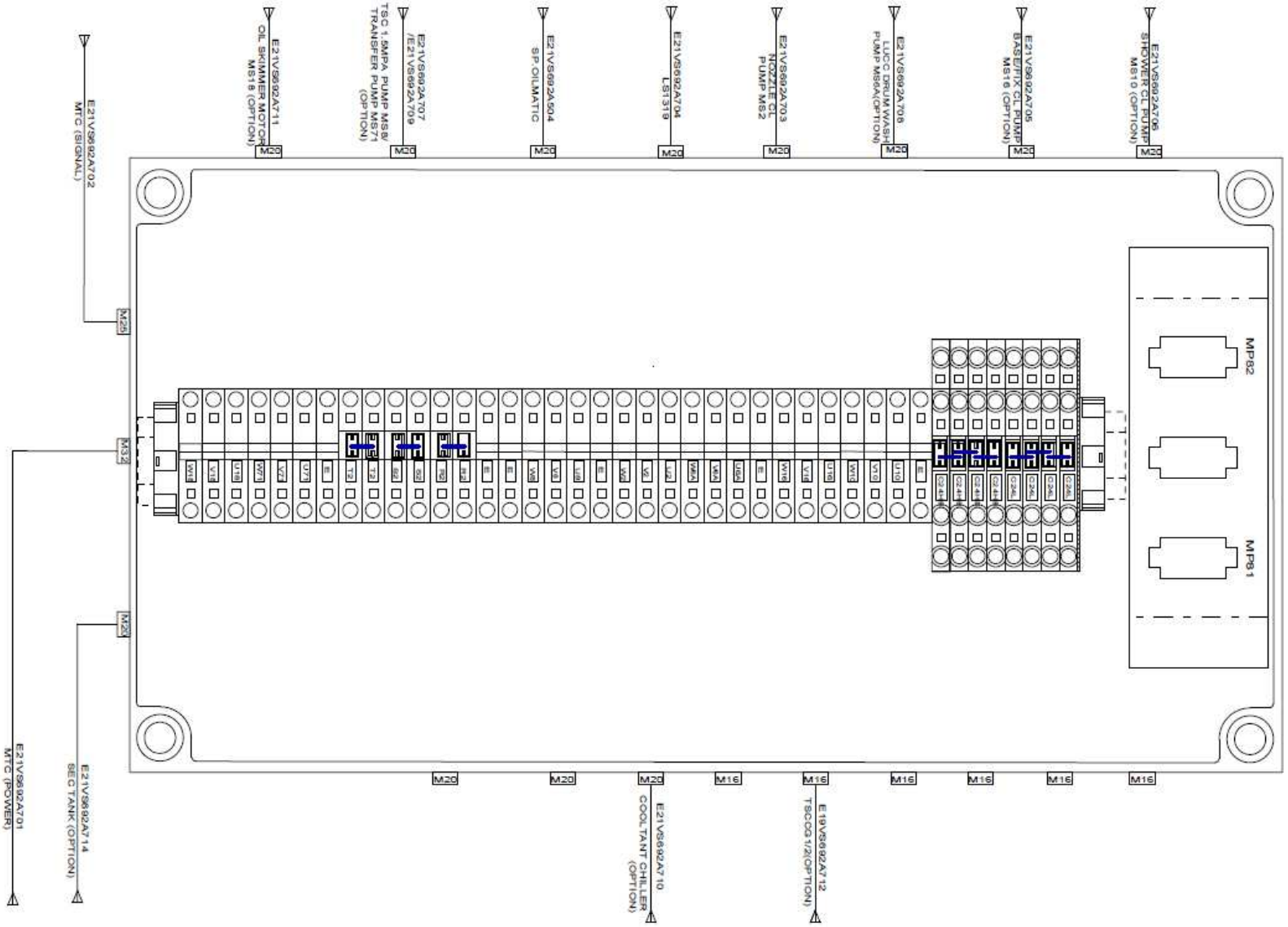


LEFT SIDE VIEW



BOTTOM SIDE VIEW

REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH		TB40 SPINDLE TERMINAL BOX	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	E008
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TB50 COOLANT TERMINAL BOX	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	E009
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					

PIN DETAILS	COLOUR CODE	CONNECTION	SYMBOL	SIGNAL NAME
V+	BROWN	TB30-C24HB	C24HB	
V-	BLUE	TB30-C24L	C24L	
PE	GREEN/YELLOW	2-PE	PE	
4-1	WHITE	1YB2-A2	SL905	Master Solenoid Valve
2-1	GREY/PINK	1YB2-B1	LS905	Air Pressure Switch
4-2	GREEN	XA2-B1	SL181	Air Blow
2-2	RED/BLUE			
4-3	YELLOW	XA2-B2	SL09	Taper Cleaning Air 1
2-3	WHITE/GREEN	XA2-B4	SL18	Spindle Air Seal
4-4	GREY	1YB8-A2		
2-4	BROWN/GREEN			
4-5	PINK	1YA6-A3	SL711A	ATC Shutter Close
2-5	WHITE/YELLOW	1YA6-B3	SL711B	ATC Shutter Open
4-6	RED	XA2-B5		
2-6	YELLOW/BROWN			
4-7	BLACK	1YA3-A2	LS1698	Auto Grease Injector Pressure UP
2-7	WHITE/GREY	1YA3-B2	LS2064	Auto Grease Supply Unit Level Switch
4-8	VIOLET	1YA3-B3	ALSXYF	Auto Grease Injector CW
2-8	GREY/BROWN			

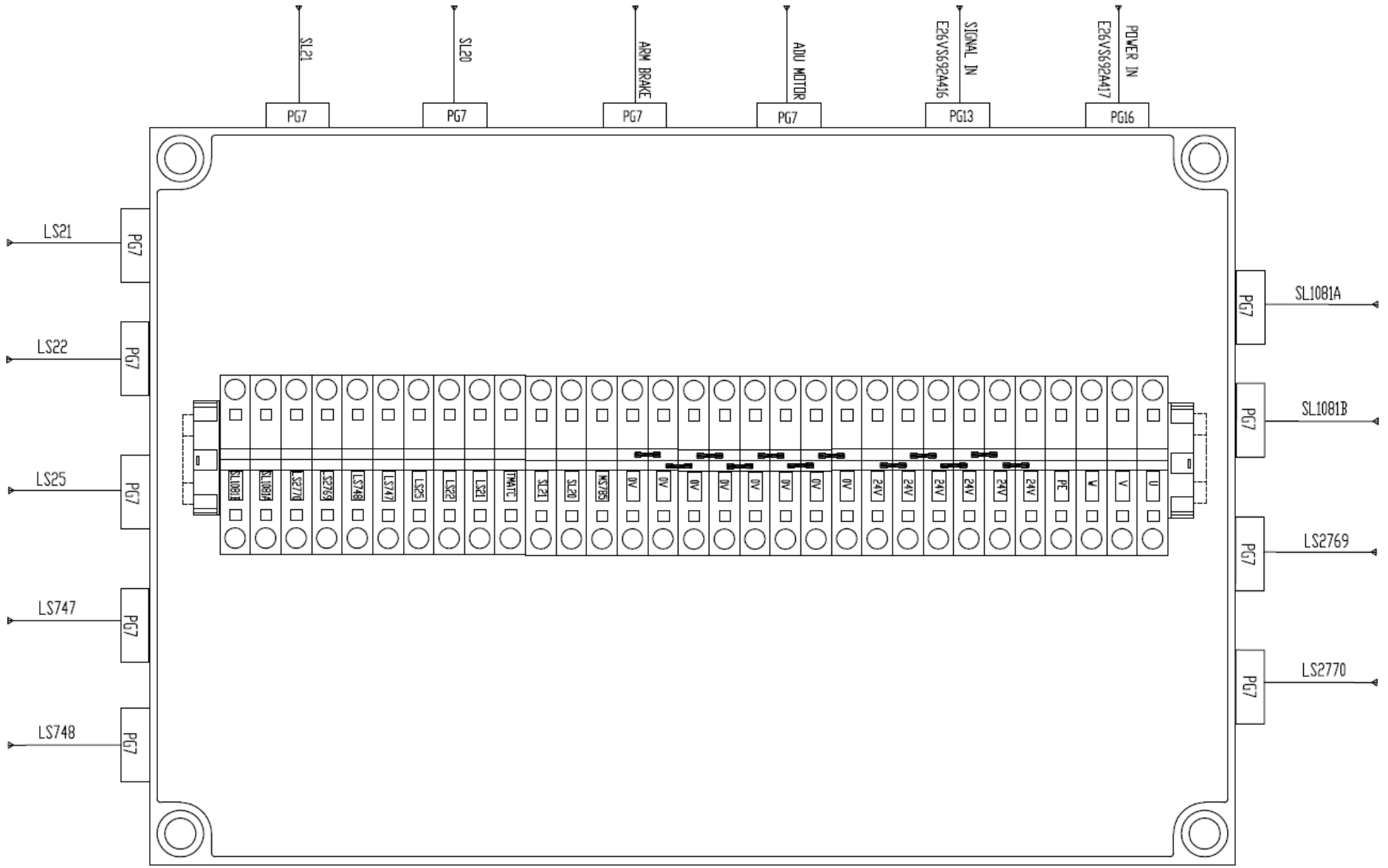
M12 JB No.1			
2	SL181	SL905	1A
		LS905	1B
4		SL09	3A
		SL18	3B
6		SL711A	5A
		SL711B	5B
8	ALSXYF	LS1698	7A
		LS2064	7B
E26VS692A508-XX			

REVISION							
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE F3/ F5 Pro 6
	----.--.--	CHECKED	AVINASH		M12 JUNCTION BOX 1	V2631I-C1E-Y	Ver. B
	----.--.--	DRAWN	ANISHA				PAGE E010
	----.--.--	DATE	28/6/2018				

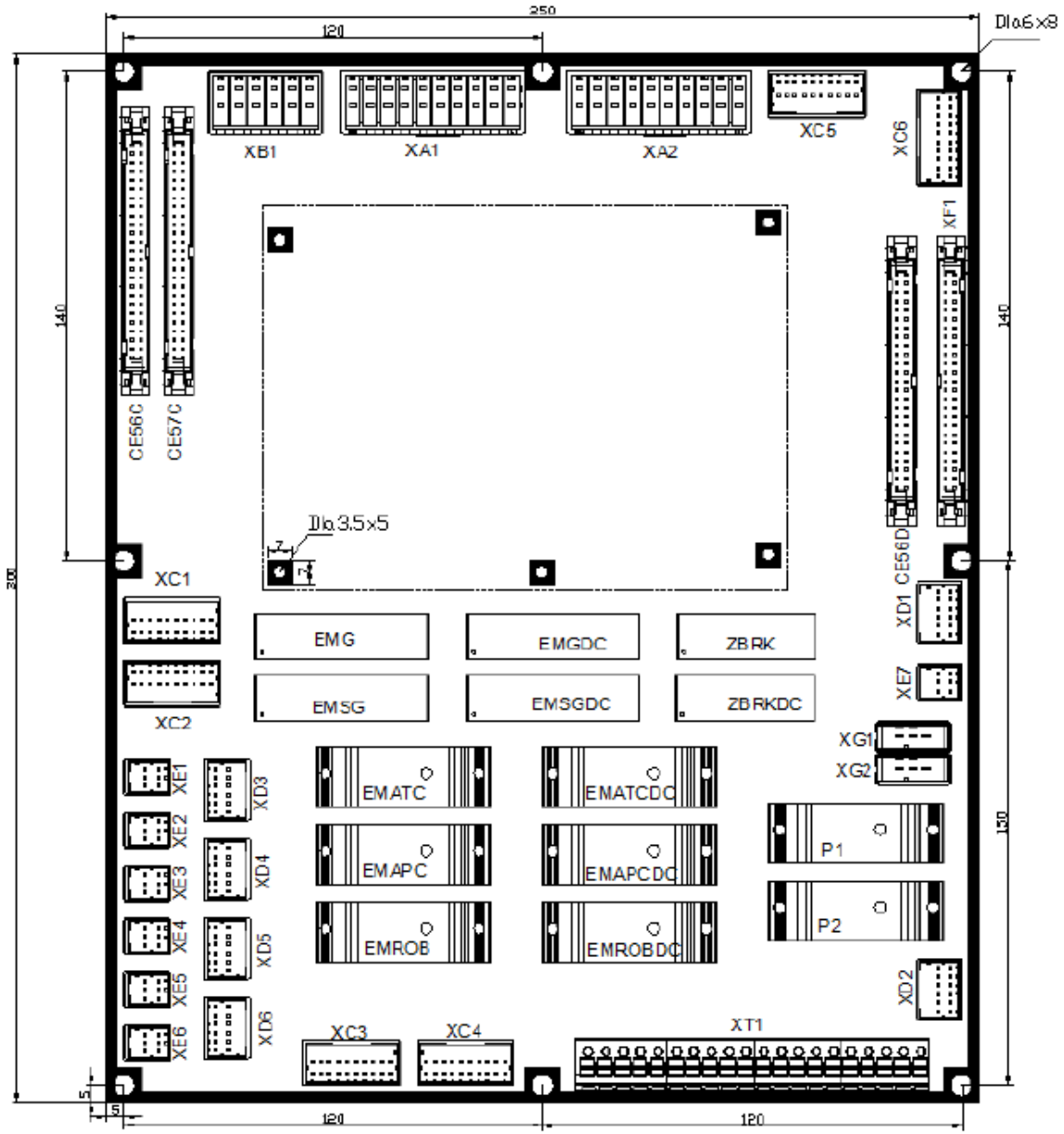
PIN DETAILS	COLOUR CODE	CONNECTION	SYMBOL	SIGNAL NAME
V+	BROWN	TB30-C24HB	C24HB	
V-	BLUE	TB30-C24L	C24L	
PE	GREEN/YELLOW	2-PE	PE	
4-1	WHITE	2YA3-A3	SL1308	TSC Union Backup Air
2-1	GREY/PINK	2YA4-B3	SL1310	TSC Suction Solenoid
4-2	GREEN	2YA7-B3	SL130A	Right Side Door Close
2-2	RED/BLUE			
4-3	YELLOW	2YA3-B3		
2-3	WHITE/GREEN	XA2-B3	SL192	Through Spindle Air
4-4	GREY	2YA7-A3	SL130B	Right Side Door Open
2-4	BROWN/GREEN			
4-5	PINK	TB100-SL302	SL302	Tool Cleaning Air (XC3-B5)
2-5	WHITE/YELLOW	TB100-SL304	SL304	Laser Shutter Open (XC3-B6)
4-6	RED	TB100-SL303	SL303	Purge Air (XC3-A6)
2-6	YELLOW/BROWN			
4-7	BLACK	TB30-E24HM	SL22	Master Solenoid-Pot UP/DWN
2-7	WHITE/GREY	TB30-E24HM	SL8022	Master Solenoid-Shutter OPN/CLS
4-8	VIOLET			
2-8	GREY/BROWN			

M12 JB No.2			
2	SL130A	SL1308	1A
		SL1310	1B
4	SL130B	SL1309	3A
		SL192	3B
6	SL303	SL302	5A
		SL304	5B
8		SL22	7A
		SL8022	7B
E26VS692A509-XX			

REVISION							
	----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE F3/ F5 Pro 6
	----.--.--	CHECKED	AVINASH		M12 JUNCTION BOX 2	V2631I-C1E-Y	Ver. B
	----.--.--	DRAWN	ANISHA				PAGE E011
	----.--.--	DATE	28/6/2018				



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TB70 ATC TERMINAL BOX	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	E012
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	6/28/2018					

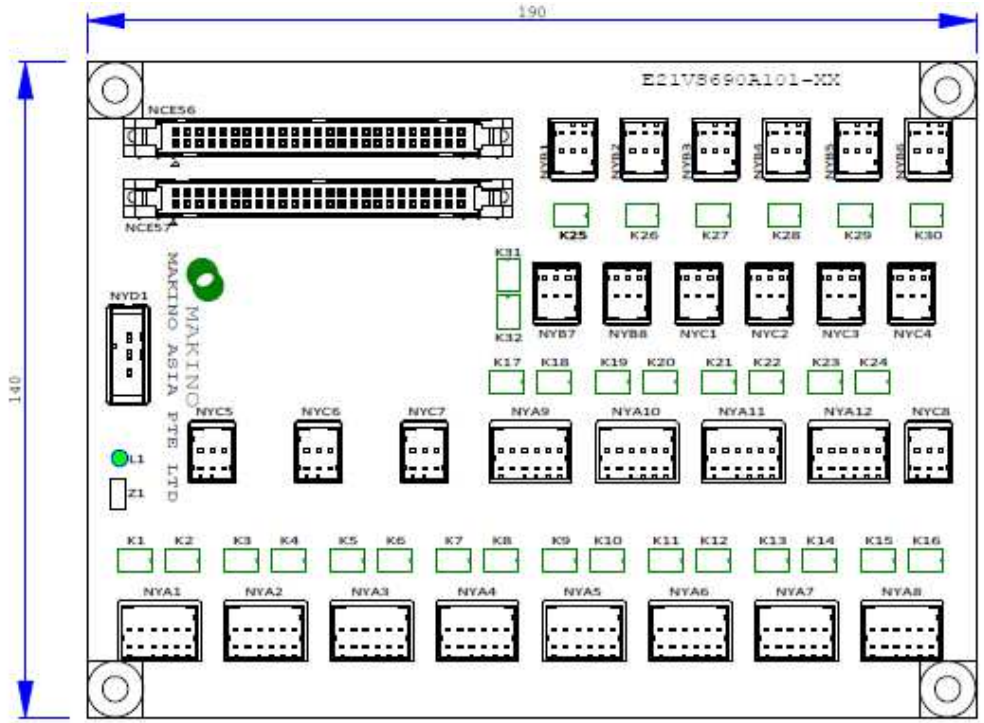


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	RELAY PCB LAYOUT	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	E013
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

MAKINO
MAKINO ASIA PTE LTD

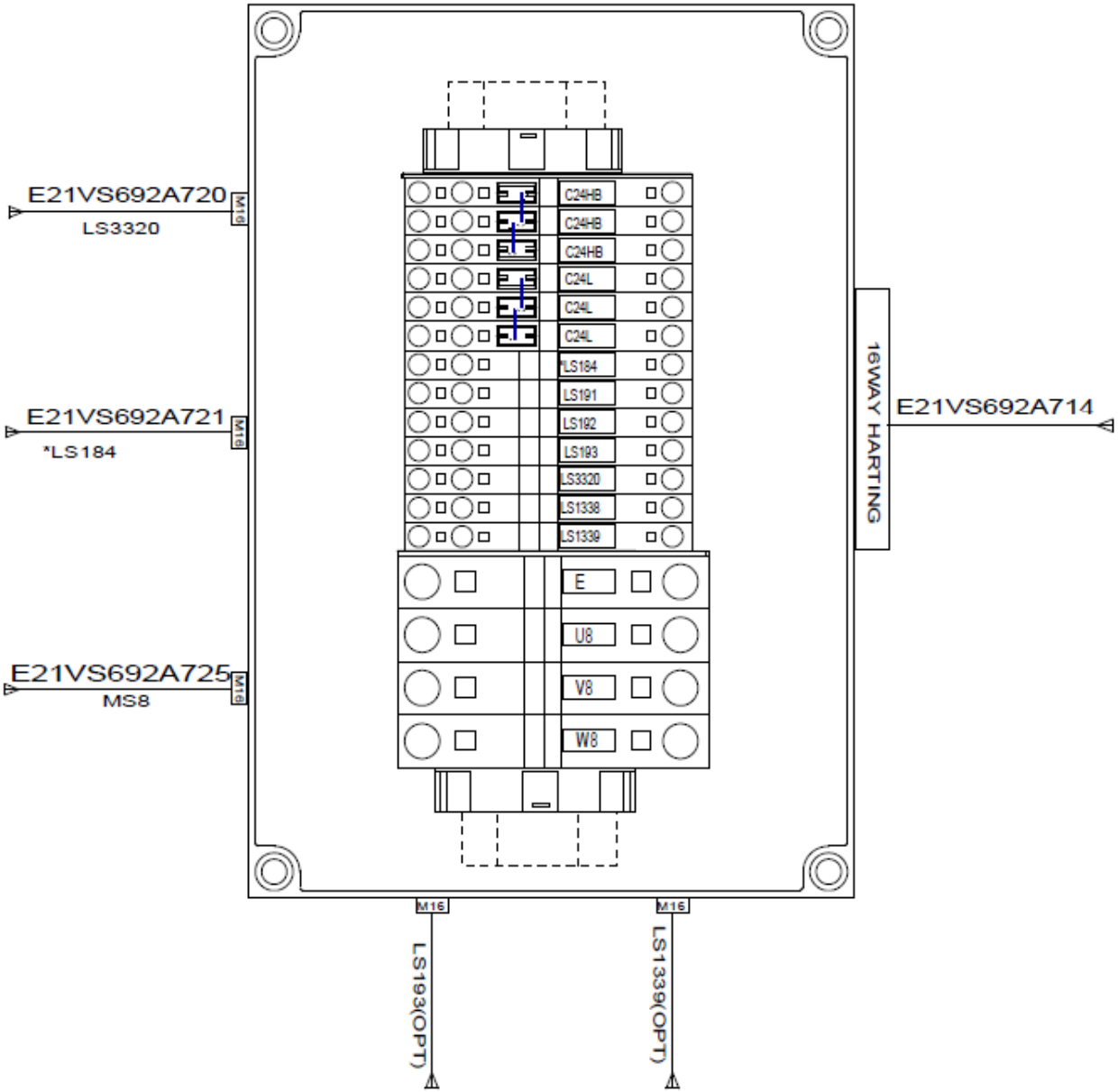
S.No	Description	Specification	Designation	Qty
1	PCB	300X250		1
2	3WAY D3100 SERIES	1-178313-5	XG1 , XG2	2
3	6 WAY X KEY D2100 SERIES	1318124-1	XE1 , XE2 , XE3 , XE4 , XE5 , XE6 , XE7	7
4	12 WAY D2100 SERIES	1318126-1	XD1 , XD2 , XD3 , XD4 , XD5 , XD6	6
5	20 WAY D2100 SERIES	1318127-1	XC1 , XC3 , XC5	3
6	20 WAY D2100 SERIES	1318127-2	XC2 , XC4 , XC6	3
7	12 WAY D3100 SERIES	316516-5	XB1	1
8	20WAY D3100 SERIES	316517-5	XA1 , XA2	2
9	50WAY IDC	LD50-H	CE56C , CE56D , CE57C , XF1	4
10	Photomos Relay	AQZ102 (1 NO)		24
11	SURGE ABSORBER	ERZV10D820	VAR	1
12	SPARK KILLER	XE0471	RC	1
13	SAFETY RELAY 3A+1B	SFS3-L-DC24V-D (Safety Relay)	ZBRK , ZBRKDC	2
14	RELAY SOCKET	SFS4-PS (Safety Relay Base)	EMAPC , EMAPCDC , EMATC , EMATCDC ,EMROB , EMROBDC , P1 , P2	8
15	SAFETY RELAY , SAFETY RELAY 5A+1B	SFS5-L-DC24V-D (Safety Relay)	EMG , EMGDC , EMSG , EMSGDC	4
16	Jumper	2 WAY JUMPER	J1-J14	14
17	Resistor	3K Ohms, 1/4W, +/- 5% TOL ,THRU HOLE	R1-R33	33
18	TVS Diodes - Transient Voltage Suppressors 1500W 33V Unidirect	1.5KE33A (Zener Diode)	Z1	1
19	PCB Terminal Block with Switch, 5 Pole	804-105	XT1	4

REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH			RELAY PCB PARTS LIST	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH					PAGE	E014
	----.--.--	DRAWN	ANISHA						
	----.--.--	DATE	28/6/2018						

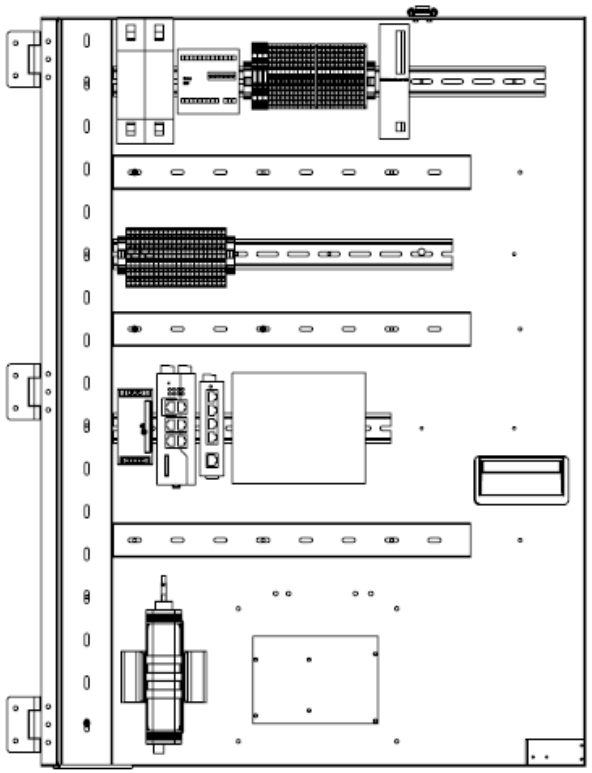


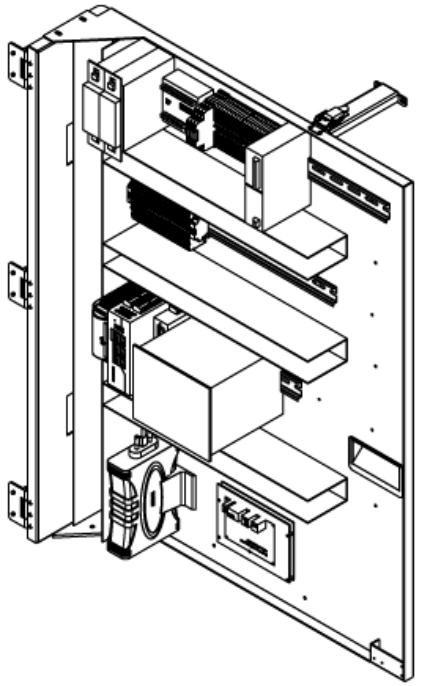
S.NO	Description	Specification	Designation	Qty
1	PCB	190 x 140mm		1
2	LED	DC 2.1V,Green 3mm	L1	1
3	Panasonic Photomos	AQV252G	K1 - K32	32
4	Resistor	3KOhms , 1/4W, +/-5%	R1 - R33	33
5	50 WAY FLAT CABLE CONNECTOR	HIF3BB-50PA-2.54DSA (HIROSE) LD50 (AUK)	NCE56, NCE57	2
6	12 WAY D2100 Series	1318126-1	NYA1 - NYA2	12
7	6 WAY X-KEY D2100 Series	1318124-1	NYB1 - NYB8	8
8	6 WAY Y-KEY D2100 Series	1318124-2	NYC1-NYC8	8
9	3 WAY D3100 Series	1-178313-5	NYD1	1
10	Zener Overvoltage Transient Suppressors	ST - 1.5KE33A	Z1	1

REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH		OPTION PCB LAYOUT & PARTS DETAILS	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	E015
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					

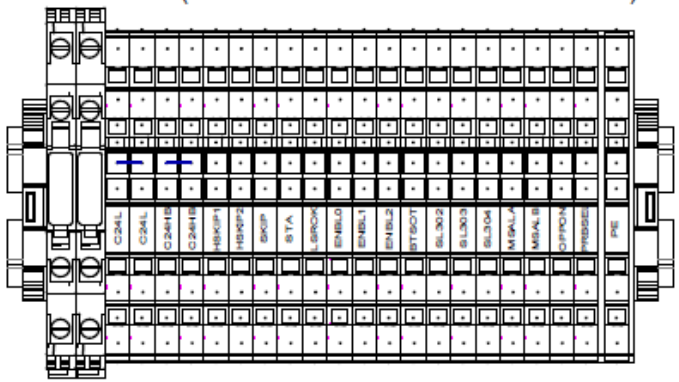


REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	TB90 SEC TANK TERMINAL BOX	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	E016
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					

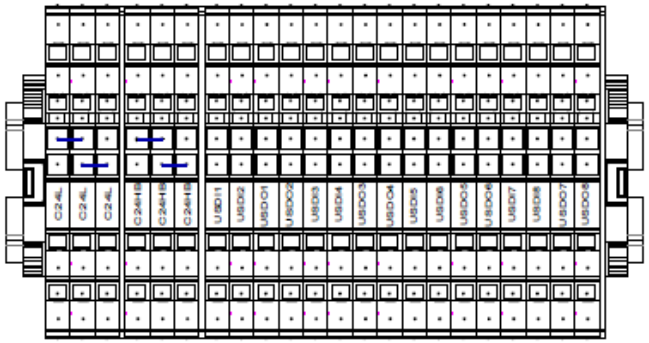




TB100 (MEASUREMENT SYSTEM INTERFACE)



TB110 (USER M-CODE INTERFACE)



REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH		SWING PANEL TB100/TB110 LAYOUT	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	E017
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					

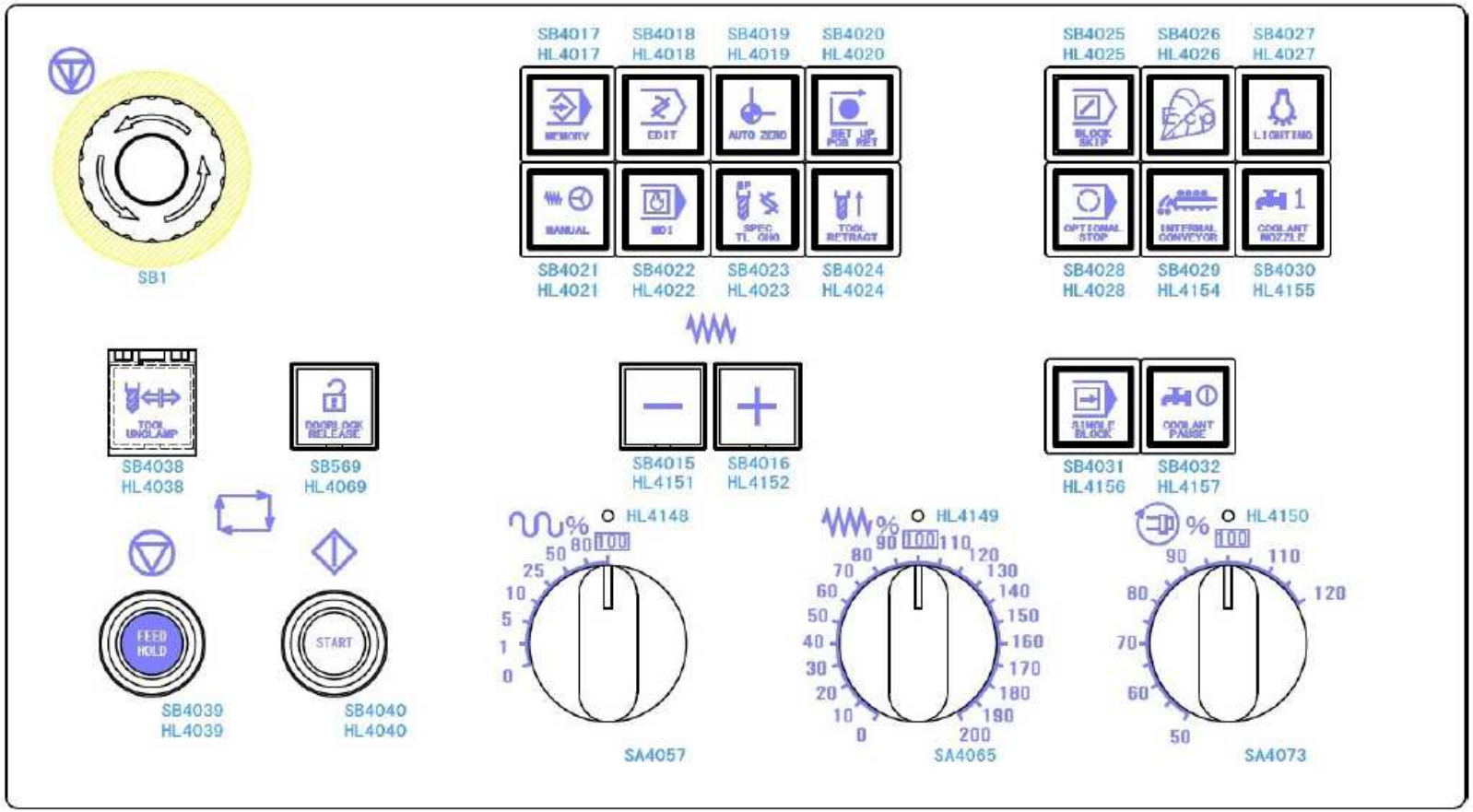




REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH		OPERATOR PANEL A LAYOUT	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	E018
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	28/6/2018					

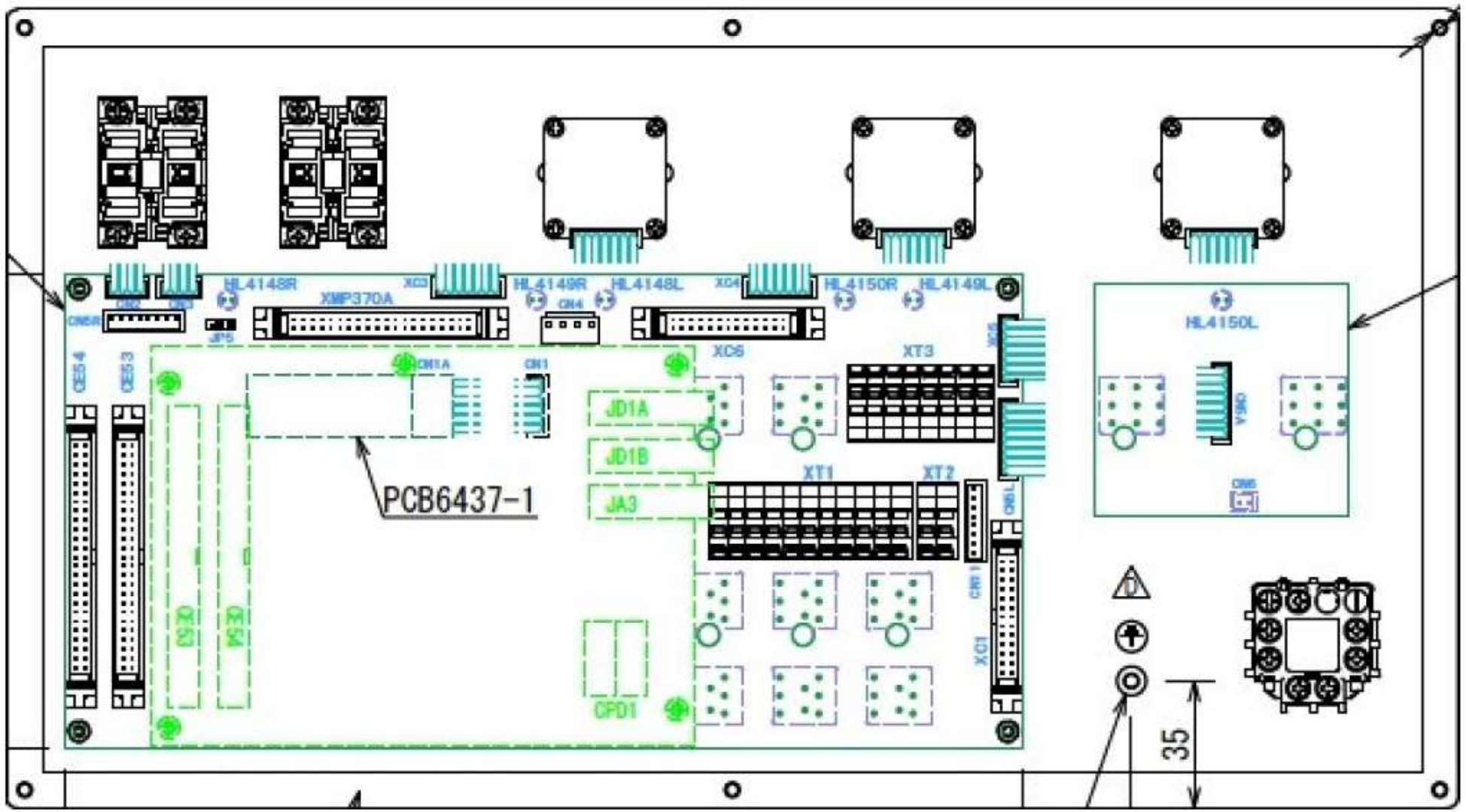
Sno.	Part No.	Part description	Type	Qty	Manufacturer
1	SB4036/HL4036 SB4174/HL4146 SB4041/HL4041 SB4042/HL4042	Illuminated Pushbutton(W)	LBW7L-M1T14VPW	4	IDEC
2	SB1093/HL1022	Illuminated Pushbutton(W)	LBW7L-M1T24VPW	1	IDEC
3	XC1	26-pin Connector	XG4A-2631	1	OMRON
4	CN7,7A	8-pin Connector	B8B-XH-A	2	JST
5	CN10	10-pin Connector (Option)	B10B-XH-A	1	JST
6		Metal Panel	SPCC 1.6t	1	texture-paint
7		Main PCB	PCB6442	1	
8		Sub-1 PCB	PCB6442-1	1	
9		Harness 9 Harness 7 (Flat Cable)	SS3990-0Y09 SS3990-0Y07	1	CN7-CN7A PCB6442 XC1-PCB6437 XC1
10		Earth Seal	*	1	
11		Gasket	SCF401	1	

REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH		OPERATOR PANEL A PARTS LIST	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	E019
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	6/28/2018					



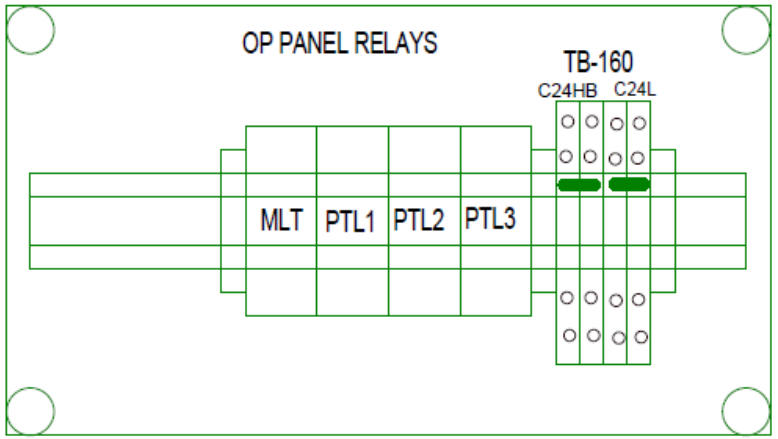
REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	OPERATOR PANEL FRONT B LAYOUT	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	E020
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	6/28/2018				

MAKINO
MAKINO ASIA PTE LTD



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	OPERATOR PANEL B REAR LAYOUT	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH			PAGE	E021
		----.--.--	DRAWN	ANISHA				
		----.--.--	DATE	28/6/2018				

MAKINO
MAKINO ASIA PTE LTD



REVISION					TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH	MAKINO ASIA PTE LTD	OP PANEL REALYS & TB-160	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH				PAGE	E022
		----.--.--	DRAWN	ANISHA					
		----.--.--	DATE	6/28/2018					

S.No	PART No.	PART DESCRIPTION	TYPE	QTY	MANUFACTURER
1	SB4017~4028 HL4017~4028 SB4029/HL4154 SB4030/HL4155 SB4031/HL41546	Illuminated push button(W)	LBW7L-M1T14VPW	16	IDEC
2	SB4015/HL4151				
3	SB4016/HL4152	Illuminated push button(W)	HA2L-M1T24VPW	2	IDEC
4	SB569/HL4069	Illuminated push button(W)	HA2L-M1T14VPW	1	IDEC
5		Illuminated push button(W)	HA2L-M1T14VPW	1	IDEC
6	SB4038/HL4038	Switch Guard	HA9Z-KW1	1	IDEC
7		LED(S)	SLR-343BC4T3F	3	IDEC
8		LED Mouting Spacer	LDN-510	3	IDEC
9	HL4145~4150	LED Lens	S33990-DOUKOU-LENS	3	IDEC
10	SB4039/HL4039 SB4040/HL4040	Full Shroud Illuminated Push button(W)	ALFW2B2211DPW-TK2955	2	IDEC
11	SB1	Emergency Stop Switch	XW1E-BV402MR	1	IDEC
12		Coad switch	DPP01-07J20RCB	1	TOKYO SOKUTEIKIZAI
13	SA4057	Knob	*	1	supplies
14		Coad switch	DPP01-020J20RCB	1	TOKYO SOKUTEIKIZAI
15	SA4065	Knob	*	1	supplies
16		Coad switch	DPP01-014J20RCB	1	TOKYO SOKUTEIKIZAI
17	SA4073	Knob	*	1	supplies
18	CE53,54	50-pin Connector	XG4A-5031	2	OMRON
19	XMP370A	40-pin Connector	XG4A-4031	1	OMRON
20	XC1~6	26-pin Connector	XG4A-2631	2	OMRON
21	XT1	22-pin Connector	ZFKKDS1.5C-5.0-11P-LR	1	PHOENIX CONTACT
22	XT2	16-pin Connector	ZFKKDS1.5C-5.0-8P-LR	1	PHOENIX CONTACT
23	XT3	14-pin Connector	ZFKKDS1.5C-5.0-2P-LR	1	PHOENIX CONTACT
24	HA1	Buzzer	PKB24SPCH3601-B0	1	MURATA
25	D1~54	Diode	1SS3E5 TE-17	54	ROHM
26	R1	Resistor for Buzzer	620Ω/1W	1	
27	R2~4	Resistor	9.1kΩ/1W	3	
28	C1	Capacitor	UUD1HO1OMCL1GS	1	NIPPON CHEMI-CON
29	JP5	Jumper Connector	XGBS-0331	1	OMRON

REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
	----.--.--	APPROVED	AVINASH			OPERATOR PANEL B PARTS LIST	V2631I-C1E-Y	Ver.	B
	----.--.--	CHECKED	AVINASH					PAGE	E023
	----.--.--	DRAWN	ANISHA						
	----.--.--	DATE	28/6/2018						

S.No	PART No.	PART DESCRIPTION	TYPE	QTY	MANUFACTURER
30	CN1	Jumper Socket	XJBA-211	1	JST
31	CN1A	6-pin Connector	B6B-XH-A	1	JST
32	CN4	6-pin Connector	S6B-XH-A	1	JST
33	CN2,3	4-pin Connector	B4P-VH	1	JST
34	XC3~5	4-pin Connector	B4B-XH-A	2	JST
	CN5A,CN11	7-pin Connector	B7B-XH-A	3	JST
35	CN5A,CN11	8-pin Connector	B8B-XH-A	4	JST
36		Metal Panel	SPCC16t	1	texture-paint
37		Main PCB	PCB6437	1	Ep-Fe/Zn5/Cn2
38		Sub-1 PCB	PCB6437-1	1	HIROSUGI-KEIKI
39		Sub-2 PCB	PCB6437-2	1	HIROSUGI-KEIKI
40		Mounting Plate	SPCC 1,2t	1	
41		Metal Spacer	ASF-32DE	4	
42		Metal Spacer	BSB-31DE	5	
		Harness 1	SS3990-0Y01	1	SB1-CN4
		Harness 2 and 12	SS3990-0Y02,0Y12	2	SB4039-CN2,SB4040-CN3
		Harness 3	SS3990-0Y03	3	SA4057-XC3,SA4065-XC4,SA4073-XC5
		Harness 4	SS3990-0Y04	1	CN5R-CN5A
		Harness 5(Flat Cable)	SS3990-0Y05	1	CE53-I/OCE53
		Harness6(Flat Cable)	SS3990-0Y06	1	CE54-I/OCE54
		Harness 8	SS3990-0Y08	1	CN1-CN1A
		Harness 10	SS3990-0Y010	1	XT2-CPD1
43		Harness 11	SS3990-0Y011	1	CN11-JA3
44		Marking Film		20	
45		Earth Seal	*	1	
46		Gasket	S0F401	1	

REVISION					MAKINO ASIA PTE LTD	TITLE	DRAWING No.	MACHINE	F3/ F5 Pro 6
		----.--.--	APPROVED	AVINASH		OPERATOR PANEL B PARTS LIST	V2631I-C1E-Y	Ver.	B
		----.--.--	CHECKED	AVINASH					
		----.--.--	DRAWN	ANISHA				PAGE	E024
		----.--.--	DATE	6/28/2018					